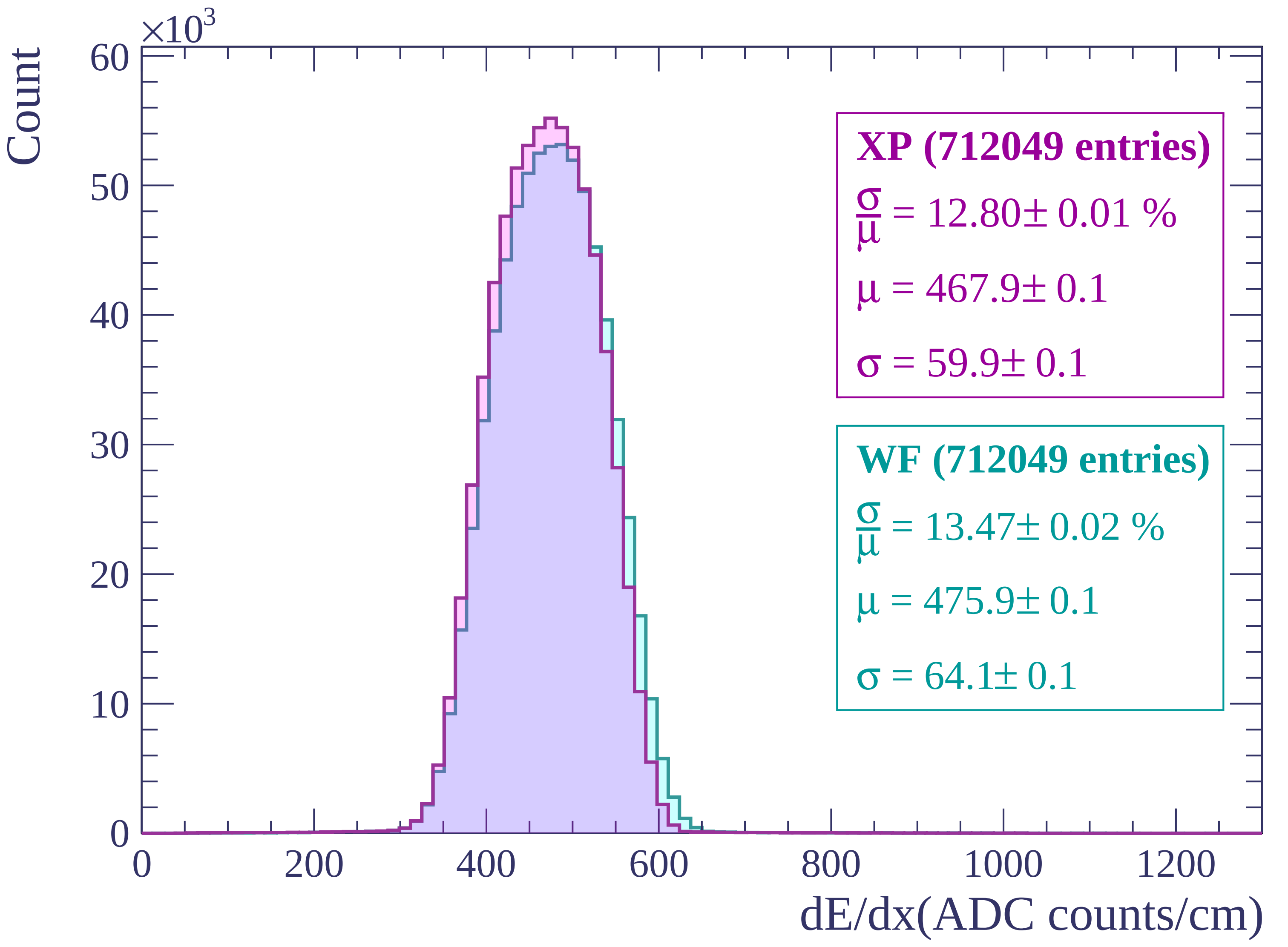
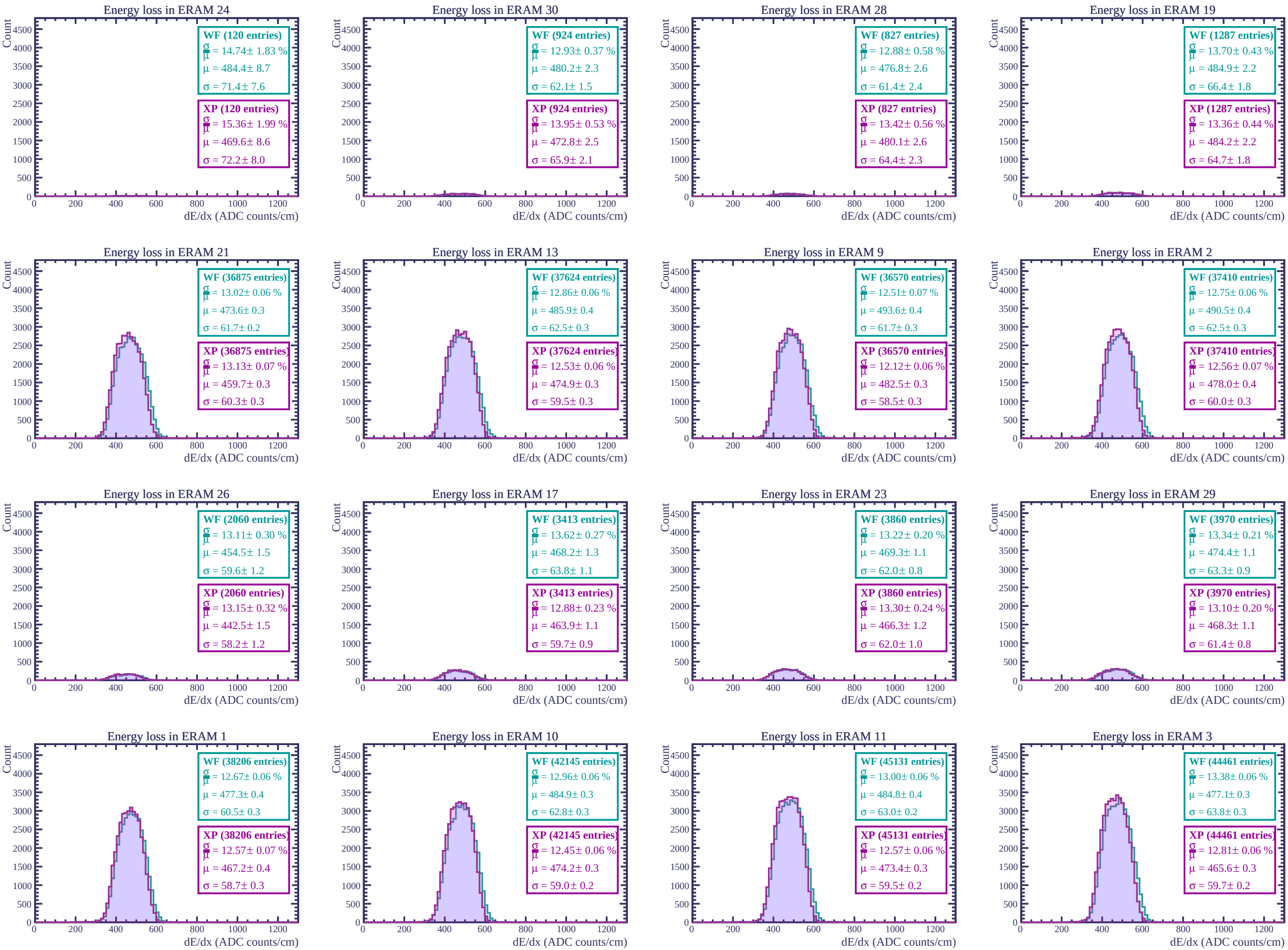
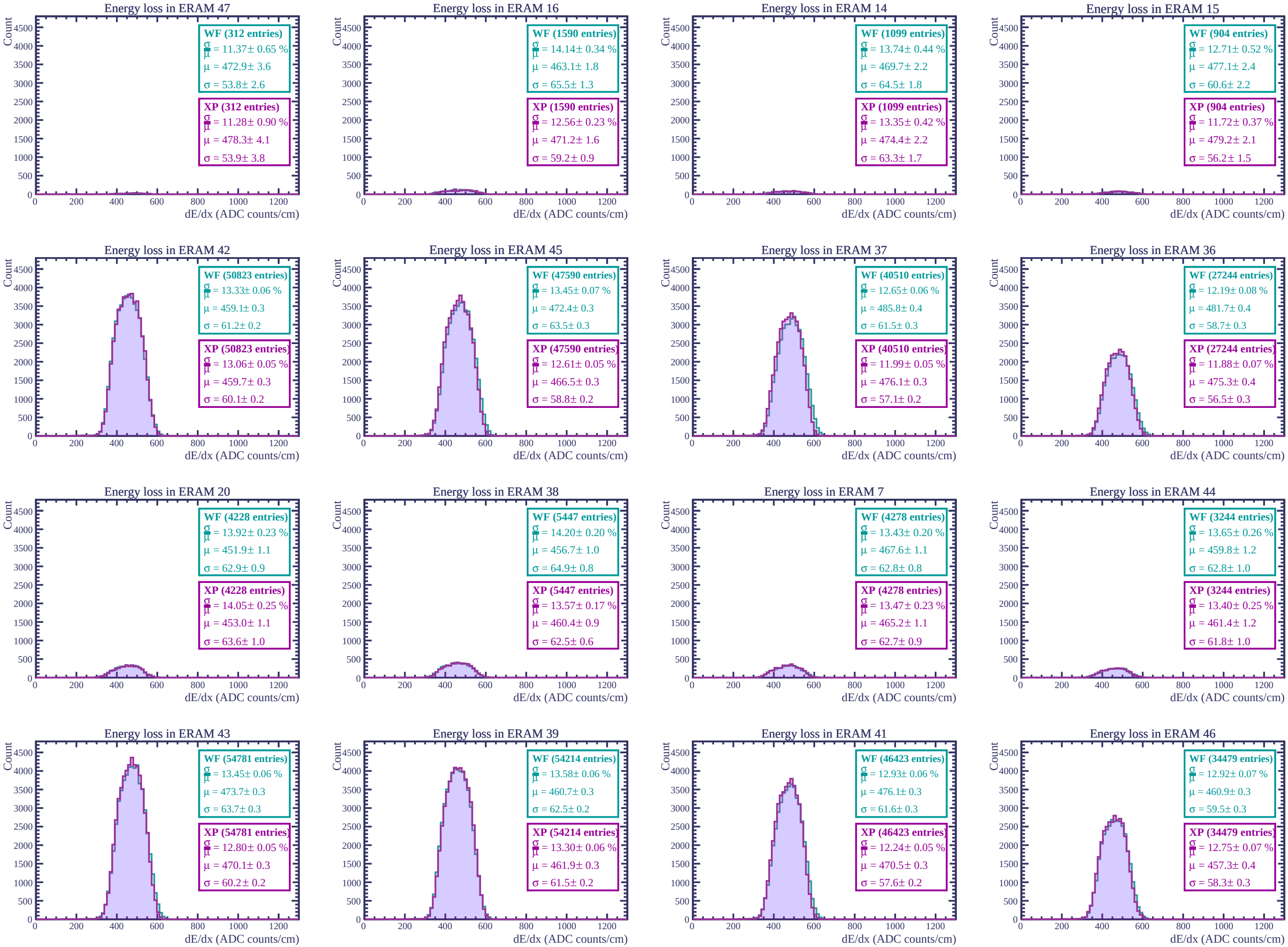


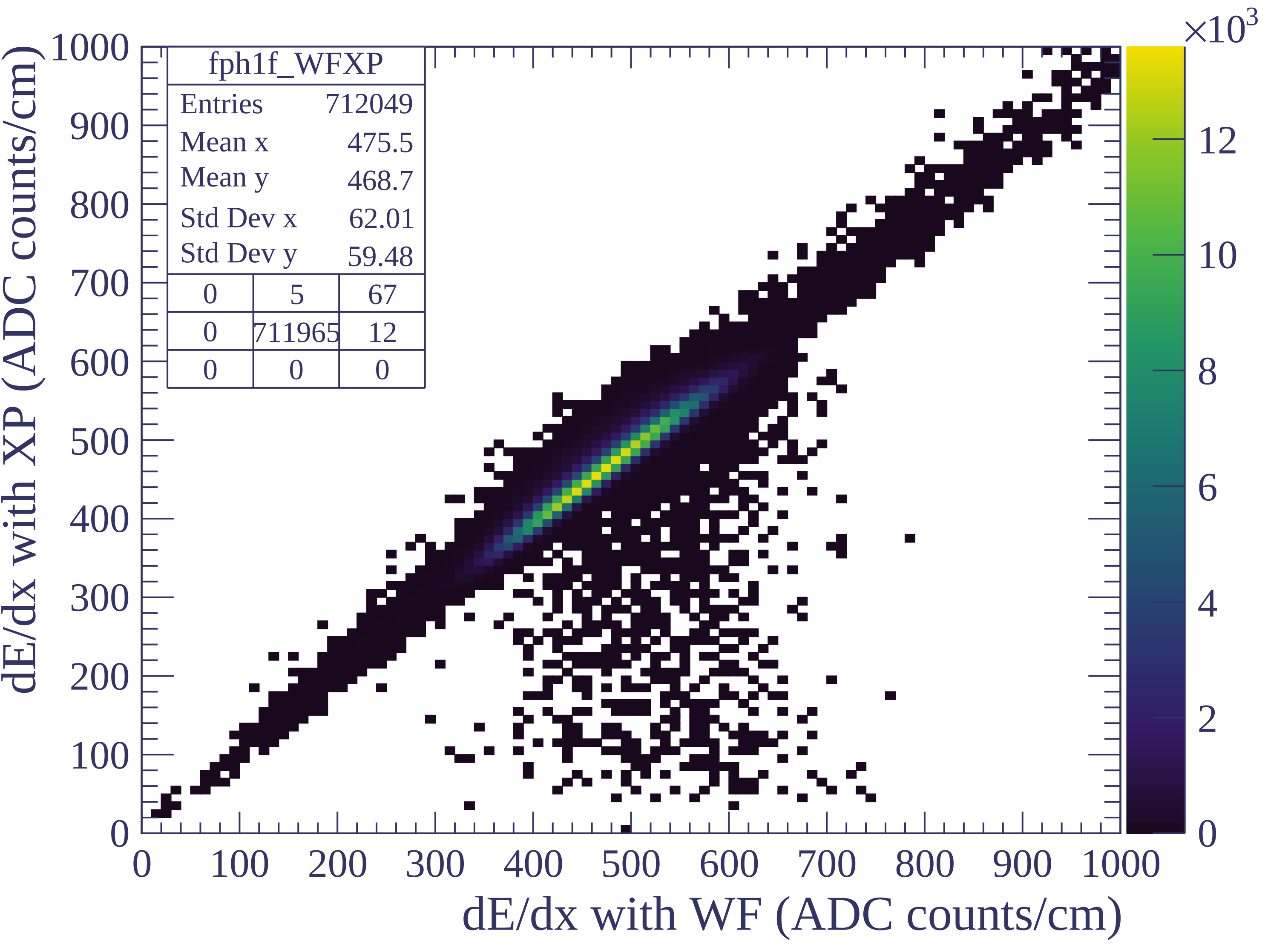
Count
×10^3
60
50
40
30
20
10
0
0
200
400
600
800
1000
1200
dE/dx(ADC counts/cm)
XP (712049 entries)
σ/μ = 12.80± 0.01 %
μ = 467.9± 0.1
σ = 59.9± 0.1
WF (712049 entries)
σ/μ = 13.47± 0.02 %
μ = 475.9± 0.1
σ = 64.1± 0.1



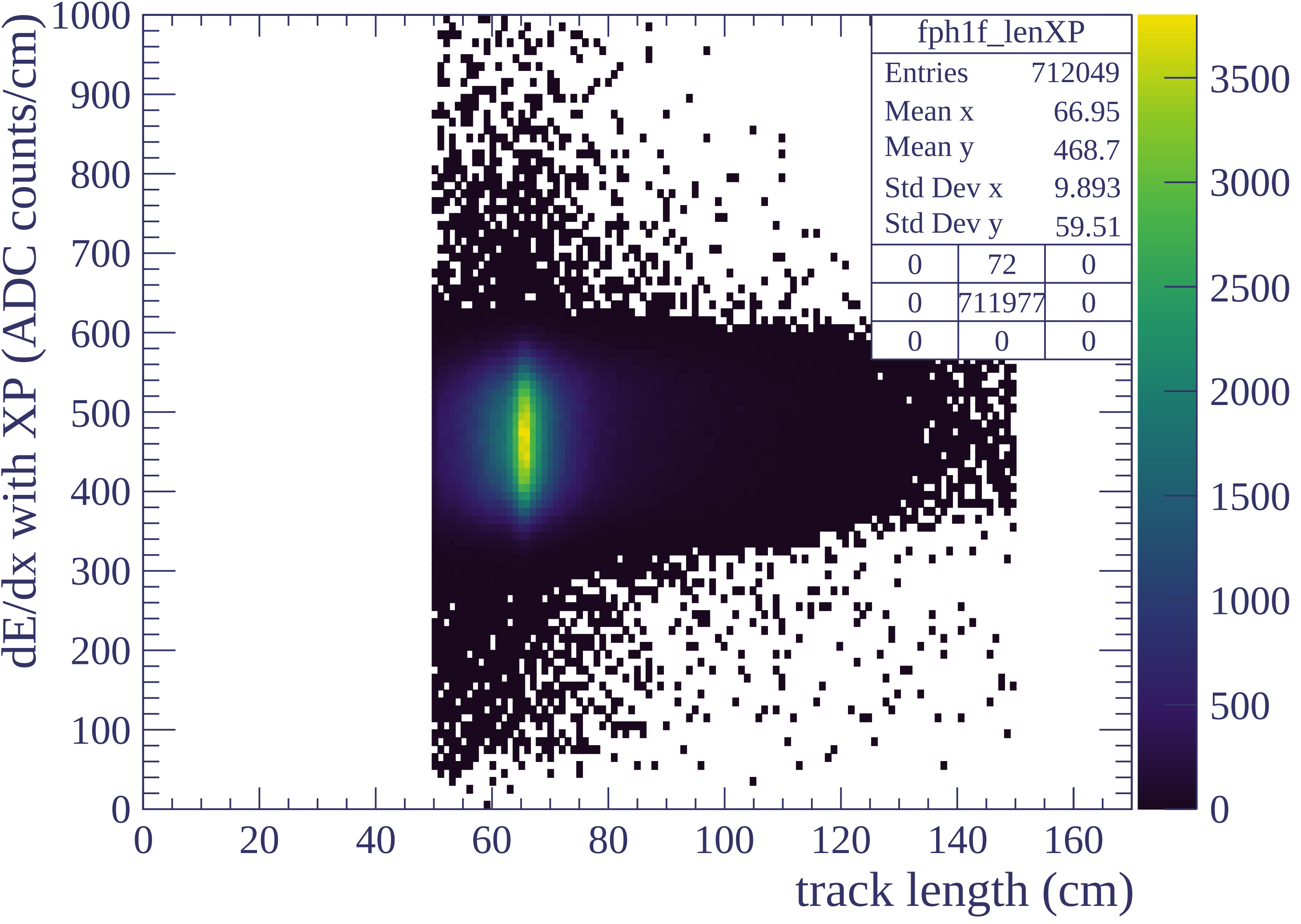
Energy loss in ERAM 24
WF (120 entries)
σ/μ = 14.74± 1.83 %
μ = 484.4± 8.7
σ = 71.4± 7.6
XP (120 entries)
σ/μ = 15.36± 1.99 %
μ = 469.6± 8.6
σ = 72.2± 8.0
Energy loss in ERAM 30
WF (924 entries)
σ/μ = 12.93± 0.37 %
μ = 480.2± 2.3
σ = 62.1± 1.5
XP (924 entries)
σ/μ = 13.95± 0.53 %
μ = 472.8± 2.5
σ = 65.9± 2.1
Energy loss in ERAM 28
WF (827 entries)
σ/μ = 12.88± 0.58 %
μ = 476.8± 2.6
σ = 61.4± 2.4
XP (827 entries)
σ/μ = 13.42± 0.56 %
μ = 480.1± 2.6
σ = 64.4± 2.3
Energy loss in ERAM 19
WF (1287 entries)
σ/μ = 13.70± 0.43 %
μ = 484.9± 2.2
σ = 66.4± 1.8
XP (1287 entries)
σ/μ = 13.36± 0.44 %
μ = 484.2± 2.2
σ = 64.7± 1.8
Energy loss in ERAM 21
WF (36875 entries)
σ/μ = 13.02± 0.06 %
μ = 473.6± 0.3
σ = 61.7± 0.2
XP (36875 entries)
σ/μ = 13.13± 0.07 %
μ = 459.7± 0.3
σ = 60.3± 0.3
Energy loss in ERAM 13
WF (37624 entries)
σ/μ = 12.86± 0.06 %
μ = 485.9± 0.4
σ = 62.5± 0.3
XP (37624 entries)
σ/μ = 12.53± 0.06 %
μ = 474.9± 0.3
σ = 59.5± 0.3
Energy loss in ERAM 9
WF (36570 entries)
σ/μ = 12.51± 0.07 %
μ = 493.6± 0.4
σ = 61.7± 0.3
XP (36570 entries)
σ/μ = 12.12± 0.06 %
μ = 482.5± 0.3
σ = 58.5± 0.3
Energy loss in ERAM 2
WF (37410 entries)
σ/μ = 12.75± 0.06 %
μ = 490.5± 0.4
σ = 62.5± 0.3
XP (37410 entries)
σ/μ = 12.56± 0.07 %
μ = 478.0± 0.4
σ = 60.0± 0.3
Energy loss in ERAM 26
WF (2060 entries)
σ/μ = 13.11± 0.30 %
μ = 454.5± 1.5
σ = 59.6± 1.2
XP (2060 entries)
σ/μ = 13.15± 0.32 %
μ = 442.5± 1.5
σ = 58.2± 1.2
Energy loss in ERAM 17
WF (3413 entries)
σ/μ = 13.62± 0.27 %
μ = 468.2± 1.3
σ = 63.8± 1.1
XP (3413 entries)
σ/μ = 12.88± 0.23 %
μ = 463.9± 1.1
σ = 59.7± 0.9
Energy loss in ERAM 23
WF (3860 entries)
σ/μ = 13.22± 0.20 %
μ = 469.3± 1.1
σ = 62.0± 0.8
XP (3860 entries)
σ/μ = 13.30± 0.24 %
μ = 466.3± 1.2
σ = 62.0± 1.0
Energy loss in ERAM 29
WF (3970 entries)
σ/μ = 13.34± 0.21 %
μ = 474.4± 1.1
σ = 63.3± 0.9
XP (3970 entries)
σ/μ = 13.10± 0.20 %
μ = 468.3± 1.1
σ = 61.4± 0.8
Energy loss in ERAM 1
WF (38206 entries)
σ/μ = 12.67± 0.06 %
μ = 477.3± 0.4
σ = 60.5± 0.3
XP (38206 entries)
σ/μ = 12.57± 0.07 %
μ = 467.2± 0.4
σ = 58.7± 0.3
Energy loss in ERAM 10
WF (42145 entries)
σ/μ = 12.96± 0.06 %
μ = 484.9± 0.3
σ = 62.8± 0.3
XP (42145 entries)
σ/μ = 12.45± 0.06 %
μ = 474.2± 0.3
σ = 59.0± 0.2
Energy loss in ERAM 11
WF (45131 entries)
σ/μ = 13.00± 0.06 %
μ = 484.8± 0.4
σ = 63.0± 0.2
XP (45131 entries)
σ/μ = 12.57± 0.06 %
μ = 473.4± 0.3
σ = 59.5± 0.2
Energy loss in ERAM 3
WF (44461 entries)
σ/μ = 13.38± 0.06 %
μ = 477.1± 0.3
σ = 63.8± 0.3
XP (44461 entries)
σ/μ = 12.81± 0.06 %
μ = 465.6± 0.3
σ = 59.7± 0.2
Count
dE/dx (ADC counts/cm)



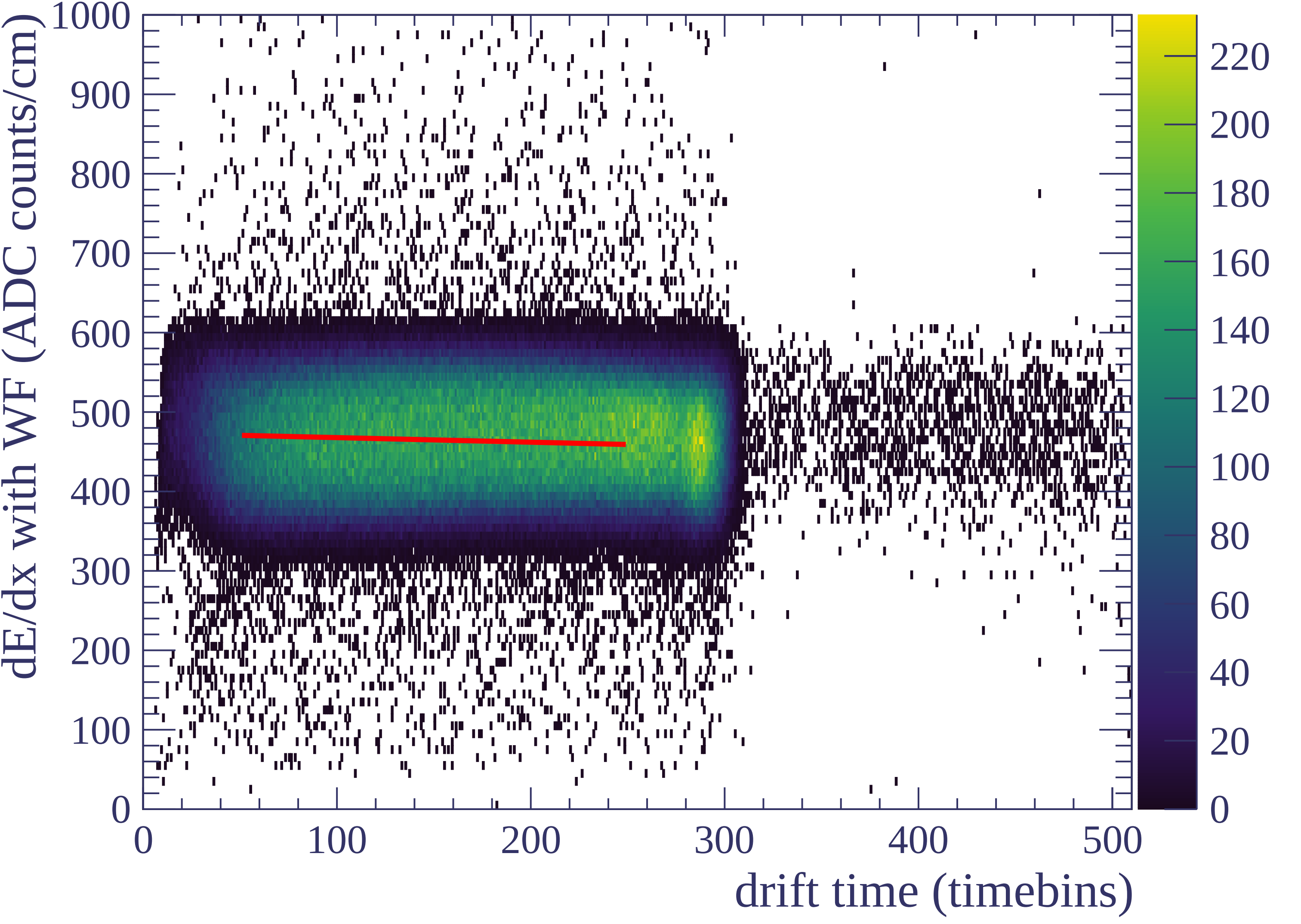
Energy loss in ERAM 47
WF (312 entries) σ/μ = 11.37± 0.65 % μ = 472.9± 3.6 σ = 53.8± 2.6
XP (312 entries) σ/μ = 11.28± 0.90 % μ = 478.3± 4.1 σ = 53.9± 3.8
Energy loss in ERAM 16
WF (1590 entries) σ/μ = 14.14± 0.34 % μ = 463.1± 1.8 σ = 65.5± 1.3
XP (1590 entries) σ/μ = 12.56± 0.23 % μ = 471.2± 1.6 σ = 59.2± 0.9
Energy loss in ERAM 14
WF (1099 entries) σ/μ = 13.74± 0.44 % μ = 469.7± 2.2 σ = 64.5± 1.8
XP (1099 entries) σ/μ = 13.35± 0.42 % μ = 474.4± 2.2 σ = 63.3± 1.7
Energy loss in ERAM 15
WF (904 entries) σ/μ = 12.71± 0.52 % μ = 477.1± 2.4 σ = 60.6± 2.2
XP (904 entries) σ/μ = 11.72± 0.37 % μ = 479.2± 2.1 σ = 56.2± 1.5
Energy loss in ERAM 42
WF (50823 entries) σ/μ = 13.33± 0.06 % μ = 459.1± 0.3 σ = 61.2± 0.2
XP (50823 entries) σ/μ = 13.06± 0.05 % μ = 459.7± 0.3 σ = 60.1± 0.2
Energy loss in ERAM 45
WF (47590 entries) σ/μ = 13.45± 0.07 % μ = 472.4± 0.3 σ = 63.5± 0.3
XP (47590 entries) σ/μ = 12.61± 0.05 % μ = 466.5± 0.3 σ = 58.8± 0.2
Energy loss in ERAM 37
WF (40510 entries) σ/μ = 12.65± 0.06 % μ = 485.8± 0.4 σ = 61.5± 0.3
XP (40510 entries) σ/μ = 11.99± 0.05 % μ = 476.1± 0.3 σ = 57.1± 0.2
Energy loss in ERAM 36
WF (27244 entries) σ/μ = 12.19± 0.08 % μ = 481.7± 0.4 σ = 58.7± 0.3
XP (27244 entries) σ/μ = 11.88± 0.07 % μ = 475.3± 0.4 σ = 56.5± 0.3
Energy loss in ERAM 20
WF (4228 entries) σ/μ = 13.92± 0.23 % μ = 451.9± 1.1 σ = 62.9± 0.9
XP (4228 entries) σ/μ = 14.05± 0.25 % μ = 453.0± 1.1 σ = 63.6± 1.0
Energy loss in ERAM 38
WF (5447 entries) σ/μ = 14.20± 0.20 % μ = 456.7± 1.0 σ = 64.9± 0.8
XP (5447 entries) σ/μ = 13.57± 0.17 % μ = 460.4± 0.9 σ = 62.5± 0.6
Energy loss in ERAM 7
WF (4278 entries) σ/μ = 13.43± 0.20 % μ = 467.6± 1.1 σ = 62.8± 0.8
XP (4278 entries) σ/μ = 13.47± 0.23 % μ = 465.2± 1.1 σ = 62.7± 0.9
Energy loss in ERAM 44
WF (3244 entries) σ/μ = 13.65± 0.26 % μ = 459.8± 1.2 σ = 62.8± 1.0
XP (3244 entries) σ/μ = 13.40± 0.25 % μ = 461.4± 1.2 σ = 61.8± 1.0
Energy loss in ERAM 43
WF (54781 entries) σ/μ = 13.45± 0.06 % μ = 473.7± 0.3 σ = 63.7± 0.3
XP (54781 entries) σ/μ = 12.80± 0.05 % μ = 470.1± 0.3 σ = 60.2± 0.2
Energy loss in ERAM 39
WF (54214 entries) σ/μ = 13.58± 0.06 % μ = 460.7± 0.3 σ = 62.5± 0.2
XP (54214 entries) σ/μ = 13.30± 0.06 % μ = 461.9± 0.3 σ = 61.5± 0.2
Energy loss in ERAM 41
WF (46423 entries) σ/μ = 12.93± 0.06 % μ = 476.1± 0.3 σ = 61.6± 0.3
XP (46423 entries) σ/μ = 12.24± 0.05 % μ = 470.5± 0.3 σ = 57.6± 0.2
Energy loss in ERAM 46
WF (34479 entries) σ/μ = 12.92± 0.07 % μ = 460.9± 0.3 σ = 59.5± 0.3
XP (34479 entries) σ/μ = 12.75± 0.07 % μ = 457.3± 0.4 σ = 58.3± 0.3
Count
dE/dx (ADC counts/cm)



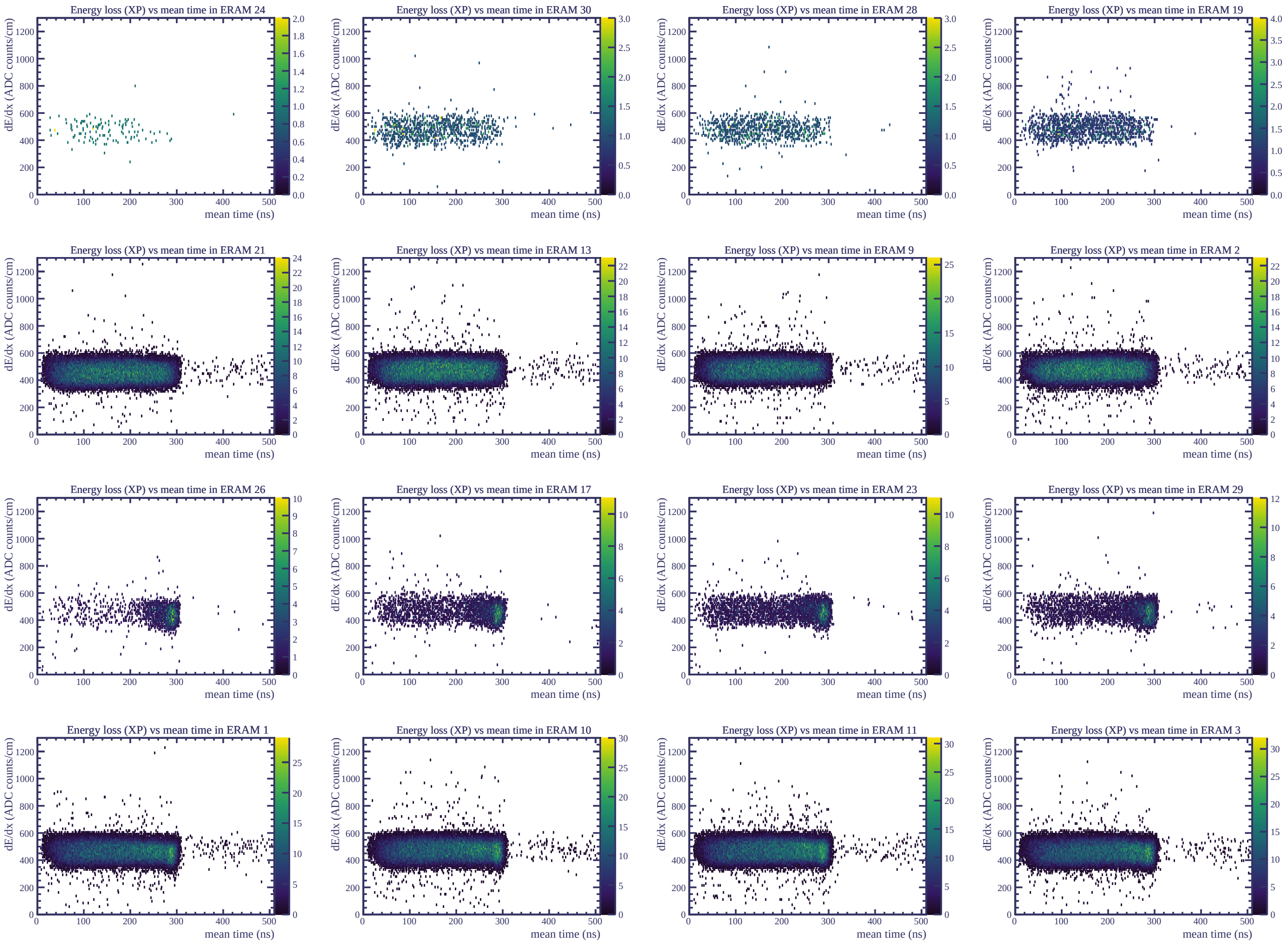
fph1f_WFXP
Entries 712049
Mean x 475.5
Mean y 468.7
Std Dev x 62.01
Std Dev y 59.48
0 5 67
0 711965 12
0 0 0
dE/dx with XP (ADC counts/cm)
dE/dx with WF (ADC counts/cm)
×10^3

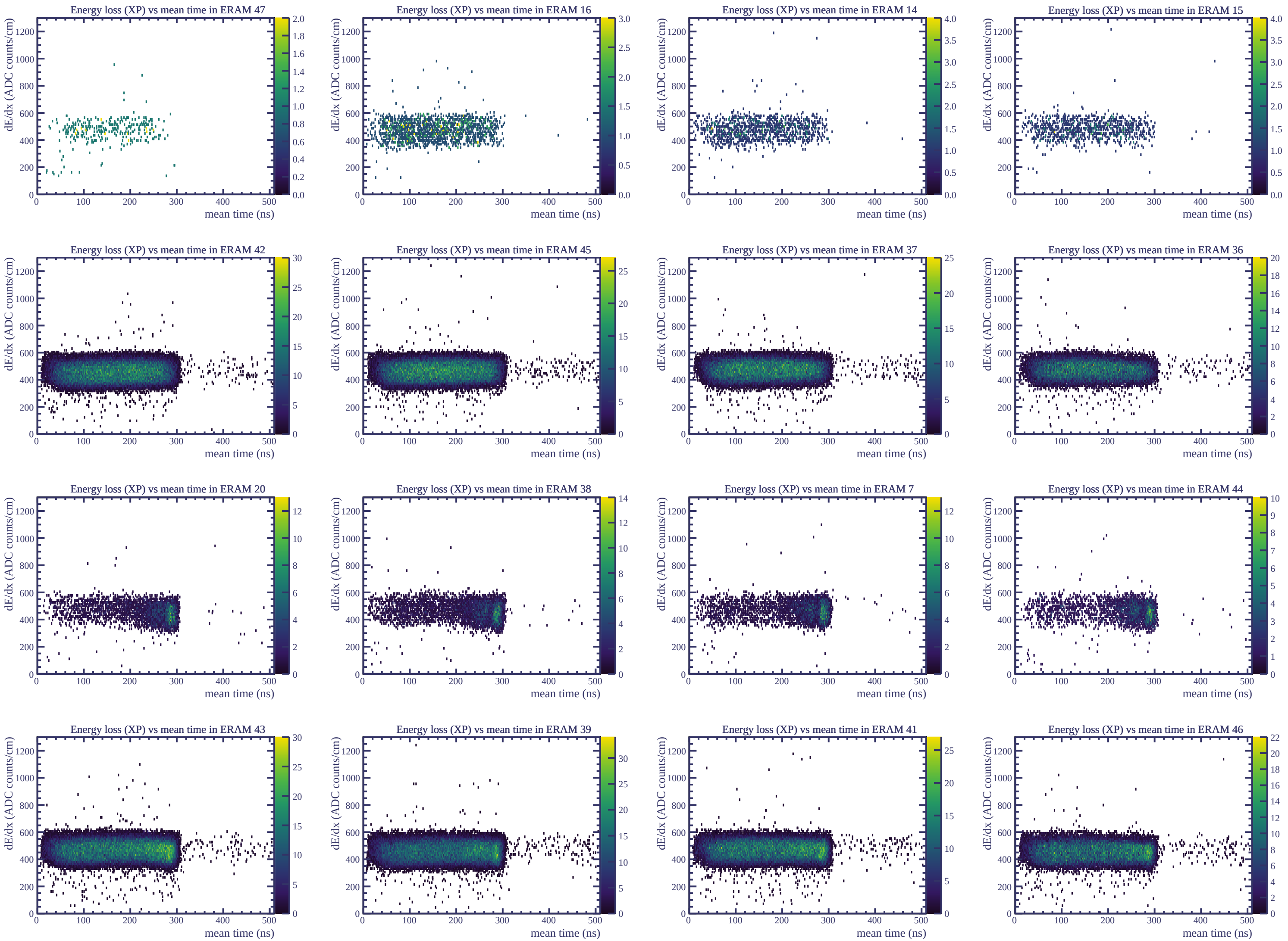


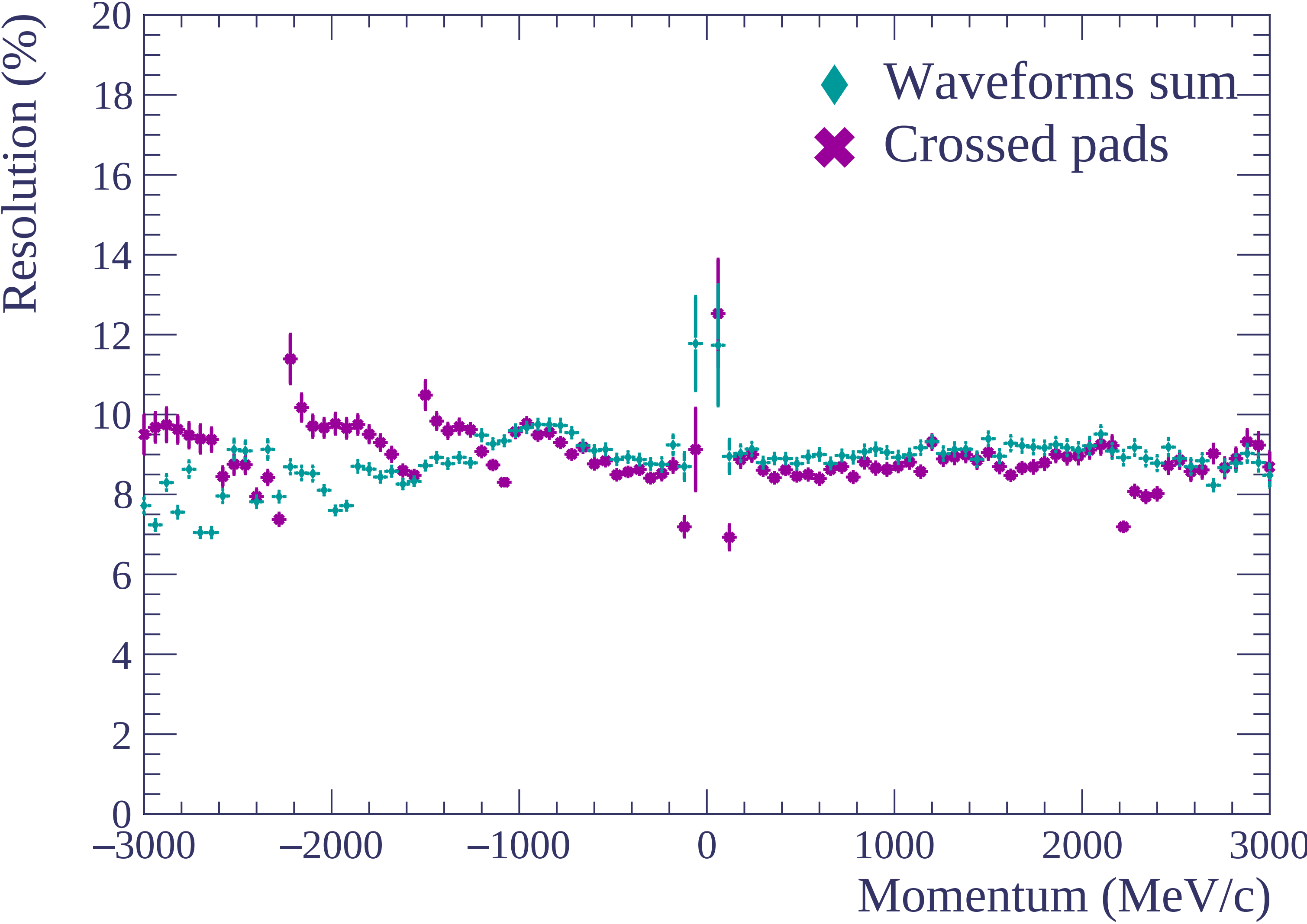
fph1f_lenXP
Entries 712049
Mean x 66.95
Mean y 468.7
Std Dev x 9.893
Std Dev y 59.51
0 72 0
0 711977 0
0 0 0
dE/dx with XP (ADC counts/cm)
track length (cm)
0
20
40
60
80
100
120
140
160
0
100
200
300
400
500
600
700
800
900
1000
500
1000
1500
2000
2500
3000
3500

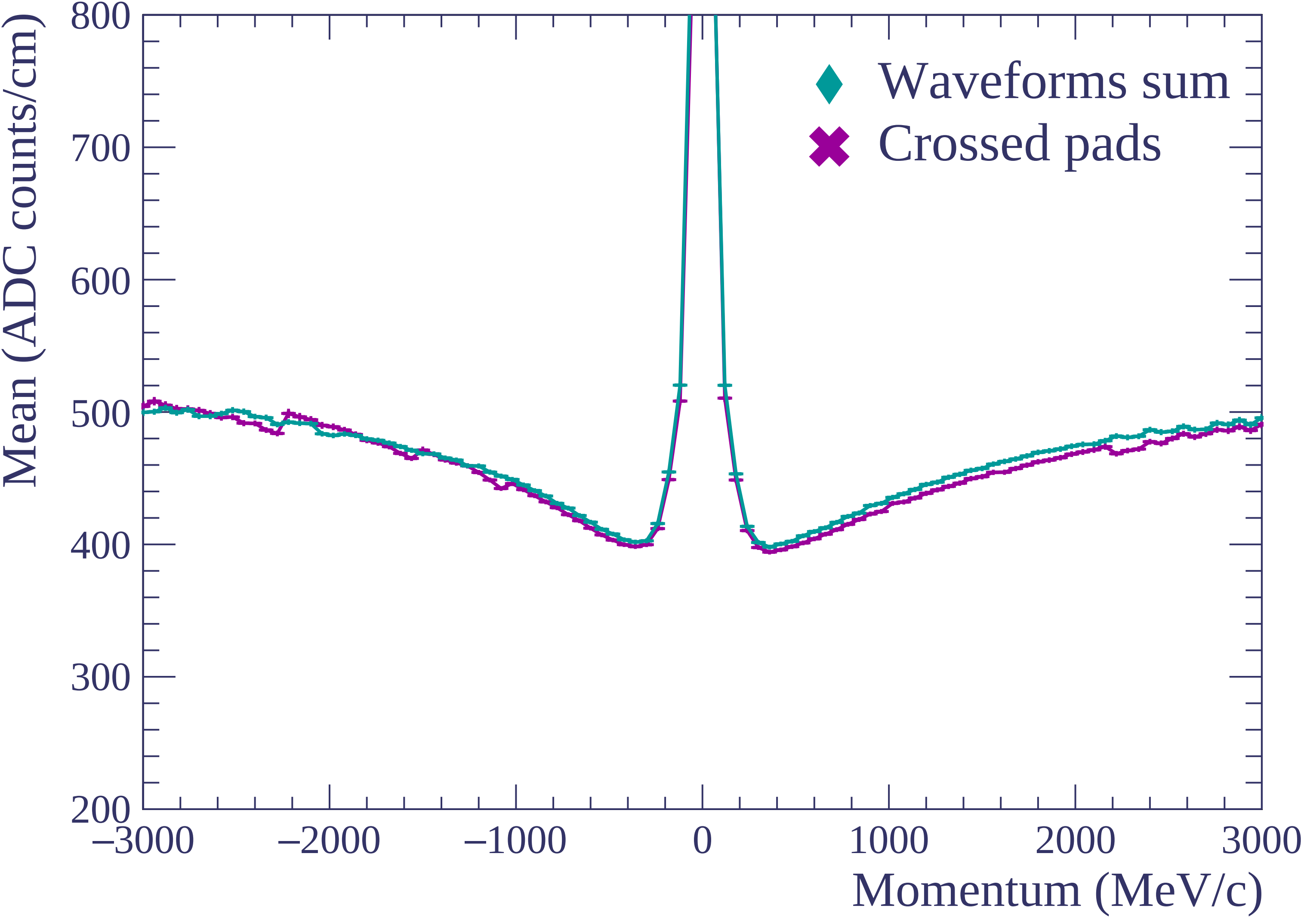


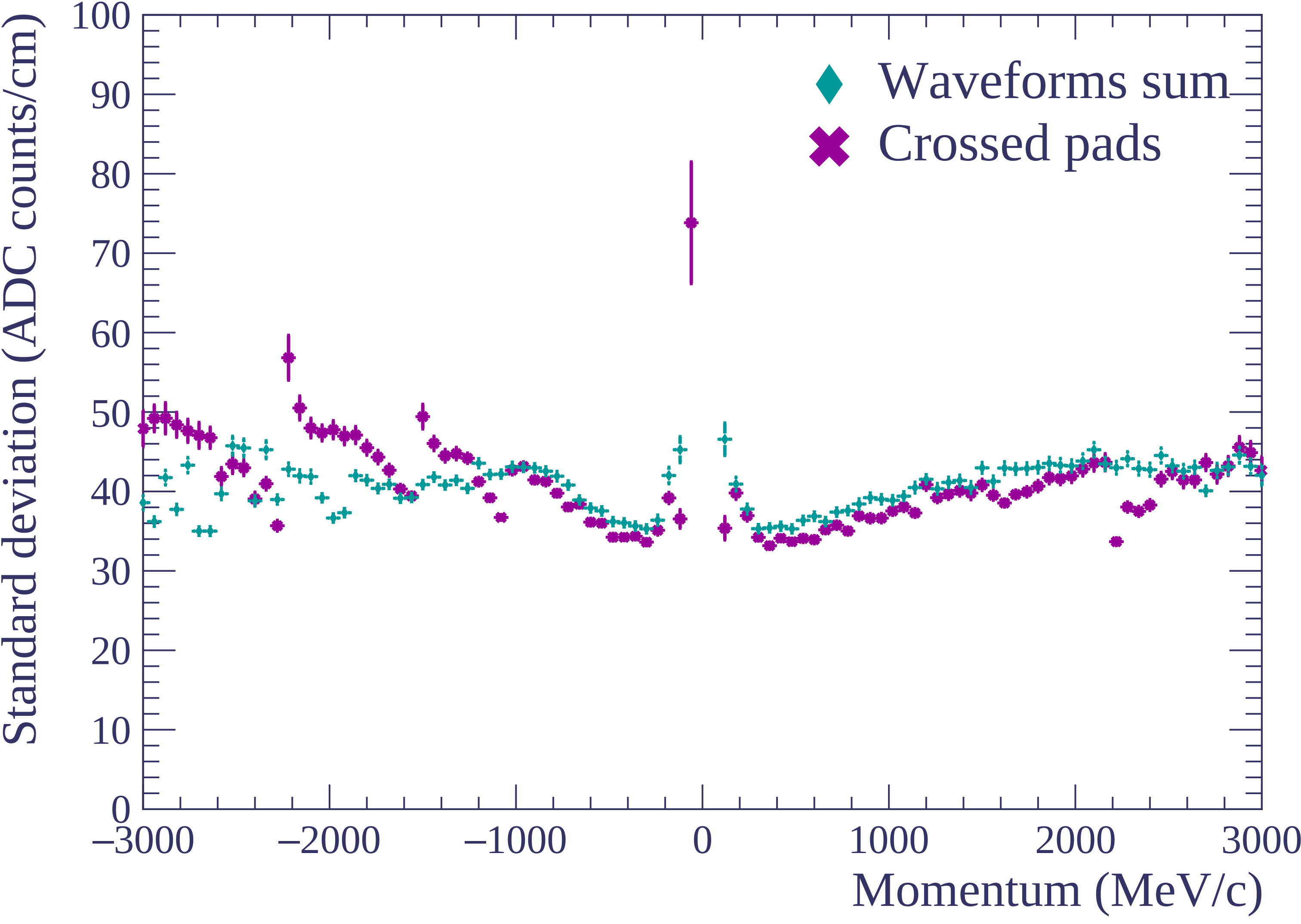
dE/dx with WF (ADC counts/cm)
0
100
200
300
400
500
600
700
800
900
1000
drift time (timebins)
0
100
200
300
400
500
0
20
40
60
80
100
120
140
160
180
200
220


Energy loss (XP) vs mean time in ERAM 24
Energy loss (XP) vs mean time in ERAM 30
Energy loss (XP) vs mean time in ERAM 28
Energy loss (XP) vs mean time in ERAM 19
Energy loss (XP) vs mean time in ERAM 21
Energy loss (XP) vs mean time in ERAM 13
Energy loss (XP) vs mean time in ERAM 9
Energy loss (XP) vs mean time in ERAM 2
Energy loss (XP) vs mean time in ERAM 26
Energy loss (XP) vs mean time in ERAM 17
Energy loss (XP) vs mean time in ERAM 23
Energy loss (XP) vs mean time in ERAM 29
Energy loss (XP) vs mean time in ERAM 1
Energy loss (XP) vs mean time in ERAM 10
Energy loss (XP) vs mean time in ERAM 11
Energy loss (XP) vs mean time in ERAM 3
dE/dx (ADC counts/cm)
mean time (ns)

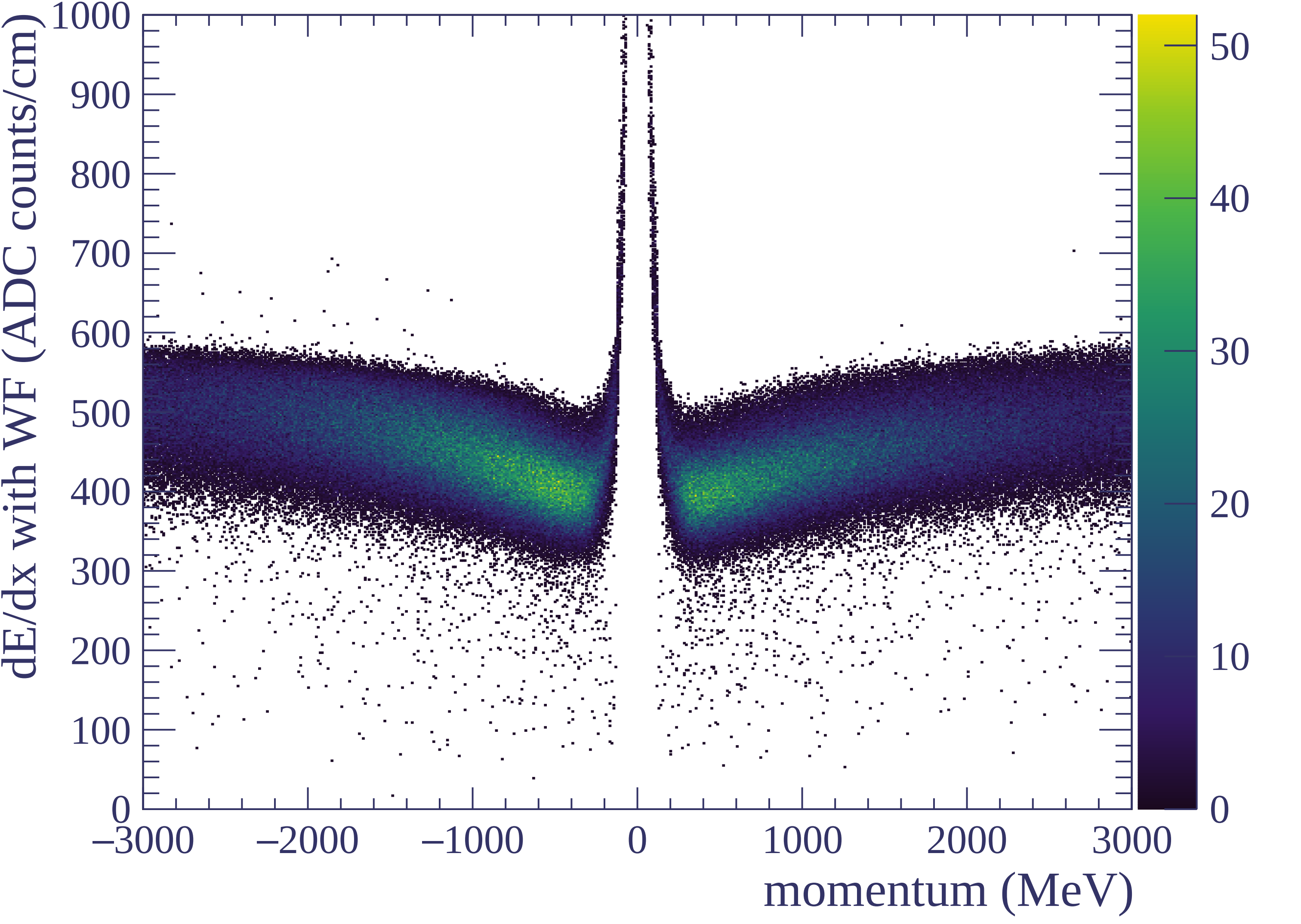

Energy loss (XP) vs mean time in ERAM 47
Energy loss (XP) vs mean time in ERAM 16
Energy loss (XP) vs mean time in ERAM 14
Energy loss (XP) vs mean time in ERAM 15
Energy loss (XP) vs mean time in ERAM 42
Energy loss (XP) vs mean time in ERAM 45
Energy loss (XP) vs mean time in ERAM 37
Energy loss (XP) vs mean time in ERAM 36
Energy loss (XP) vs mean time in ERAM 20
Energy loss (XP) vs mean time in ERAM 38
Energy loss (XP) vs mean time in ERAM 7
Energy loss (XP) vs mean time in ERAM 44
Energy loss (XP) vs mean time in ERAM 43
Energy loss (XP) vs mean time in ERAM 39
Energy loss (XP) vs mean time in ERAM 41
Energy loss (XP) vs mean time in ERAM 46
dE/dx (ADC counts/cm)
mean time (ns)


Waveforms sum
Crossed pads
Resolution (%)
Momentum (MeV/c)
0
2
4
6
8
10
12
14
16
18
20
−3000
−2000
−1000
0
1000
2000
3000

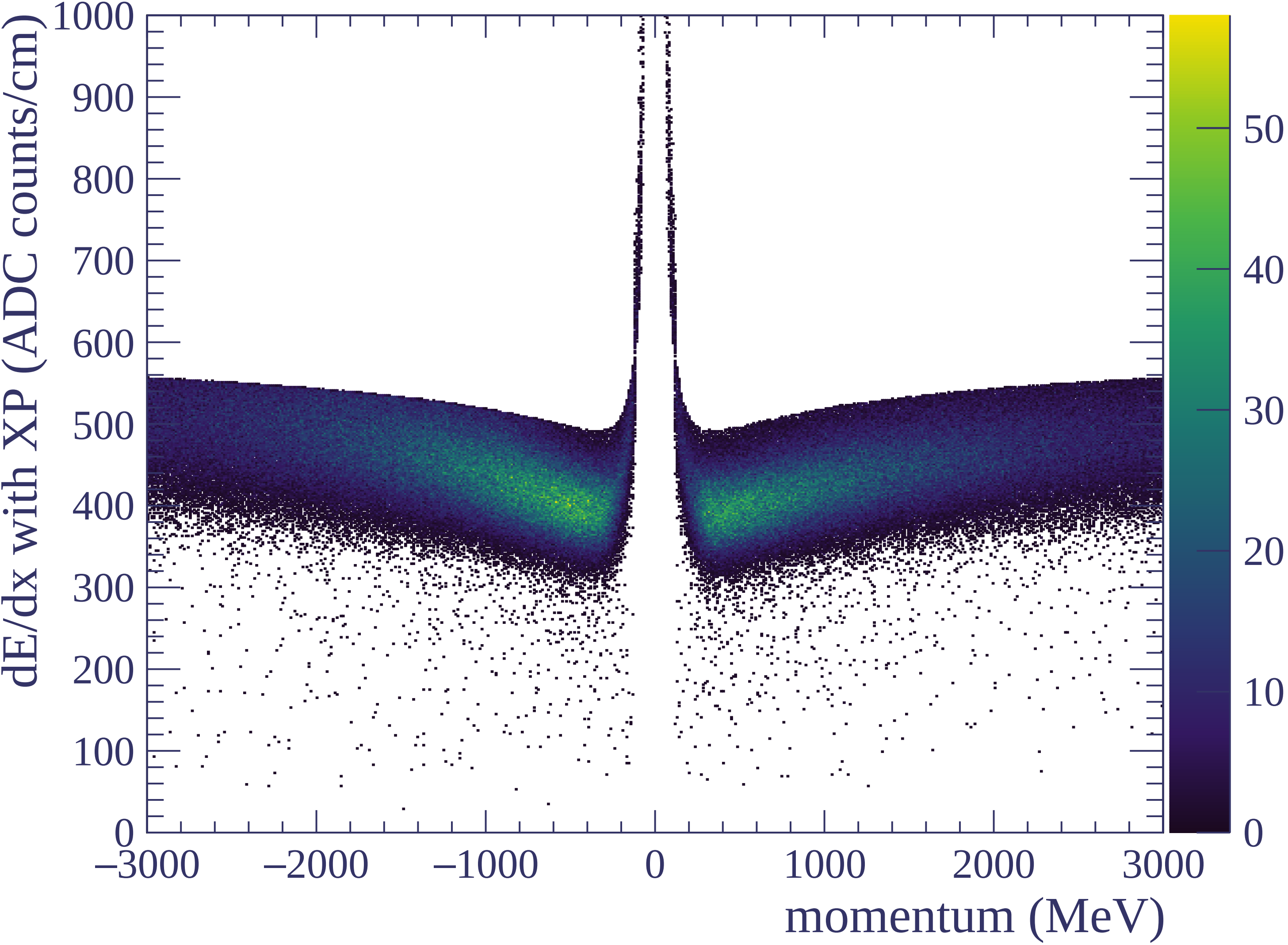

Waveforms sum
Crossed pads
Mean (ADC counts/cm)
800
700
600
500
400
300
200
−3000
−2000
−1000
0
1000
2000
3000
Momentum (MeV/c)

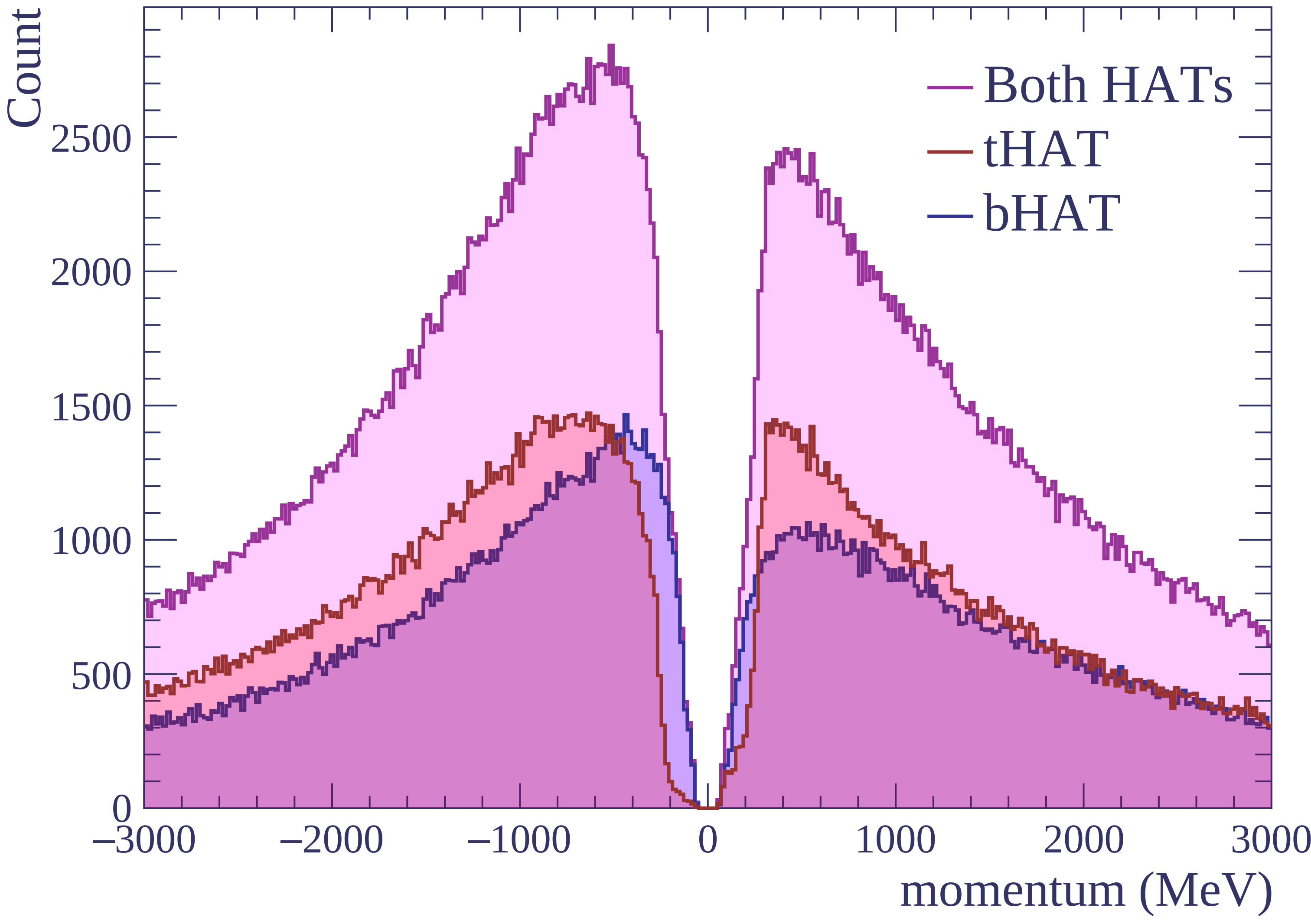


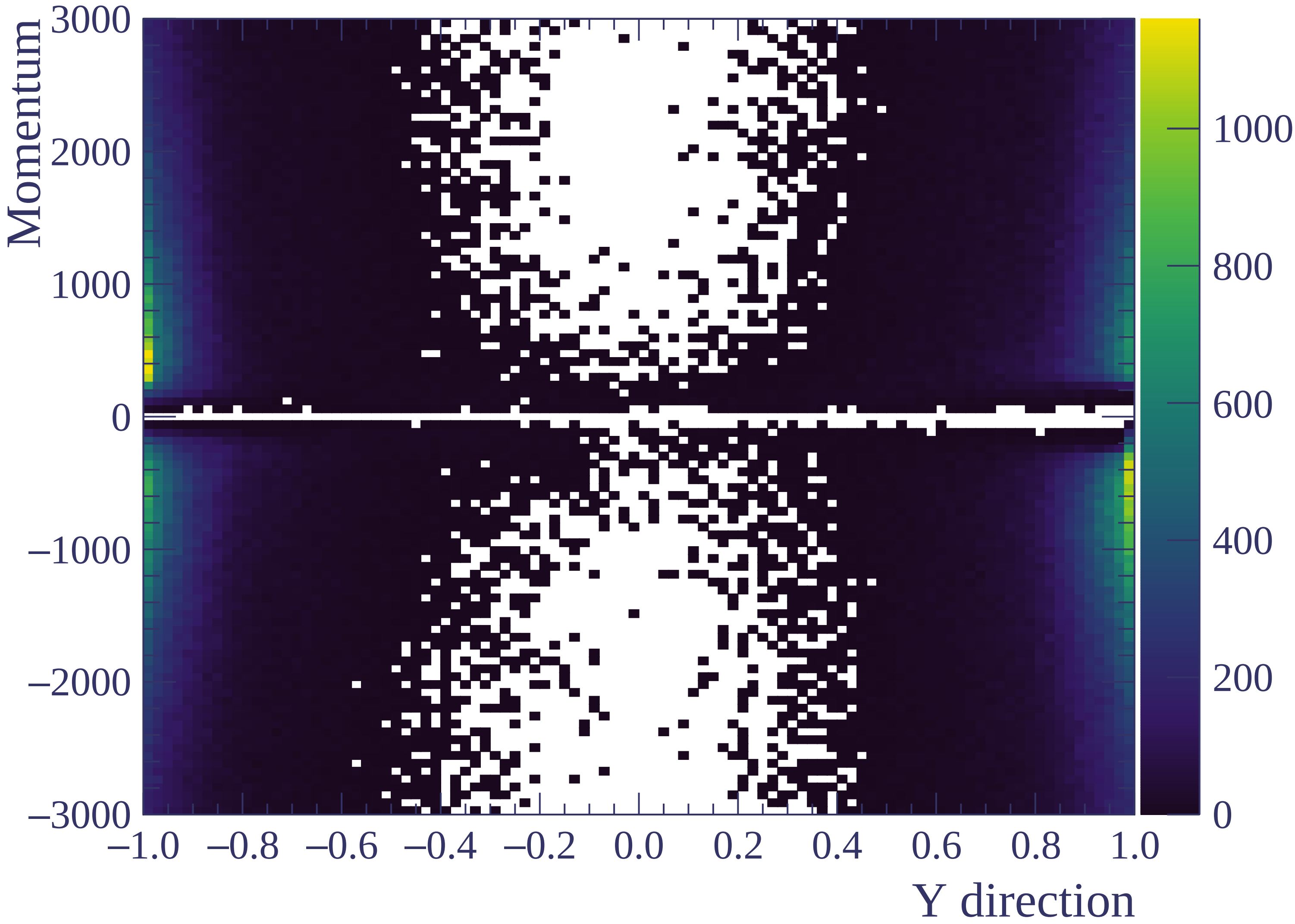
Standard deviation (ADC counts/cm)
Momentum (MeV/c)
Waveforms sum
Crossed pads
100
90
80
70
60
50
40
30
20
10
0
−3000
−2000
−1000
0
1000
2000
3000

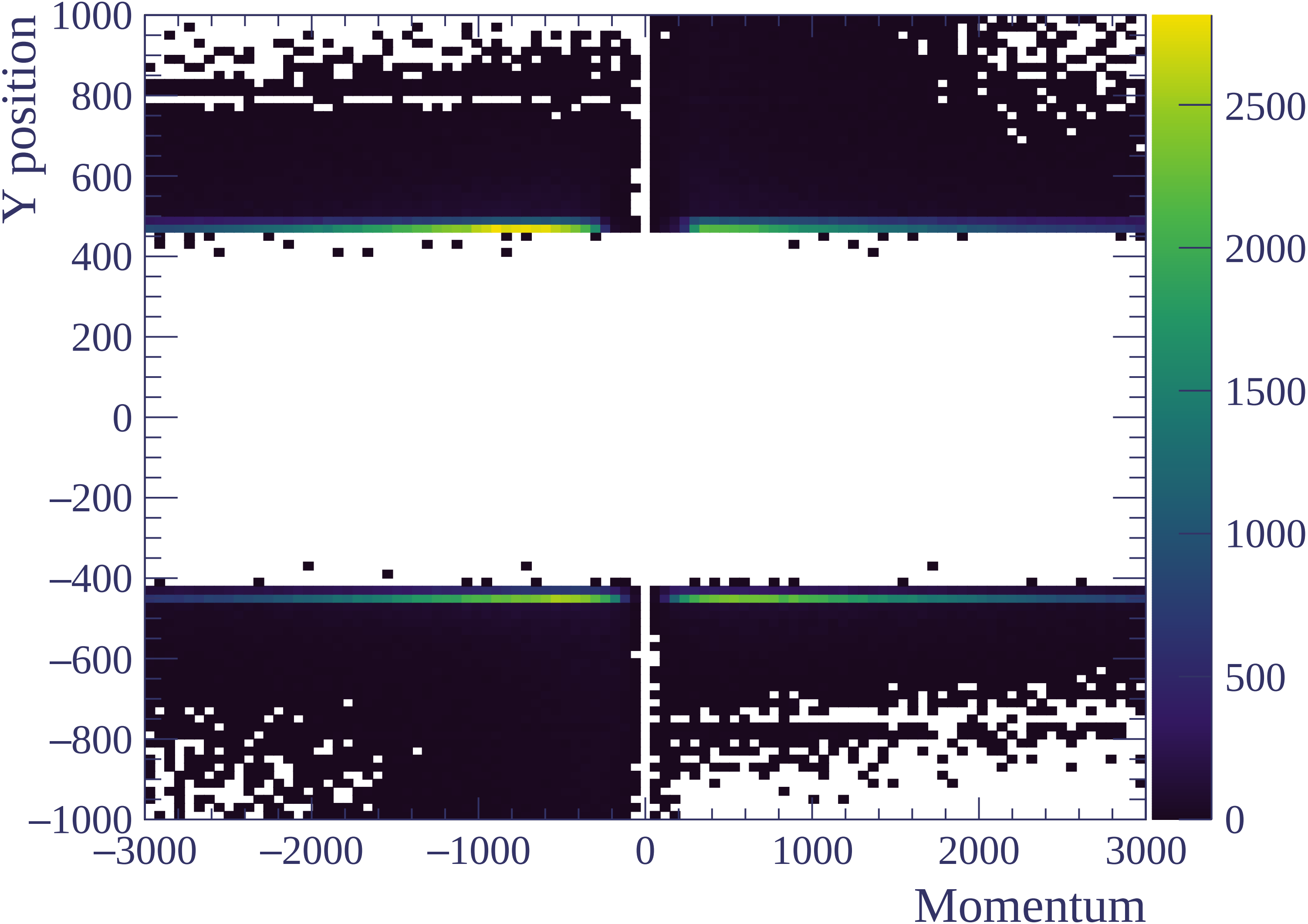


dE/dx with WF (ADC counts/cm)
momentum (MeV)
1000
900
800
700
600
500
400
300
200
100
0
−3000
−2000
−1000
0
1000
2000
3000
50
40
30
20
10
0

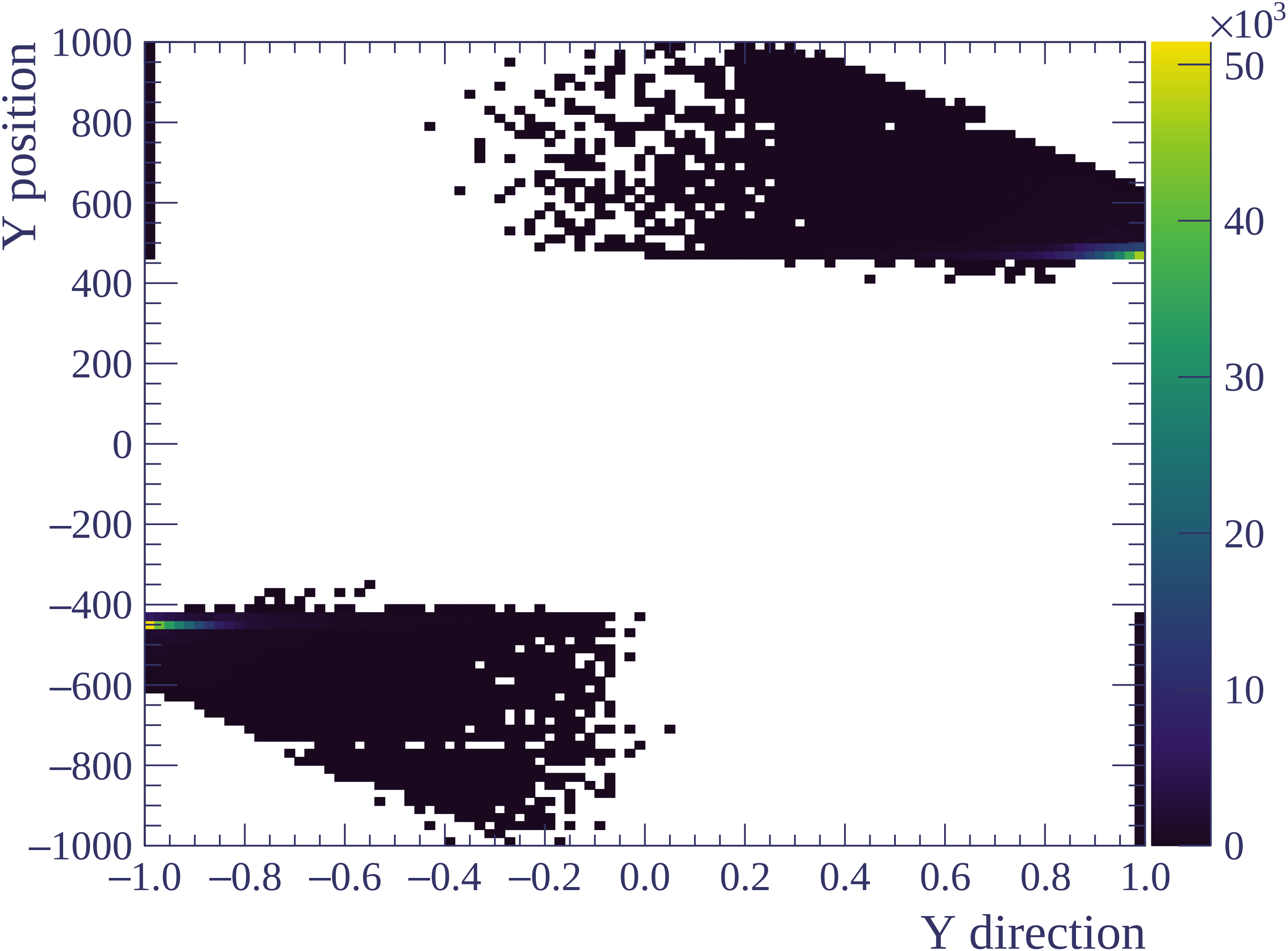

dE/dx with XP (ADC counts/cm)
1000
900
800
700
600
500
400
300
200
100
0
−3000
−2000
−1000
0
1000
2000
3000
momentum (MeV)
50
40
30
20
10
0


Count
2500
2000
1500
1000
500
0
−3000
−2000
−1000
0
1000
2000
3000
momentum (MeV)
Both HATs
tHAT
bHAT

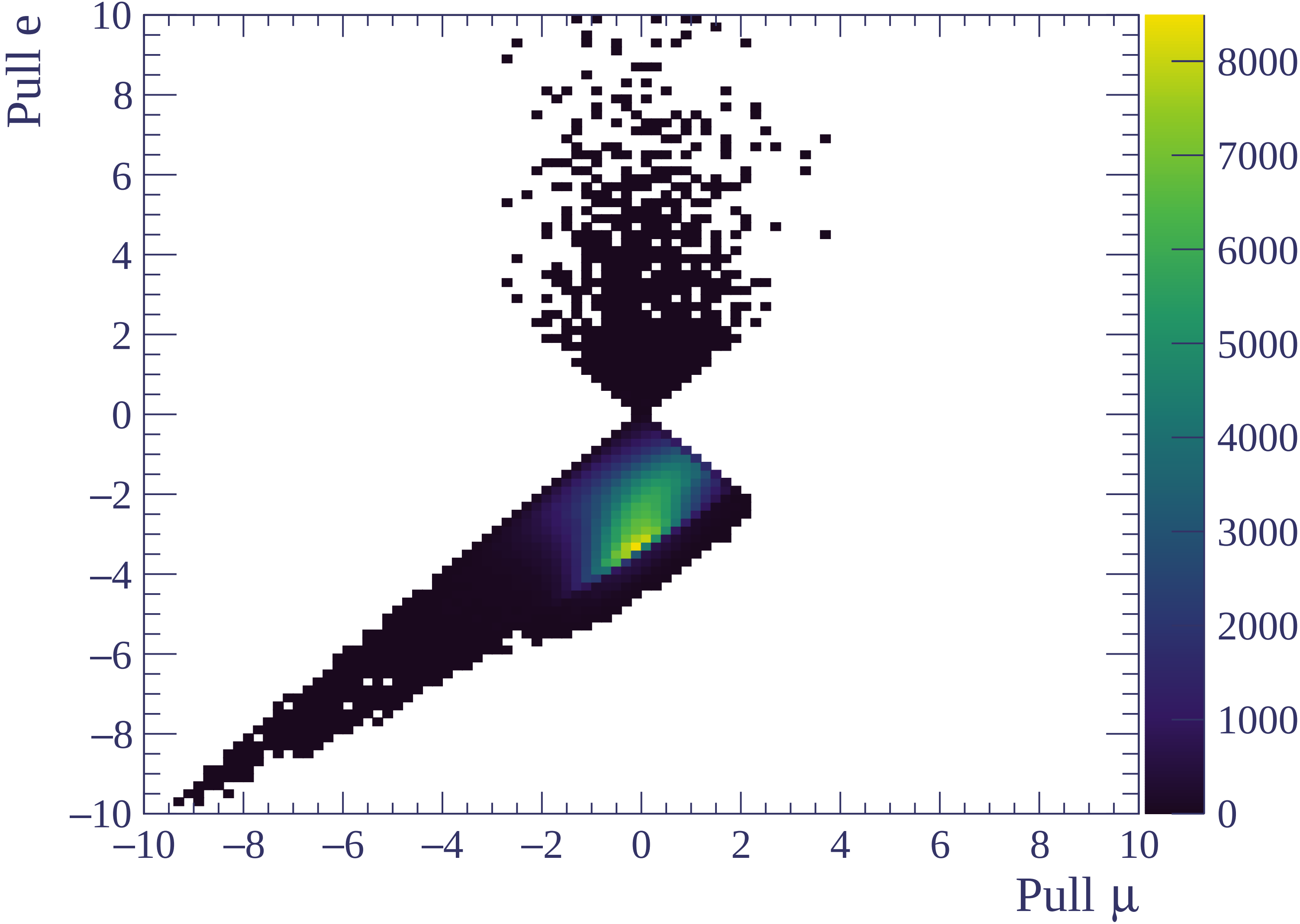

Momentum
3000
2000
1000
0
−1000
−2000
−3000
−1.0
−0.8
−0.6
−0.4
−0.2
0.0
0.2
0.4
0.6
0.8
1.0
Y direction
1000
800
600
400
200
0



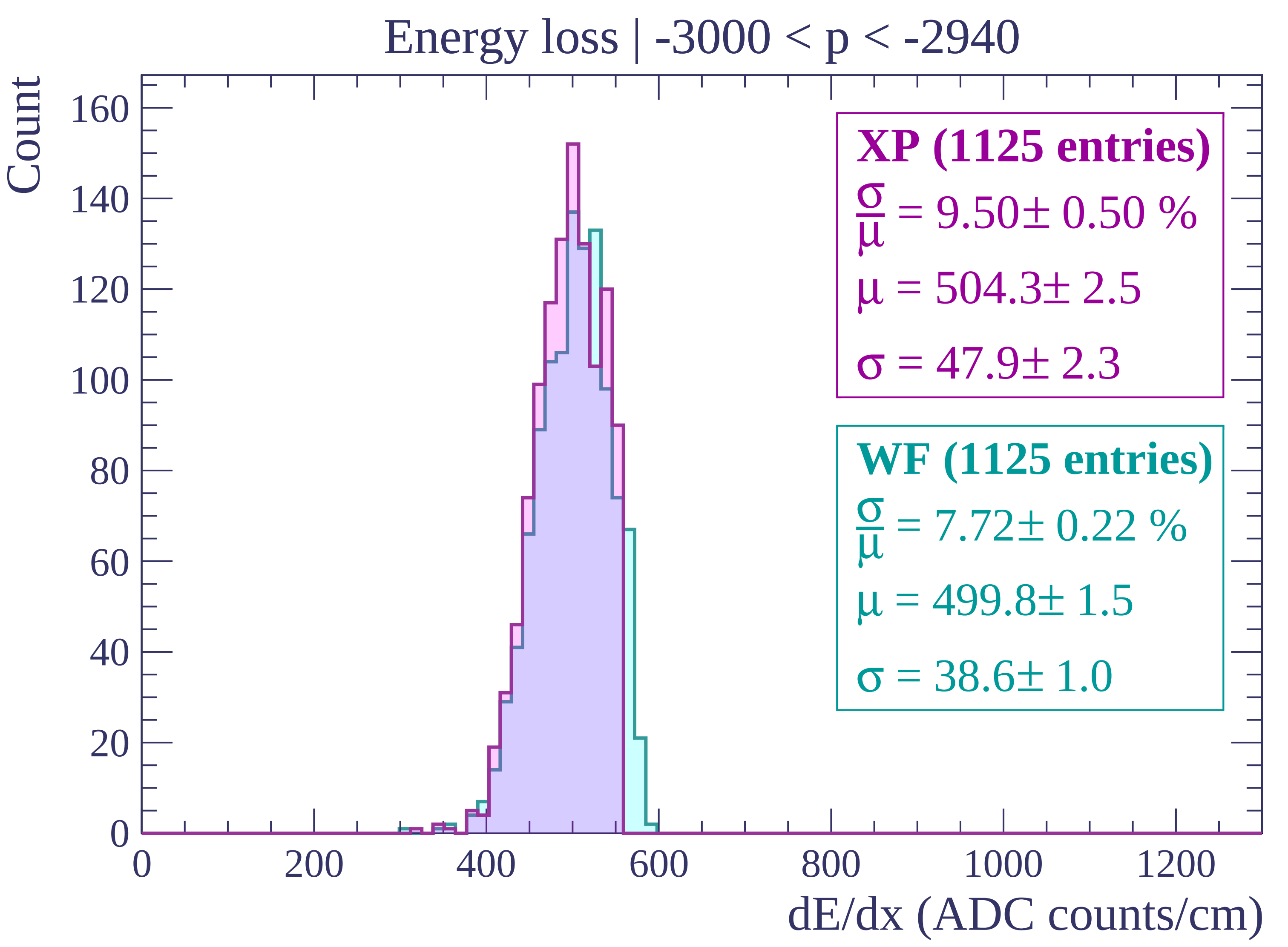
Y position
1000
800
600
400
200
0
−200
−400
−600
−800
−1000
−3000
−2000
−1000
0
1000
2000
3000
Momentum
2500
2000
1500
1000
500
0



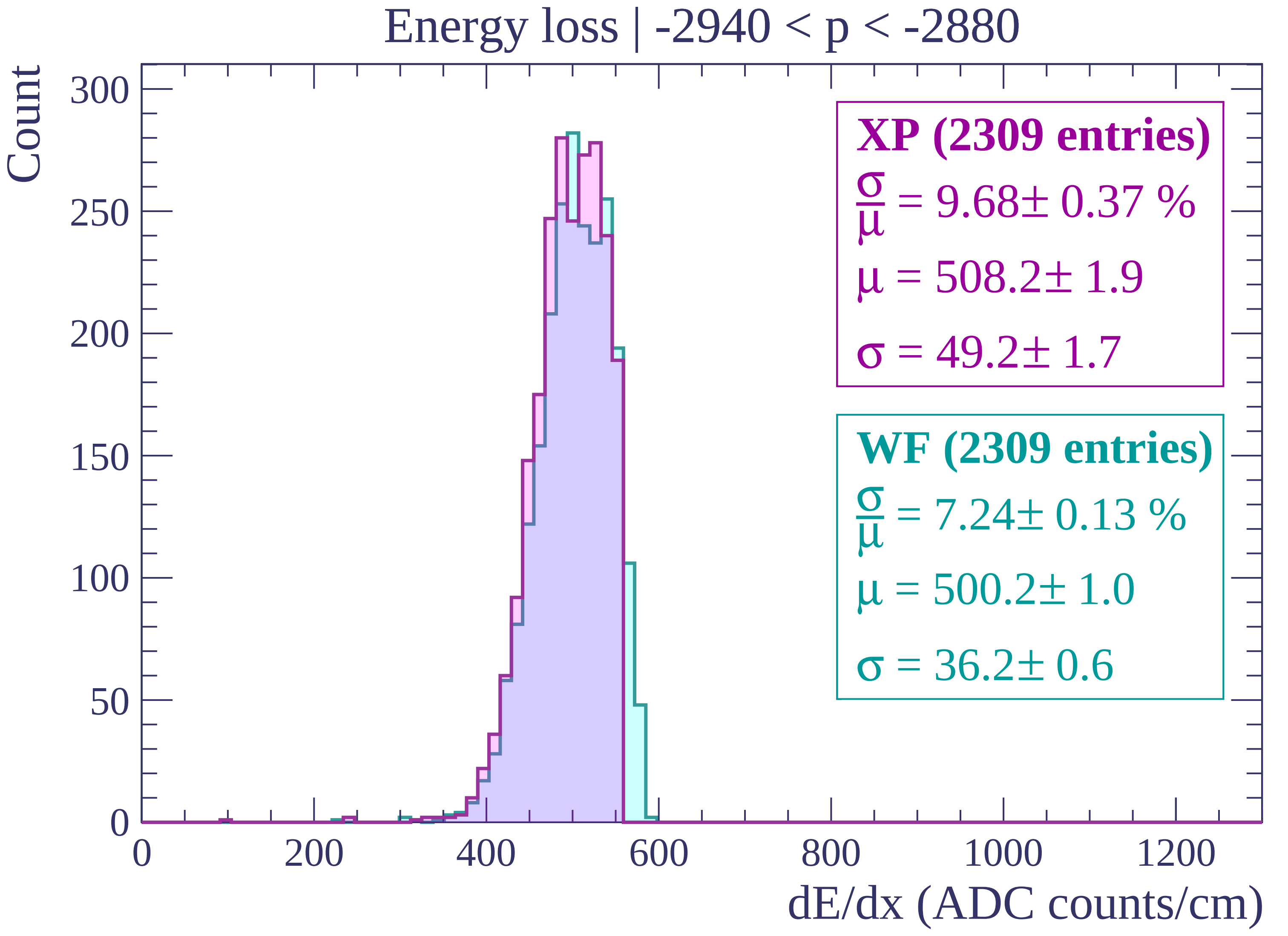
Y position
1000
800
600
400
200
0
−200
−400
−600
−800
−1000
−1.0
−0.8
−0.6
−0.4
−0.2
0.0
0.2
0.4
0.6
0.8
1.0
Y direction
$\times 10^3$
50
40
30
20
10
0

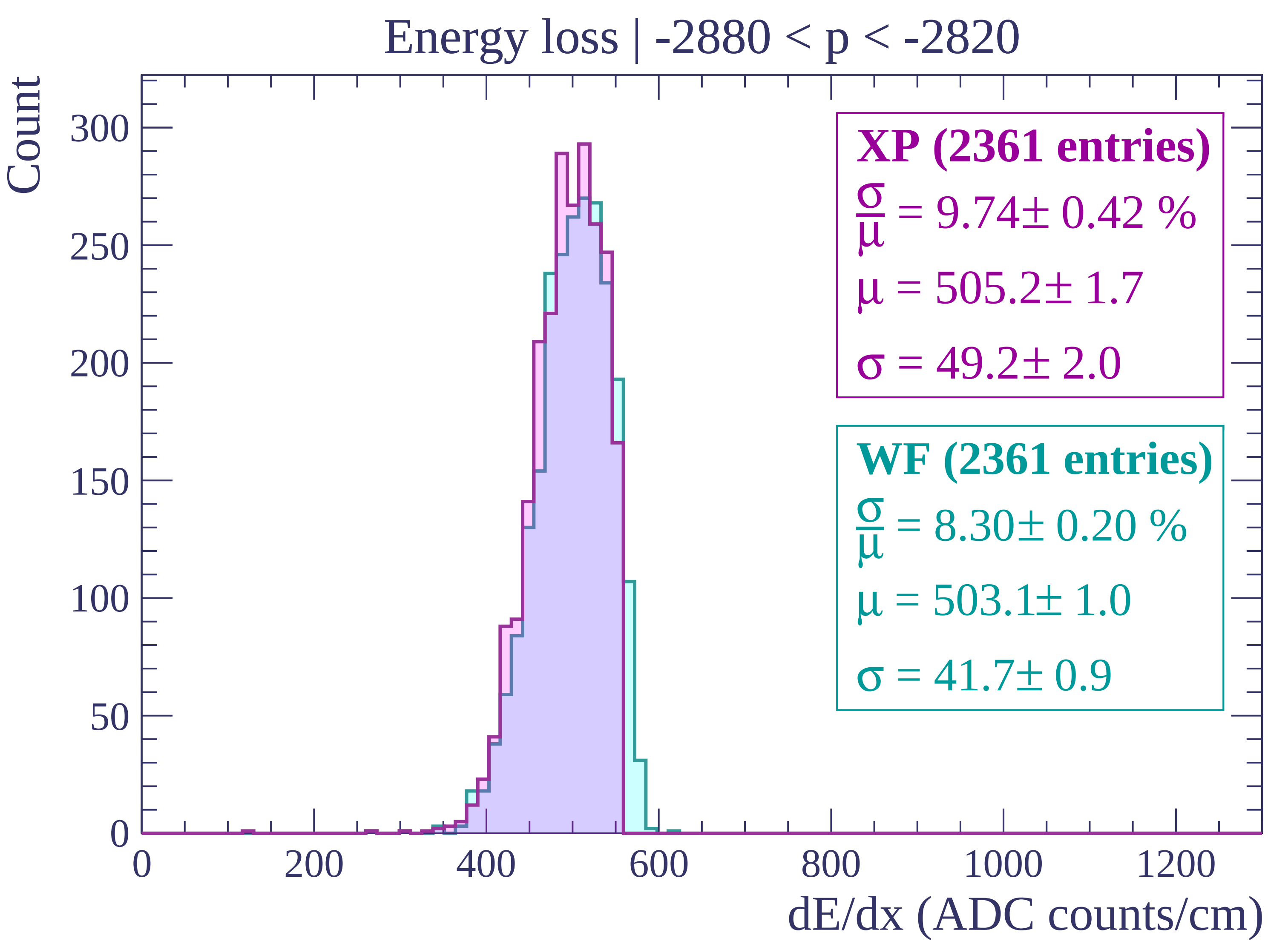


Pull e
Pull μ
10
8
6
4
2
0
−2
−4
−6
−8
−10
−10
−8
−6
−4
−2
0
2
4
6
8
10
8000
7000
6000
5000
4000
3000
2000
1000
0

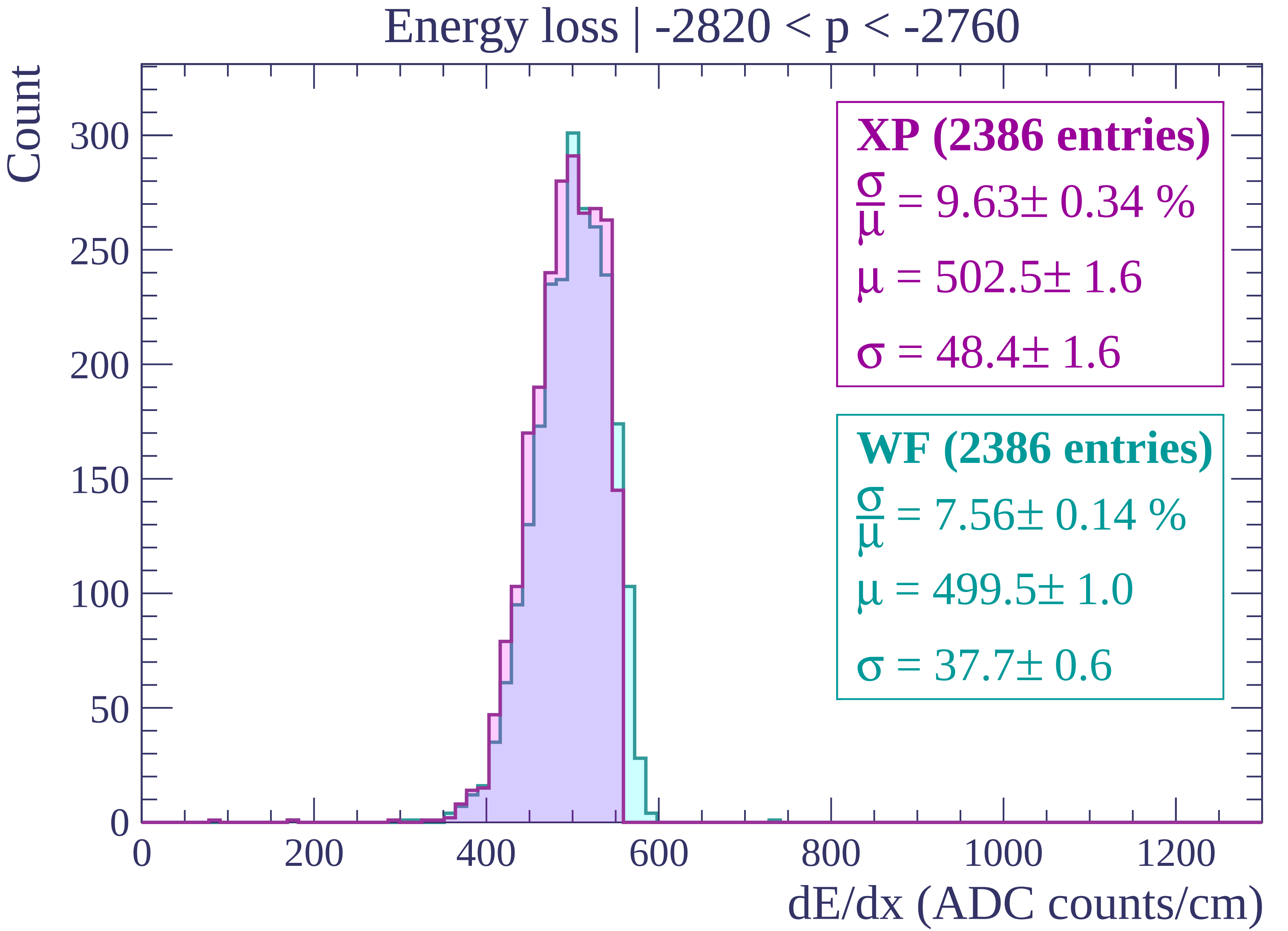


Energy loss | -3000 < p < -2940
Count
0
20
40
60
80
100
120
140
160
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP (1125 entries)
σ/μ = 9.50± 0.50 %
μ = 504.3± 2.5
σ = 47.9± 2.3
WF (1125 entries)
σ/μ = 7.72± 0.22 %
μ = 499.8± 1.5
σ = 38.6± 1.0

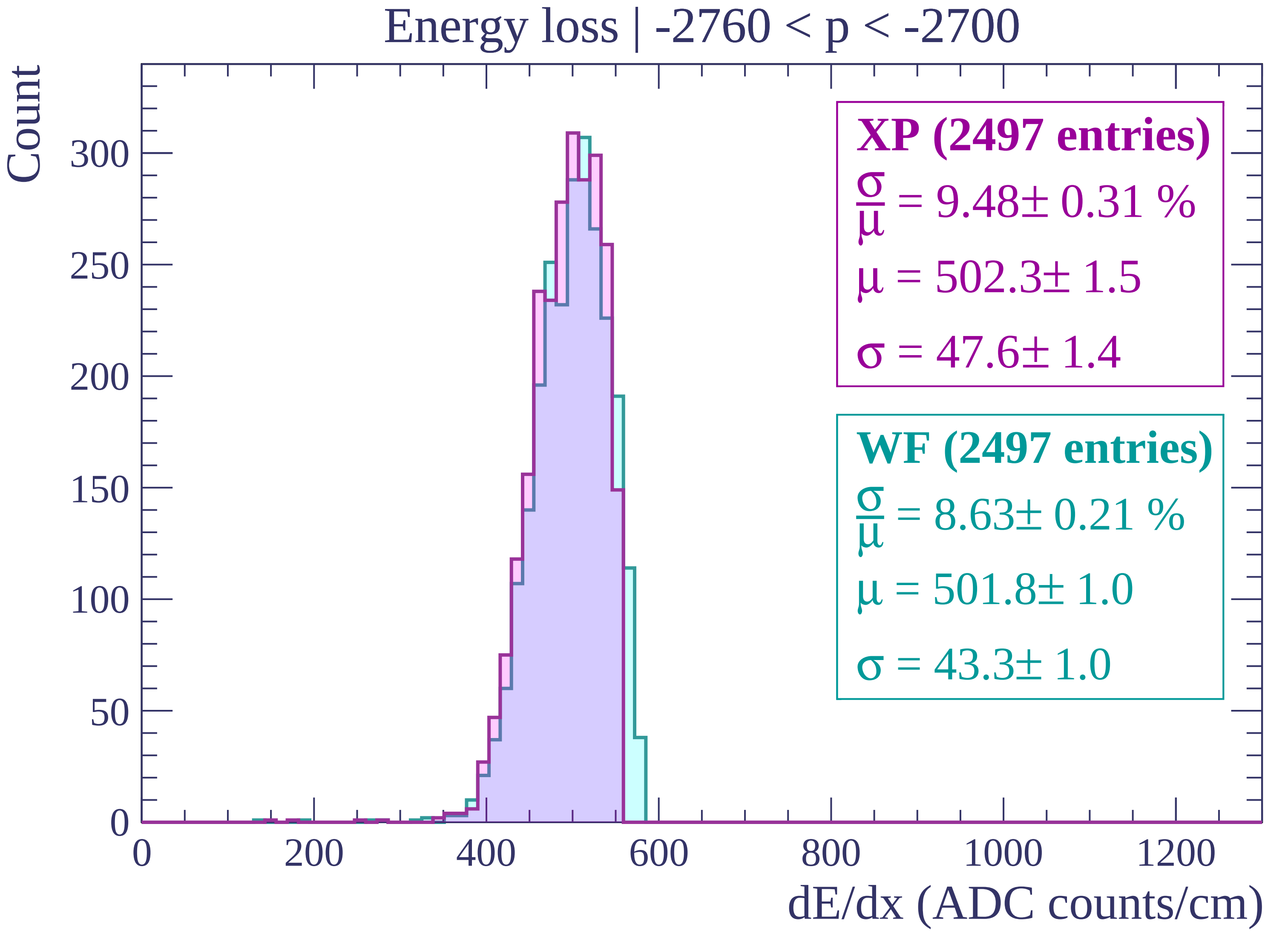

Energy loss | -2940 < p < -2880
Count
300
250
200
150
100
50
0
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP (2309 entries)
σ/μ = 9.68± 0.37 %
μ = 508.2± 1.9
σ = 49.2± 1.7
WF (2309 entries)
σ/μ = 7.24± 0.13 %
μ = 500.2± 1.0
σ = 36.2± 0.6



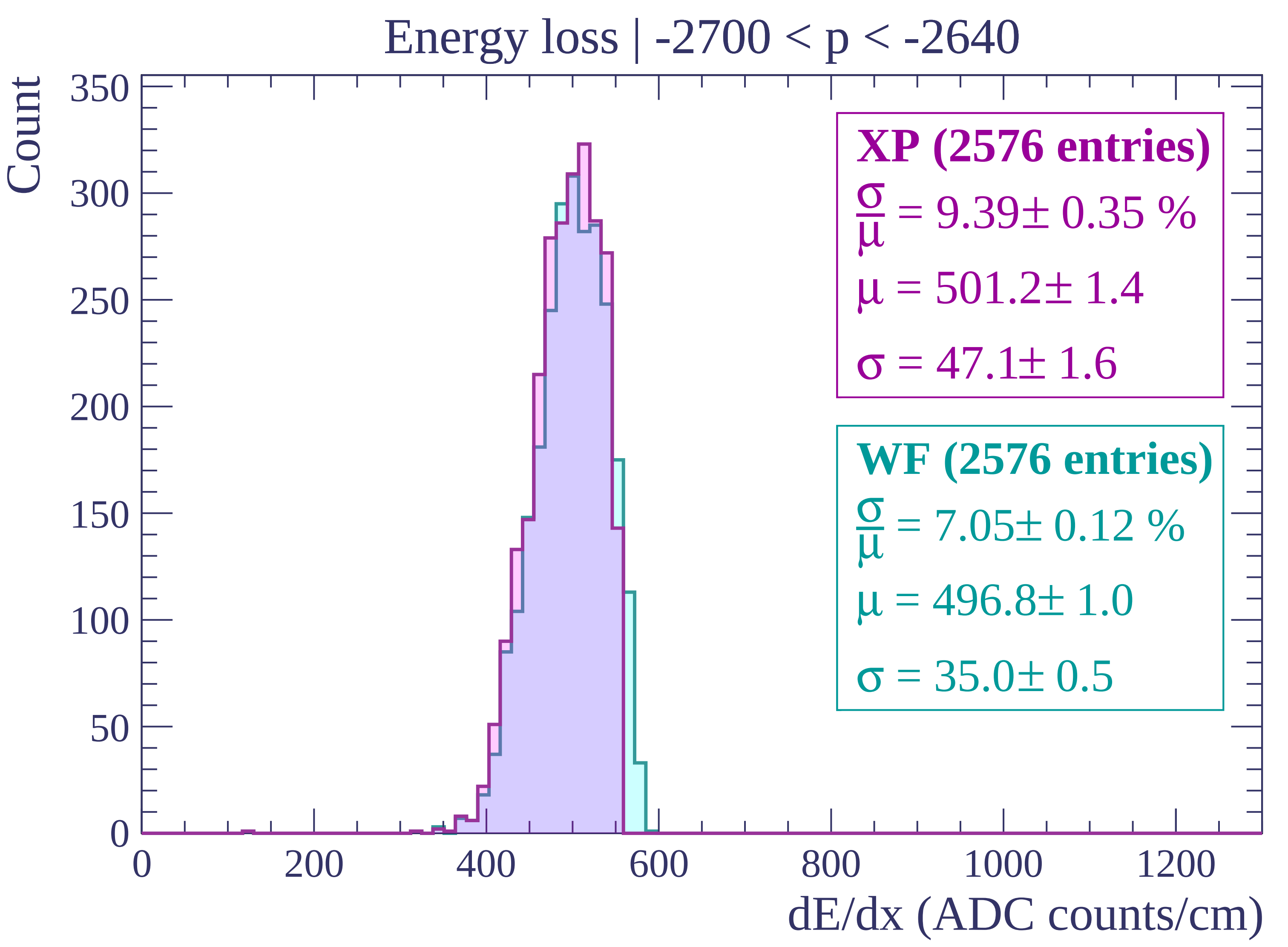
Energy loss | -2880 < p < -2820
Count
dE/dx (ADC counts/cm)
0
50
100
150
200
250
300
0
200
400
600
800
1000
1200
XP (2361 entries)
σ/μ = 9.74± 0.42 %
μ = 505.2± 1.7
σ = 49.2± 2.0
WF (2361 entries)
σ/μ = 8.30± 0.20 %
μ = 503.1± 1.0
σ = 41.7± 0.9

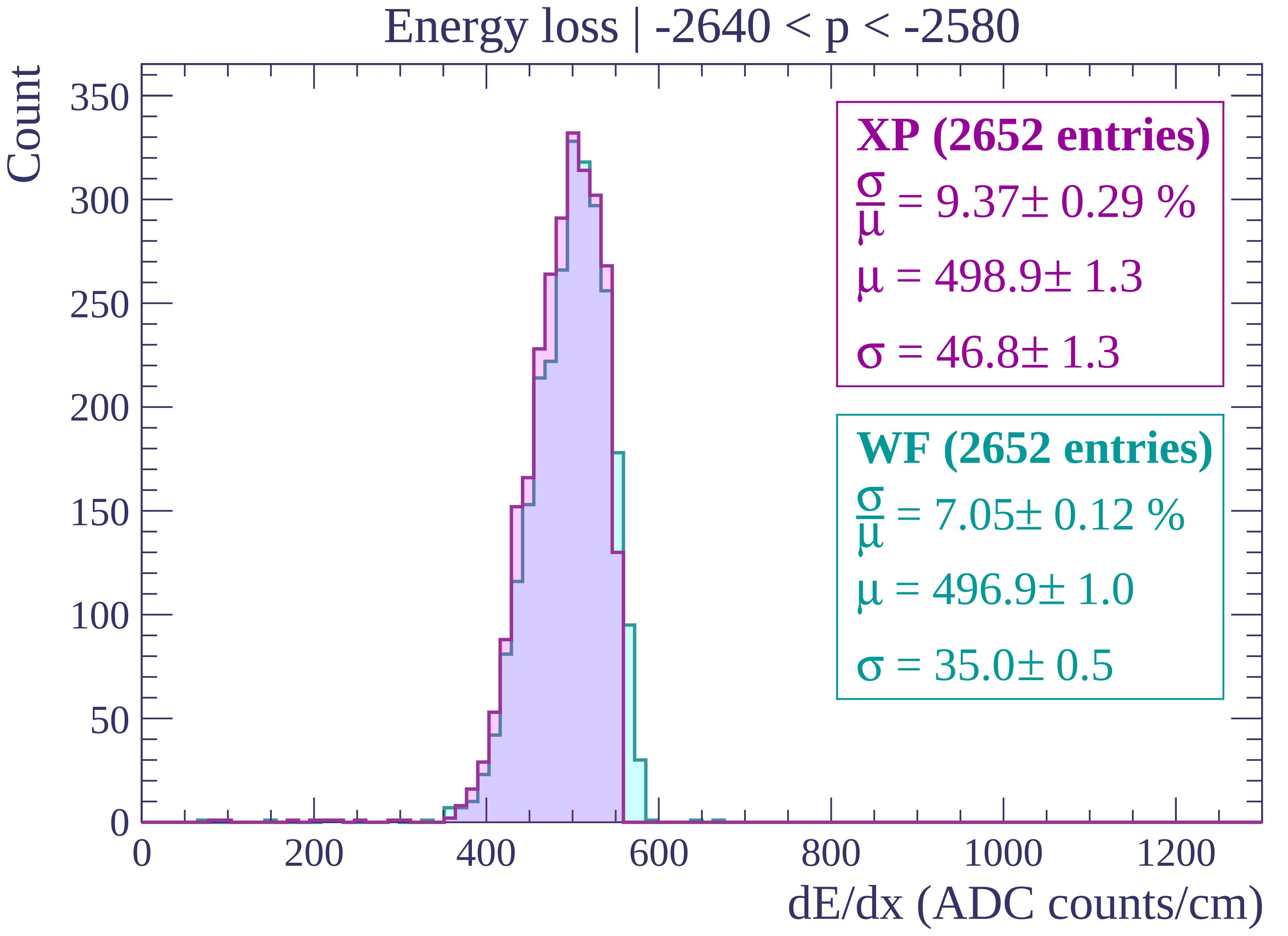


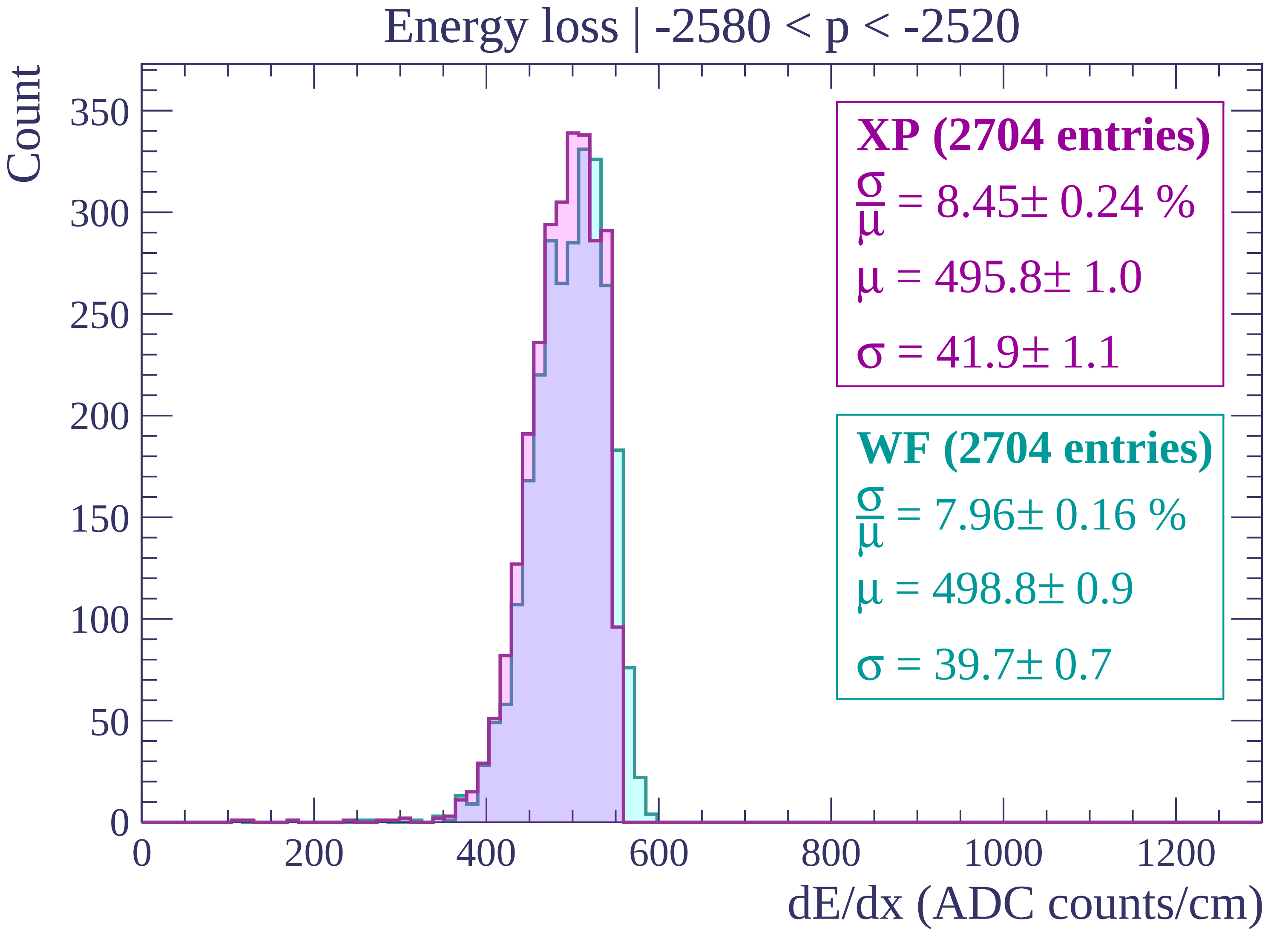
Energy loss | -2820 < p < -2760
Count
dE/dx (ADC counts/cm)
0
50
100
150
200
250
300
0
200
400
600
800
1000
1200
XP (2386 entries)
σ/μ = 9.63± 0.34 %
μ = 502.5± 1.6
σ = 48.4± 1.6
WF (2386 entries)
σ/μ = 7.56± 0.14 %
μ = 499.5± 1.0
σ = 37.7± 0.6

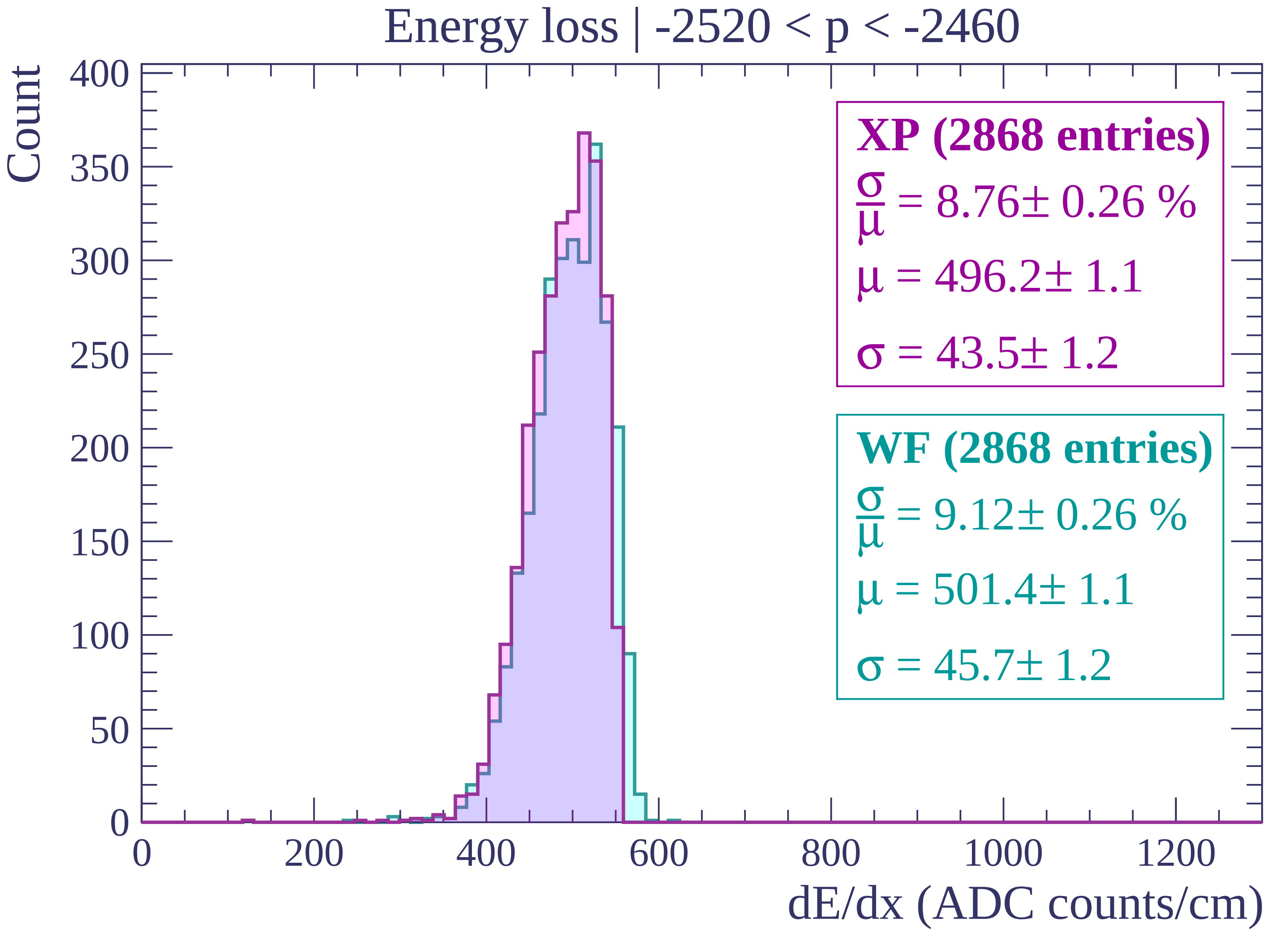


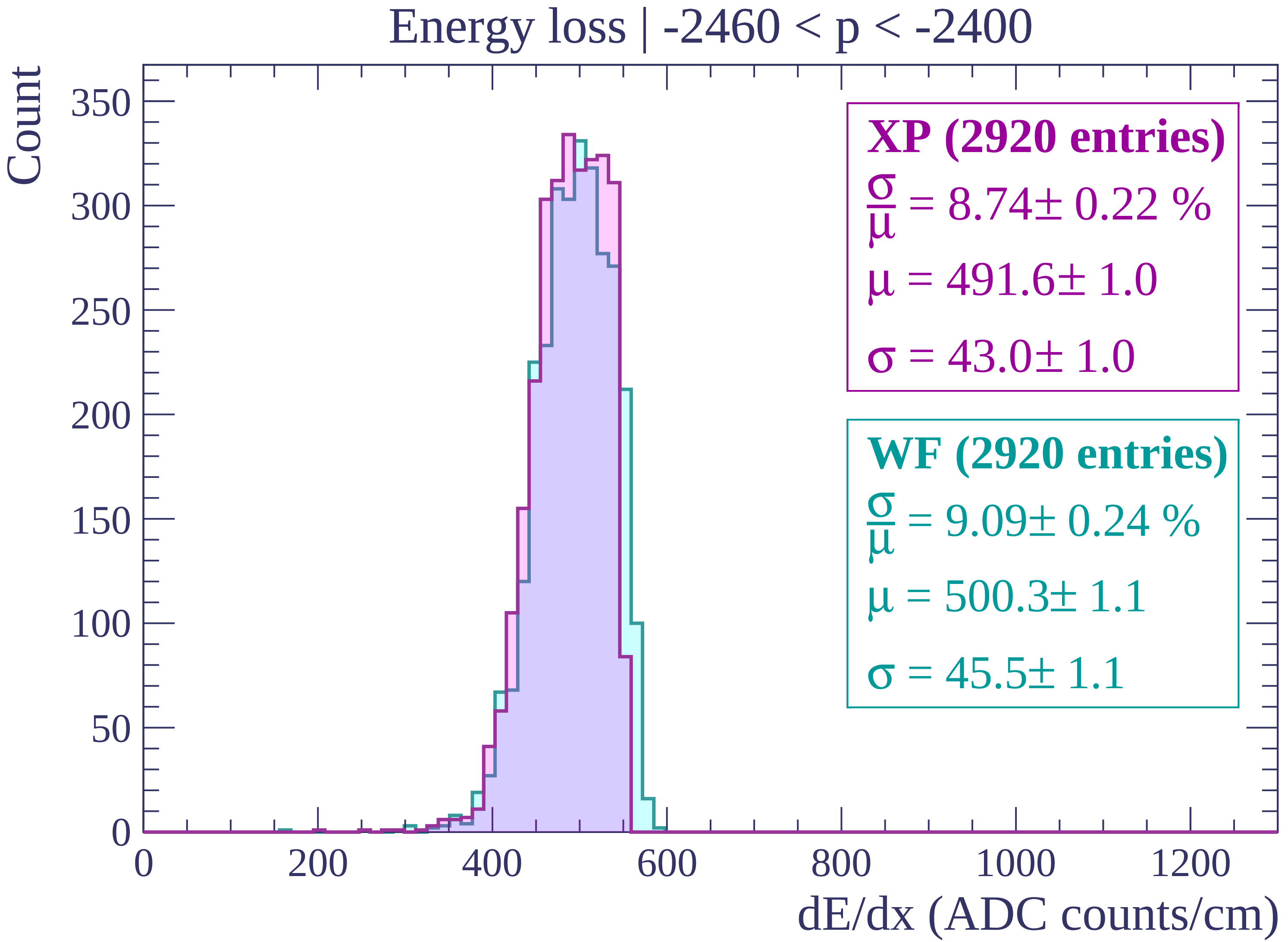
Energy loss | -2760 < p < -2700
Count
dE/dx (ADC counts/cm)
XP (2497 entries)
$\frac{\sigma}{\mu}$ = 9.48± 0.31 %
μ = 502.3± 1.5
σ = 47.6± 1.4
WF (2497 entries)
$\frac{\sigma}{\mu}$ = 8.63± 0.21 %
μ = 501.8± 1.0
σ = 43.3± 1.0
0
50
100
150
200
250
300
0
200
400
600
800
1000
1200

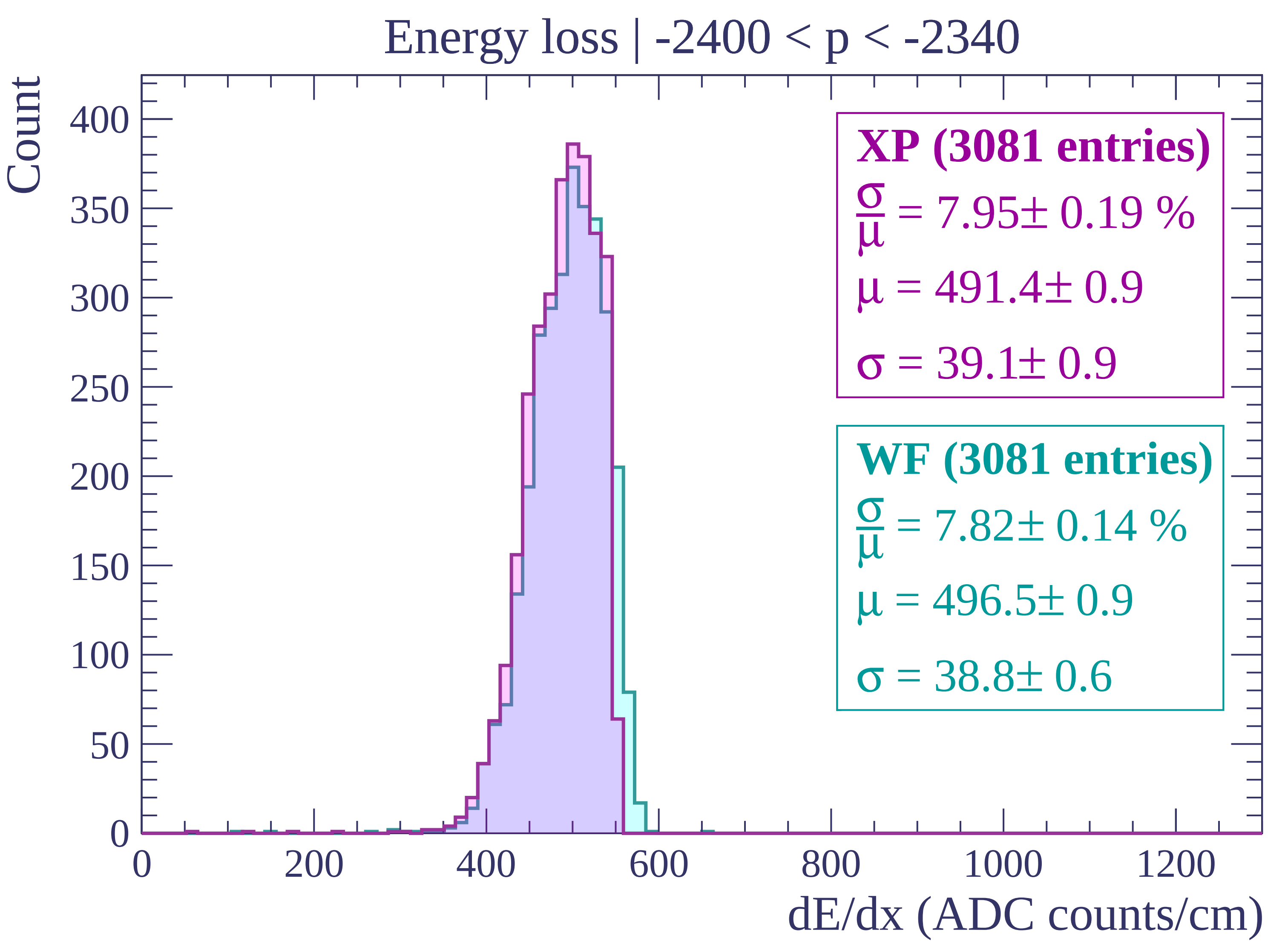

Energy loss | -2700 < p < -2640
Count
dE/dx (ADC counts/cm)
0
50
100
150
200
250
300
350
0
200
400
600
800
1000
1200
XP (2576 entries)
σ/μ = 9.39± 0.35 %
μ = 501.2± 1.4
σ = 47.1± 1.6
WF (2576 entries)
σ/μ = 7.05± 0.12 %
μ = 496.8± 1.0
σ = 35.0± 0.5

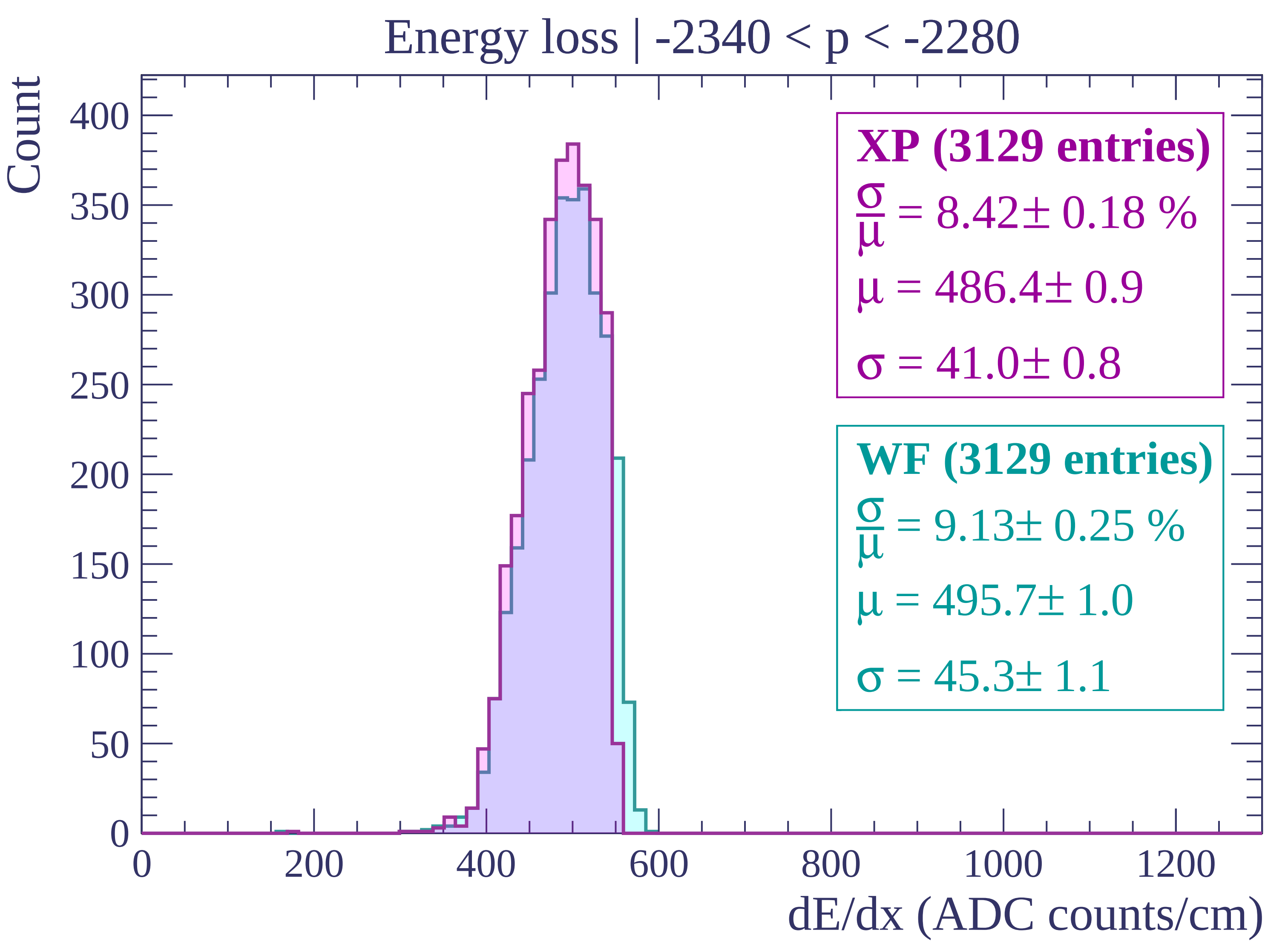

Energy loss | -2640 < p < -2580
Count
dE/dx (ADC counts/cm)
XP (2652 entries)
σ/μ = 9.37± 0.29 %
μ = 498.9± 1.3
σ = 46.8± 1.3
WF (2652 entries)
σ/μ = 7.05± 0.12 %
μ = 496.9± 1.0
σ = 35.0± 0.5
0
50
100
150
200
250
300
350
0
200
400
600
800
1000
1200


Energy loss | -2580 < p < -2520
Count
dE/dx (ADC counts/cm)
XP (2704 entries)
σ/μ = 8.45± 0.24 %
μ = 495.8± 1.0
σ = 41.9± 1.1
WF (2704 entries)
σ/μ = 7.96± 0.16 %
μ = 498.8± 0.9
σ = 39.7± 0.7

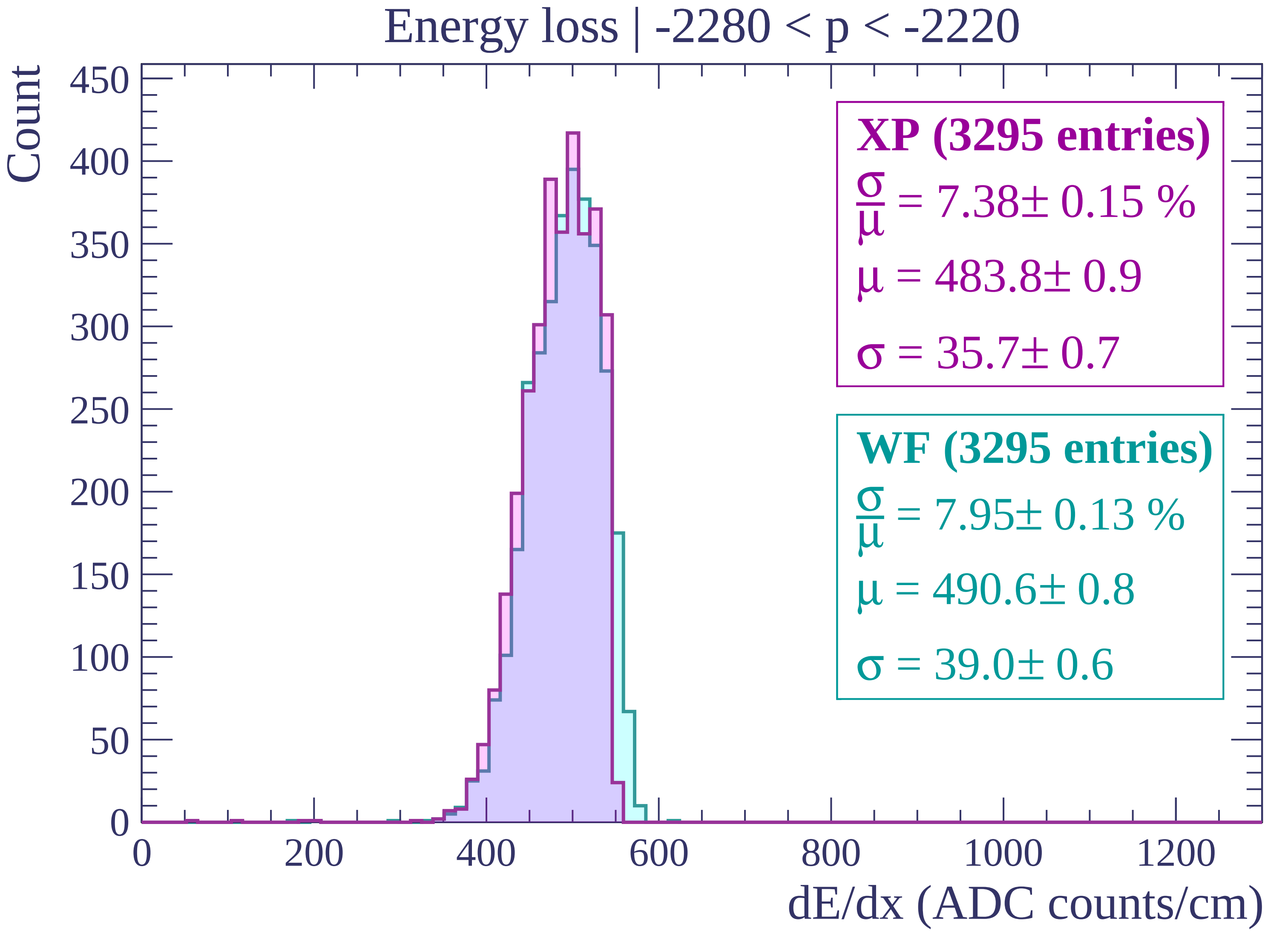

Energy loss | -2520 < p < -2460
Count
dE/dx (ADC counts/cm)
XP (2868 entries)
σ/μ = 8.76± 0.26 %
μ = 496.2± 1.1
σ = 43.5± 1.2
WF (2868 entries)
σ/μ = 9.12± 0.26 %
μ = 501.4± 1.1
σ = 45.7± 1.2
0
50
100
150
200
250
300
350
400
0
200
400
600
800
1000
1200

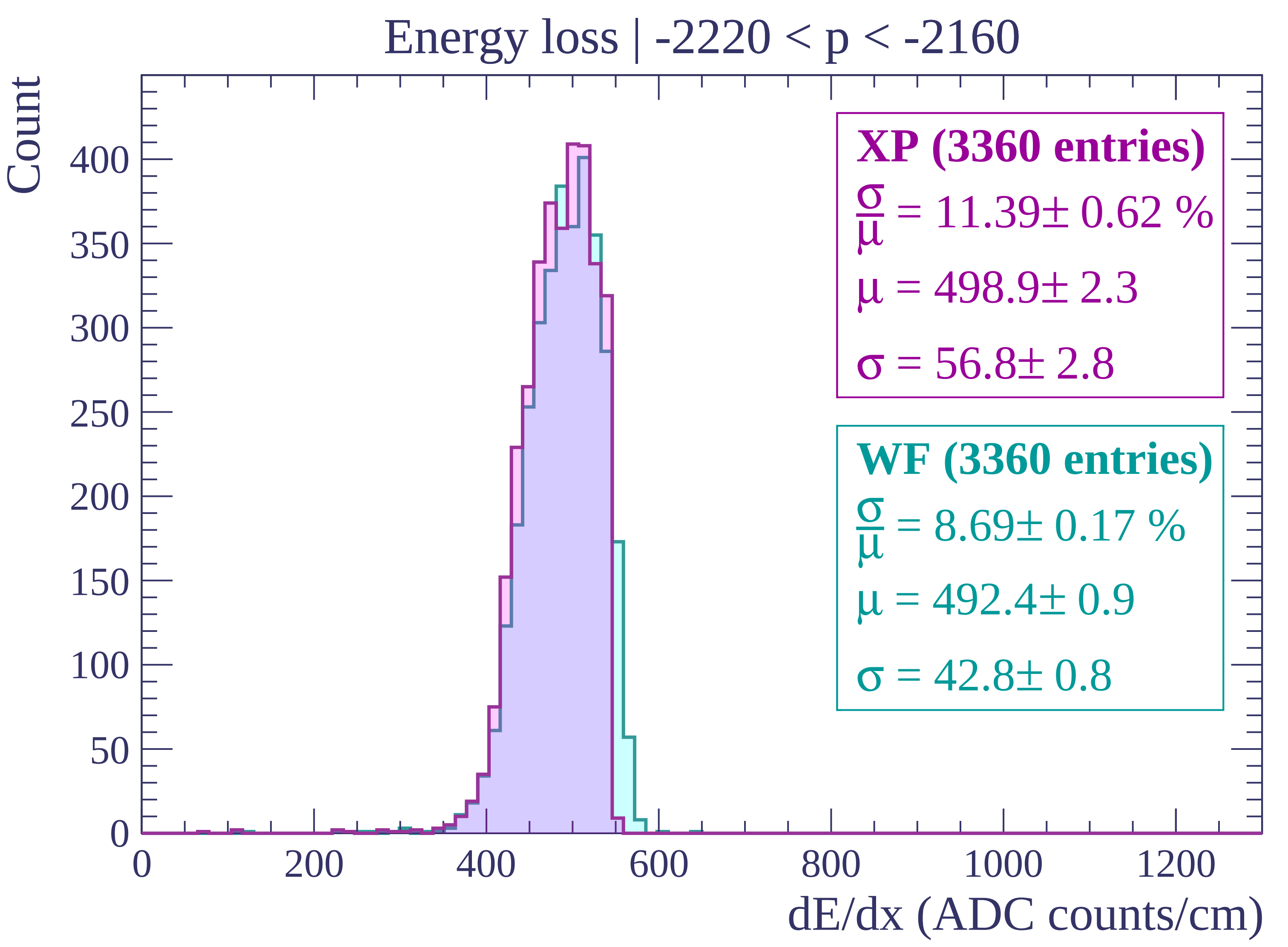

Energy loss | -2460 < p < -2400
Count
dE/dx (ADC counts/cm)
XP (2920 entries)
σ/μ = 8.74± 0.22 %
μ = 491.6± 1.0
σ = 43.0± 1.0
WF (2920 entries)
σ/μ = 9.09± 0.24 %
μ = 500.3± 1.1
σ = 45.5± 1.1
0
50
100
150
200
250
300
350
0
200
400
600
800
1000
1200

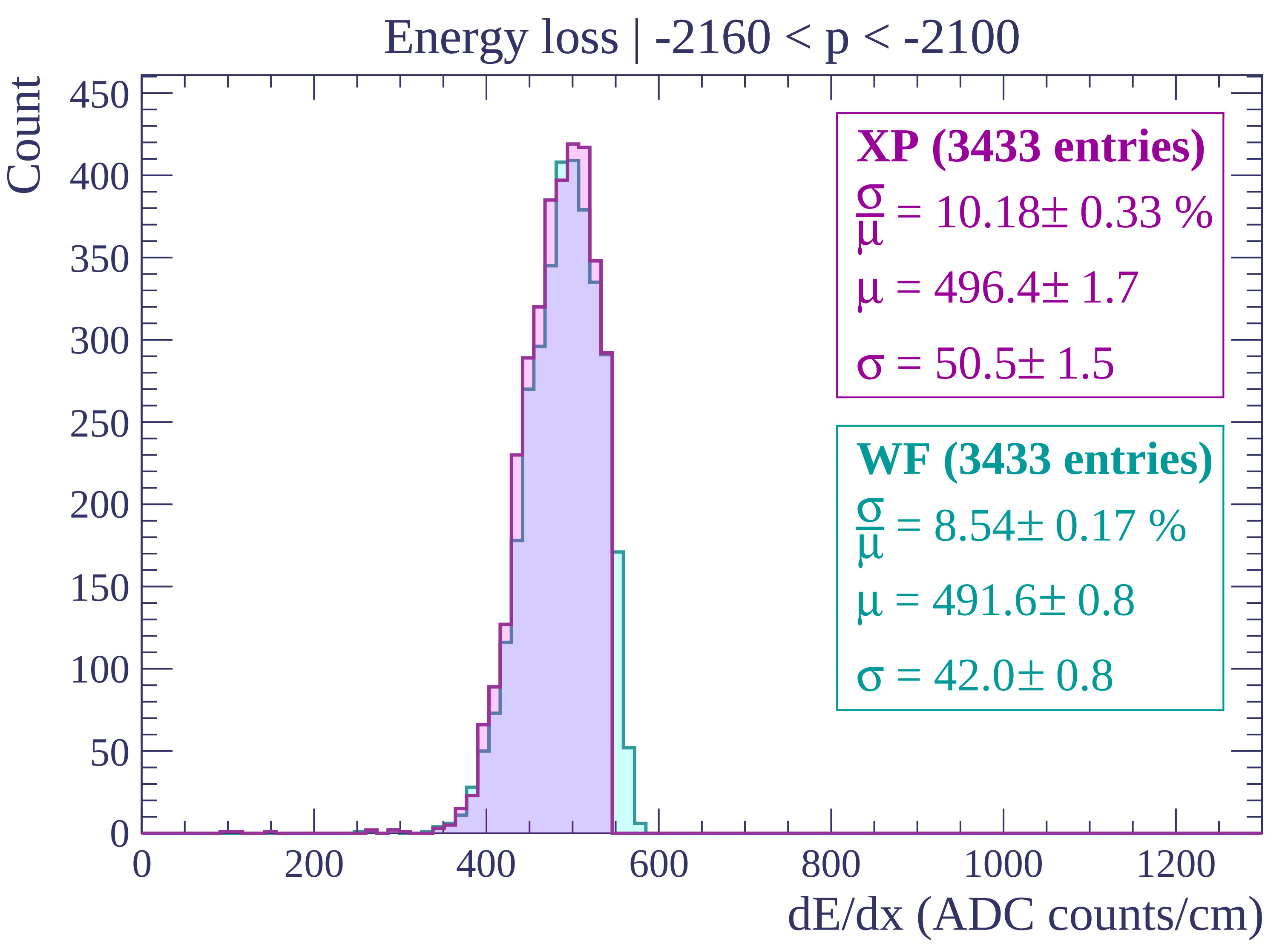

Energy loss | -2400 < p < -2340
Count
dE/dx (ADC counts/cm)
XP (3081 entries)
$\frac{\sigma}{\mu}$ = 7.95± 0.19 %
μ = 491.4± 0.9
σ = 39.1± 0.9
WF (3081 entries)
$\frac{\sigma}{\mu}$ = 7.82± 0.14 %
μ = 496.5± 0.9
σ = 38.8± 0.6
0
50
100
150
200
250
300
350
400
0
200
400
600
800
1000
1200

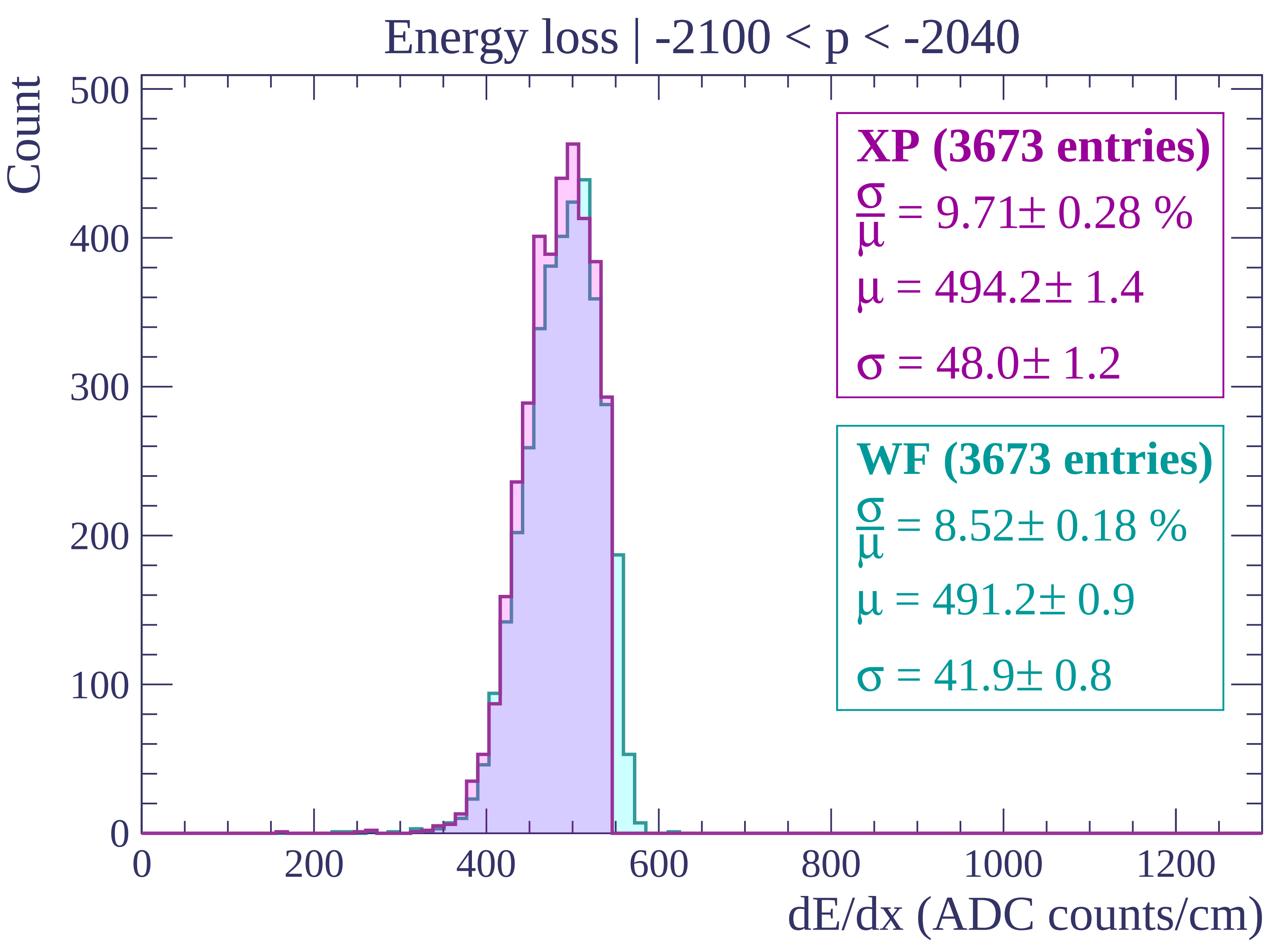


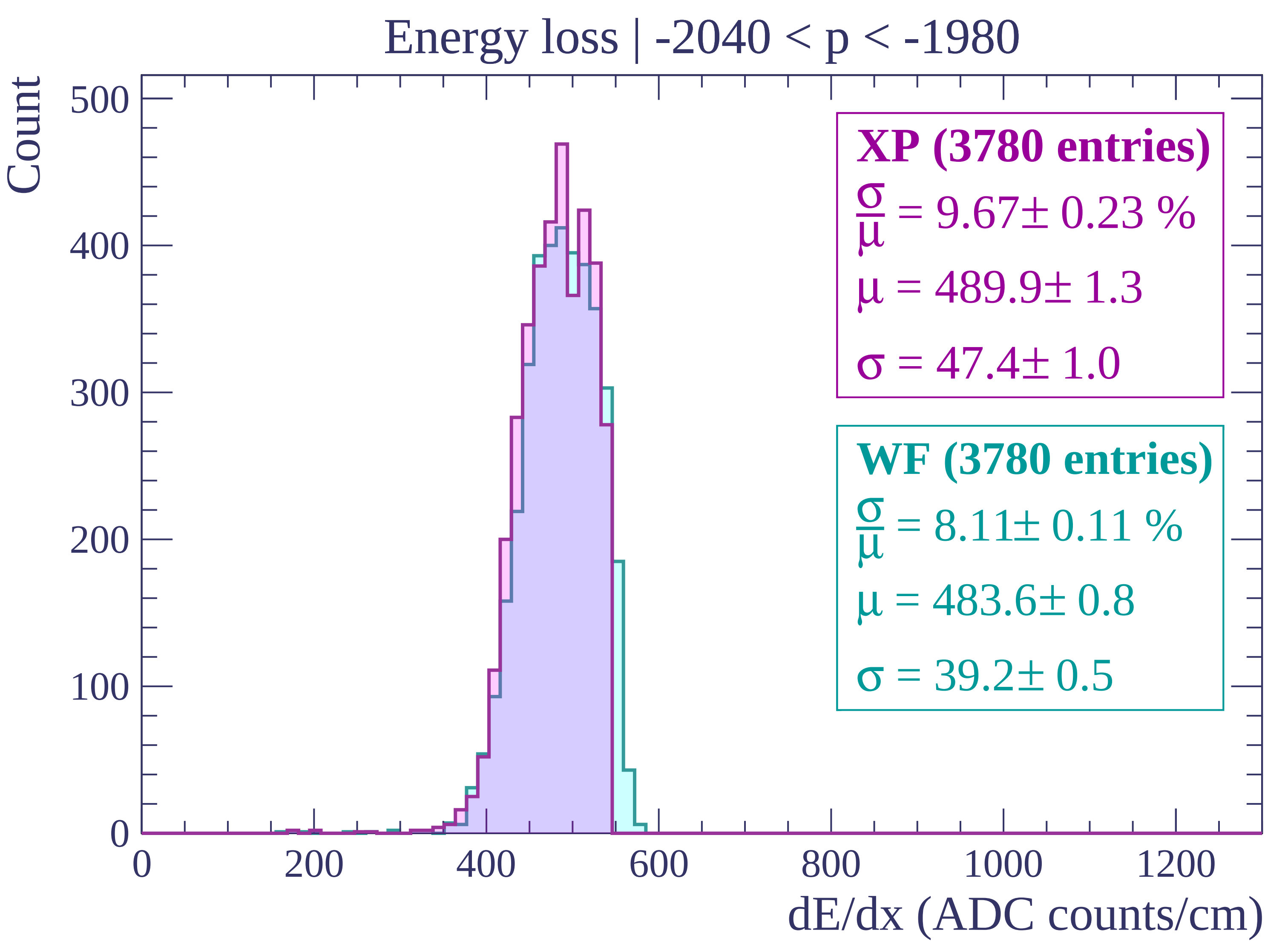
Energy loss | -2340 < p < -2280
Count
dE/dx (ADC counts/cm)
0
50
100
150
200
250
300
350
400
0
200
400
600
800
1000
1200
XP (3129 entries)
σ/μ = 8.42± 0.18 %
μ = 486.4± 0.9
σ = 41.0± 0.8
WF (3129 entries)
σ/μ = 9.13± 0.25 %
μ = 495.7± 1.0
σ = 45.3± 1.1


Energy loss | -2280 < p < -2220
Count
dE/dx (ADC counts/cm)
0
50
100
150
200
250
300
350
400
450
0
200
400
600
800
1000
1200
XP (3295 entries)
$\frac{\sigma}{\mu}$ = 7.38± 0.15 %
μ = 483.8± 0.9
σ = 35.7± 0.7
WF (3295 entries)
$\frac{\sigma}{\mu}$ = 7.95± 0.13 %
μ = 490.6± 0.8
σ = 39.0± 0.6

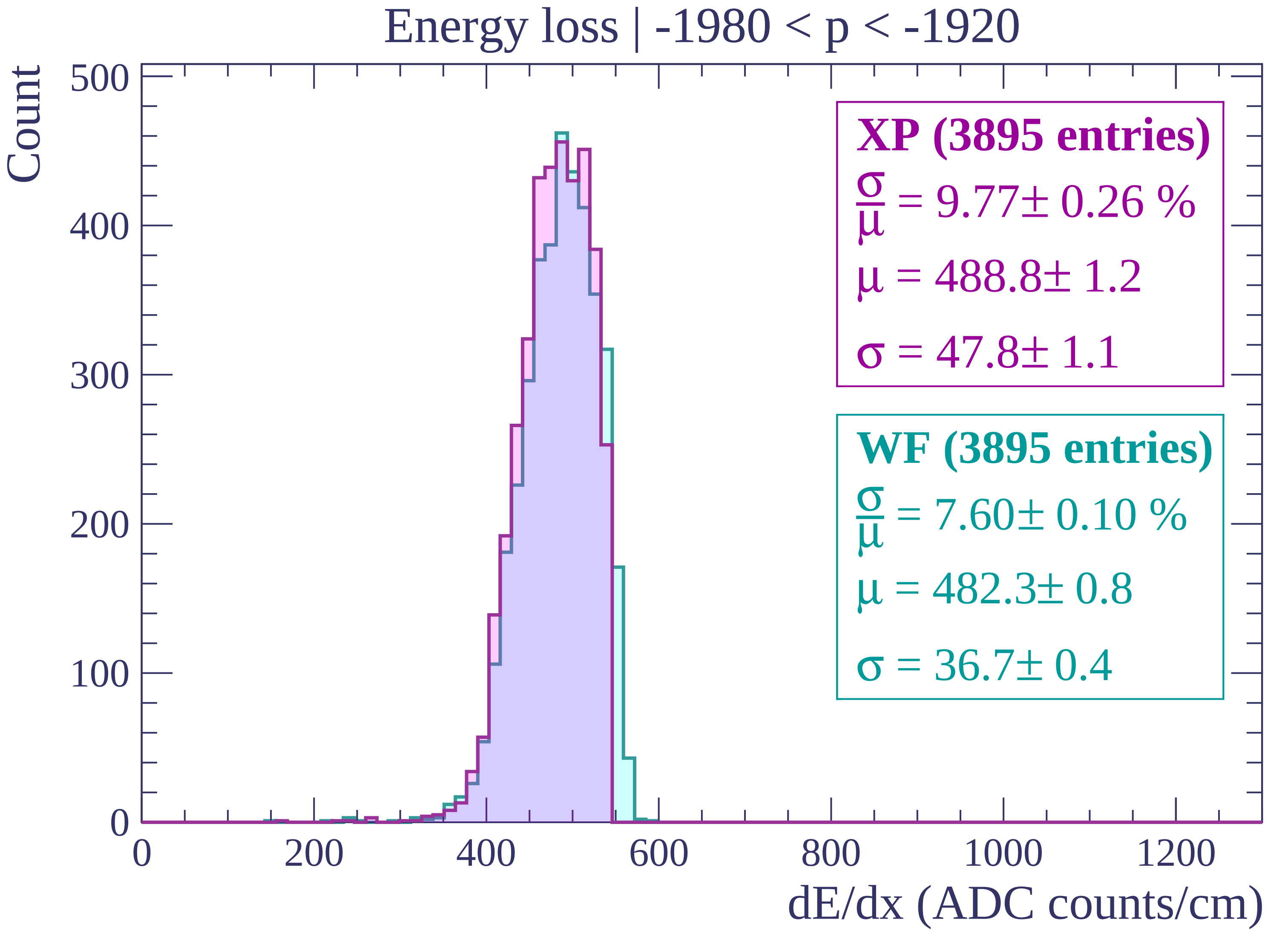

Energy loss | -2220 < p < -2160
Count
dE/dx (ADC counts/cm)
0
50
100
150
200
250
300
350
400
0
200
400
600
800
1000
1200
XP (3360 entries)
$\frac{\sigma}{\mu}$ = 11.39± 0.62 %
μ = 498.9± 2.3
σ = 56.8± 2.8
WF (3360 entries)
$\frac{\sigma}{\mu}$ = 8.69± 0.17 %
μ = 492.4± 0.9
σ = 42.8± 0.8

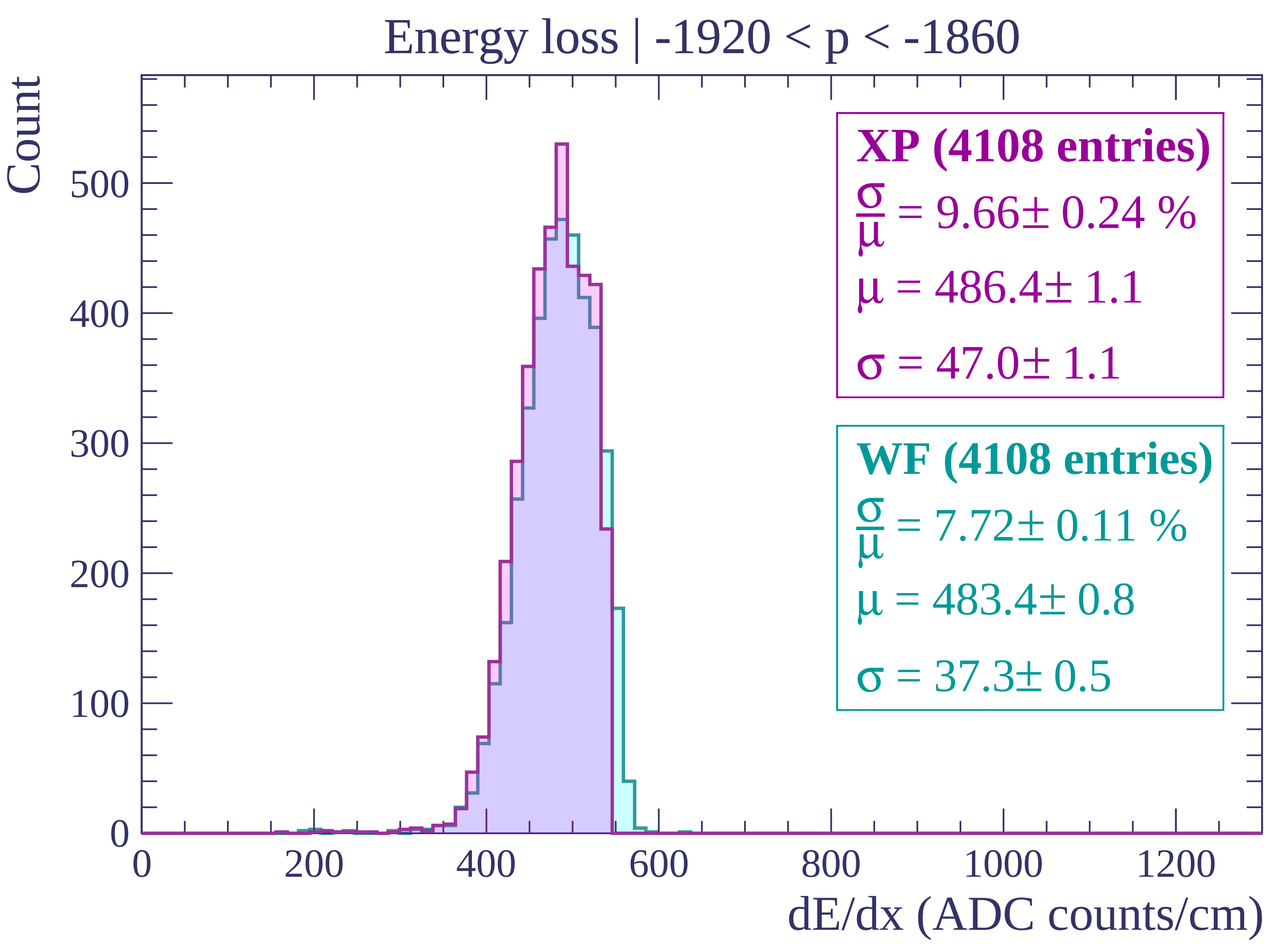

Energy loss | -2160 < p < -2100
Count
dE/dx (ADC counts/cm)
0
50
100
150
200
250
300
350
400
450
0
200
400
600
800
1000
1200
XP (3433 entries)
σ/μ = 10.18± 0.33 %
μ = 496.4± 1.7
σ = 50.5± 1.5
WF (3433 entries)
σ/μ = 8.54± 0.17 %
μ = 491.6± 0.8
σ = 42.0± 0.8

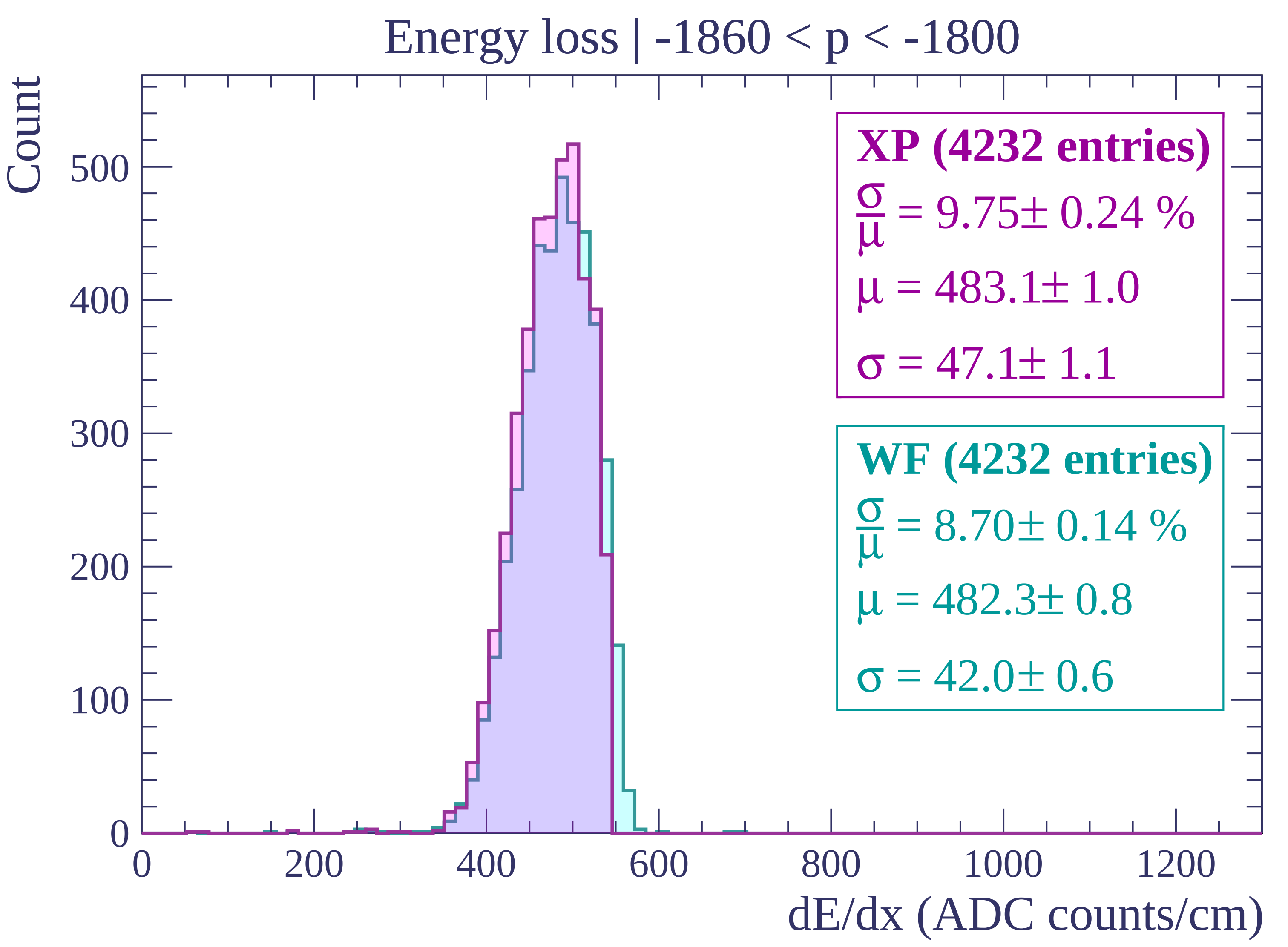

Energy loss | -2100 < p < -2040
Count
XP (3673 entries)
σ/μ = 9.71± 0.28 %
μ = 494.2± 1.4
σ = 48.0± 1.2
WF (3673 entries)
σ/μ = 8.52± 0.18 %
μ = 491.2± 0.9
σ = 41.9± 0.8
0
100
200
300
400
500
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)



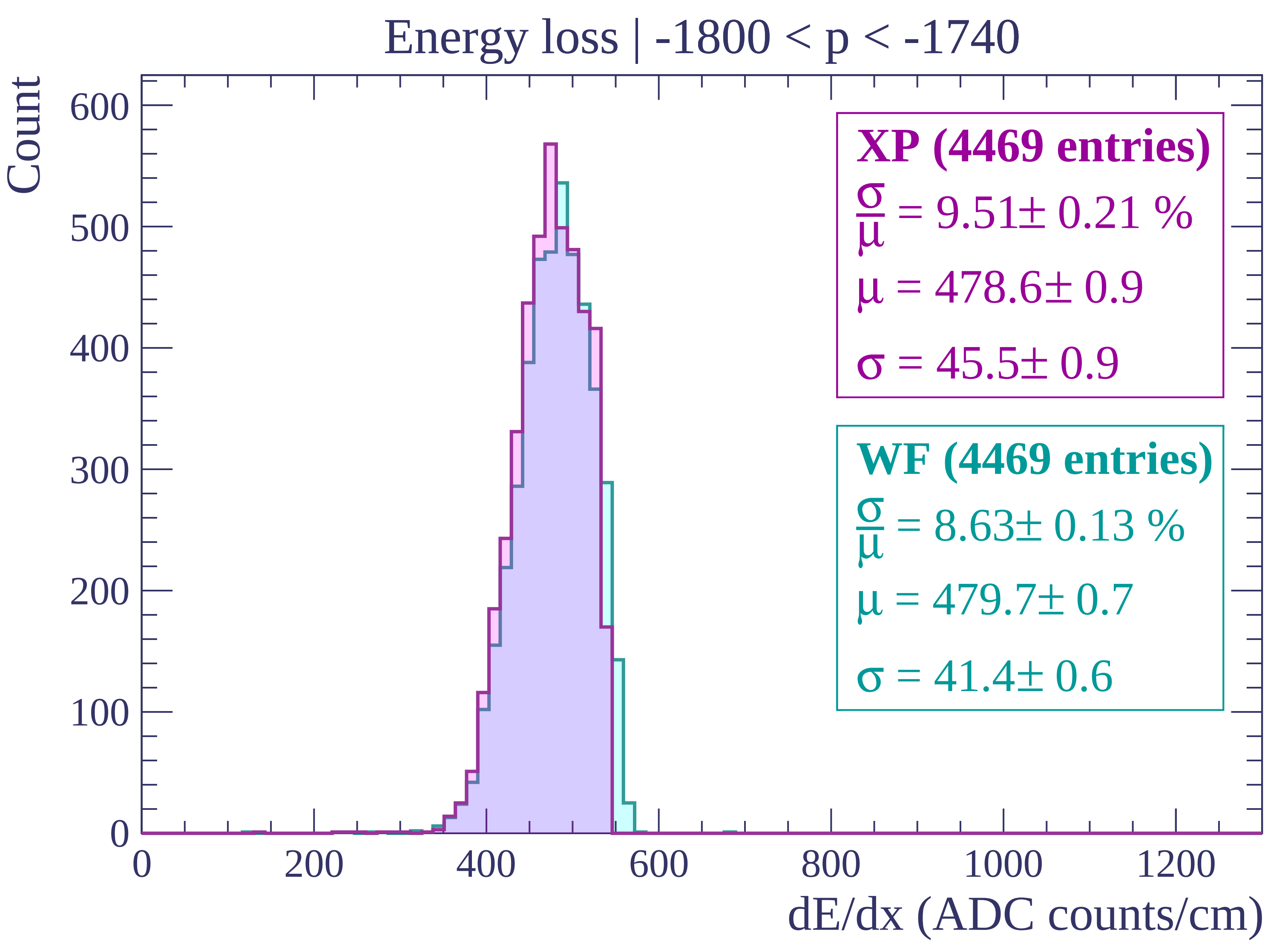
Energy loss | -2040 < p < -1980
Count
500
400
300
200
100
0
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP (3780 entries)
σ/μ = 9.67± 0.23 %
μ = 489.9± 1.3
σ = 47.4± 1.0
WF (3780 entries)
σ/μ = 8.11± 0.11 %
μ = 483.6± 0.8
σ = 39.2± 0.5

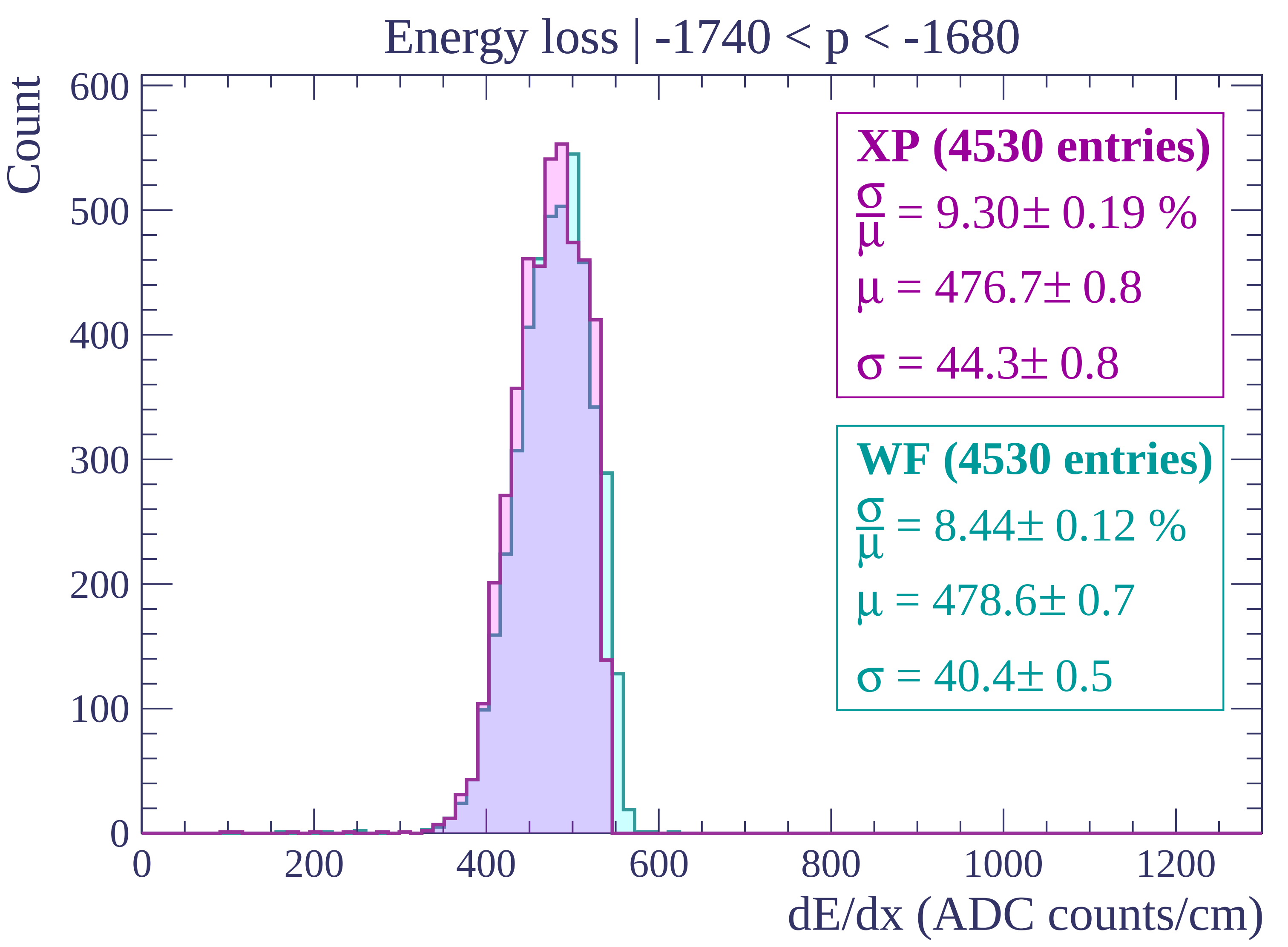

Energy loss | -1980 < p < -1920
Count
0
100
200
300
400
500
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP (3895 entries)
$\frac{\sigma}{\mu}$ = 9.77± 0.26 %
μ = 488.8± 1.2
σ = 47.8± 1.1
WF (3895 entries)
$\frac{\sigma}{\mu}$ = 7.60± 0.10 %
μ = 482.3± 0.8
σ = 36.7± 0.4


Energy loss | -1920 < p < -1860
Count
dE/dx (ADC counts/cm)
0
100
200
300
400
500
0
200
400
600
800
1000
1200
XP (4108 entries)
$\frac{\sigma}{\mu}$ = 9.66± 0.24 %
μ = 486.4± 1.1
σ = 47.0± 1.1
WF (4108 entries)
$\frac{\sigma}{\mu}$ = 7.72± 0.11 %
μ = 483.4± 0.8
σ = 37.3± 0.5

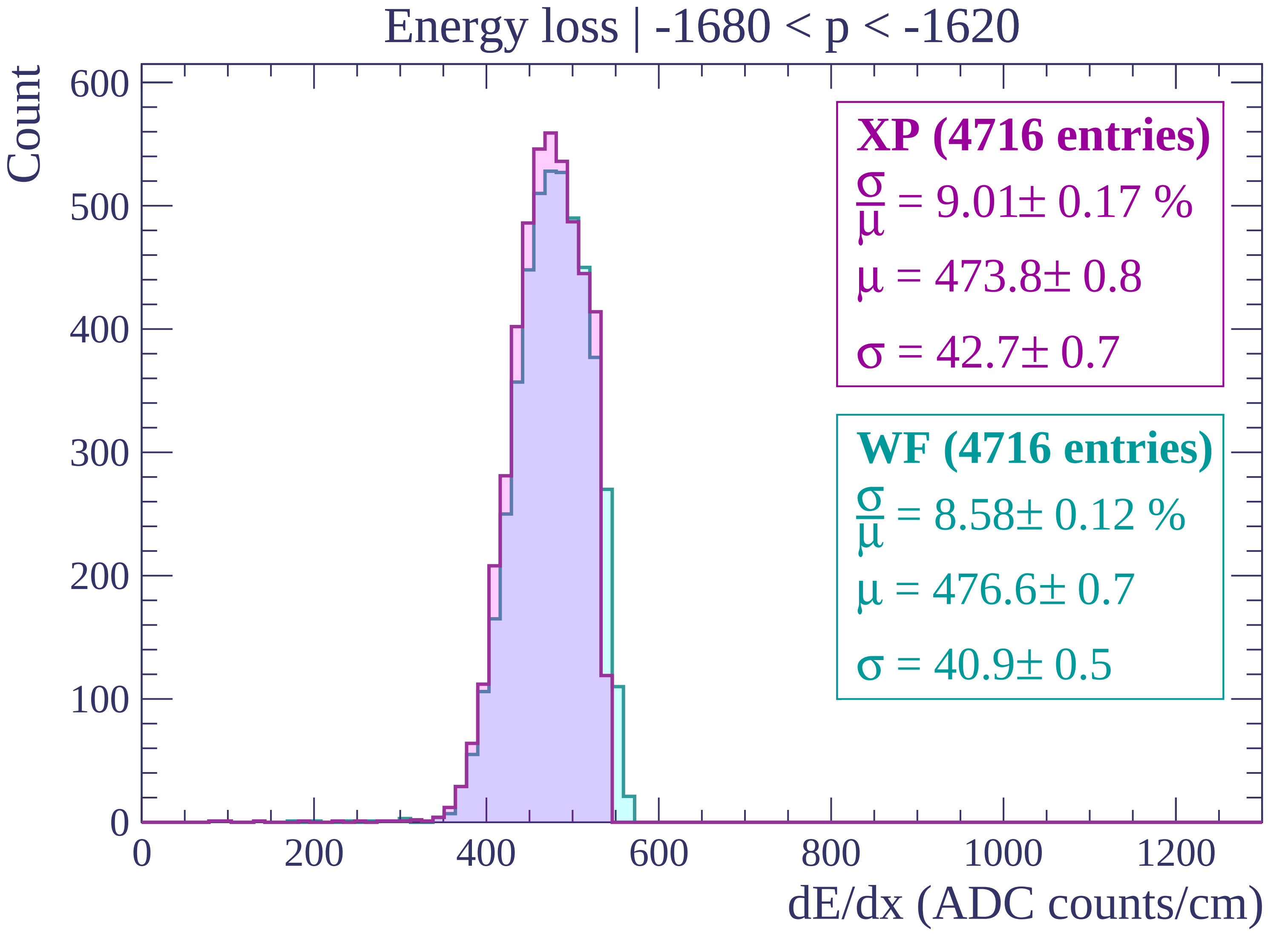


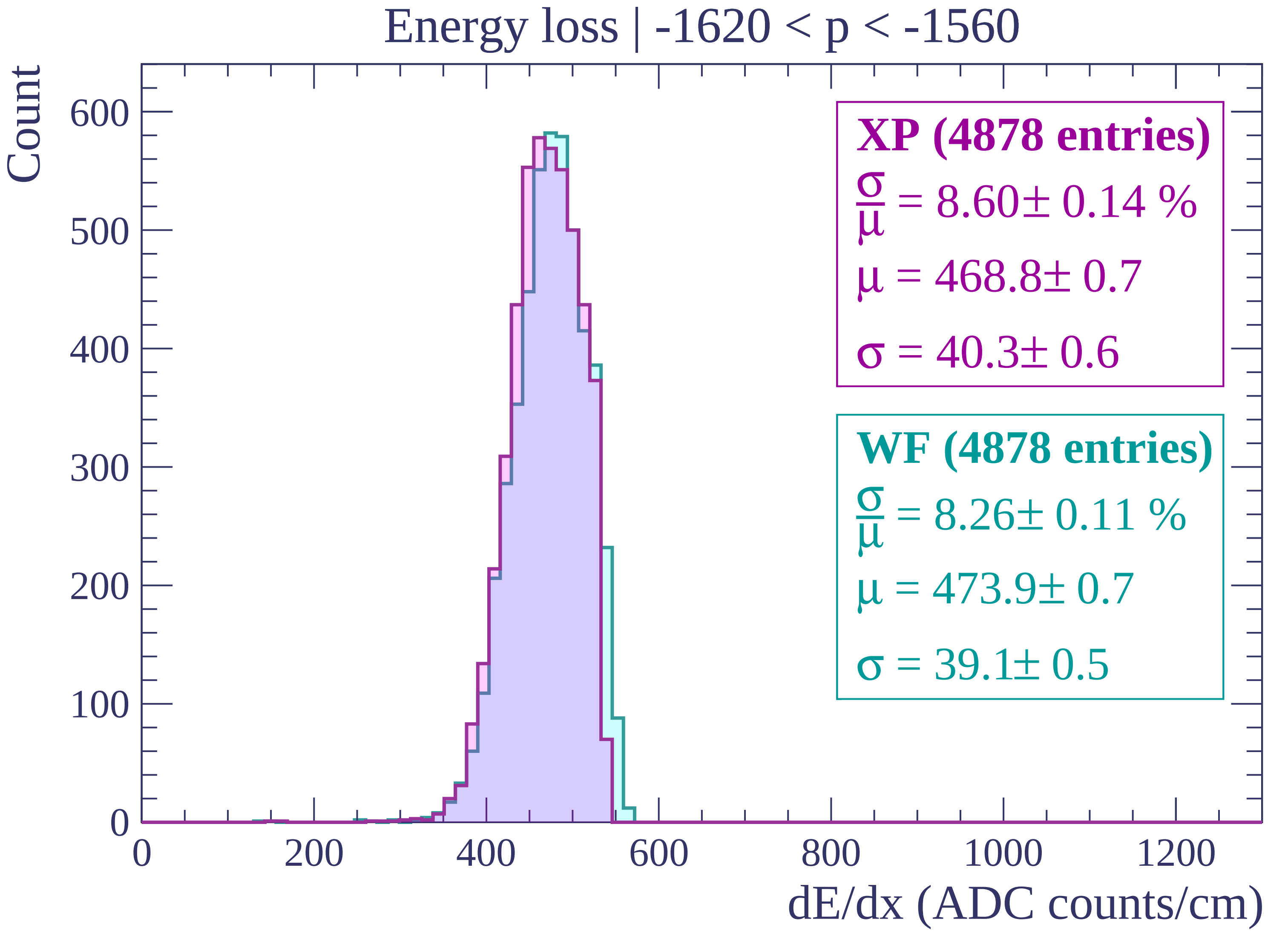
Energy loss | -1860 < p < -1800
Count
0
100
200
300
400
500
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP (4232 entries)
$\frac{\sigma}{\mu}$ = 9.75± 0.24 %
μ = 483.1± 1.0
σ = 47.1± 1.1
WF (4232 entries)
$\frac{\sigma}{\mu}$ = 8.70± 0.14 %
μ = 482.3± 0.8
σ = 42.0± 0.6

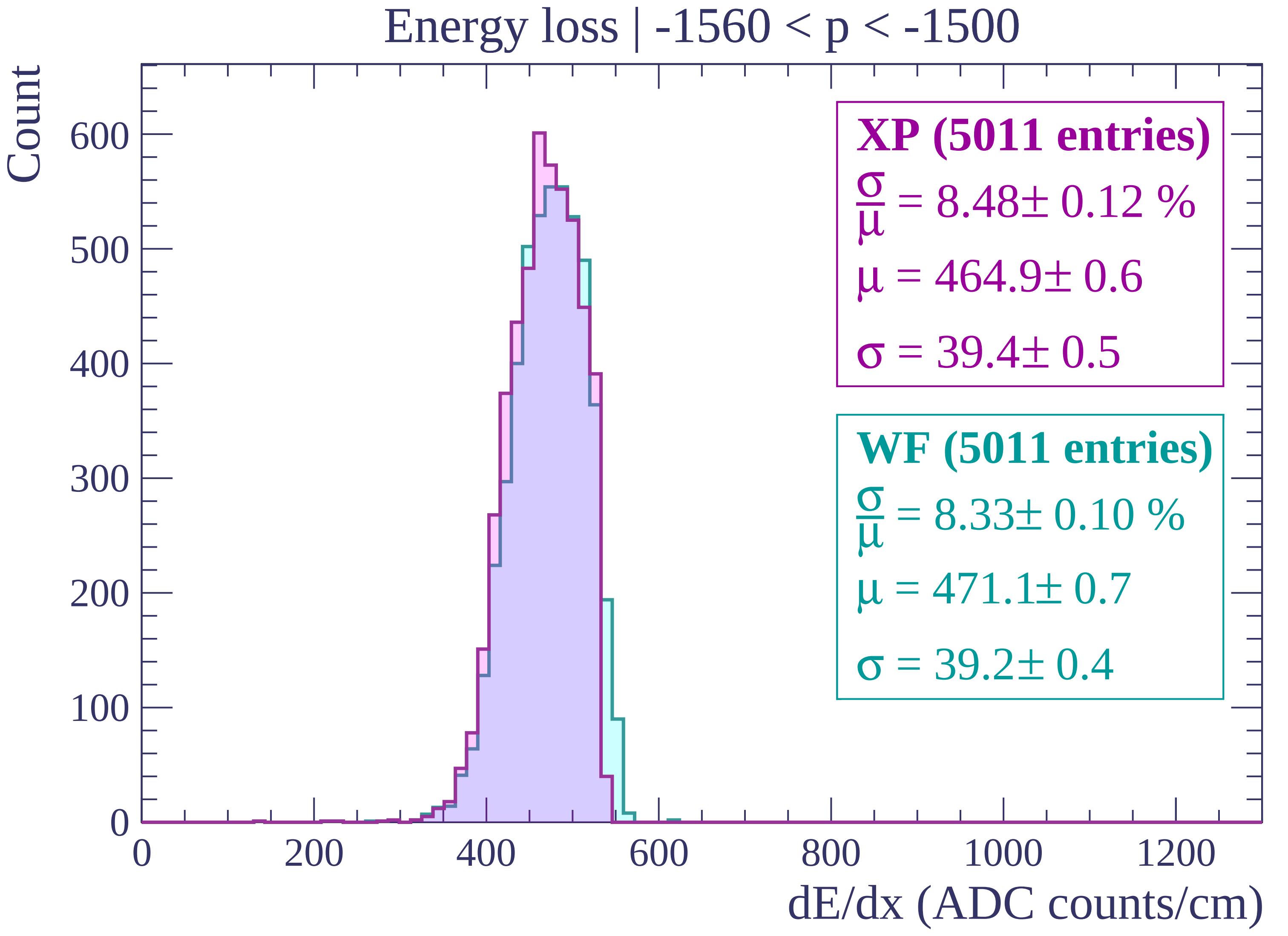

Energy loss | -1800 < p < -1740
Count
600
500
400
300
200
100
0
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP (4469 entries)
σ/μ = 9.51± 0.21 %
μ = 478.6± 0.9
σ = 45.5± 0.9
WF (4469 entries)
σ/μ = 8.63± 0.13 %
μ = 479.7± 0.7
σ = 41.4± 0.6

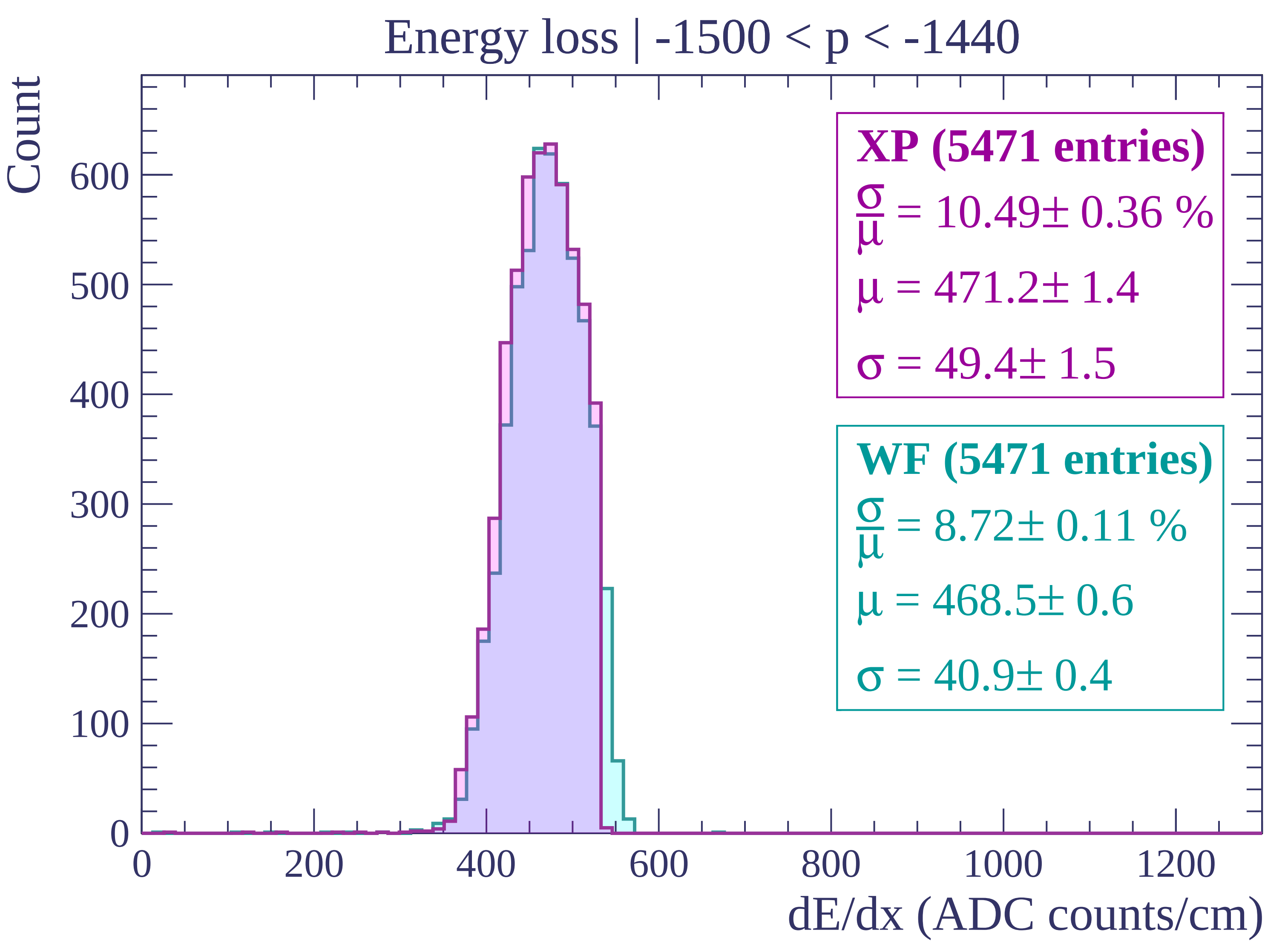


Energy loss | -1740 < p < -1680
Count
dE/dx (ADC counts/cm)
0
100
200
300
400
500
600
0
200
400
600
800
1000
1200
XP (4530 entries)
$\frac{\sigma}{\mu}$ = 9.30± 0.19 %
μ = 476.7± 0.8
σ = 44.3± 0.8
WF (4530 entries)
$\frac{\sigma}{\mu}$ = 8.44± 0.12 %
μ = 478.6± 0.7
σ = 40.4± 0.5

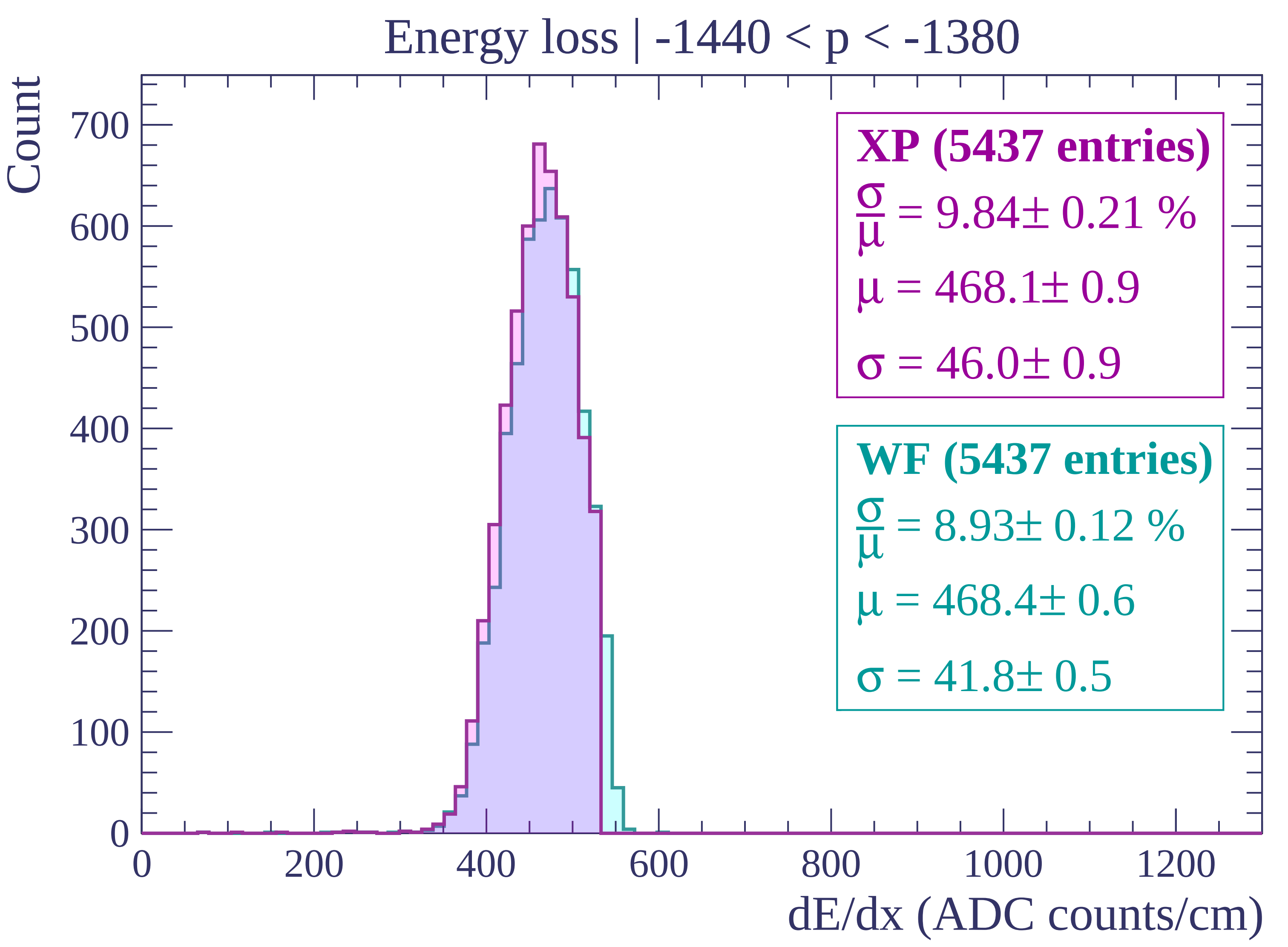

Energy loss | -1680 < p < -1620
Count
dE/dx (ADC counts/cm)
0
100
200
300
400
500
600
0
200
400
600
800
1000
1200
XP (4716 entries)
$\frac{\sigma}{\mu}$ = 9.01± 0.17 %
μ = 473.8± 0.8
σ = 42.7± 0.7
WF (4716 entries)
$\frac{\sigma}{\mu}$ = 8.58± 0.12 %
μ = 476.6± 0.7
σ = 40.9± 0.5


Energy loss | -1620 < p < -1560
Count
dE/dx (ADC counts/cm)
0
100
200
300
400
500
600
0
200
400
600
800
1000
1200
XP (4878 entries)
$\frac{\sigma}{\mu}$ = 8.60± 0.14 %
μ = 468.8± 0.7
σ = 40.3± 0.6
WF (4878 entries)
$\frac{\sigma}{\mu}$ = 8.26± 0.11 %
μ = 473.9± 0.7
σ = 39.1± 0.5

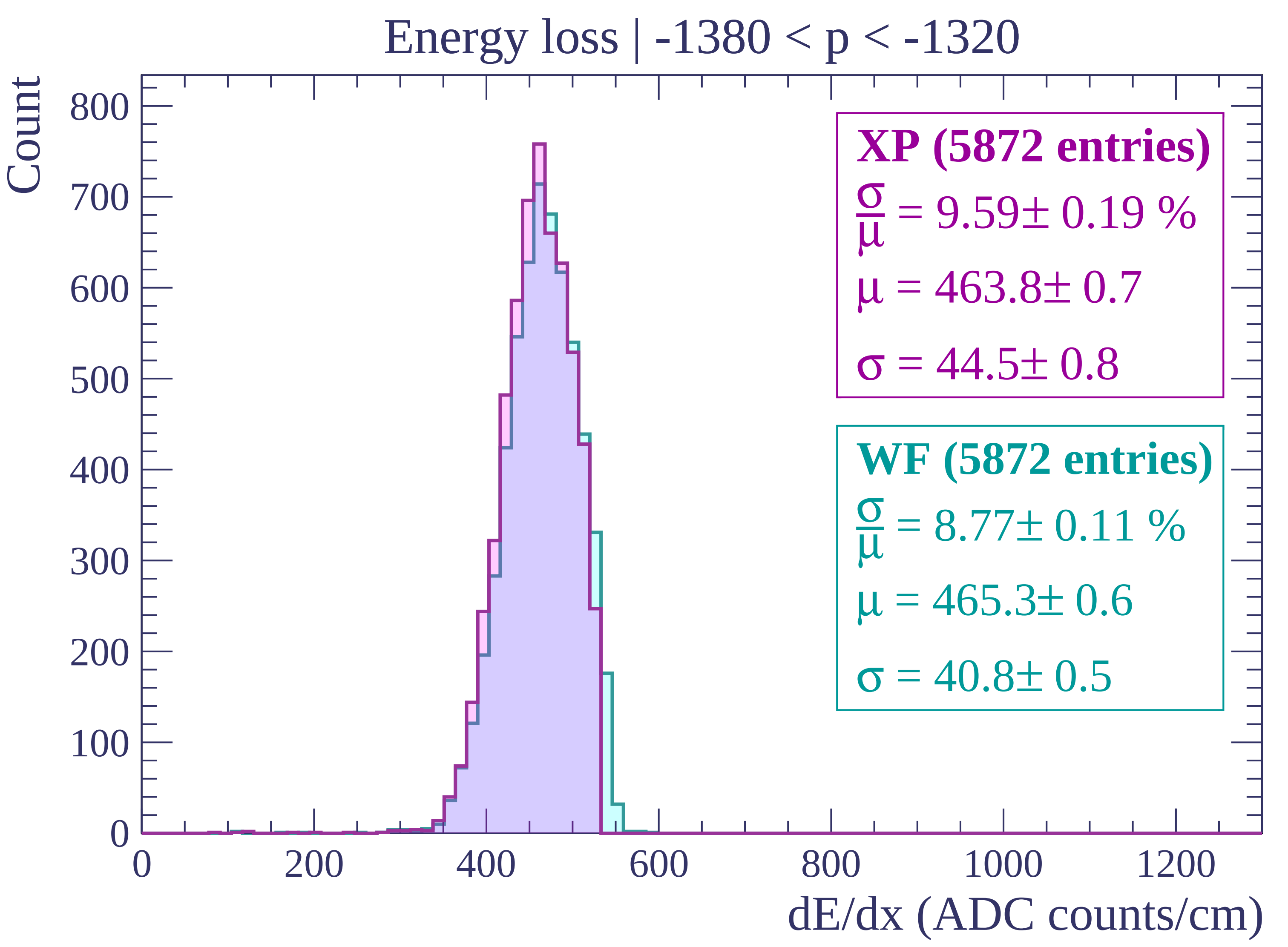

Energy loss | -1560 < p < -1500
Count
dE/dx (ADC counts/cm)
0
100
200
300
400
500
600
0
200
400
600
800
1000
1200
XP (5011 entries)
σ/μ = 8.48± 0.12 %
μ = 464.9± 0.6
σ = 39.4± 0.5
WF (5011 entries)
σ/μ = 8.33± 0.10 %
μ = 471.1± 0.7
σ = 39.2± 0.4

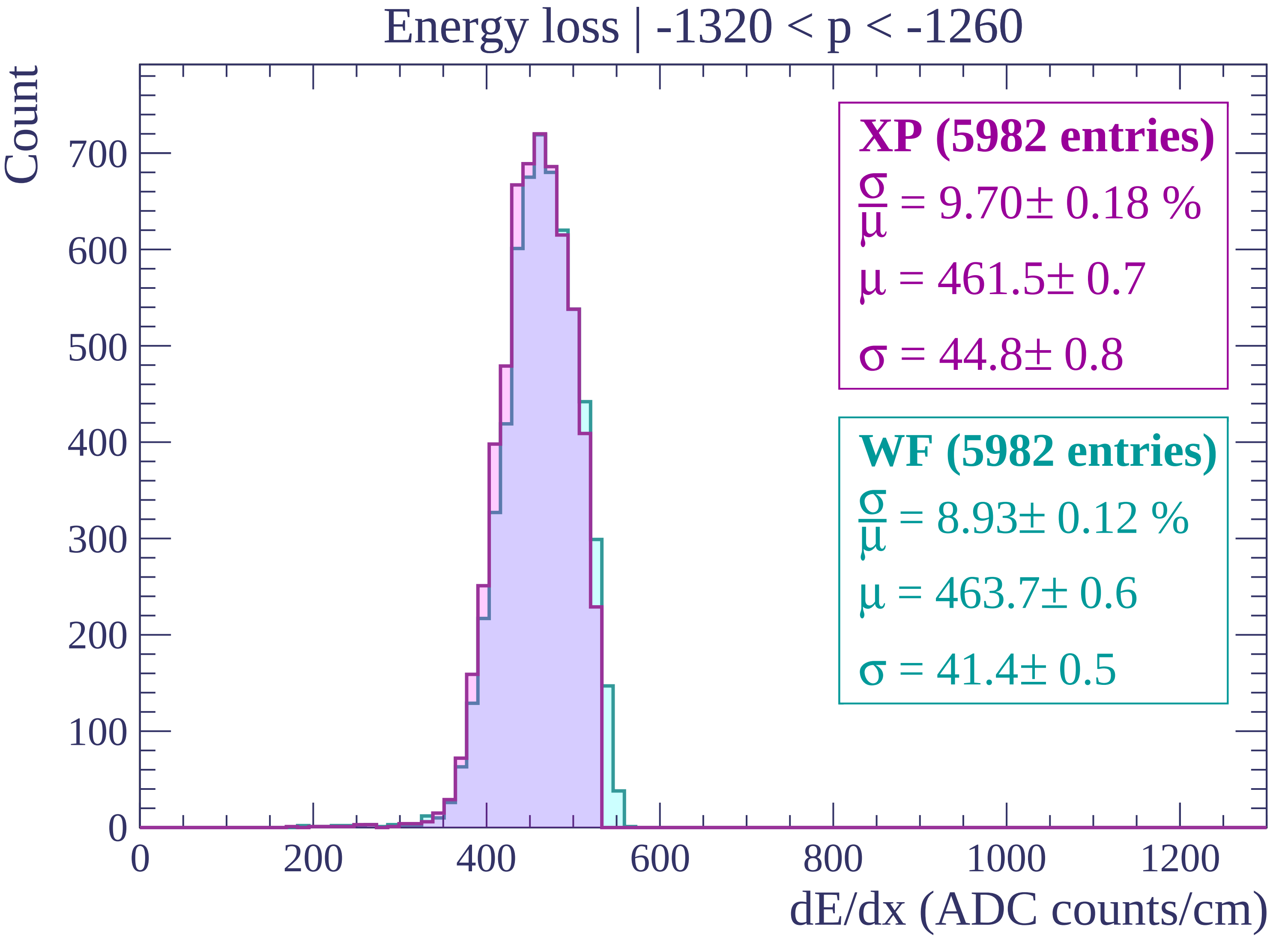


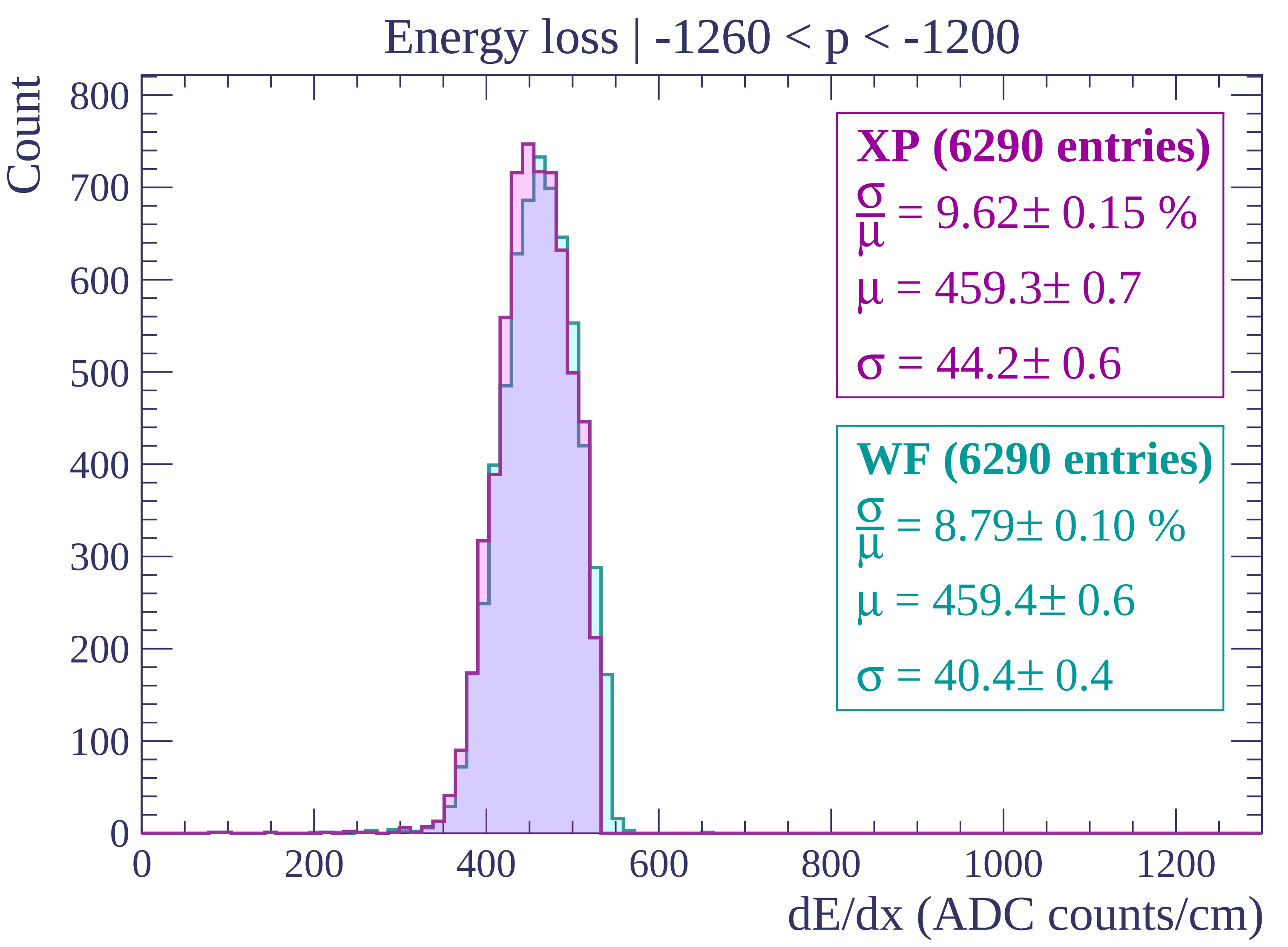
Energy loss | -1500 < p < -1440
Count
dE/dx (ADC counts/cm)
0
100
200
300
400
500
600
0
200
400
600
800
1000
1200
XP (5471 entries)
$\frac{\sigma}{\mu}$ = 10.49± 0.36 %
μ = 471.2± 1.4
σ = 49.4± 1.5
WF (5471 entries)
$\frac{\sigma}{\mu}$ = 8.72± 0.11 %
μ = 468.5± 0.6
σ = 40.9± 0.4

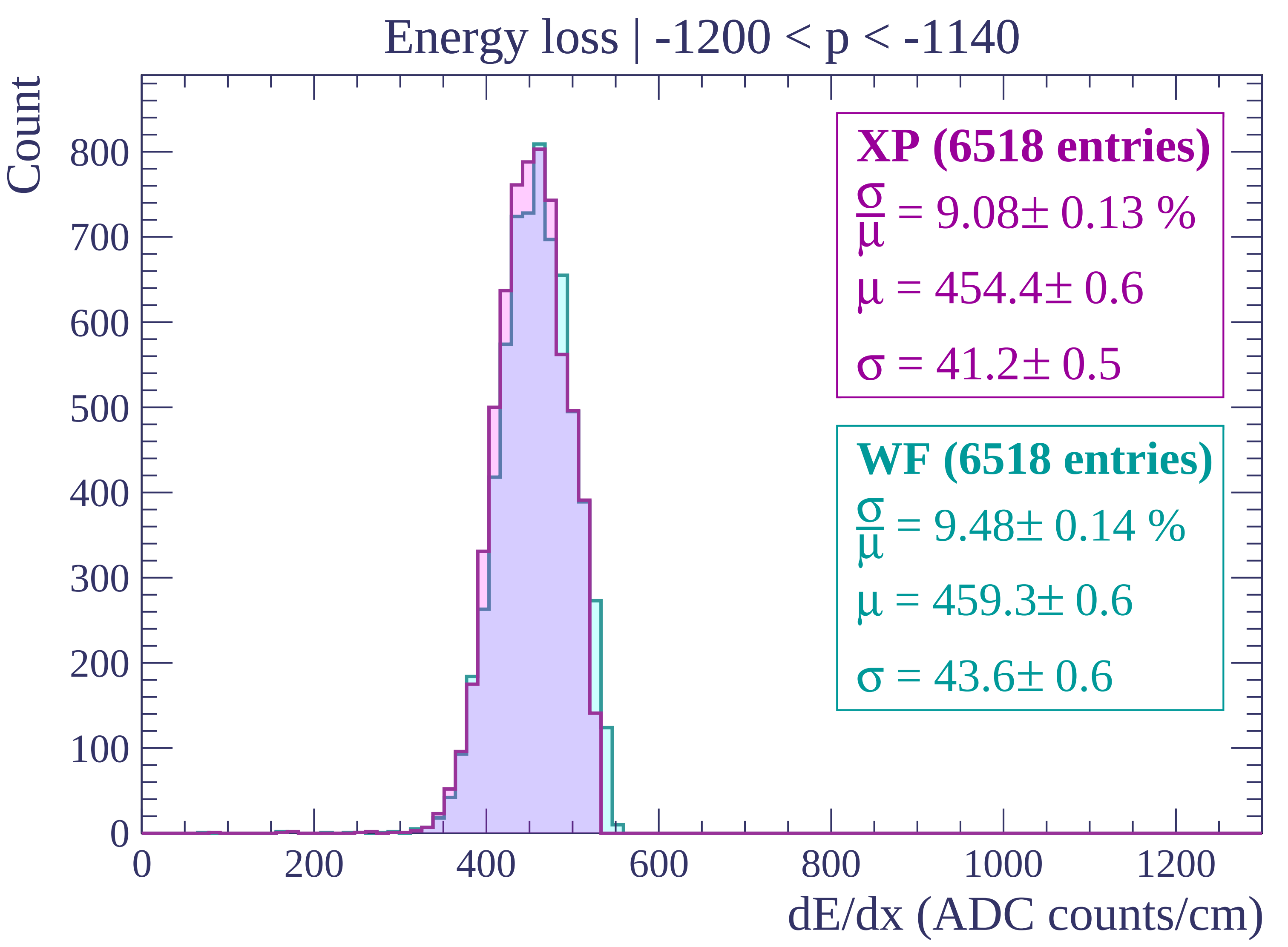


Energy loss | -1440 < p < -1380
Count
dE/dx (ADC counts/cm)
XP (5437 entries)
σ/μ = 9.84± 0.21 %
μ = 468.1± 0.9
σ = 46.0± 0.9
WF (5437 entries)
σ/μ = 8.93± 0.12 %
μ = 468.4± 0.6
σ = 41.8± 0.5
0
100
200
300
400
500
600
700
0
200
400
600
800
1000
1200

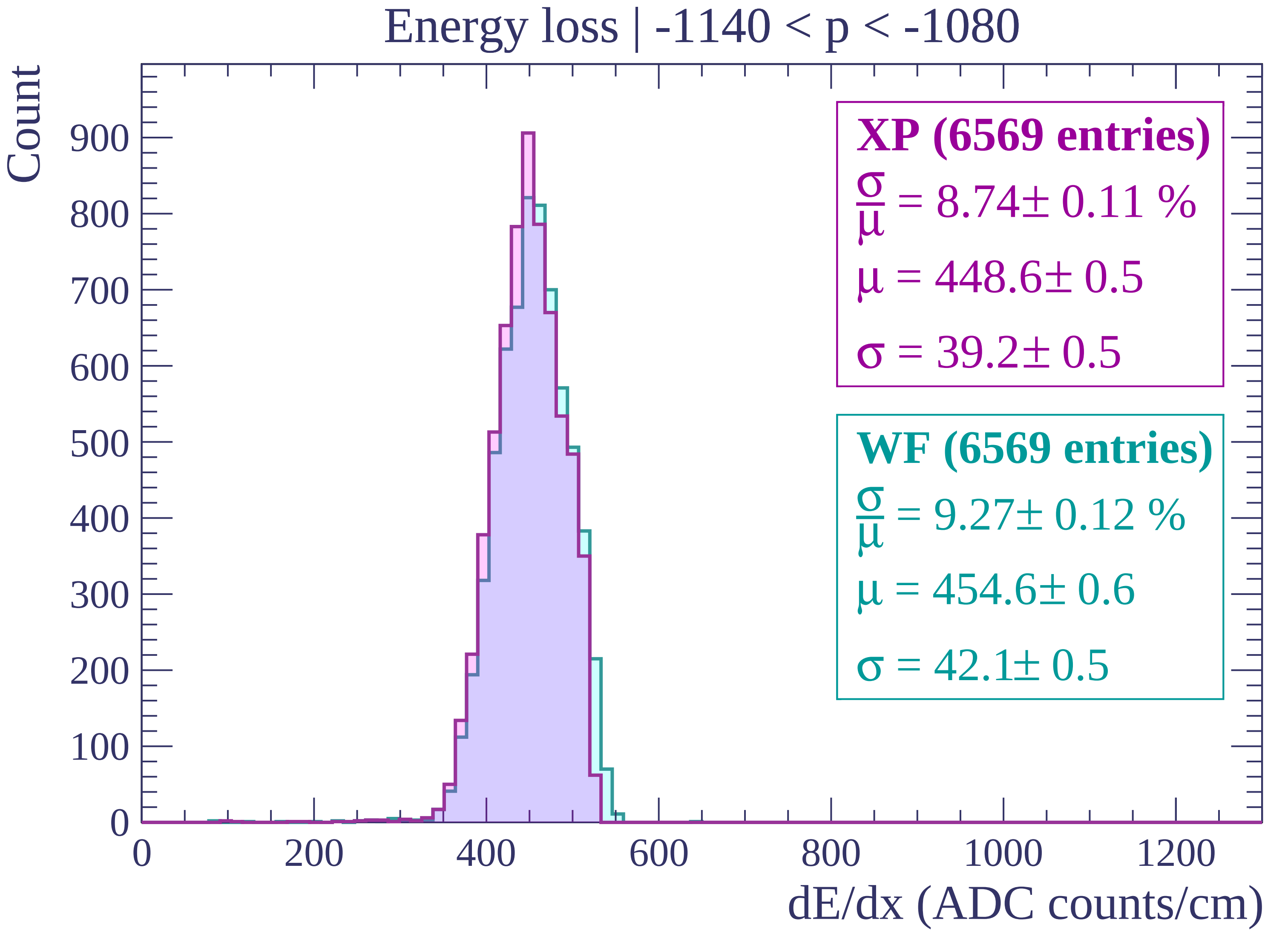

Energy loss | -1380 < p < -1320
Count
dE/dx (ADC counts/cm)
0
200
400
600
800
1000
1200
100
300
500
700
XP (5872 entries)
σ/μ = 9.59± 0.19 %
μ = 463.8± 0.7
σ = 44.5± 0.8
WF (5872 entries)
σ/μ = 8.77± 0.11 %
μ = 465.3± 0.6
σ = 40.8± 0.5


Energy loss | -1320 < p < -1260
Count
dE/dx (ADC counts/cm)
XP (5982 entries)
$\frac{\sigma}{\mu}$ = 9.70± 0.18 %
μ = 461.5± 0.7
σ = 44.8± 0.8
WF (5982 entries)
$\frac{\sigma}{\mu}$ = 8.93± 0.12 %
μ = 463.7± 0.6
σ = 41.4± 0.5
0
100
200
300
400
500
600
700
0
200
400
600
800
1000
1200

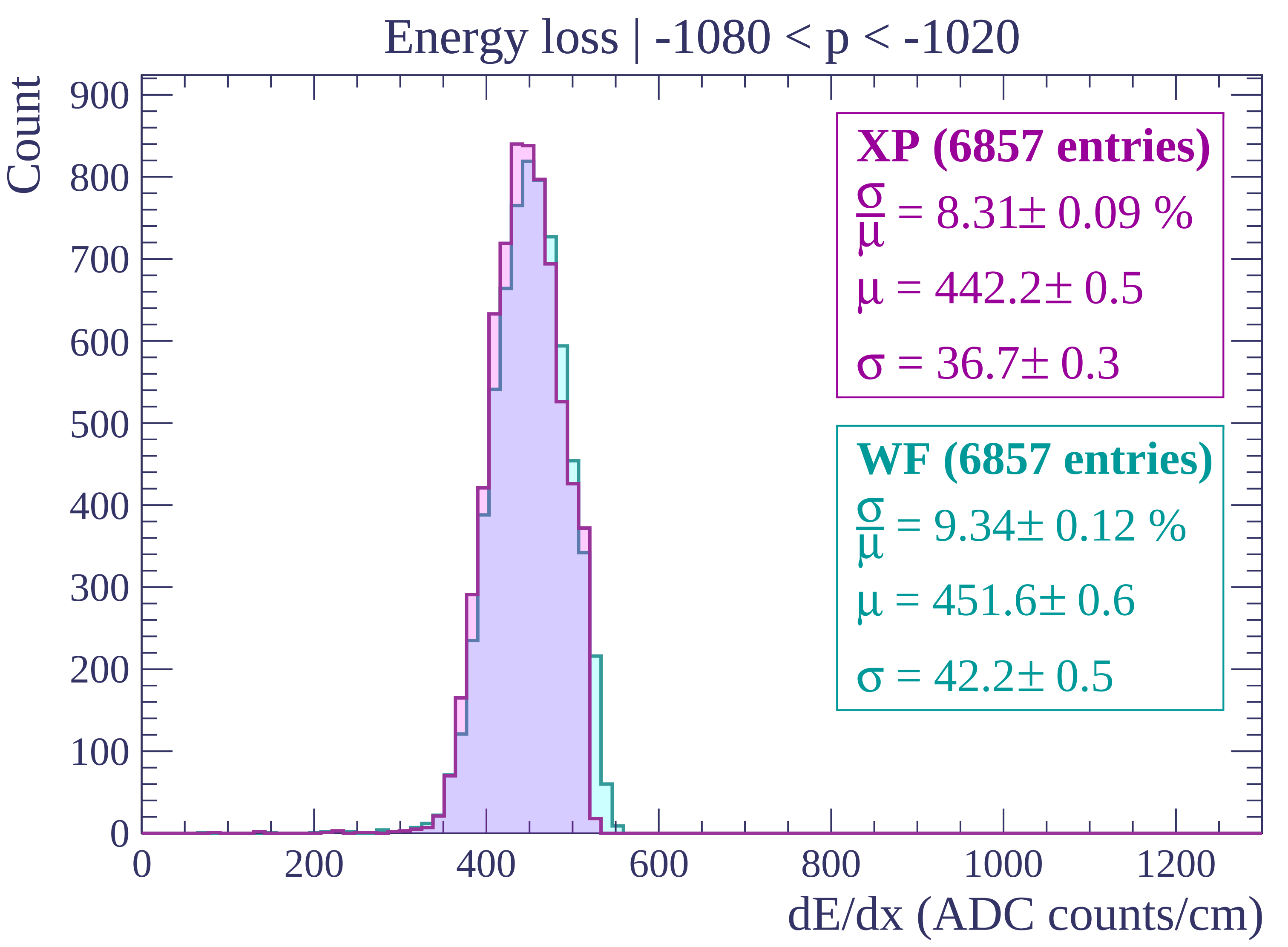

Energy loss | -1260 < p < -1200
Count
dE/dx (ADC counts/cm)
XP (6290 entries)
σ/μ = 9.62± 0.15 %
μ = 459.3± 0.7
σ = 44.2± 0.6
WF (6290 entries)
σ/μ = 8.79± 0.10 %
μ = 459.4± 0.6
σ = 40.4± 0.4
0
100
200
300
400
500
600
700
800
1000
1200



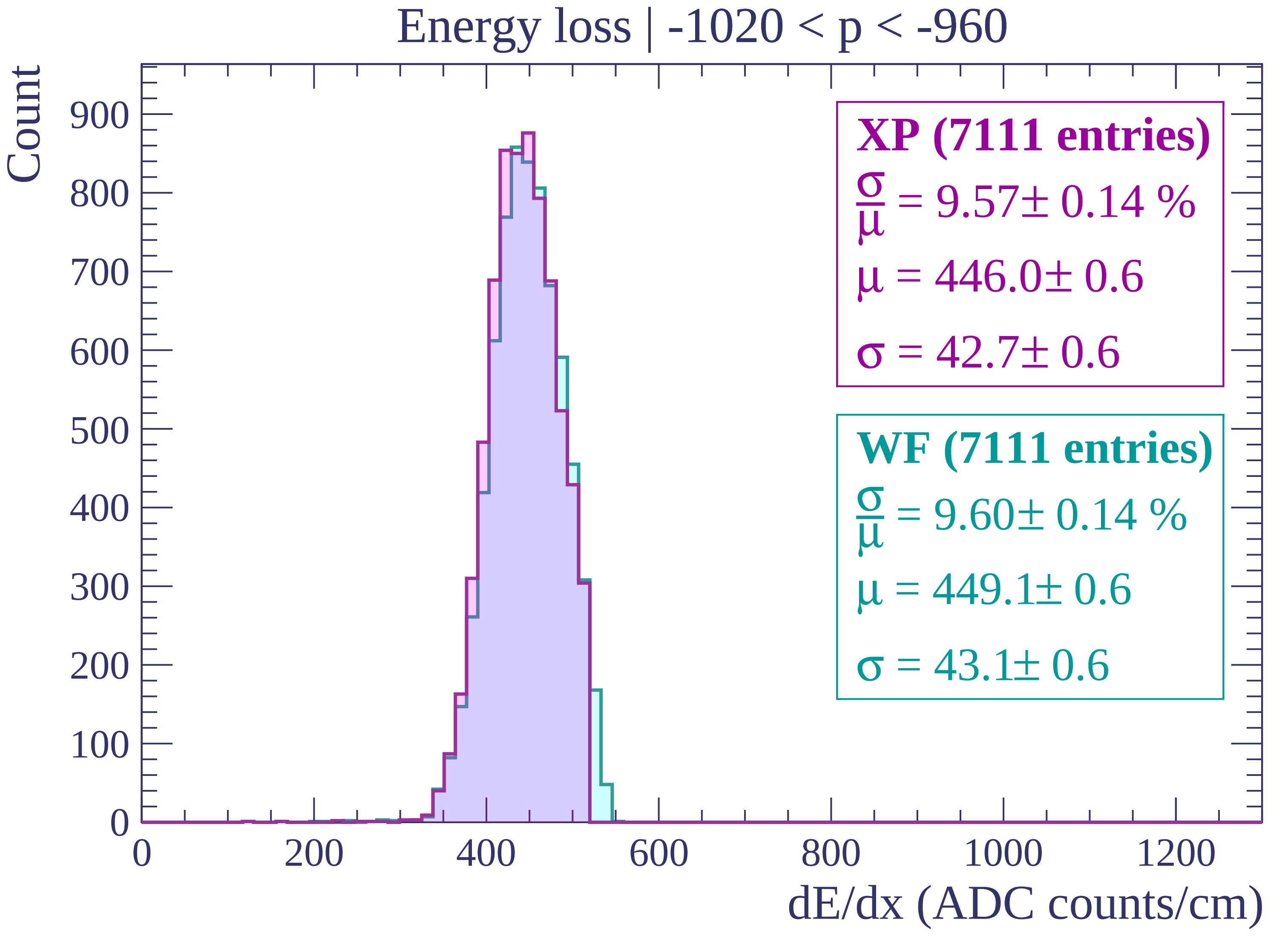
Energy loss | -1200 < p < -1140
Count
800
700
600
500
400
300
200
100
0
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP (6518 entries)
σ/μ = 9.08± 0.13 %
μ = 454.4± 0.6
σ = 41.2± 0.5
WF (6518 entries)
σ/μ = 9.48± 0.14 %
μ = 459.3± 0.6
σ = 43.6± 0.6

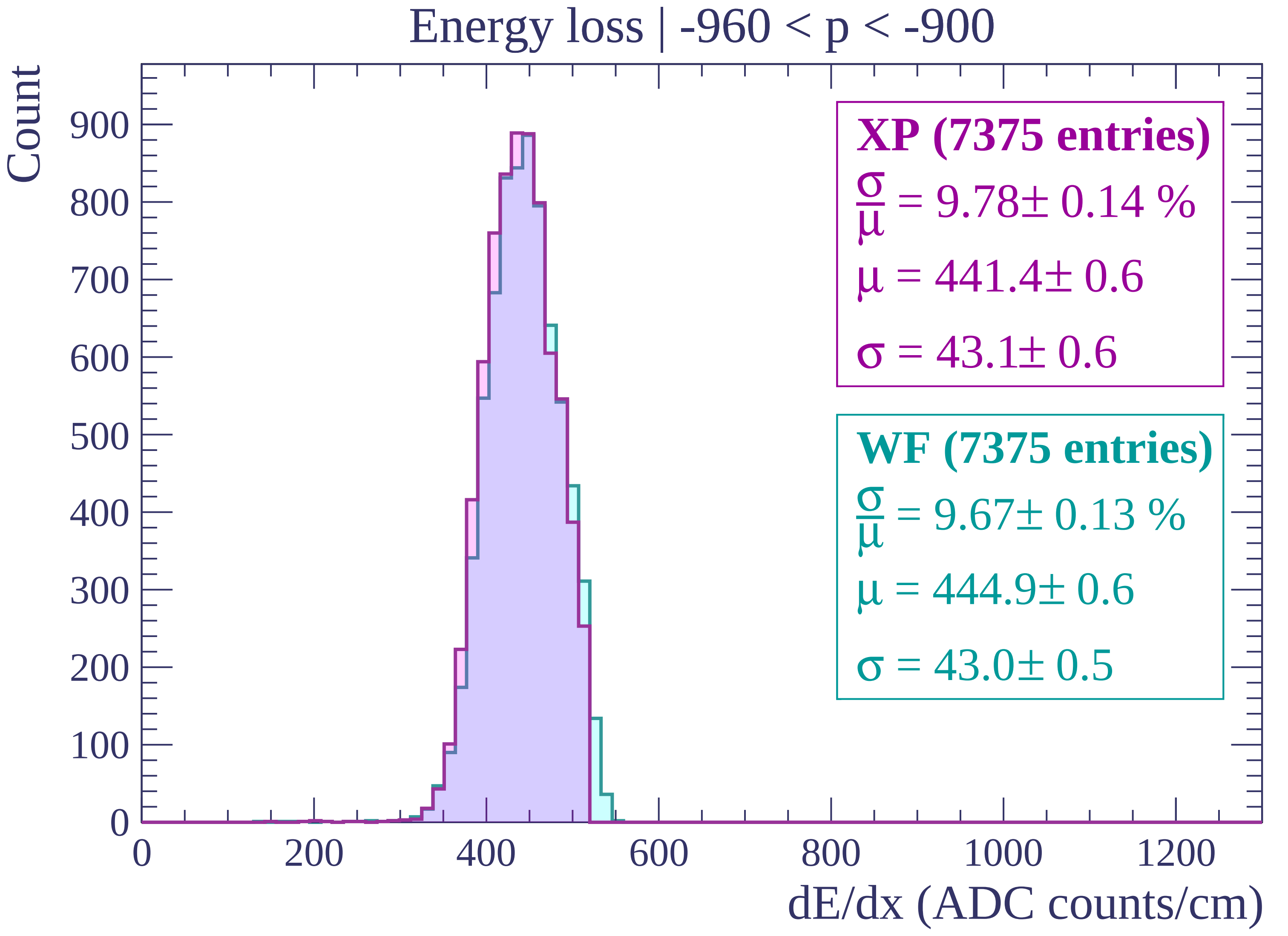


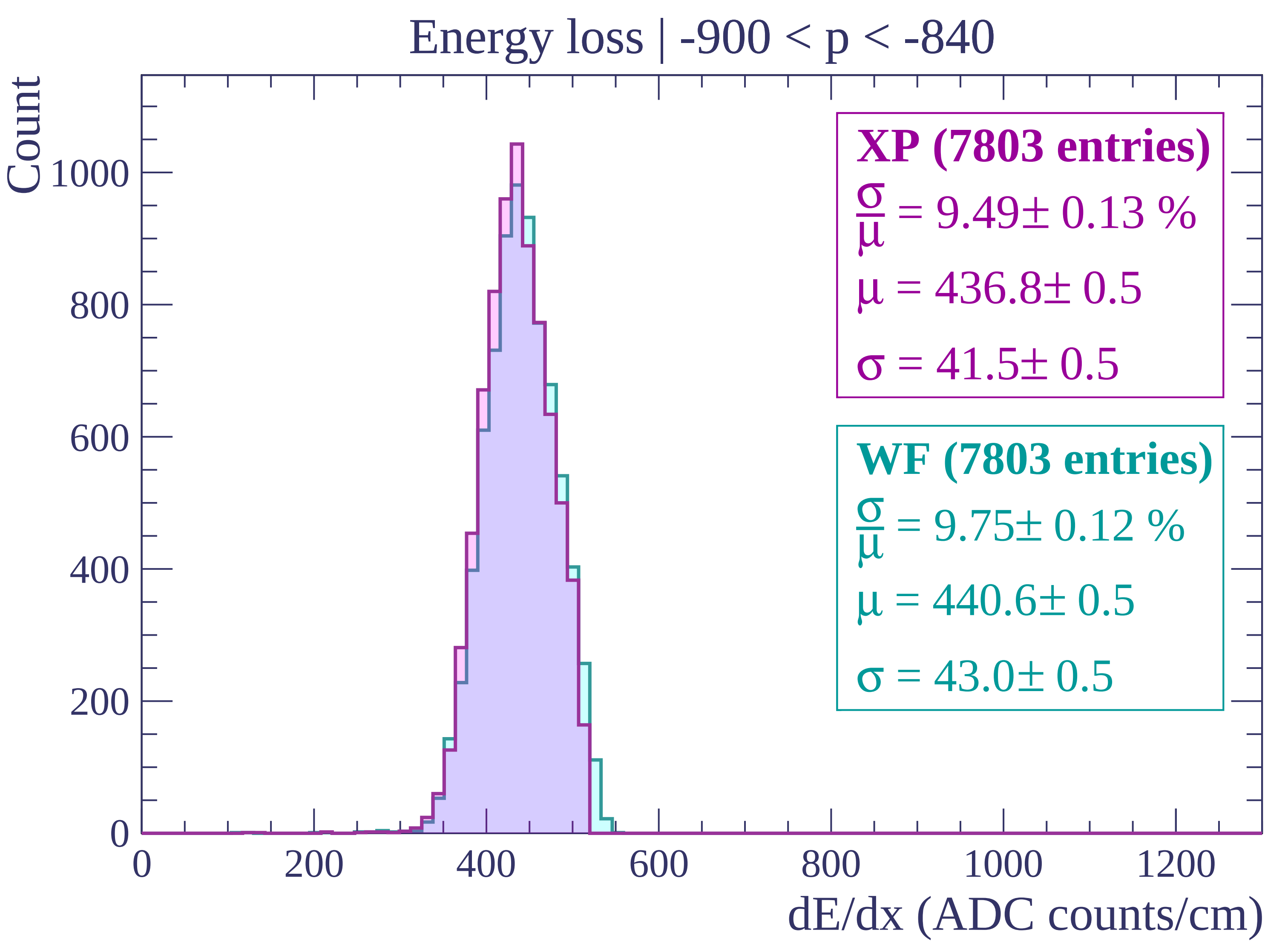
Energy loss | -1140 < p < -1080
Count
dE/dx (ADC counts/cm)
0
100
200
300
400
500
600
700
800
900
0
200
400
600
800
1000
1200
XP (6569 entries)
$\frac{\sigma}{\mu}$ = 8.74± 0.11 %
μ = 448.6± 0.5
σ = 39.2± 0.5
WF (6569 entries)
$\frac{\sigma}{\mu}$ = 9.27± 0.12 %
μ = 454.6± 0.6
σ = 42.1± 0.5

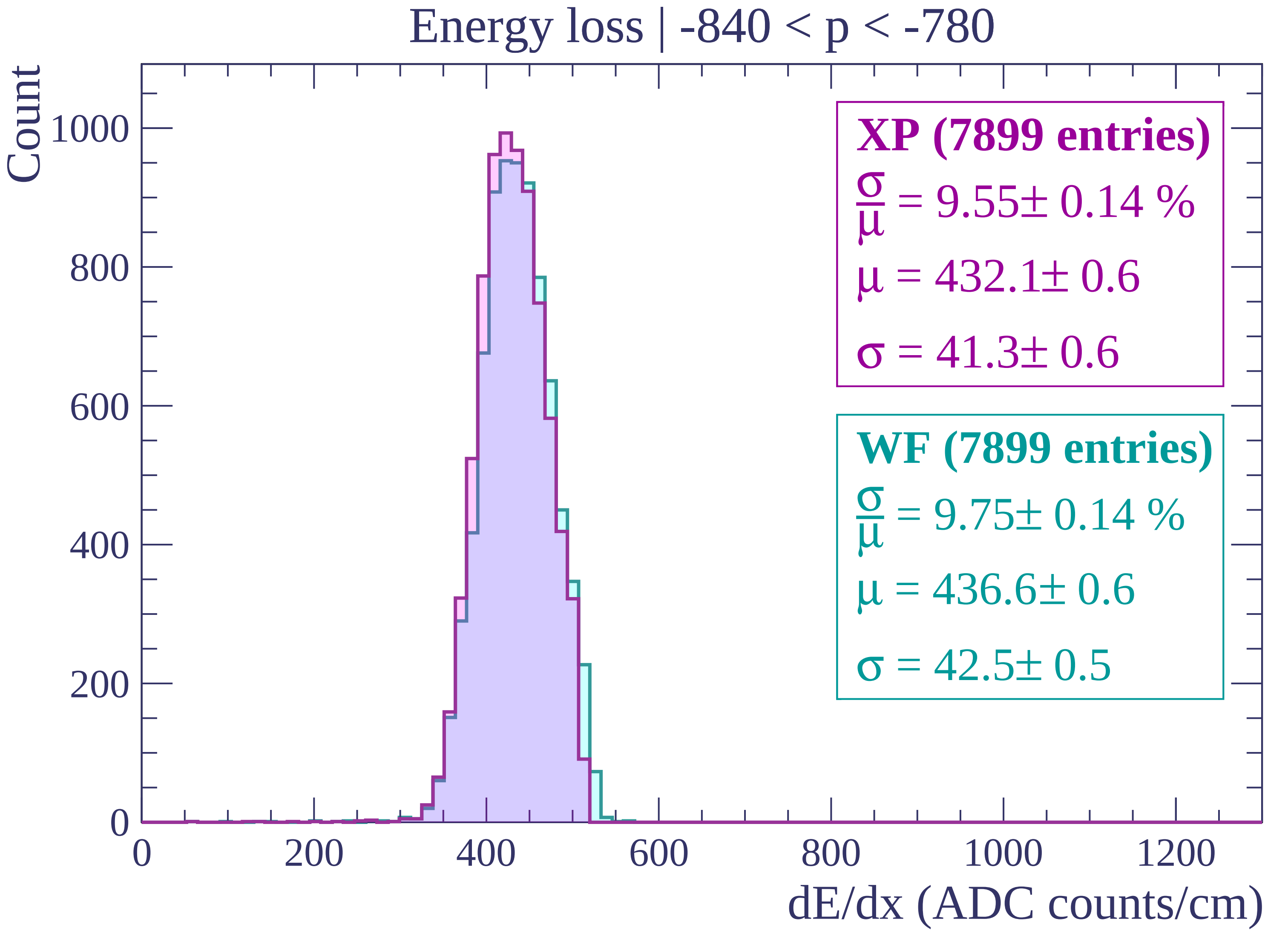


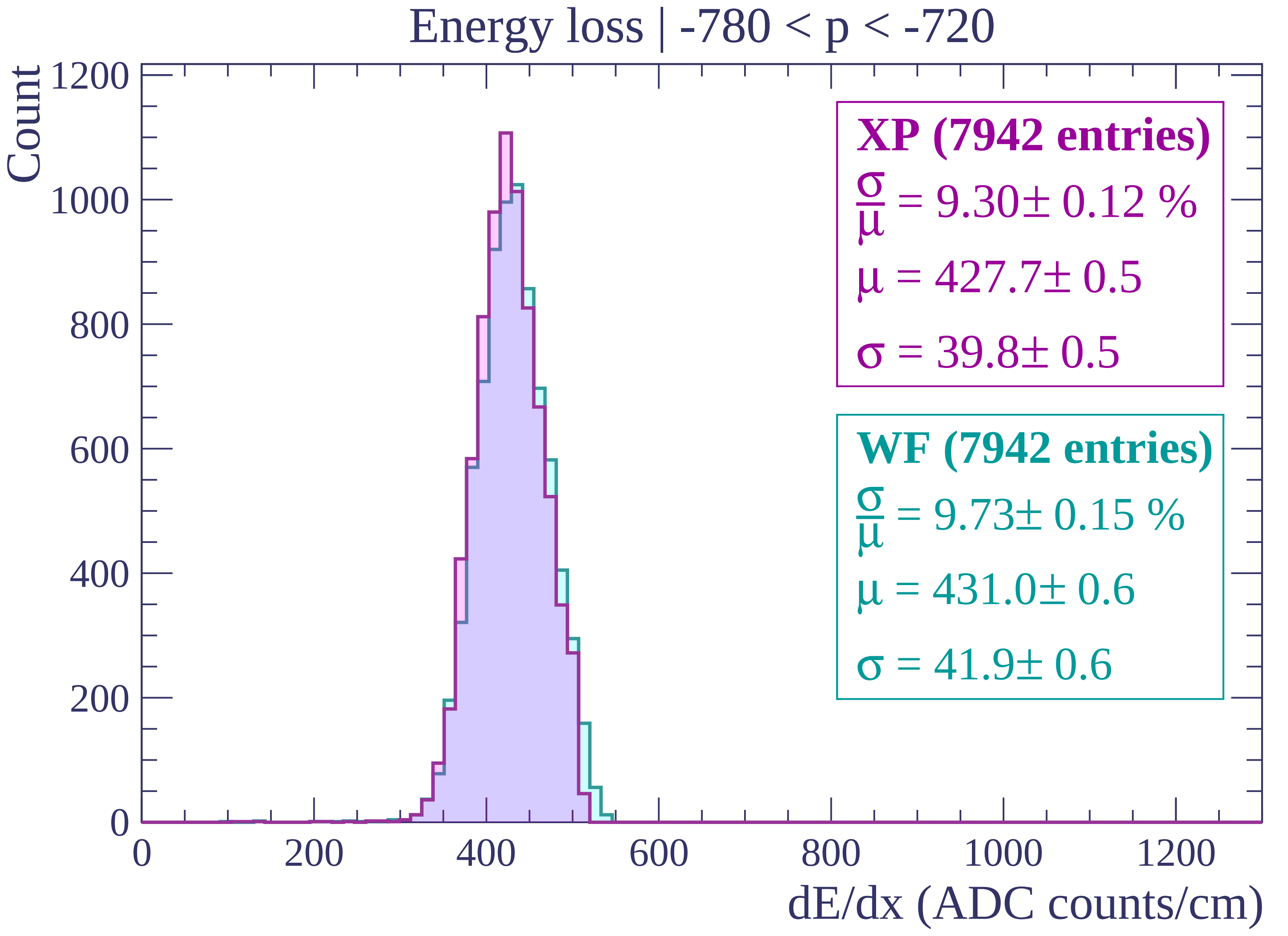
Energy loss | -1080 < p < -1020
Count
0
100
200
300
400
500
600
700
800
900
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP (6857 entries)
σ/μ = 8.31± 0.09 %
μ = 442.2± 0.5
σ = 36.7± 0.3
WF (6857 entries)
σ/μ = 9.34± 0.12 %
μ = 451.6± 0.6
σ = 42.2± 0.5

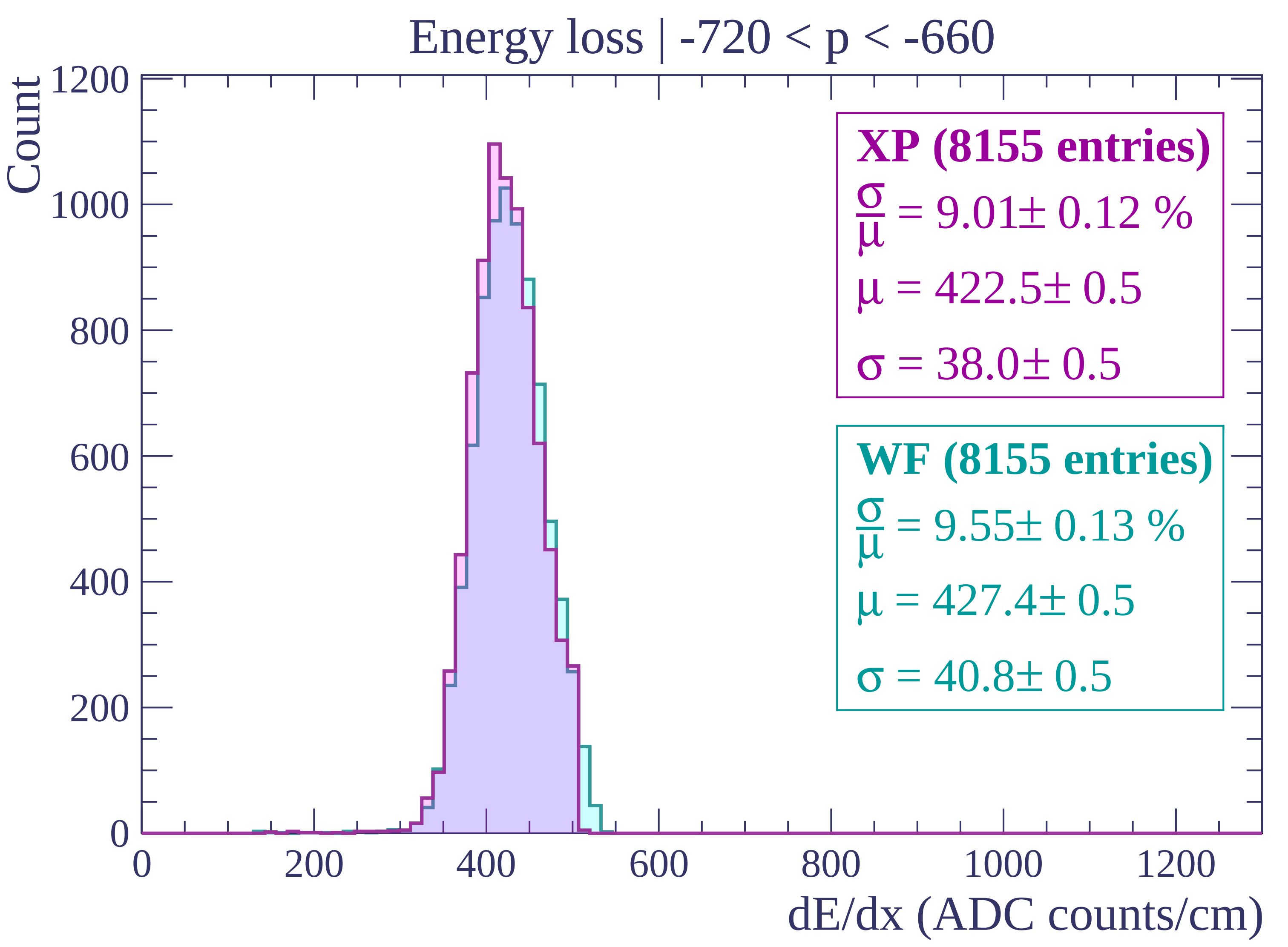

Energy loss | -1020 < p < -960
Count
dE/dx (ADC counts/cm)
0
200
400
600
800
1000
1200
100
300
500
700
900
XP (7111 entries)
$\frac{\sigma}{\mu}$ = 9.57± 0.14 %
μ = 446.0± 0.6
σ = 42.7± 0.6
WF (7111 entries)
$\frac{\sigma}{\mu}$ = 9.60± 0.14 %
μ = 449.1± 0.6
σ = 43.1± 0.6

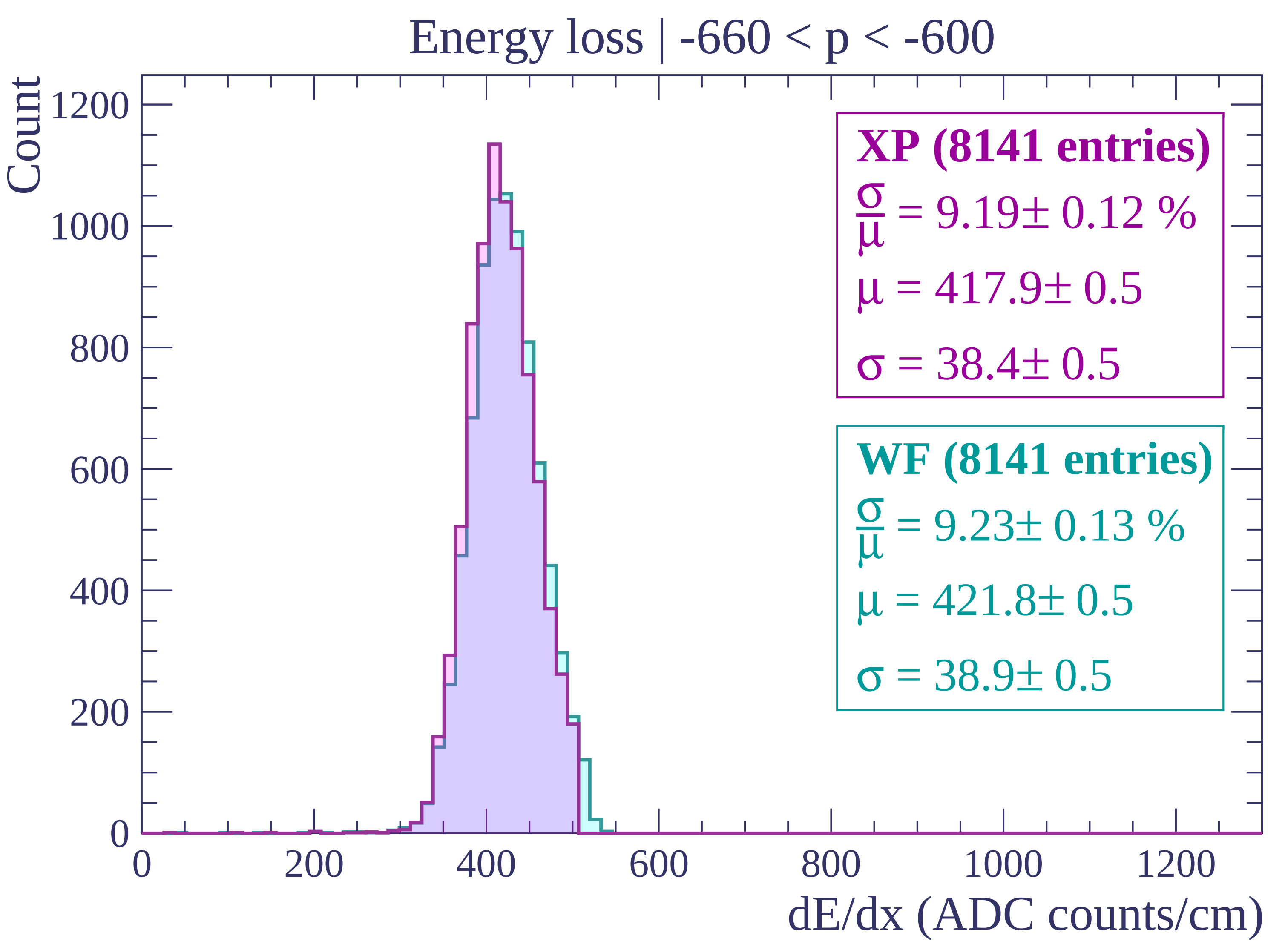

Energy loss | -960 < p < -900
Count
900
800
700
600
500
400
300
200
100
0
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP (7375 entries)
σ/μ = 9.78± 0.14 %
μ = 441.4± 0.6
σ = 43.1± 0.6
WF (7375 entries)
σ/μ = 9.67± 0.13 %
μ = 444.9± 0.6
σ = 43.0± 0.5


Energy loss | -900 < p < -840
Count
dE/dx (ADC counts/cm)
XP (7803 entries)
σ/μ = 9.49± 0.13 %
μ = 436.8± 0.5
σ = 41.5± 0.5
WF (7803 entries)
σ/μ = 9.75± 0.12 %
μ = 440.6± 0.5
σ = 43.0± 0.5
0
200
400
600
800
1000
1200

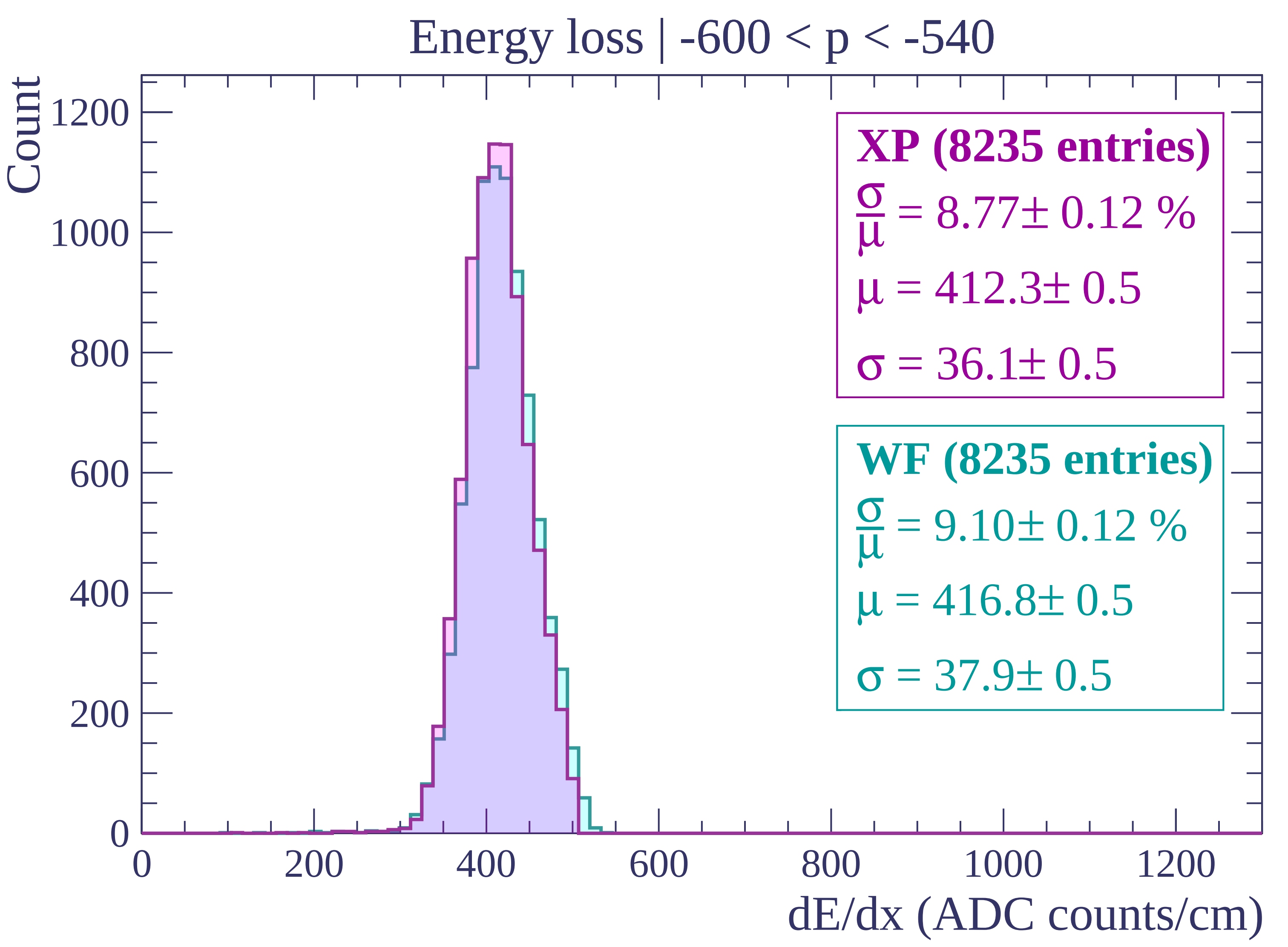

Energy loss | -840 < p < -780
Count
dE/dx (ADC counts/cm)
0
200
400
600
800
1000
1200
XP (7899 entries)
σ/μ = 9.55± 0.14 %
μ = 432.1± 0.6
σ = 41.3± 0.6
WF (7899 entries)
σ/μ = 9.75± 0.14 %
μ = 436.6± 0.6
σ = 42.5± 0.5

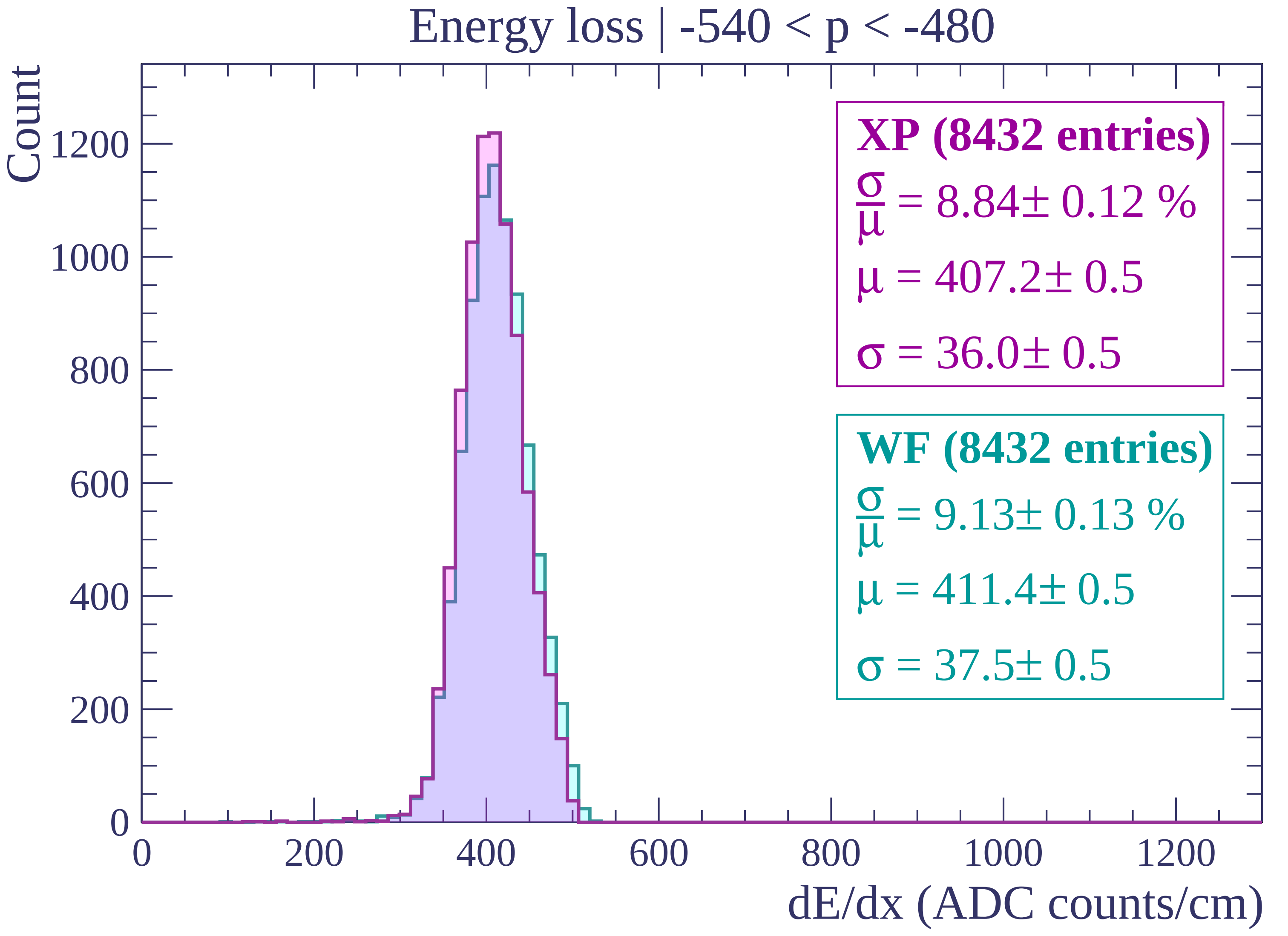

Energy loss | -780 < p < -720
Count
dE/dx (ADC counts/cm)
0
200
400
600
800
1000
1200
XP (7942 entries)
σ/μ = 9.30± 0.12 %
μ = 427.7± 0.5
σ = 39.8± 0.5
WF (7942 entries)
σ/μ = 9.73± 0.15 %
μ = 431.0± 0.6
σ = 41.9± 0.6

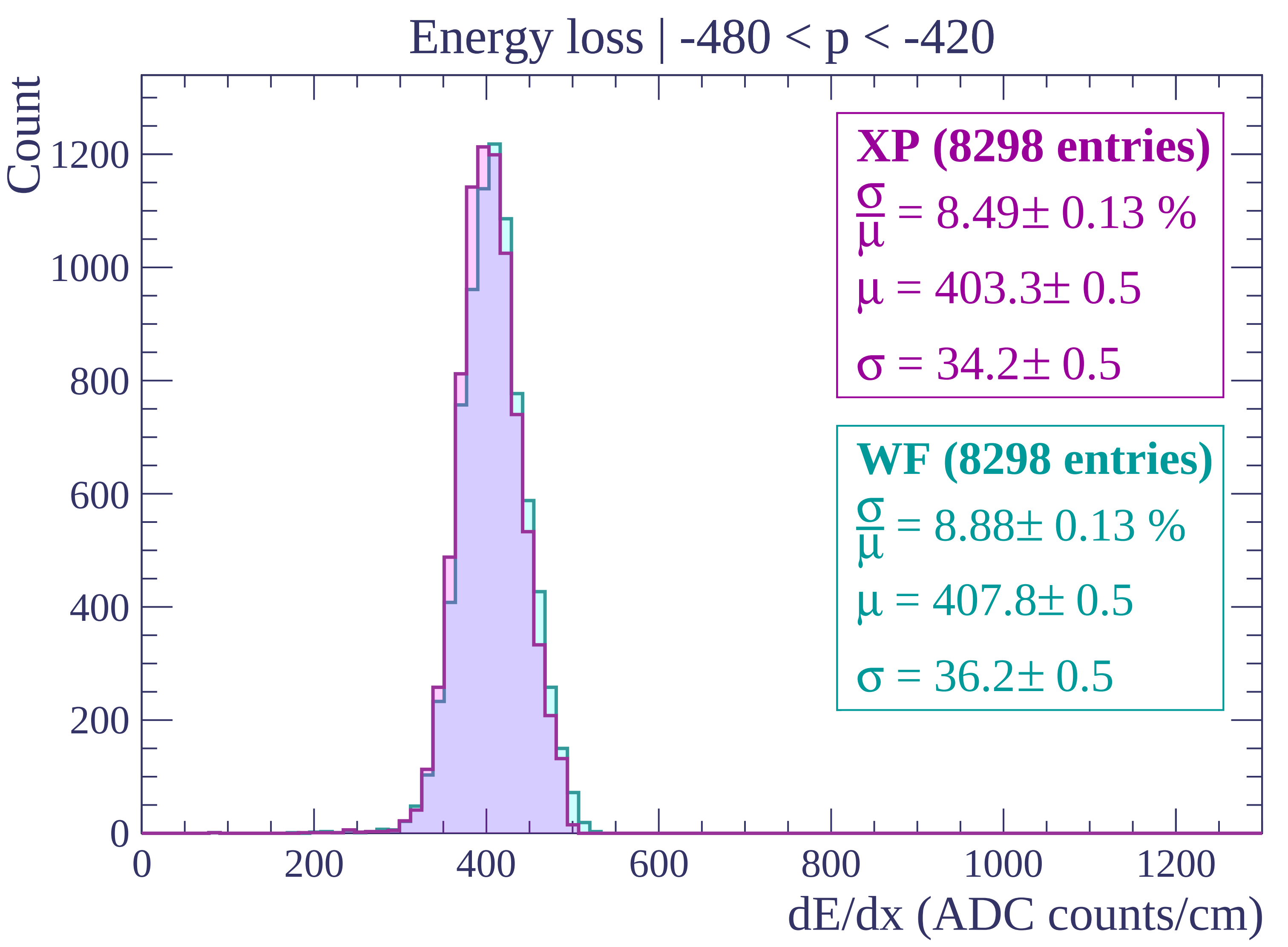

Energy loss | -720 < p < -660
Count
dE/dx (ADC counts/cm)
0
200
400
600
800
1000
1200
XP (8155 entries)
σ/μ = 9.01± 0.12 %
μ = 422.5± 0.5
σ = 38.0± 0.5
WF (8155 entries)
σ/μ = 9.55± 0.13 %
μ = 427.4± 0.5
σ = 40.8± 0.5

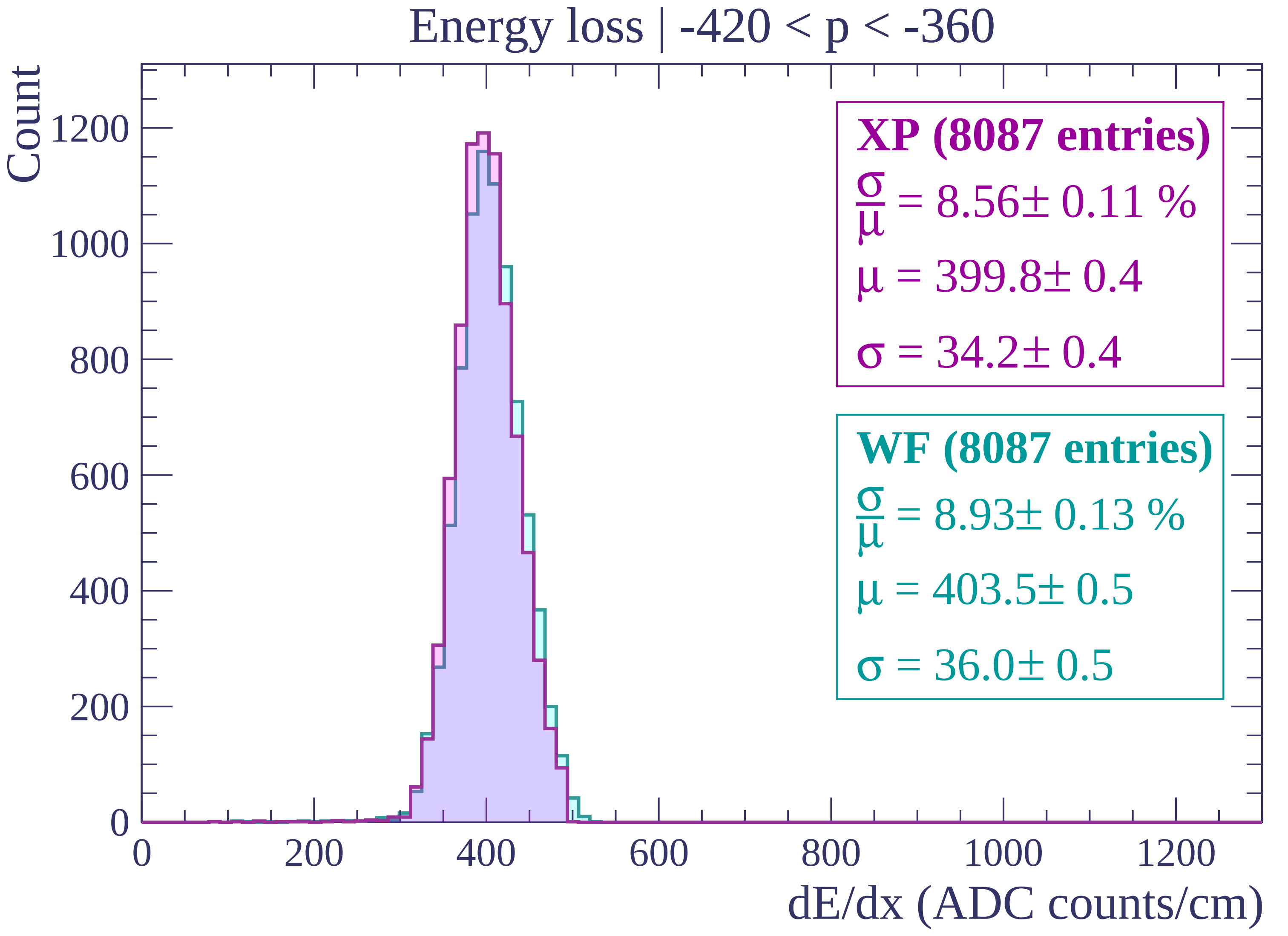


Energy loss | -660 < p < -600
Count
dE/dx (ADC counts/cm)
0
200
400
600
800
1000
1200
XP (8141 entries)
$\frac{\sigma}{\mu}$ = 9.19± 0.12 %
μ = 417.9± 0.5
σ = 38.4± 0.5
WF (8141 entries)
$\frac{\sigma}{\mu}$ = 9.23± 0.13 %
μ = 421.8± 0.5
σ = 38.9± 0.5

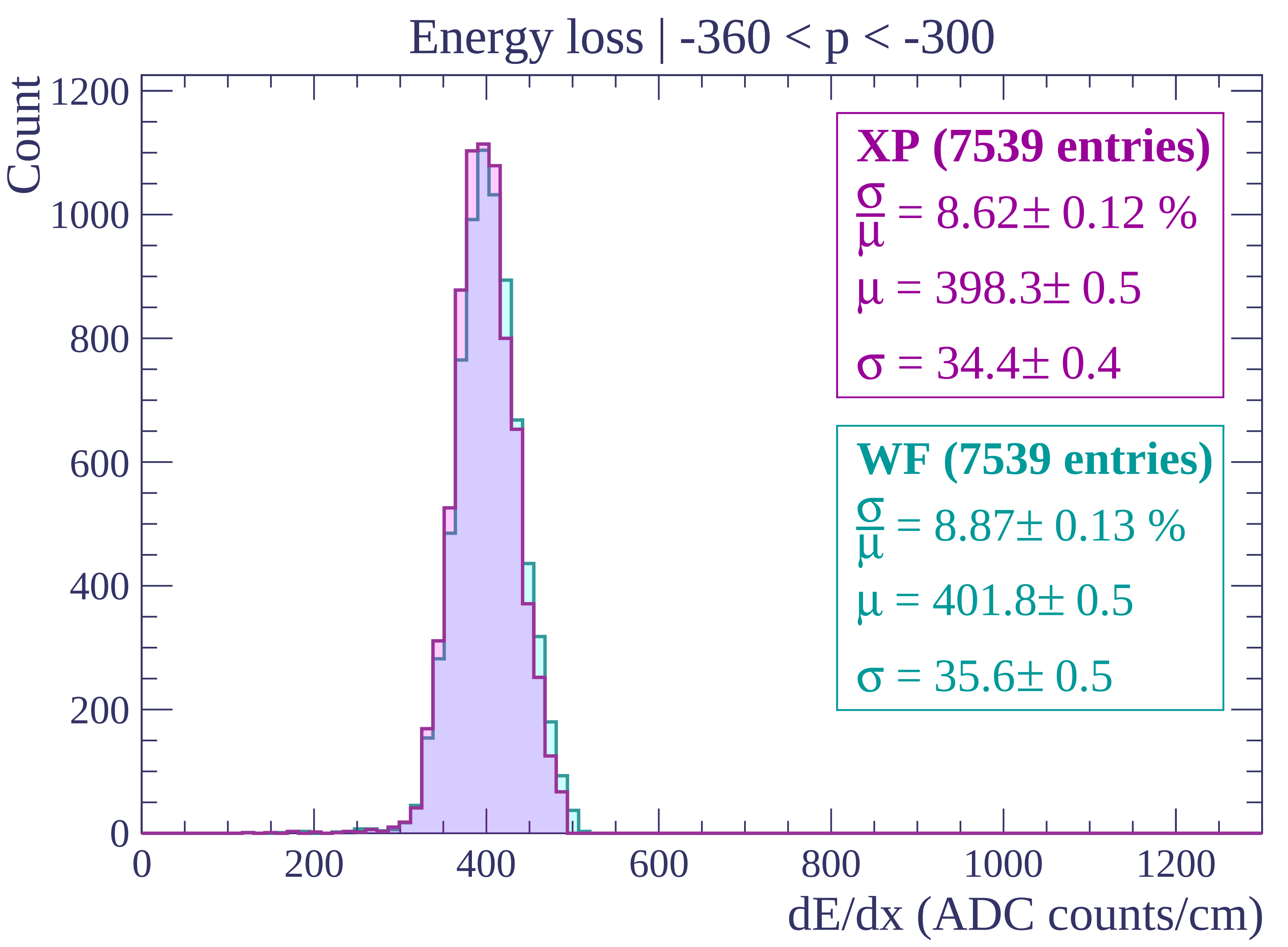

Energy loss | -600 < p < -540
Count
dE/dx (ADC counts/cm)
XP (8235 entries)
σ/μ = 8.77± 0.12 %
μ = 412.3± 0.5
σ = 36.1± 0.5
WF (8235 entries)
σ/μ = 9.10± 0.12 %
μ = 416.8± 0.5
σ = 37.9± 0.5
0
200
400
600
800
1000
1200


Energy loss | -540 < p < -480
Count
dE/dx (ADC counts/cm)
XP (8432 entries)
σ/μ = 8.84± 0.12 %
μ = 407.2± 0.5
σ = 36.0± 0.5
WF (8432 entries)
σ/μ = 9.13± 0.13 %
μ = 411.4± 0.5
σ = 37.5± 0.5

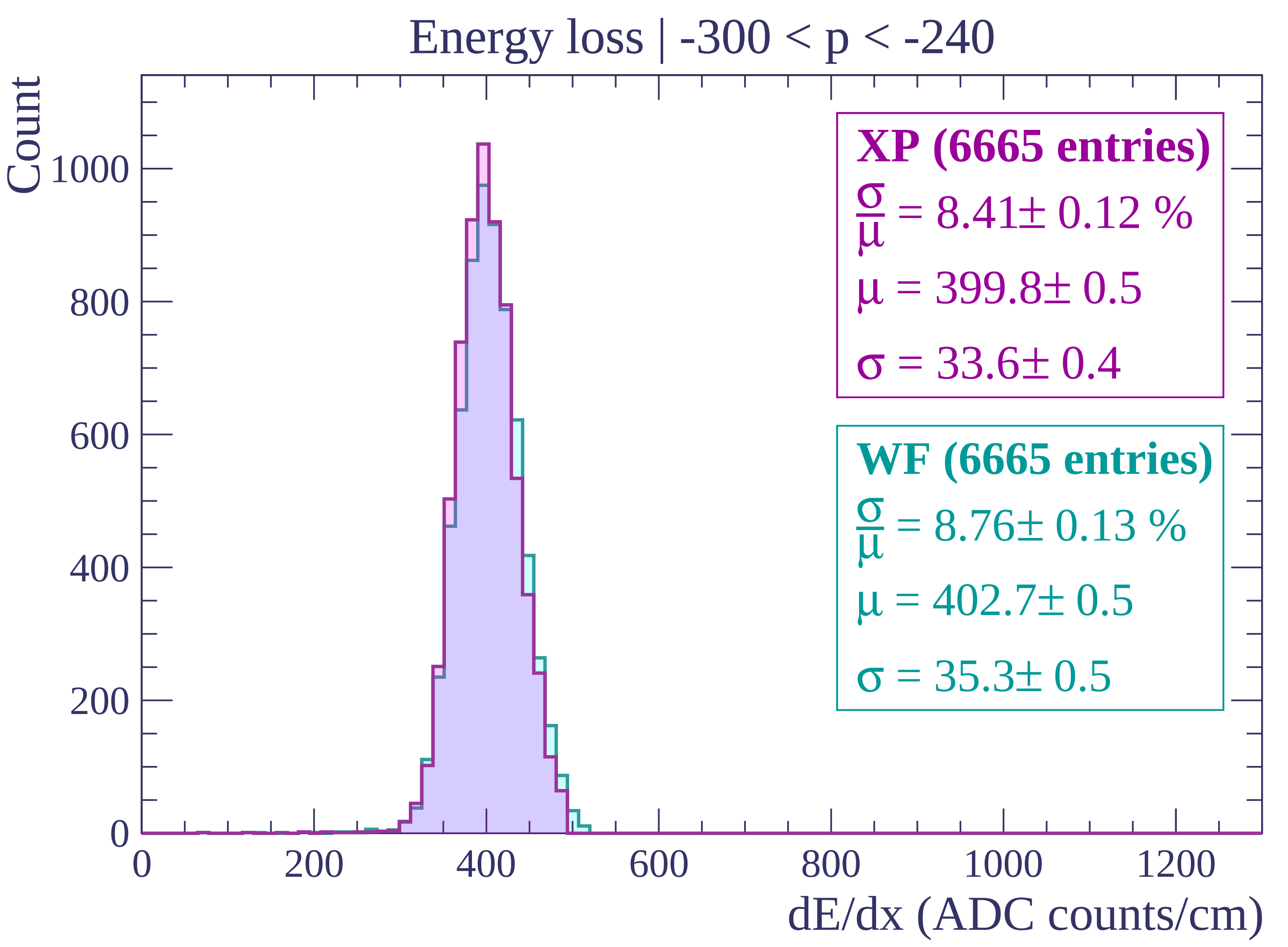

Energy loss | -480 < p < -420
Count
dE/dx (ADC counts/cm)
XP (8298 entries)
σ/μ = 8.49± 0.13 %
μ = 403.3± 0.5
σ = 34.2± 0.5
WF (8298 entries)
σ/μ = 8.88± 0.13 %
μ = 407.8± 0.5
σ = 36.2± 0.5

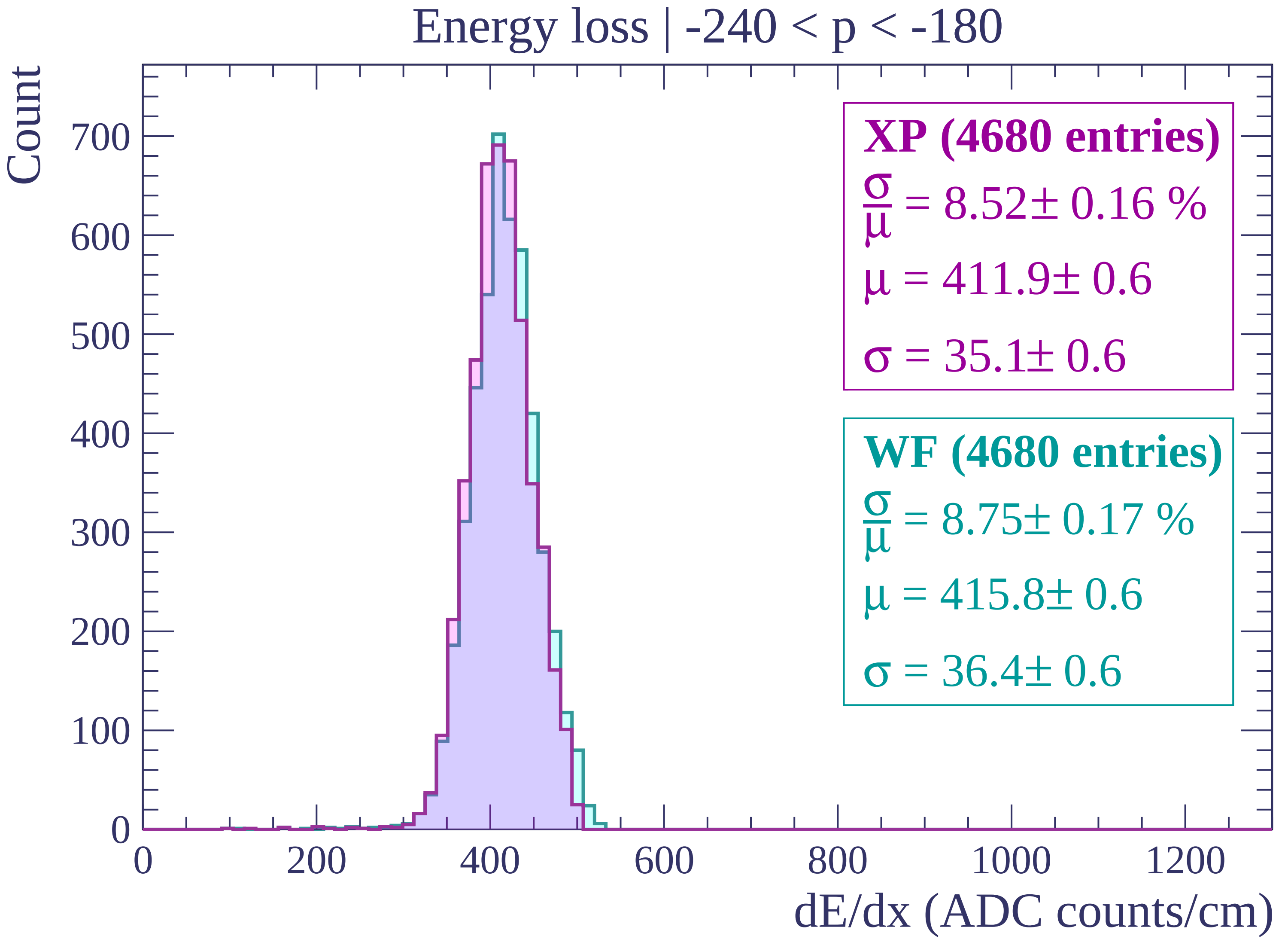


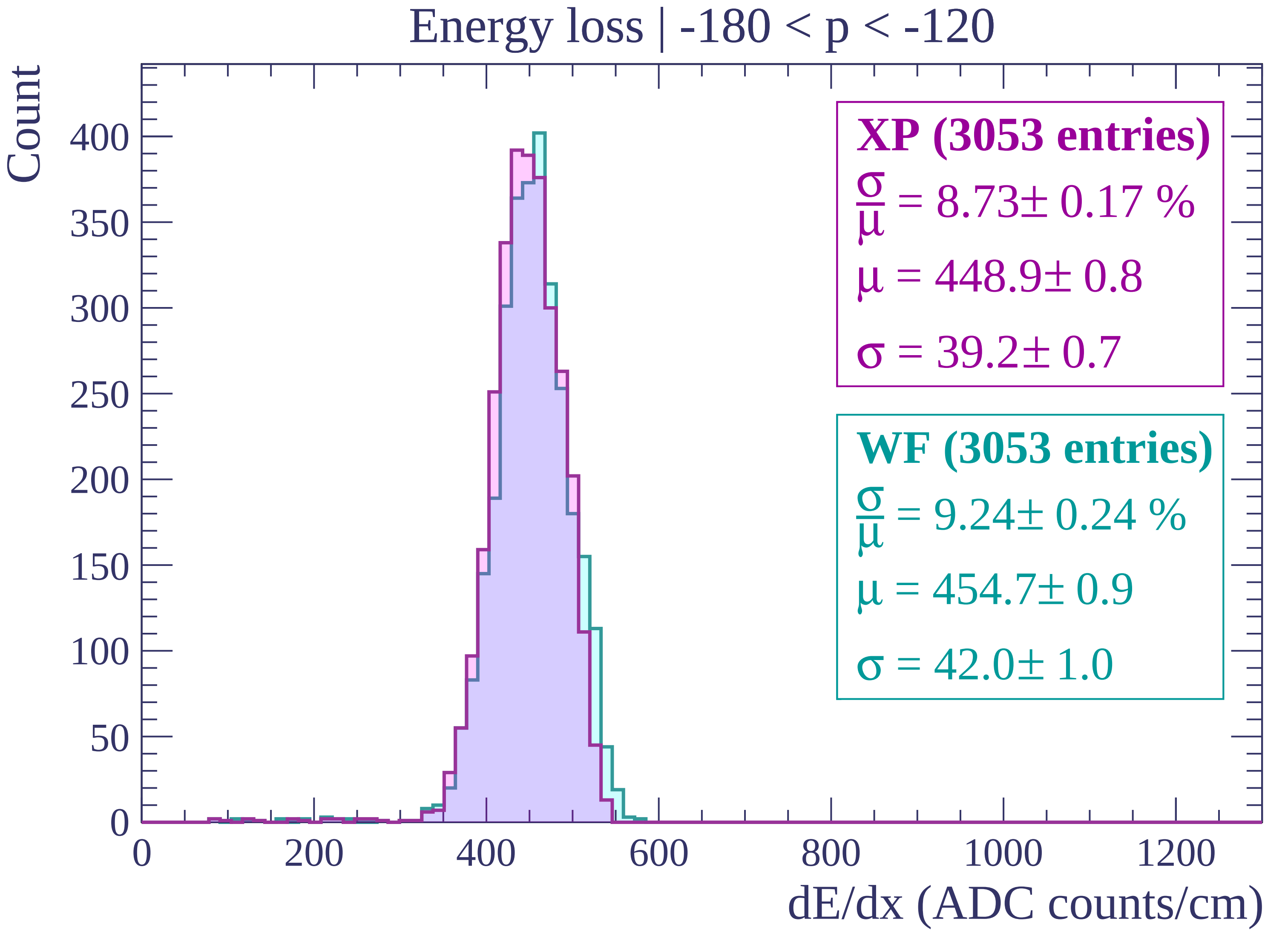
Energy loss | -420 < p < -360
Count
dE/dx (ADC counts/cm)
0
200
400
600
800
1000
1200
XP (8087 entries)
$\frac{\sigma}{\mu}$ = 8.56± 0.11 %
μ = 399.8± 0.4
σ = 34.2± 0.4
WF (8087 entries)
$\frac{\sigma}{\mu}$ = 8.93± 0.13 %
μ = 403.5± 0.5
σ = 36.0± 0.5



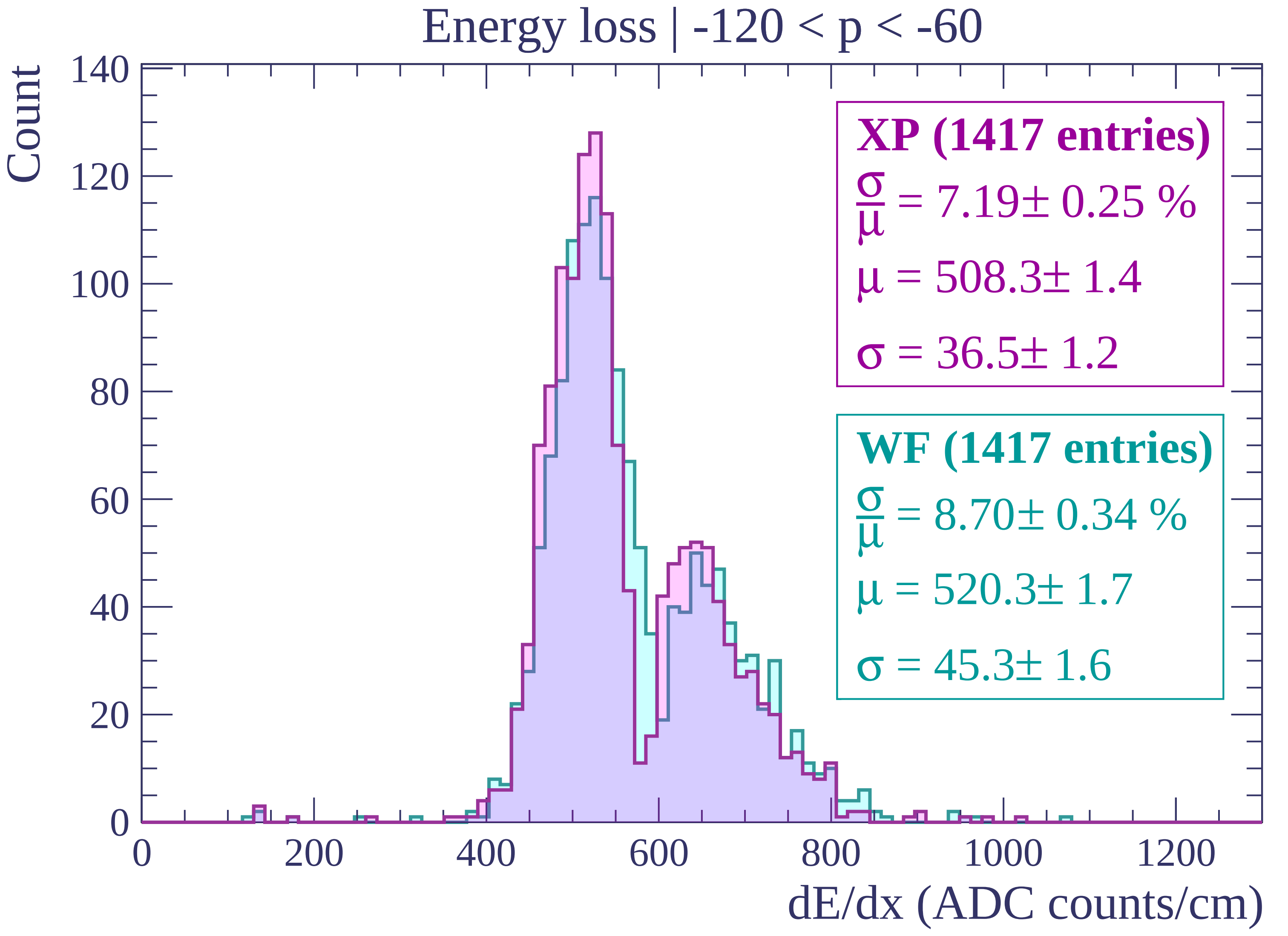
Energy loss | -360 < p < -300
Count
dE/dx (ADC counts/cm)
XP (7539 entries)
$\frac{\sigma}{\mu}$ = 8.62± 0.12 %
μ = 398.3± 0.5
σ = 34.4± 0.4
WF (7539 entries)
$\frac{\sigma}{\mu}$ = 8.87± 0.13 %
μ = 401.8± 0.5
σ = 35.6± 0.5


Energy loss | -300 < p < -240
Count
dE/dx (ADC counts/cm)
0
200
400
600
800
1000
1200
XP (6665 entries)
$\frac{\sigma}{\mu}$ = 8.41± 0.12 %
μ = 399.8± 0.5
σ = 33.6± 0.4
WF (6665 entries)
$\frac{\sigma}{\mu}$ = 8.76± 0.13 %
μ = 402.7± 0.5
σ = 35.3± 0.5

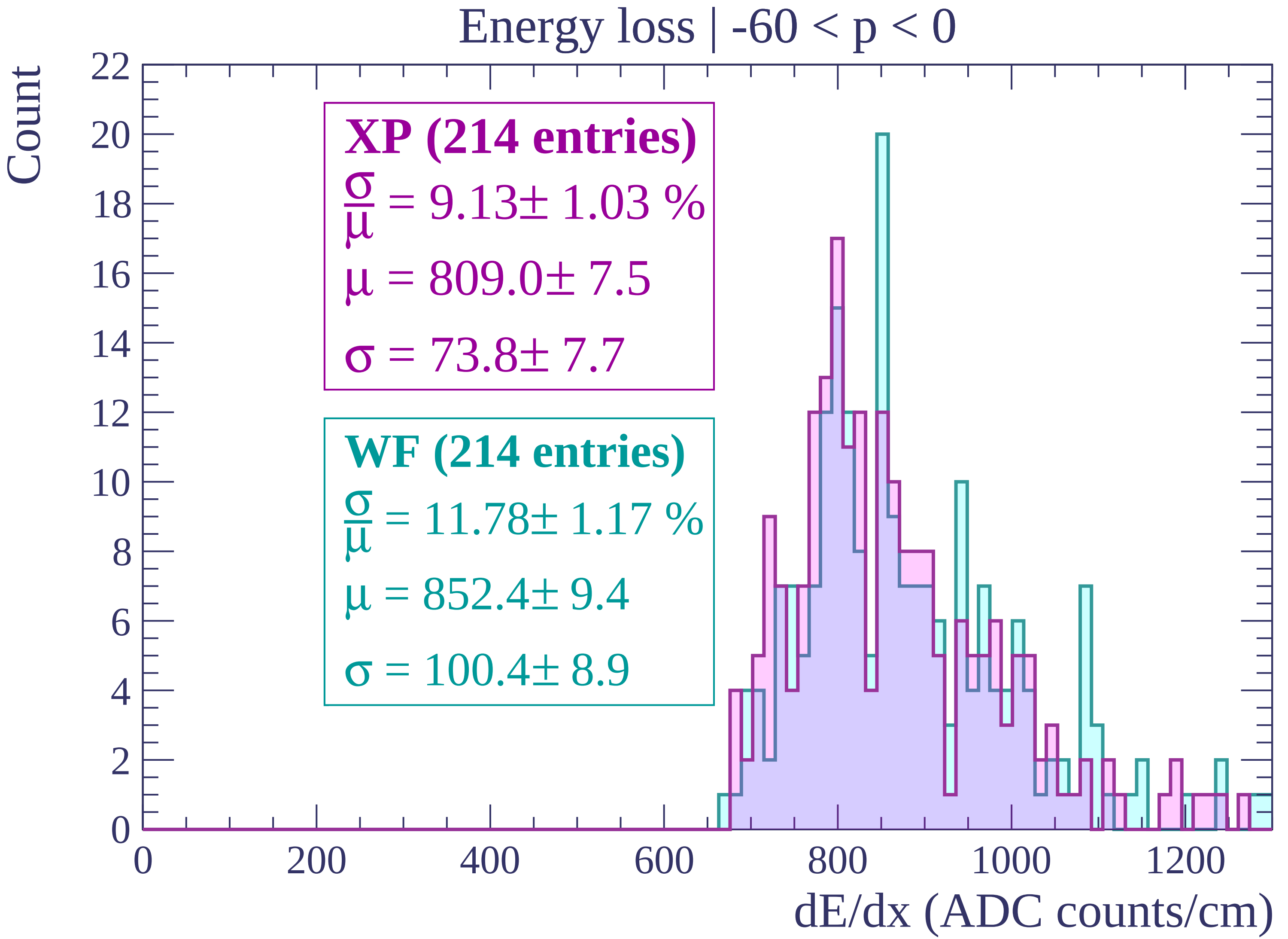

Energy loss | -240 < p < -180
Count
dE/dx (ADC counts/cm)
0
100
200
300
400
500
600
700
0
200
400
600
800
1000
1200
XP (4680 entries)
σ/μ = 8.52± 0.16 %
μ = 411.9± 0.6
σ = 35.1± 0.6
WF (4680 entries)
σ/μ = 8.75± 0.17 %
μ = 415.8± 0.6
σ = 36.4± 0.6



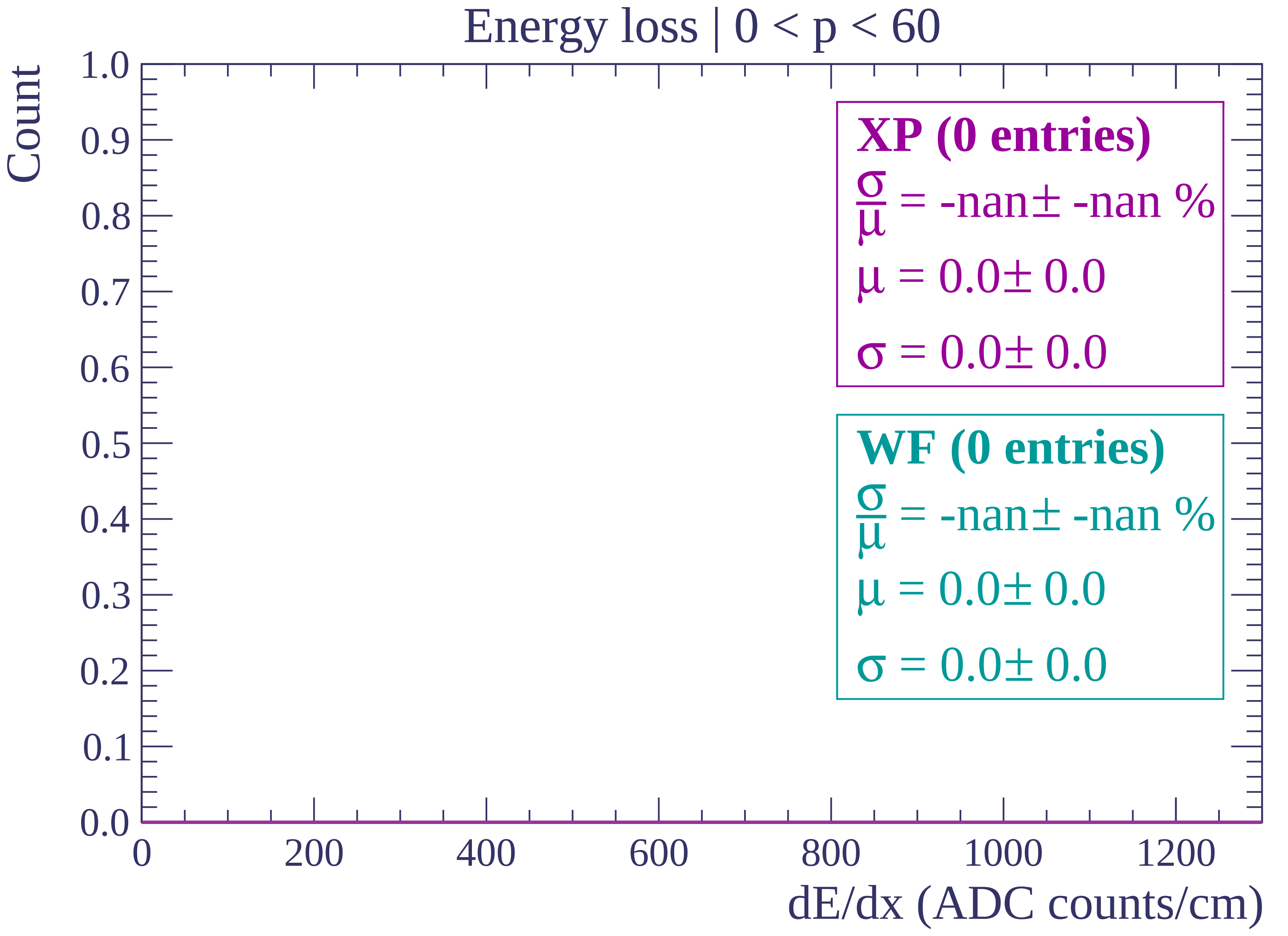
Energy loss | -180 < p < -120
Count
dE/dx (ADC counts/cm)
0
50
100
150
200
250
300
350
400
0
200
400
600
800
1000
1200
XP (3053 entries)
σ/μ = 8.73± 0.17 %
μ = 448.9± 0.8
σ = 39.2± 0.7
WF (3053 entries)
σ/μ = 9.24± 0.24 %
μ = 454.7± 0.9
σ = 42.0± 1.0



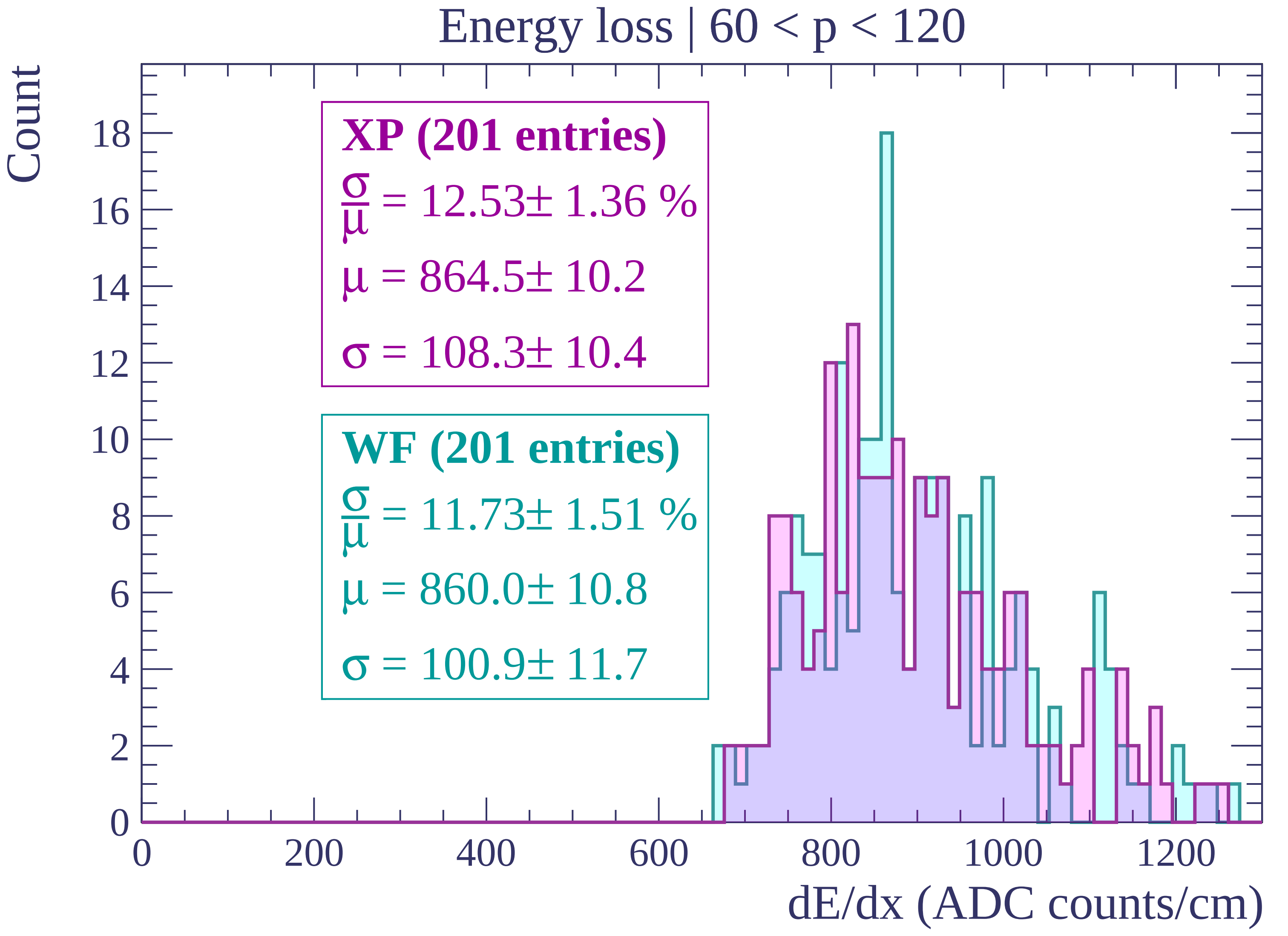
Energy loss | -120 < p < -60
Count
dE/dx (ADC counts/cm)
XP (1417 entries)
$\frac{\sigma}{\mu}$ = 7.19± 0.25 %
μ = 508.3± 1.4
σ = 36.5± 1.2
WF (1417 entries)
$\frac{\sigma}{\mu}$ = 8.70± 0.34 %
μ = 520.3± 1.7
σ = 45.3± 1.6
0
20
40
60
80
100
120
140
0
200
400
600
800
1000
1200



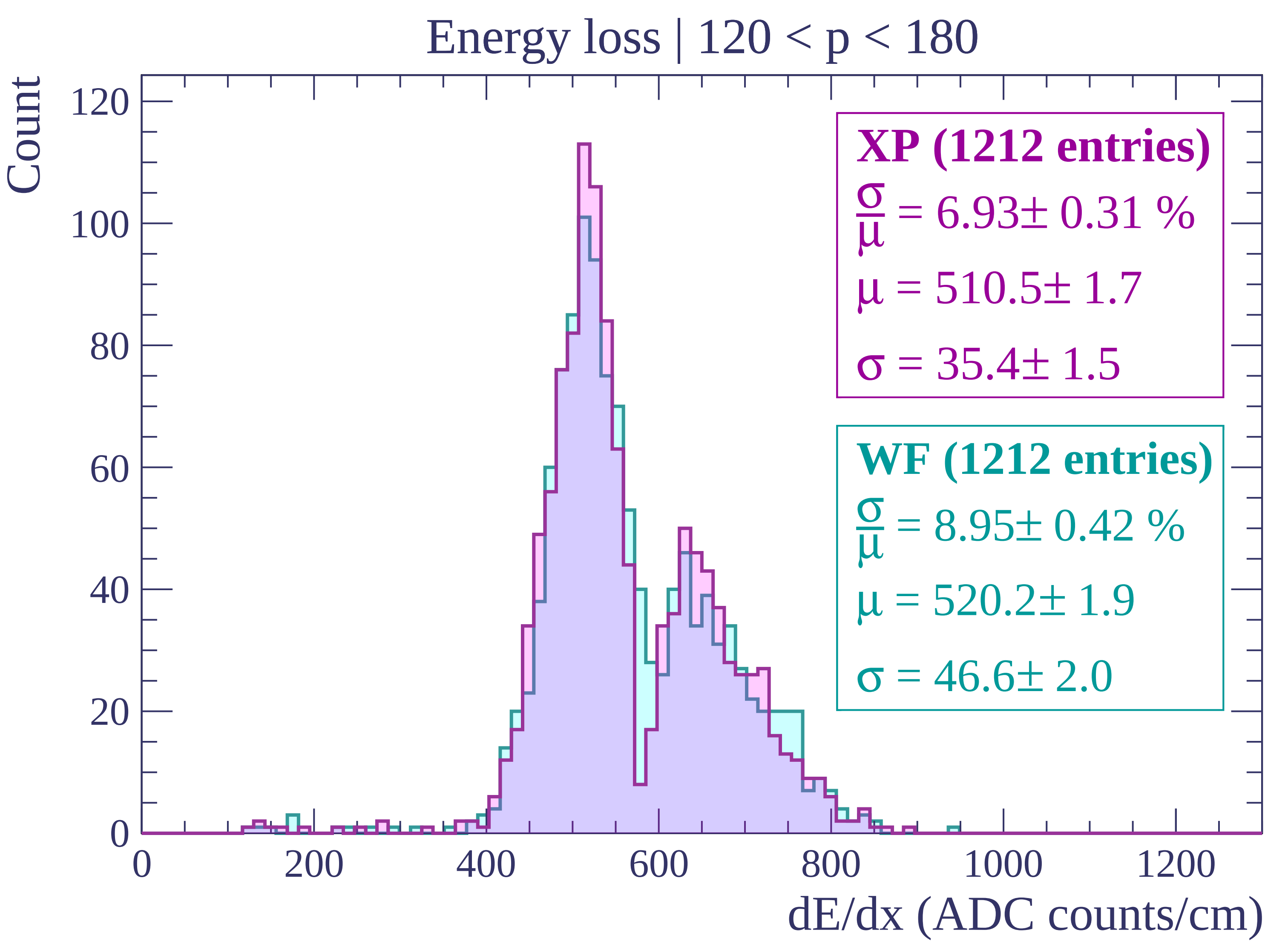
Energy loss | -60 < p < 0
Count
XP (214 entries)
σ/μ = 9.13± 1.03 %
μ = 809.0± 7.5
σ = 73.8± 7.7
WF (214 entries)
σ/μ = 11.78± 1.17 %
μ = 852.4± 9.4
σ = 100.4± 8.9
0
2
4
6
8
10
12
14
16
18
20
22
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)



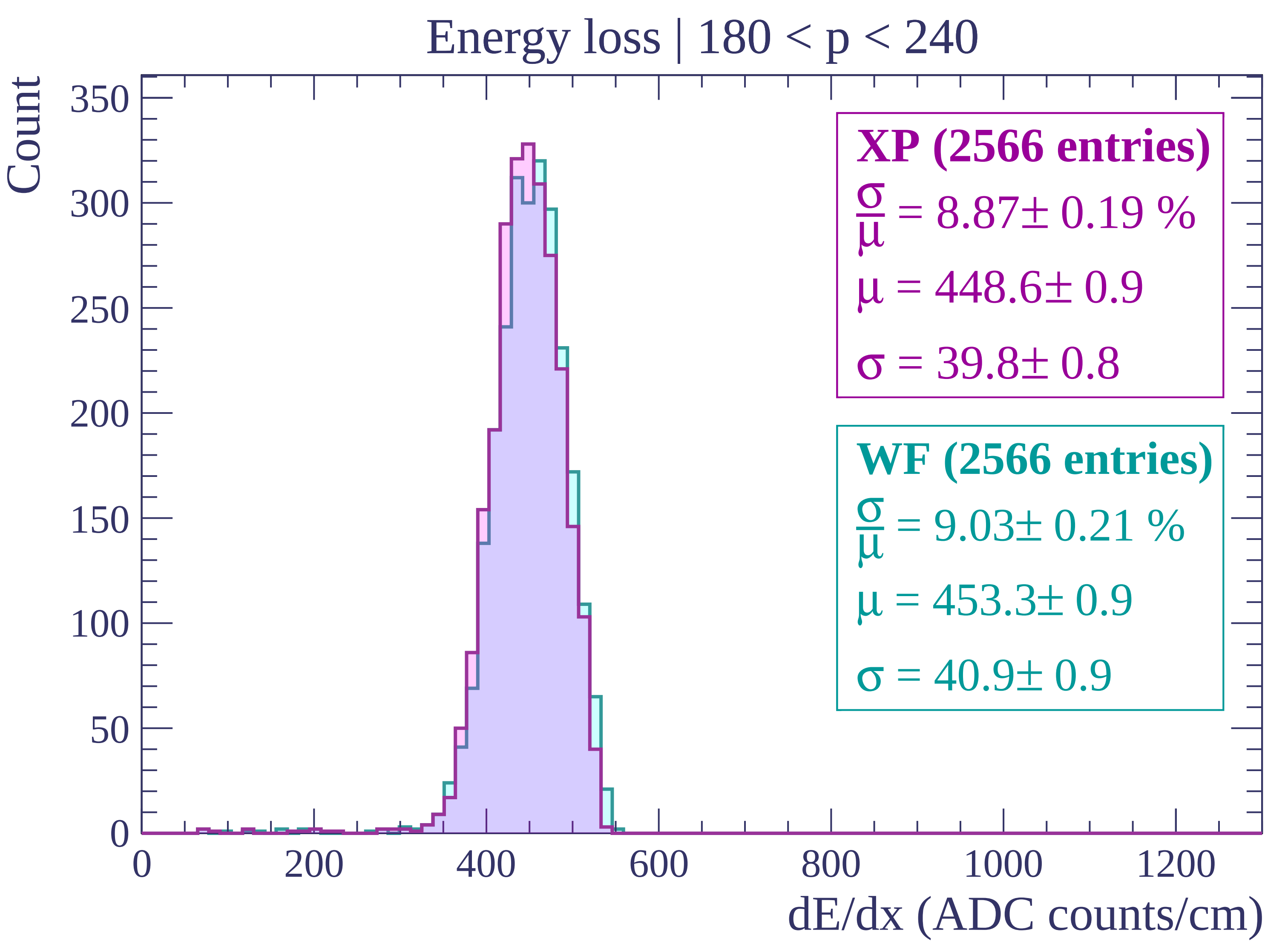
Energy loss | 0 < p < 60
Count
1.0
0.9
0.8
0.7
0.6
0.5
0.4
0.3
0.2
0.1
0.0
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP (0 entries)
σ/μ = -nan± -nan %
μ = 0.0± 0.0
σ = 0.0± 0.0
WF (0 entries)
σ/μ = -nan± -nan %
μ = 0.0± 0.0
σ = 0.0± 0.0



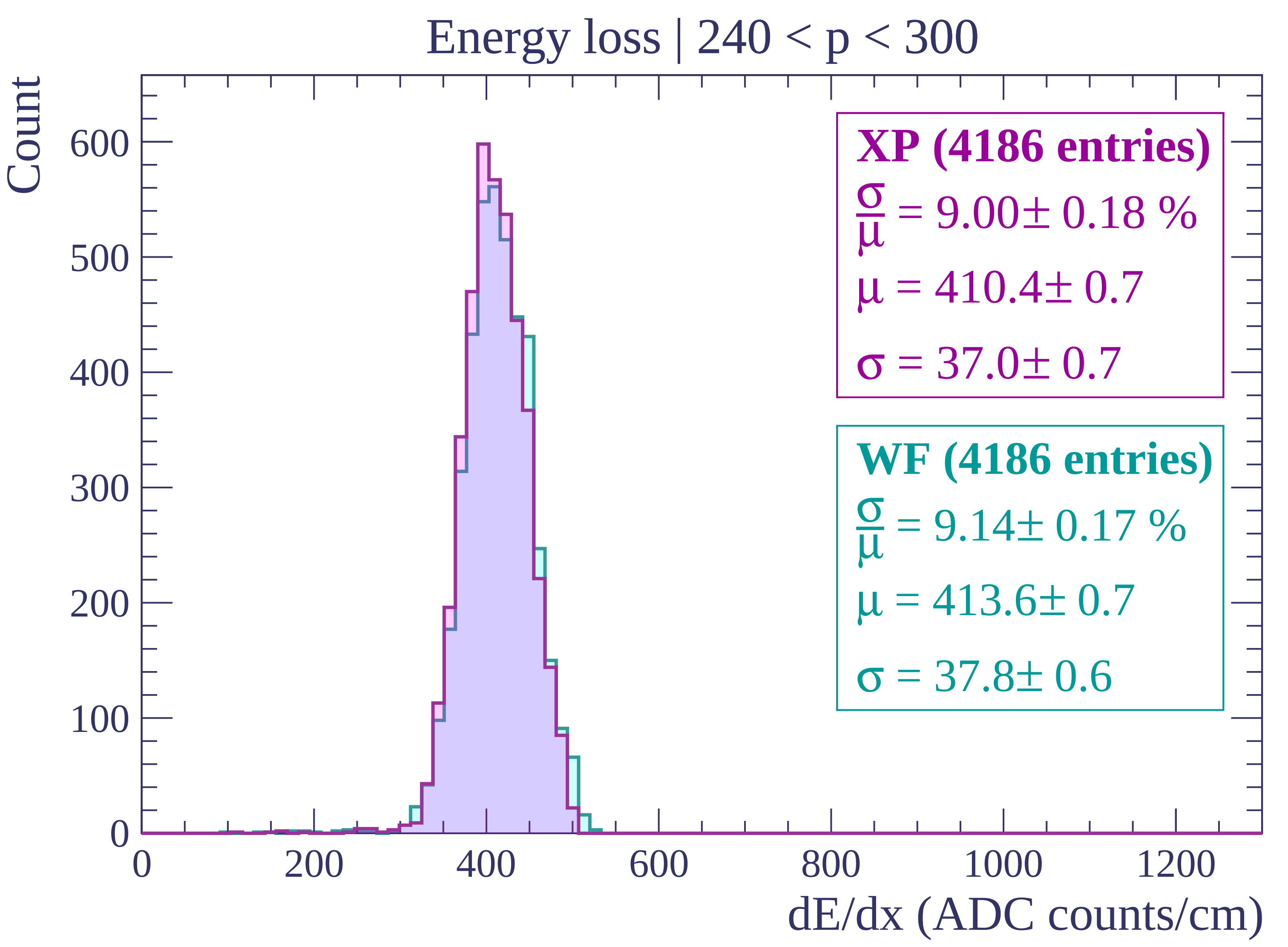
Energy loss | 60 < p < 120
Count
XP (201 entries)
$\frac{\sigma}{\mu}$ = 12.53± 1.36 %
μ = 864.5± 10.2
σ = 108.3± 10.4
WF (201 entries)
$\frac{\sigma}{\mu}$ = 11.73± 1.51 %
μ = 860.0± 10.8
σ = 100.9± 11.7
0
2
4
6
8
10
12
14
16
18
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)



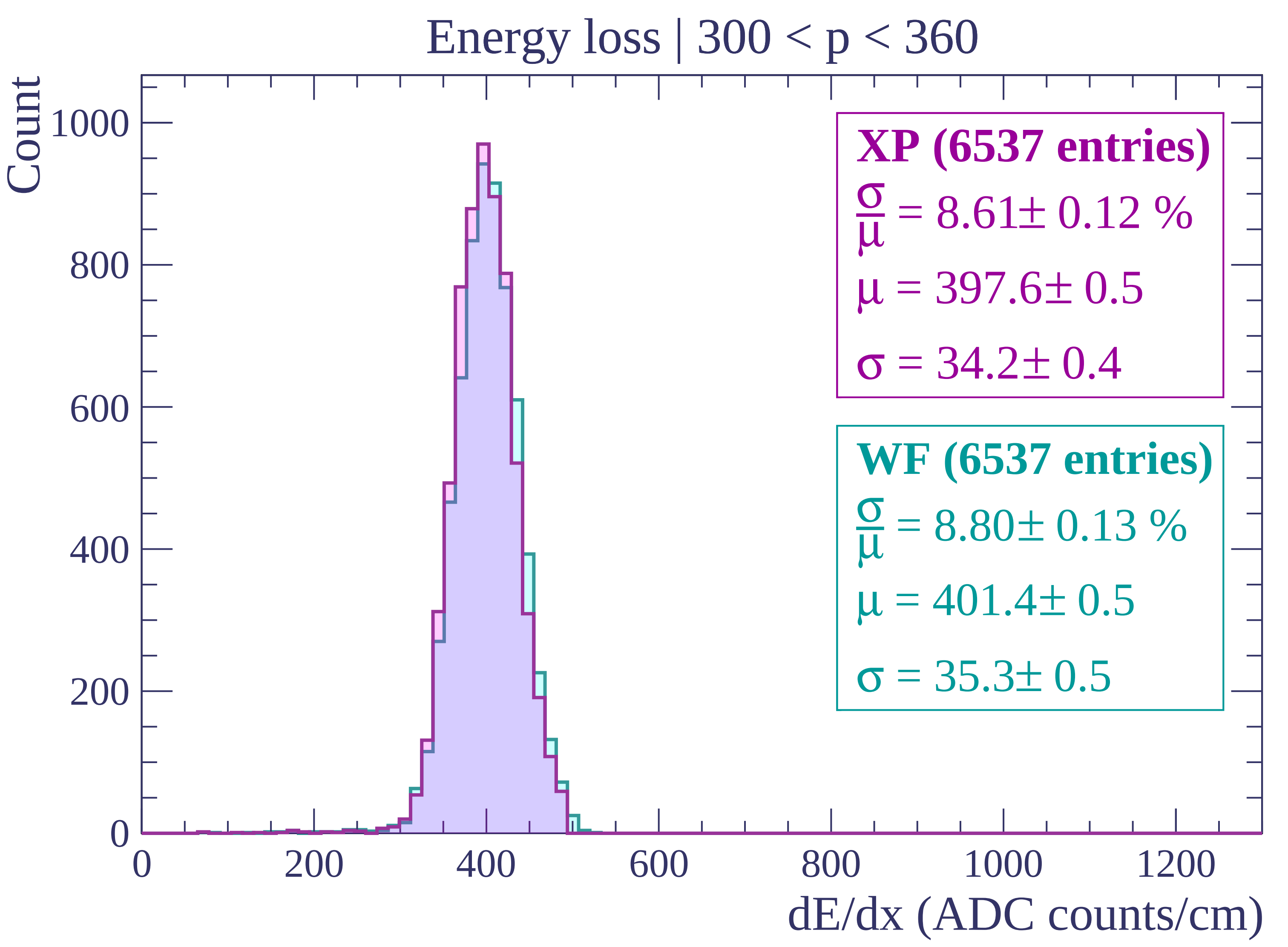
Energy loss | 120 < p < 180
Count
dE/dx (ADC counts/cm)
XP (1212 entries)
σ/μ = 6.93± 0.31 %
μ = 510.5± 1.7
σ = 35.4± 1.5
WF (1212 entries)
σ/μ = 8.95± 0.42 %
μ = 520.2± 1.9
σ = 46.6± 2.0
0
20
40
60
80
100
120
0
200
400
600
800
1000
1200

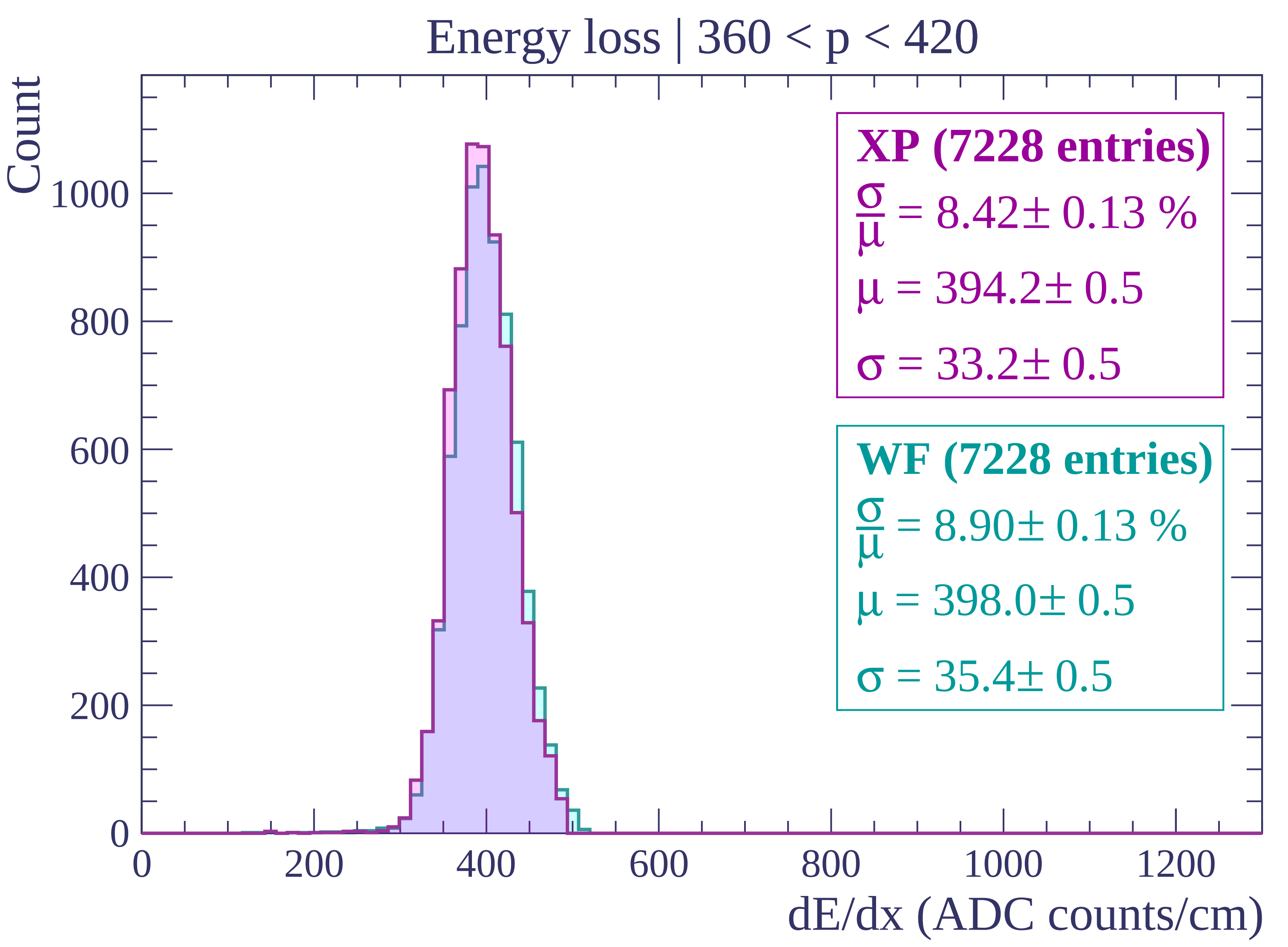


Energy loss | 180 < p < 240
Count
dE/dx (ADC counts/cm)
XP (2566 entries)
$\frac{\sigma}{\mu}$ = 8.87± 0.19 %
μ = 448.6± 0.9
σ = 39.8± 0.8
WF (2566 entries)
$\frac{\sigma}{\mu}$ = 9.03± 0.21 %
μ = 453.3± 0.9
σ = 40.9± 0.9
0
50
100
150
200
250
300
350
0
200
400
600
800
1000
1200

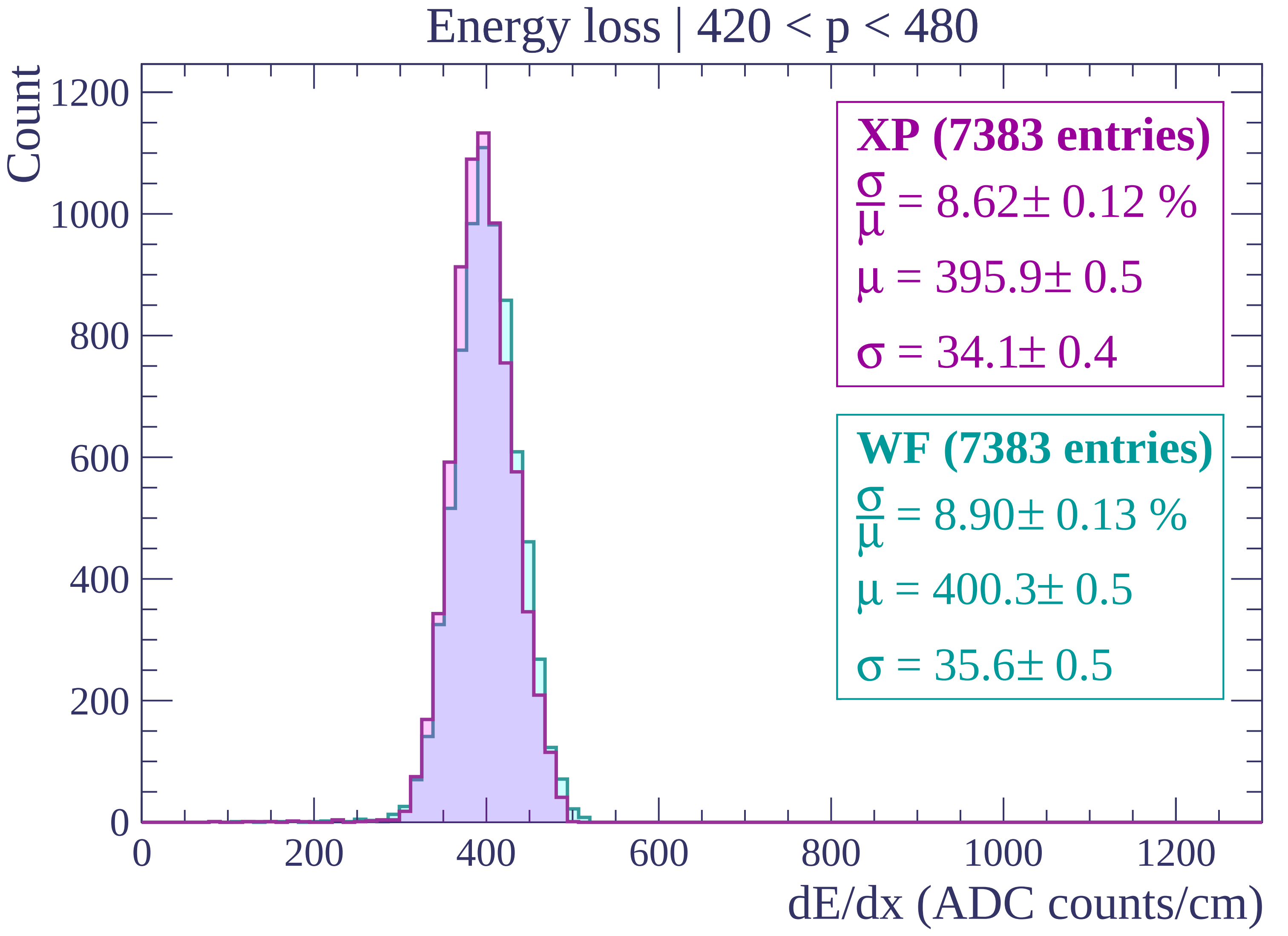


Energy loss | 240 < p < 300
Count
dE/dx (ADC counts/cm)
0
100
200
300
400
500
600
0
200
400
600
800
1000
1200
XP (4186 entries)
$\frac{\sigma}{\mu}$ = 9.00± 0.18 %
μ = 410.4± 0.7
σ = 37.0± 0.7
WF (4186 entries)
$\frac{\sigma}{\mu}$ = 9.14± 0.17 %
μ = 413.6± 0.7
σ = 37.8± 0.6

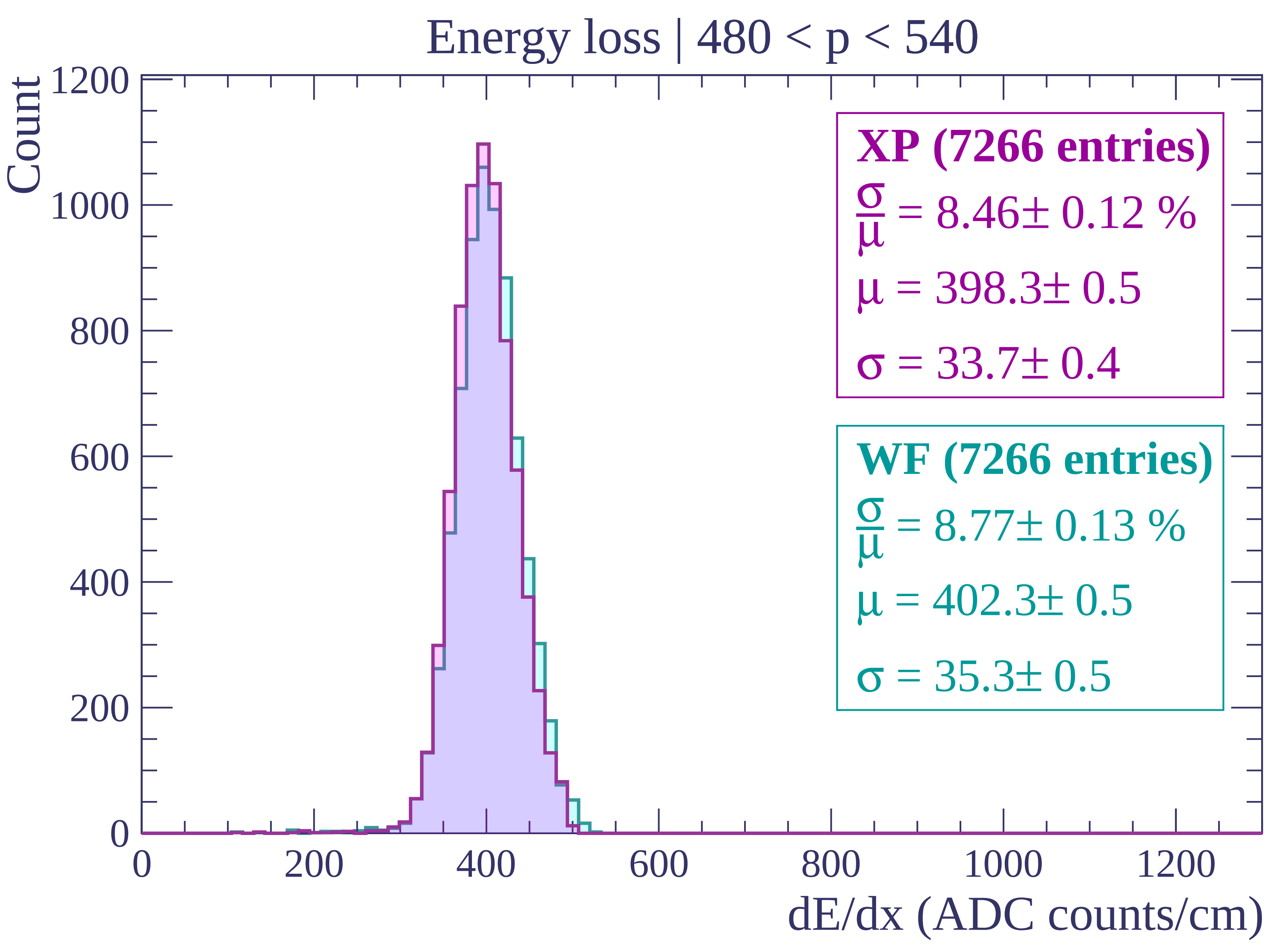

Energy loss | 300 < p < 360
Count
1000
800
600
400
200
0
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP (6537 entries)
$\frac{\sigma}{\mu}$ = 8.61± 0.12 %
μ = 397.6± 0.5
σ = 34.2± 0.4
WF (6537 entries)
$\frac{\sigma}{\mu}$ = 8.80± 0.13 %
μ = 401.4± 0.5
σ = 35.3± 0.5



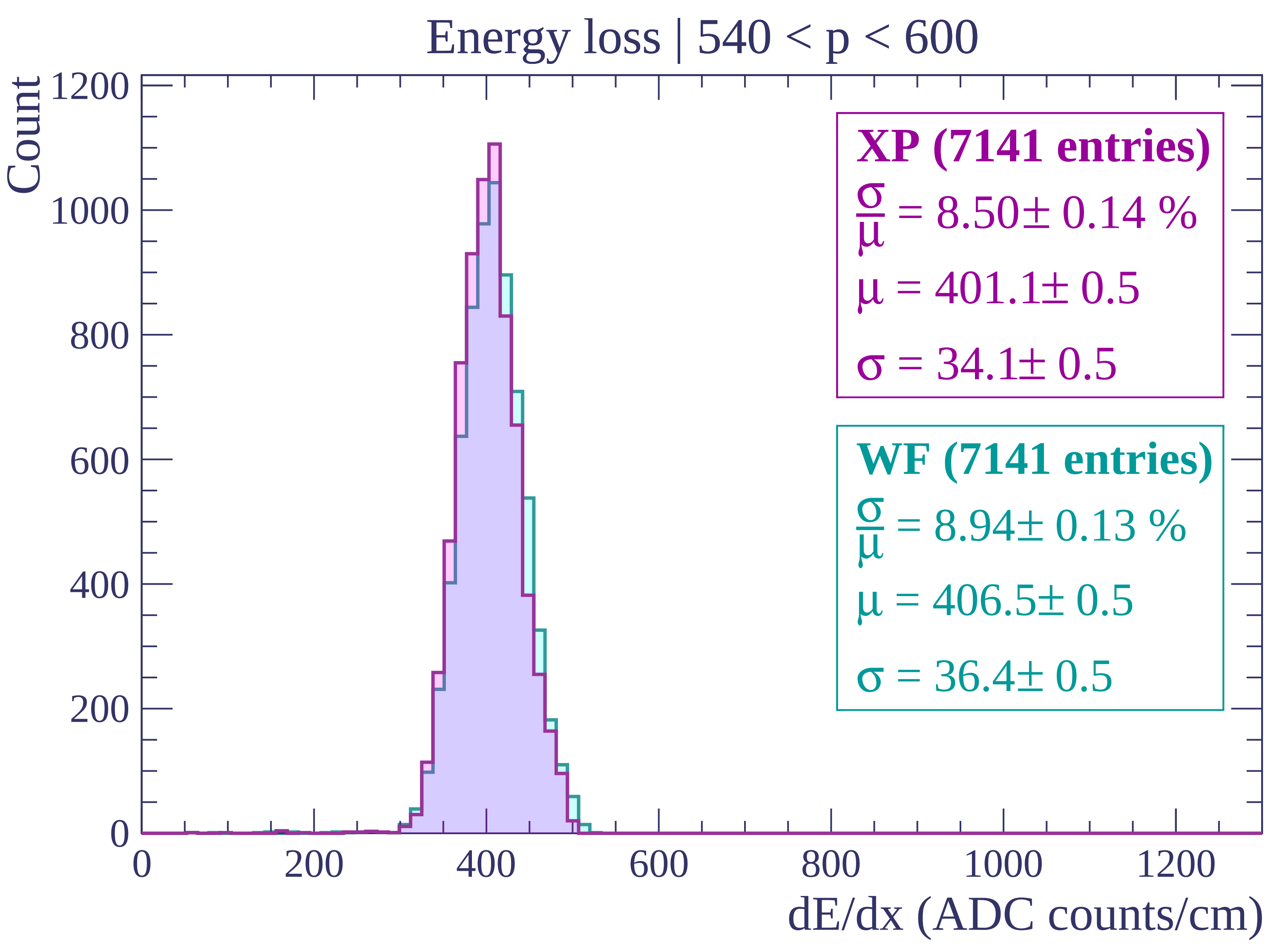
Energy loss | 360 < p < 420
Count
dE/dx (ADC counts/cm)
0
200
400
600
800
1000
1200
XP (7228 entries)
σ/μ = 8.42± 0.13 %
μ = 394.2± 0.5
σ = 33.2± 0.5
WF (7228 entries)
σ/μ = 8.90± 0.13 %
μ = 398.0± 0.5
σ = 35.4± 0.5



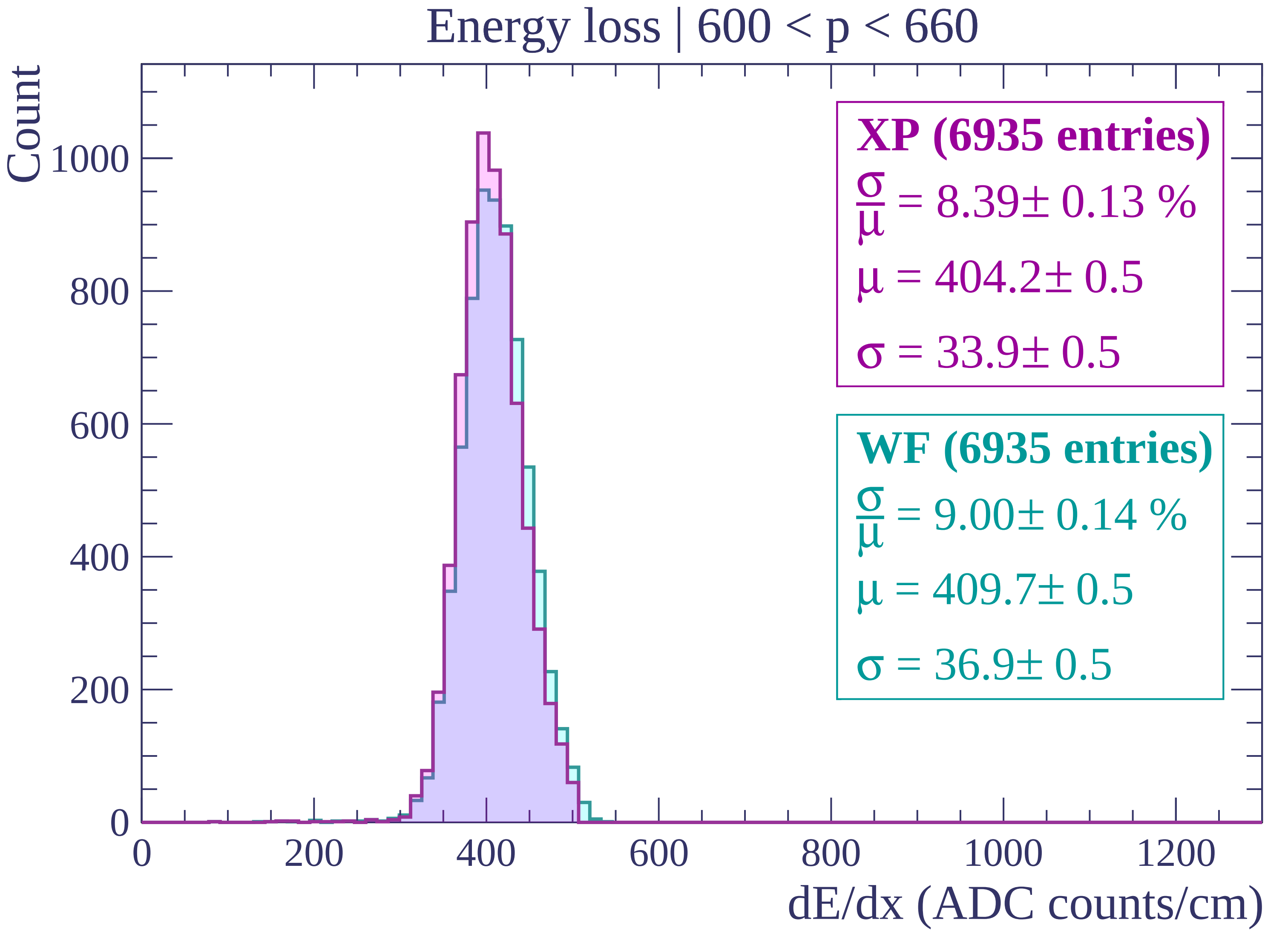
Energy loss | 420 < p < 480
Count
dE/dx (ADC counts/cm)
0
200
400
600
800
1000
1200
XP (7383 entries)
$\frac{\sigma}{\mu}$ = 8.62± 0.12 %
μ = 395.9± 0.5
σ = 34.1± 0.4
WF (7383 entries)
$\frac{\sigma}{\mu}$ = 8.90± 0.13 %
μ = 400.3± 0.5
σ = 35.6± 0.5

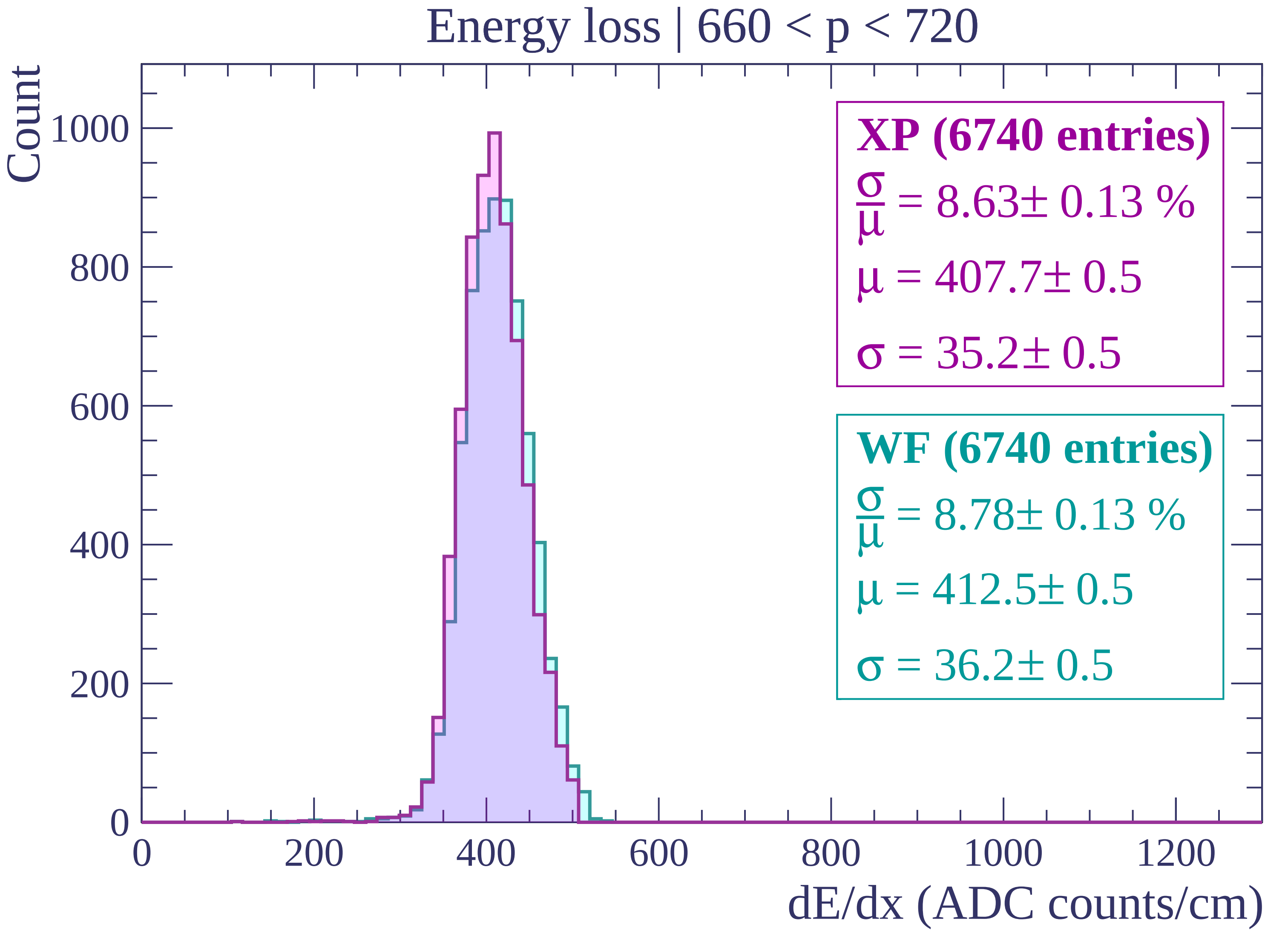


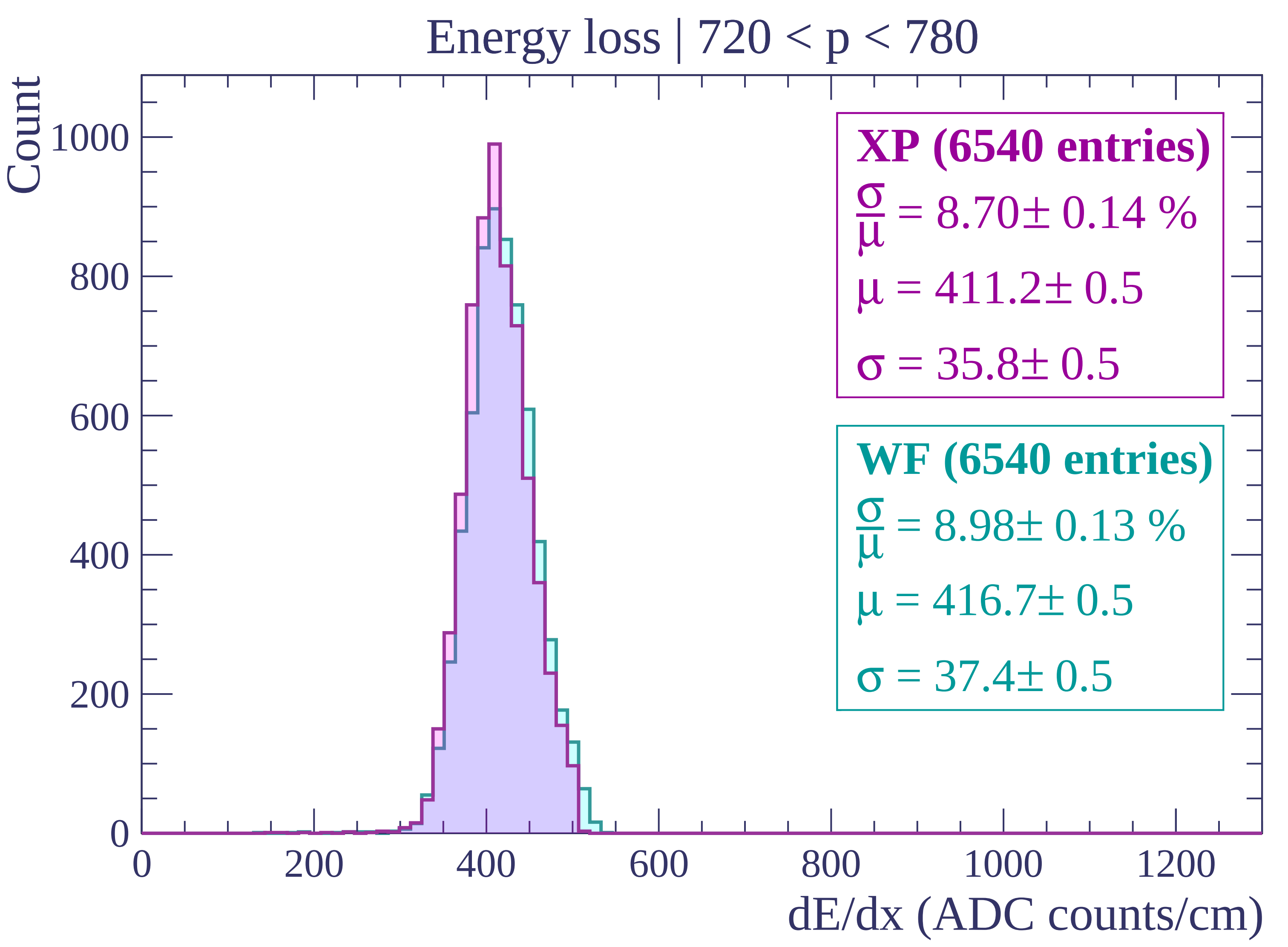
Energy loss | 480 < p < 540
Count
dE/dx (ADC counts/cm)
0
200
400
600
800
1000
1200
XP (7266 entries)
σ/μ = 8.46± 0.12 %
μ = 398.3± 0.5
σ = 33.7± 0.4
WF (7266 entries)
σ/μ = 8.77± 0.13 %
μ = 402.3± 0.5
σ = 35.3± 0.5



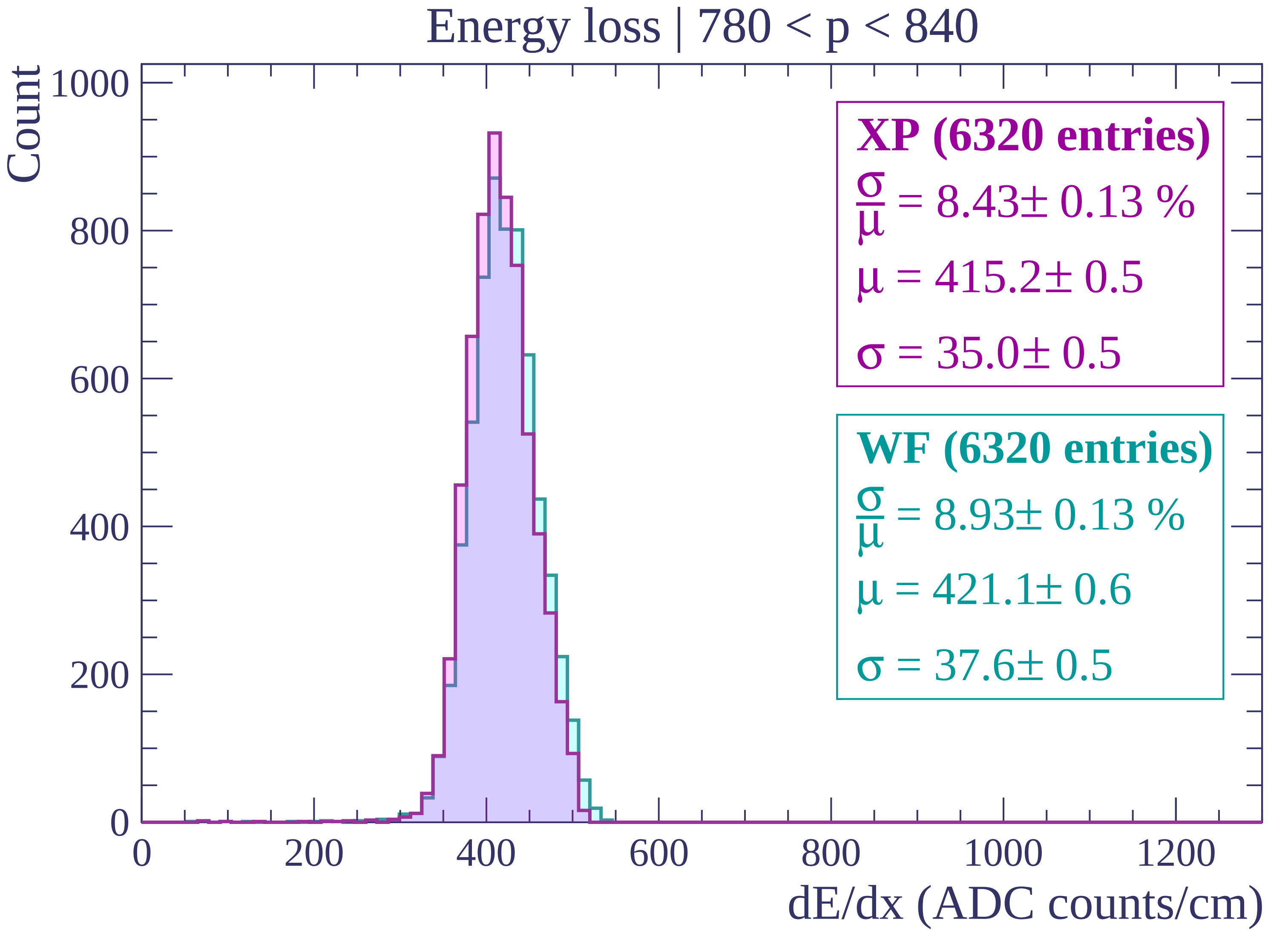
Energy loss | 540 < p < 600
Count
XP (7141 entries)
σ/μ = 8.50± 0.14 %
μ = 401.1± 0.5
σ = 34.1± 0.5
WF (7141 entries)
σ/μ = 8.94± 0.13 %
μ = 406.5± 0.5
σ = 36.4± 0.5
dE/dx (ADC counts/cm)


Energy loss | 600 < p < 660
Count
dE/dx (ADC counts/cm)
XP (6935 entries)
σ/μ = 8.39± 0.13 %
μ = 404.2± 0.5
σ = 33.9± 0.5
WF (6935 entries)
σ/μ = 9.00± 0.14 %
μ = 409.7± 0.5
σ = 36.9± 0.5
0
200
400
600
800
1000
1200

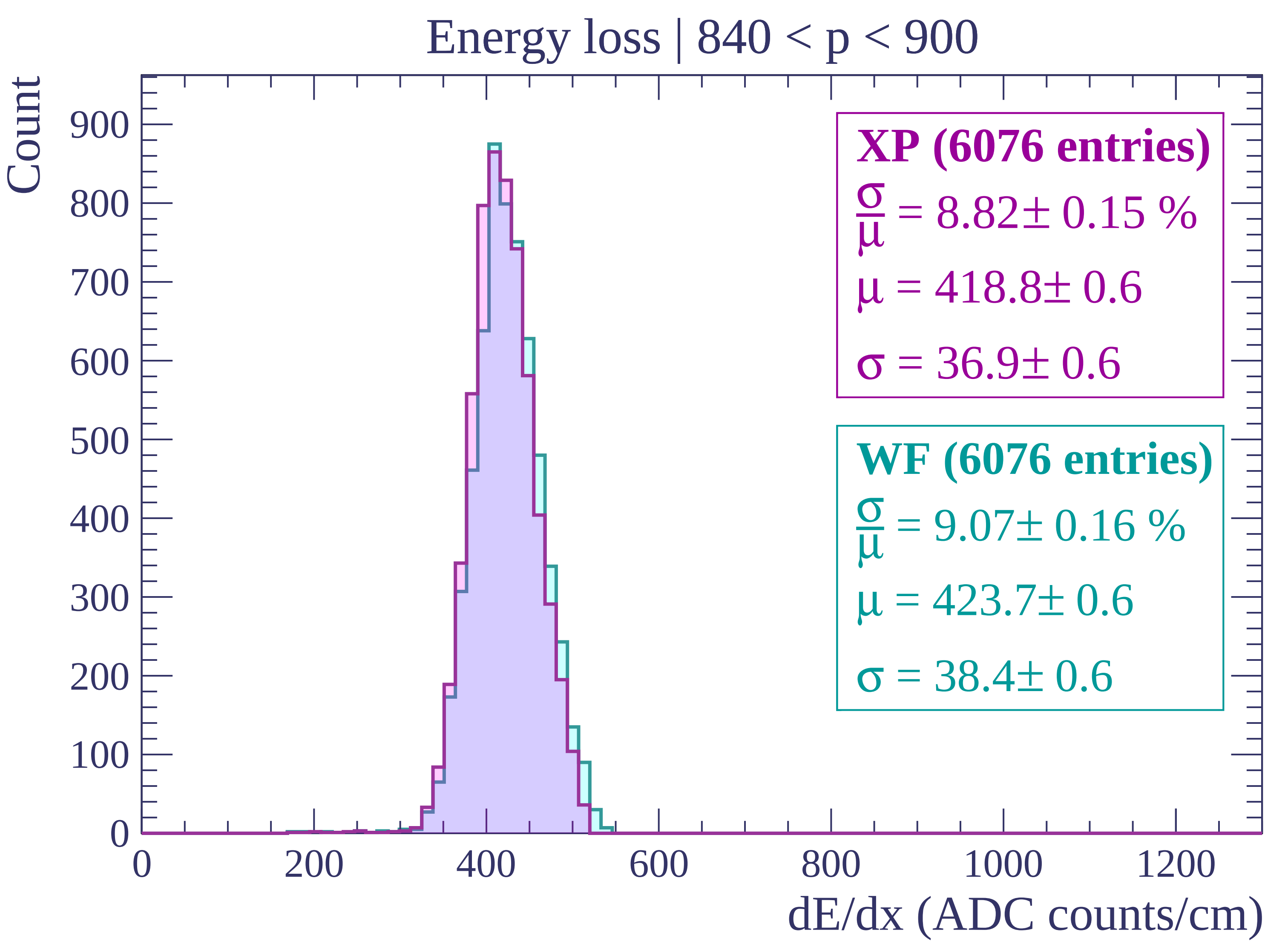

Energy loss | 660 < p < 720
Count
1000
800
600
400
200
0
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP (6740 entries)
$\frac{\sigma}{\mu}$ = 8.63± 0.13 %
μ = 407.7± 0.5
σ = 35.2± 0.5
WF (6740 entries)
$\frac{\sigma}{\mu}$ = 8.78± 0.13 %
μ = 412.5± 0.5
σ = 36.2± 0.5

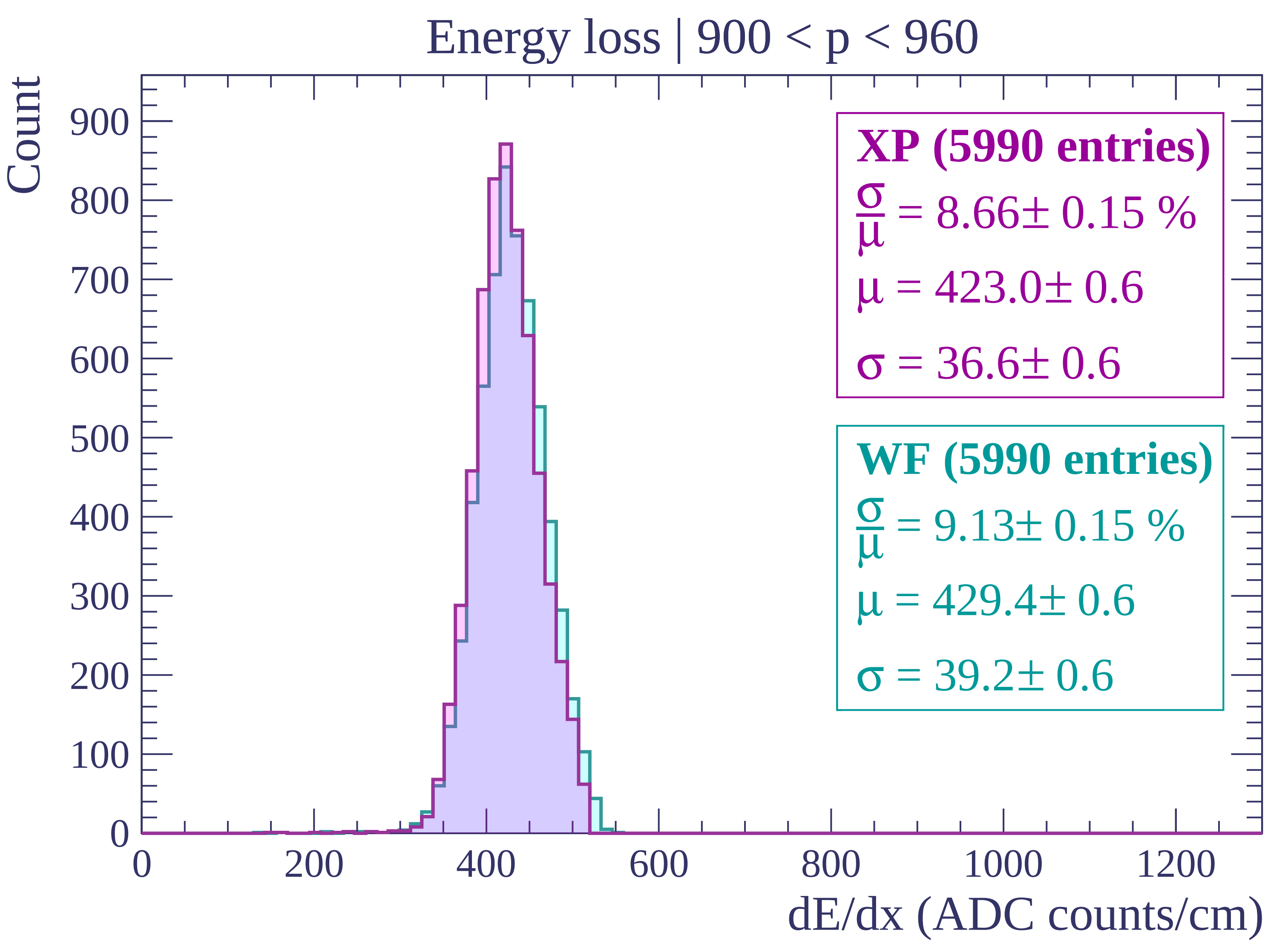


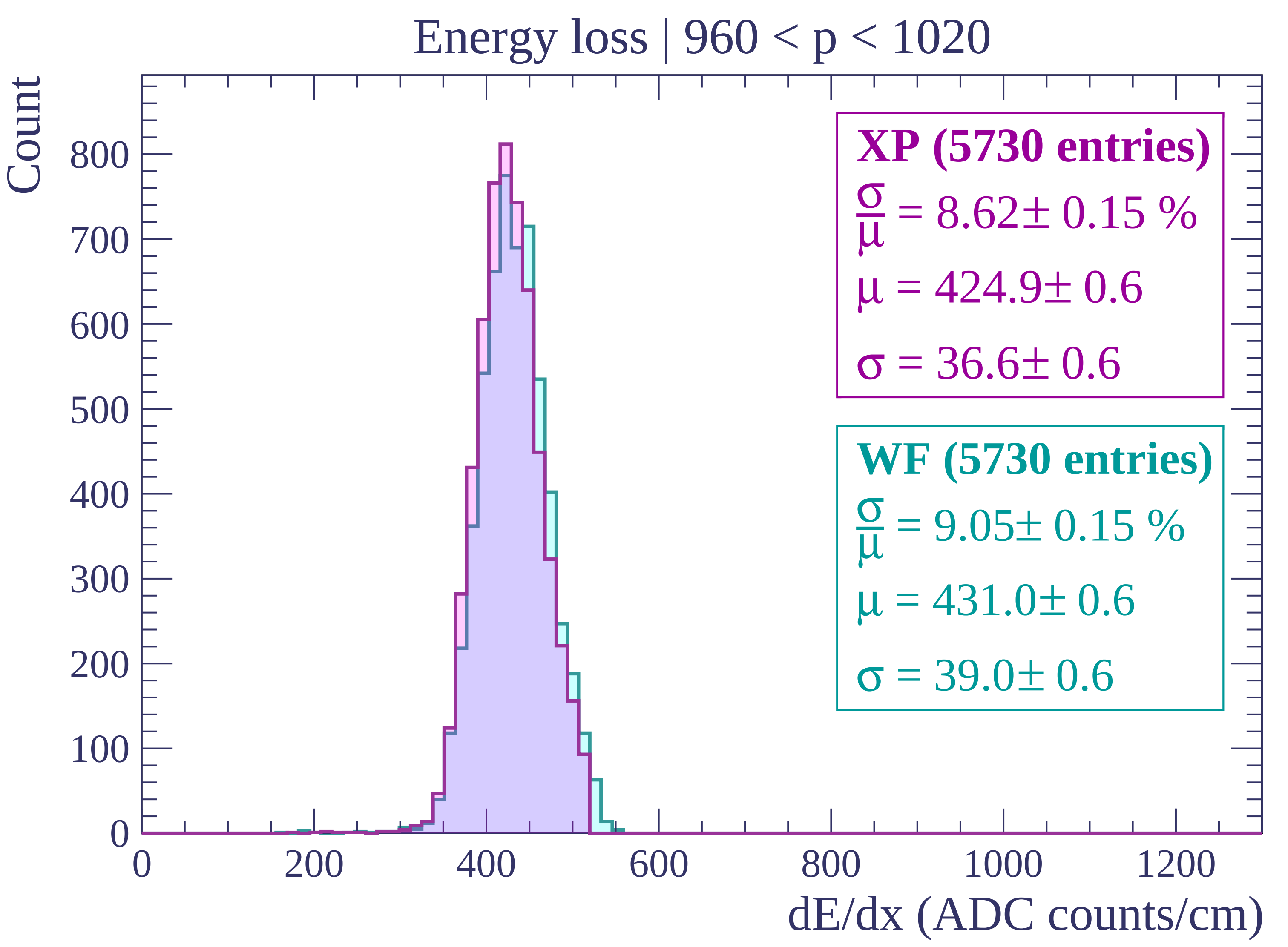
Energy loss | 720 < p < 780
Count
dE/dx (ADC counts/cm)
0
200
400
600
800
1000
1200
XP (6540 entries)
σ/μ = 8.70± 0.14 %
μ = 411.2± 0.5
σ = 35.8± 0.5
WF (6540 entries)
σ/μ = 8.98± 0.13 %
μ = 416.7± 0.5
σ = 37.4± 0.5



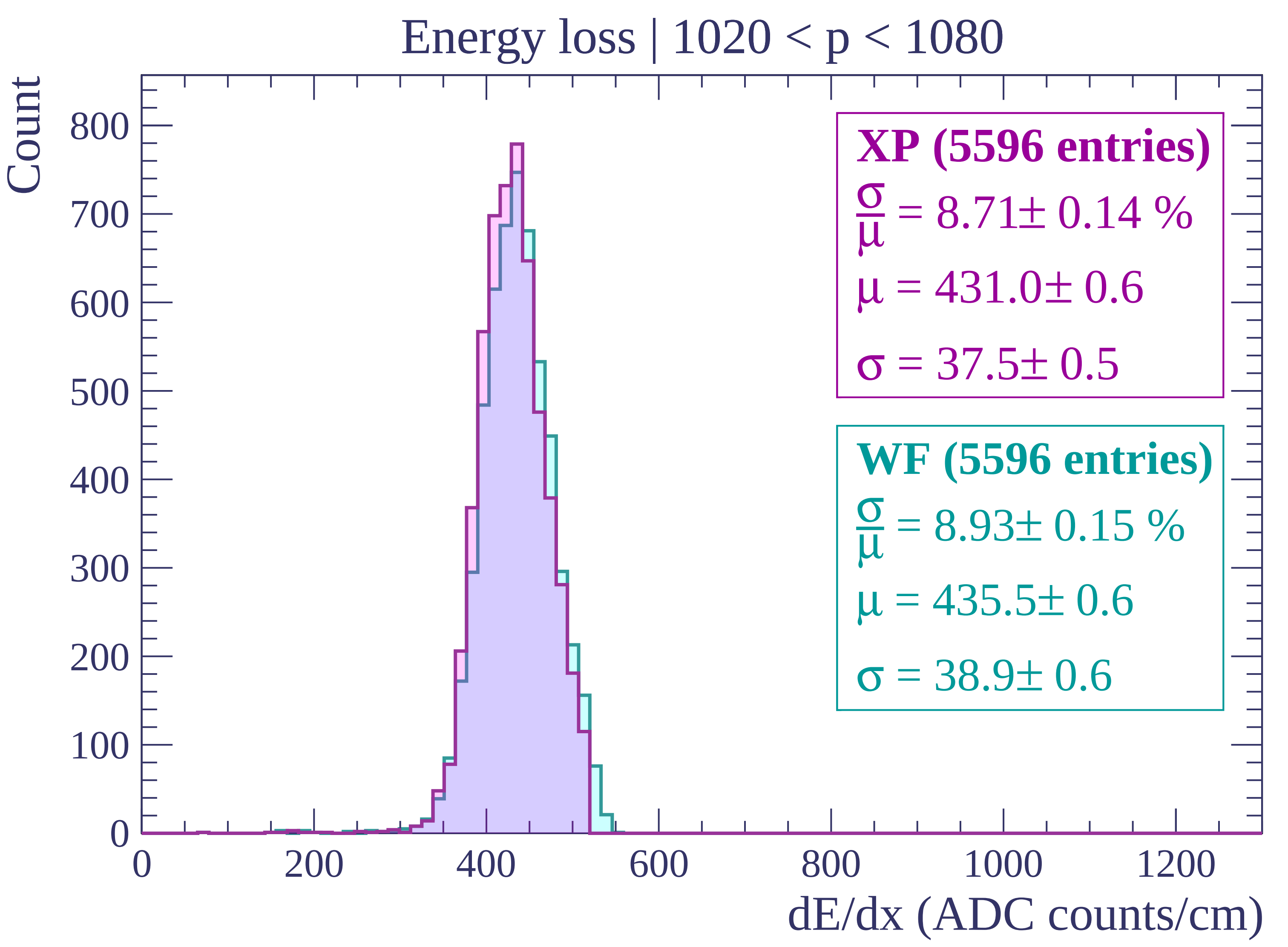
Energy loss | 780 < p < 840
Count
dE/dx (ADC counts/cm)
0
200
400
600
800
1000
1200
XP (6320 entries)
$\frac{\sigma}{\mu}$ = 8.43± 0.13 %
μ = 415.2± 0.5
σ = 35.0± 0.5
WF (6320 entries)
$\frac{\sigma}{\mu}$ = 8.93± 0.13 %
μ = 421.1± 0.6
σ = 37.6± 0.5


Energy loss | 840 < p < 900
Count
dE/dx (ADC counts/cm)
0
100
200
300
400
500
600
700
800
900
0
200
400
600
800
1000
1200
XP (6076 entries)
$\frac{\sigma}{\mu} = 8.82 \pm 0.15$ %
$\mu = 418.8 \pm 0.6$
$\sigma = 36.9 \pm 0.6$
WF (6076 entries)
$\frac{\sigma}{\mu} = 9.07 \pm 0.16$ %
$\mu = 423.7 \pm 0.6$
$\sigma = 38.4 \pm 0.6$

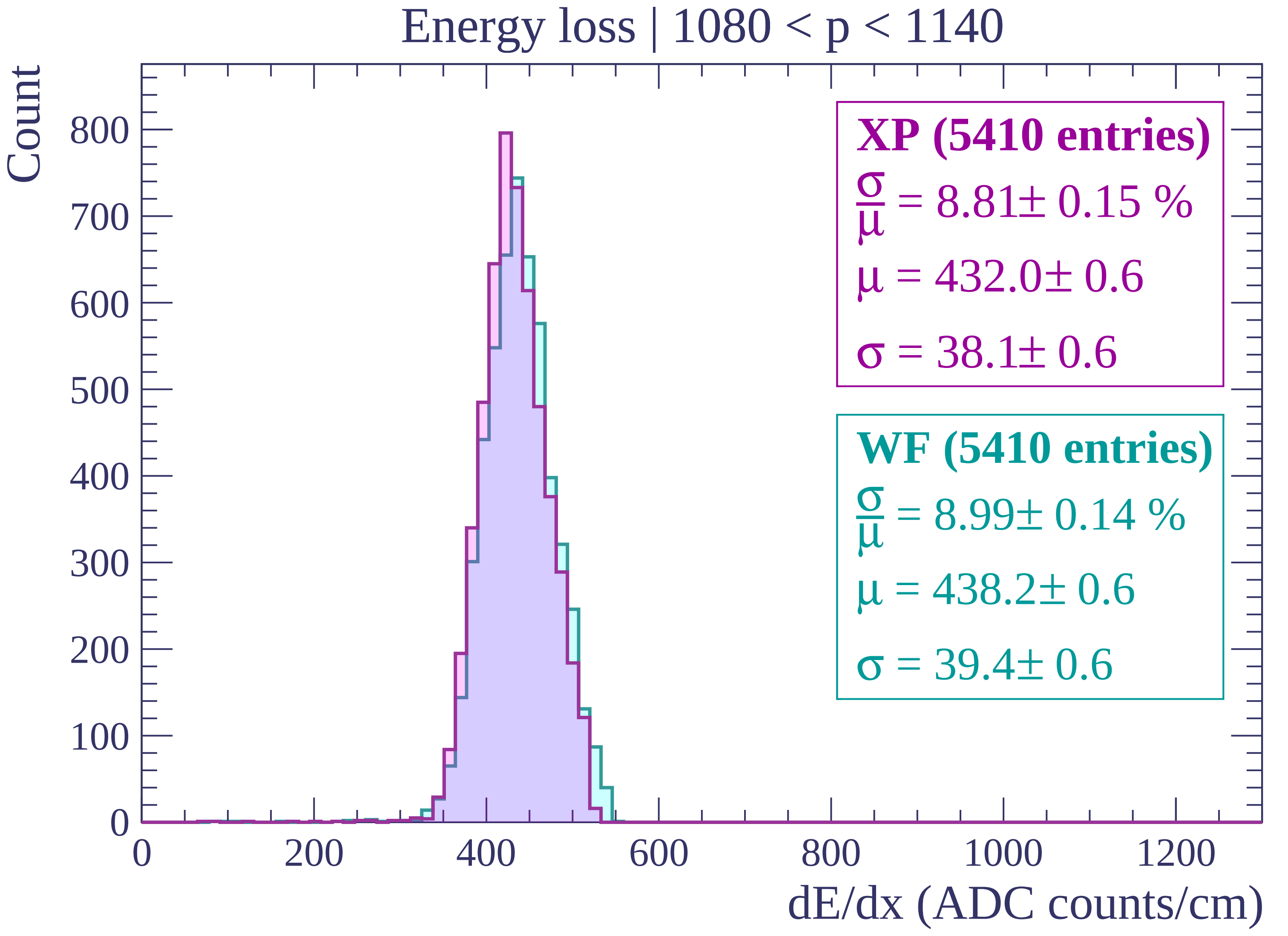

Energy loss | 900 < p < 960
Count
dE/dx (ADC counts/cm)
XP (5990 entries)
σ/μ = 8.66± 0.15 %
μ = 423.0± 0.6
σ = 36.6± 0.6
WF (5990 entries)
σ/μ = 9.13± 0.15 %
μ = 429.4± 0.6
σ = 39.2± 0.6
0
100
200
300
400
500
600
700
800
900
0
200
400
600
800
1000
1200



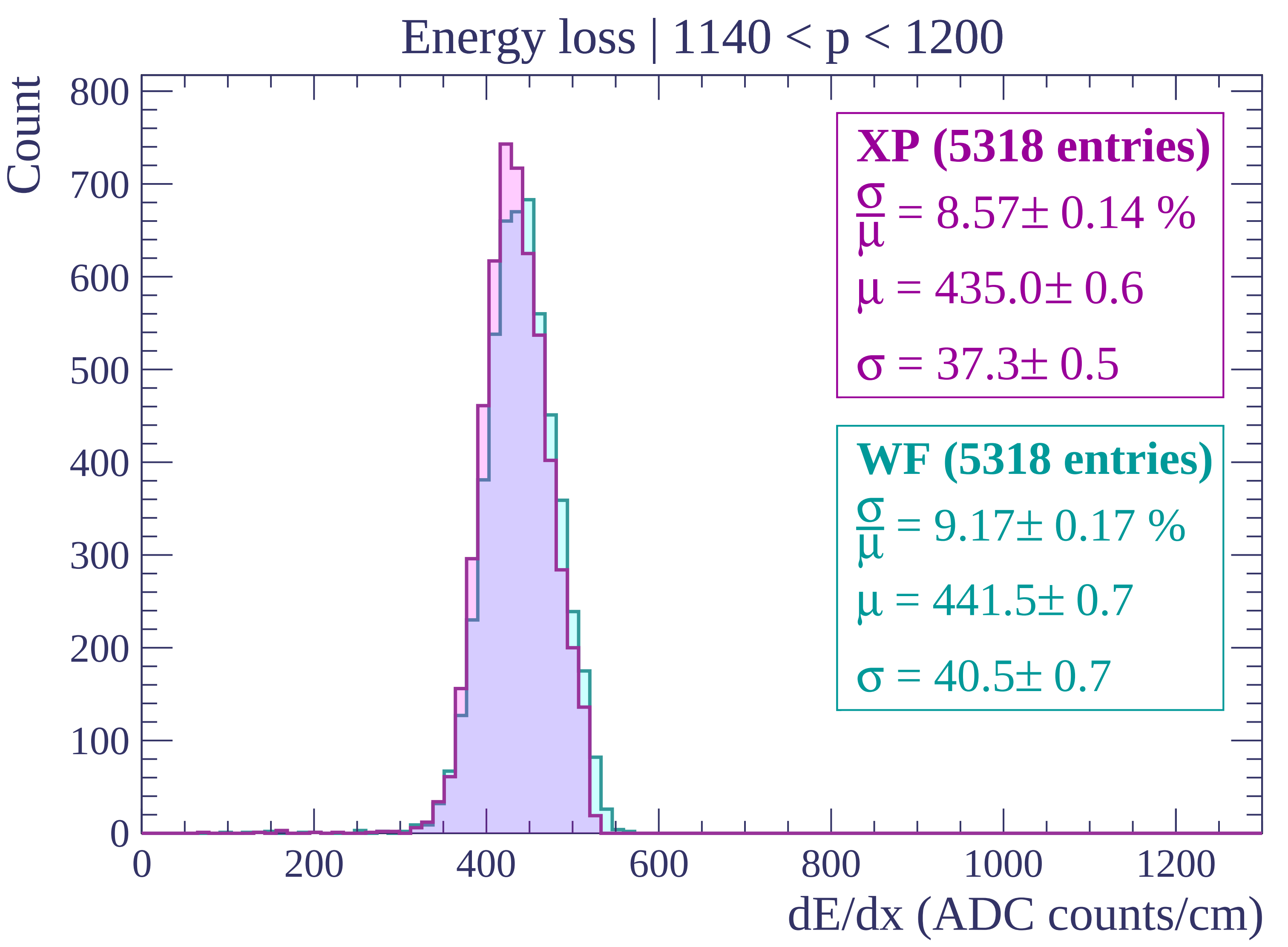
Energy loss | 960 < p < 1020
Count
dE/dx (ADC counts/cm)
0
100
200
300
400
500
600
700
800
0
200
400
600
800
1000
1200
XP (5730 entries)
σ/μ = 8.62± 0.15 %
μ = 424.9± 0.6
σ = 36.6± 0.6
WF (5730 entries)
σ/μ = 9.05± 0.15 %
μ = 431.0± 0.6
σ = 39.0± 0.6

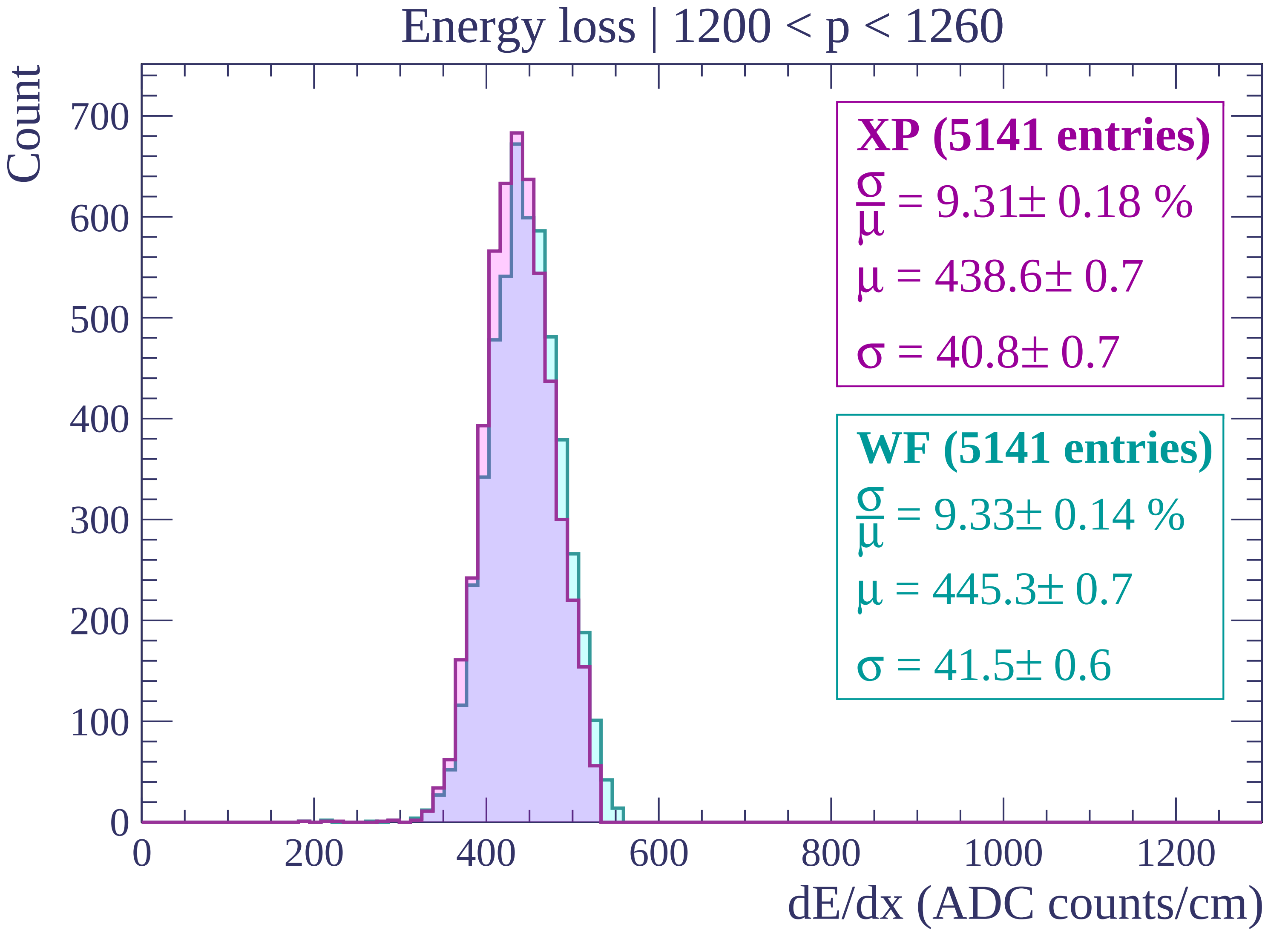


Energy loss | 1020 < p < 1080
Count
dE/dx (ADC counts/cm)
XP (5596 entries)
σ/μ = 8.71± 0.14 %
μ = 431.0± 0.6
σ = 37.5± 0.5
WF (5596 entries)
σ/μ = 8.93± 0.15 %
μ = 435.5± 0.6
σ = 38.9± 0.6

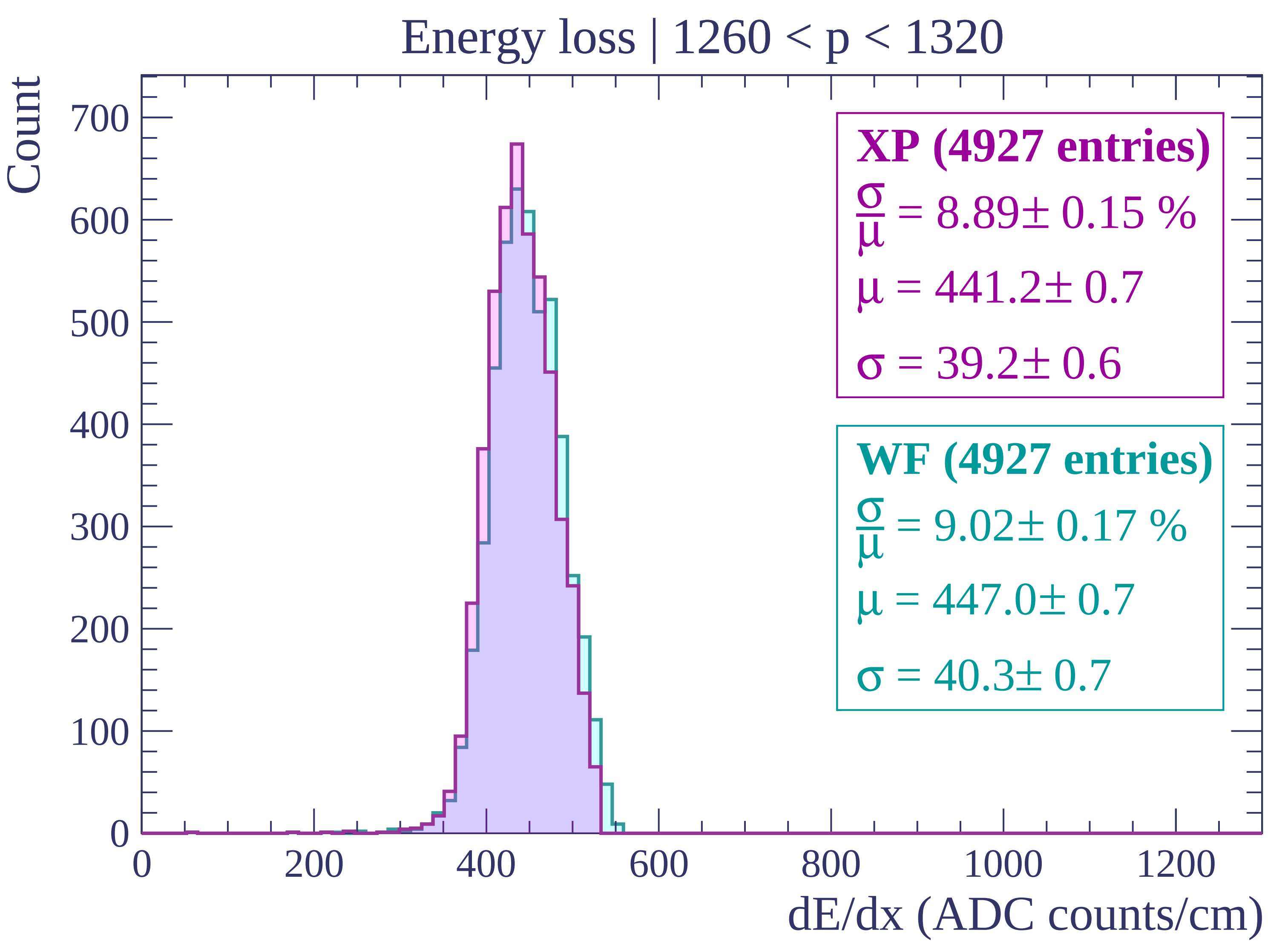


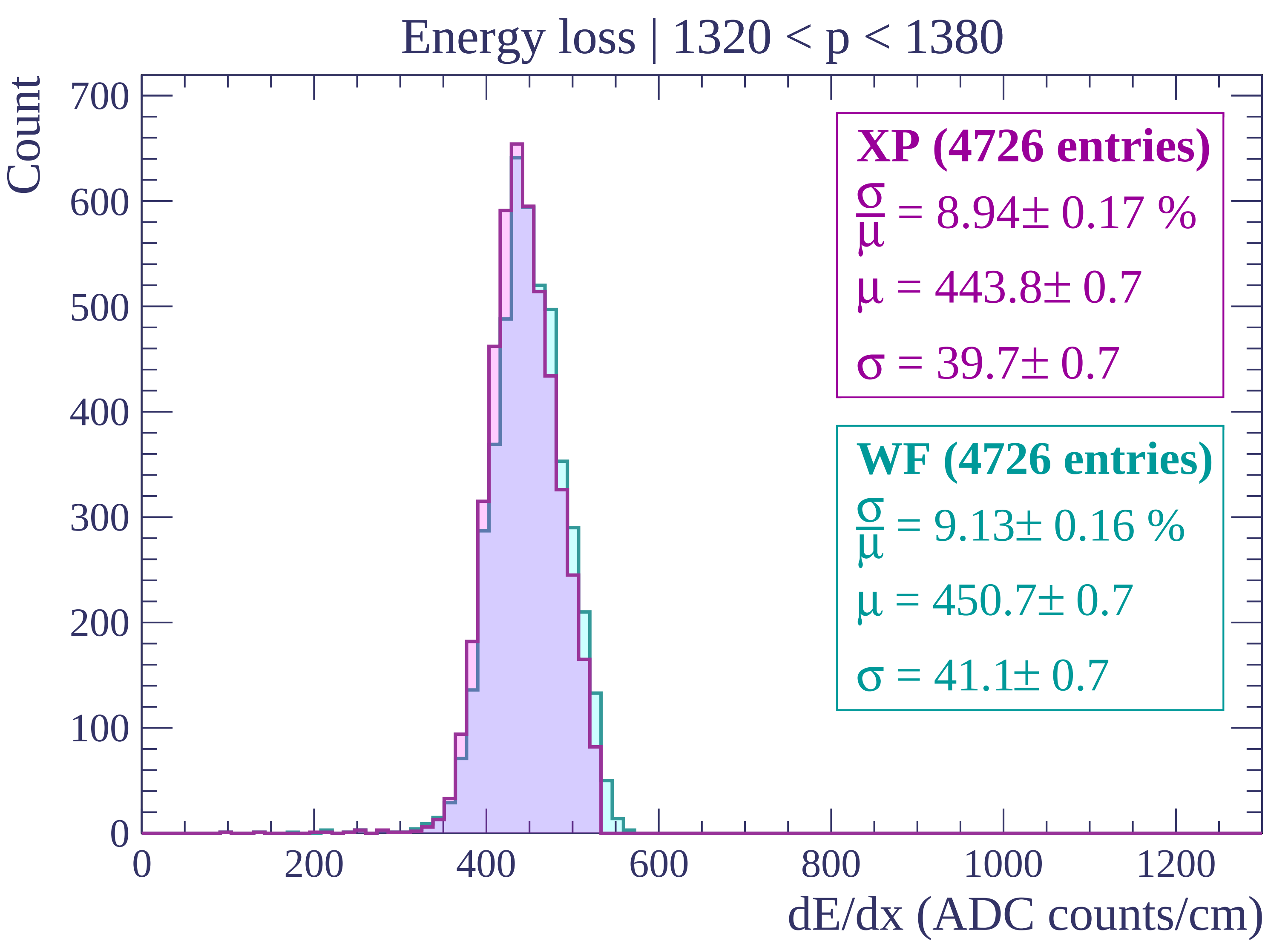
Energy loss | 1080 < p < 1140
Count
dE/dx (ADC counts/cm)
0
100
200
300
400
500
600
700
800
0
200
400
600
800
1000
1200
XP (5410 entries)
$\frac{\sigma}{\mu}$ = 8.81± 0.15 %
μ = 432.0± 0.6
σ = 38.1± 0.6
WF (5410 entries)
$\frac{\sigma}{\mu}$ = 8.99± 0.14 %
μ = 438.2± 0.6
σ = 39.4± 0.6


Energy loss | 1140 < p < 1200
Count
800
700
600
500
400
300
200
100
0
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP (5318 entries)
σ/μ = 8.57± 0.14 %
μ = 435.0± 0.6
σ = 37.3± 0.5
WF (5318 entries)
σ/μ = 9.17± 0.17 %
μ = 441.5± 0.7
σ = 40.5± 0.7

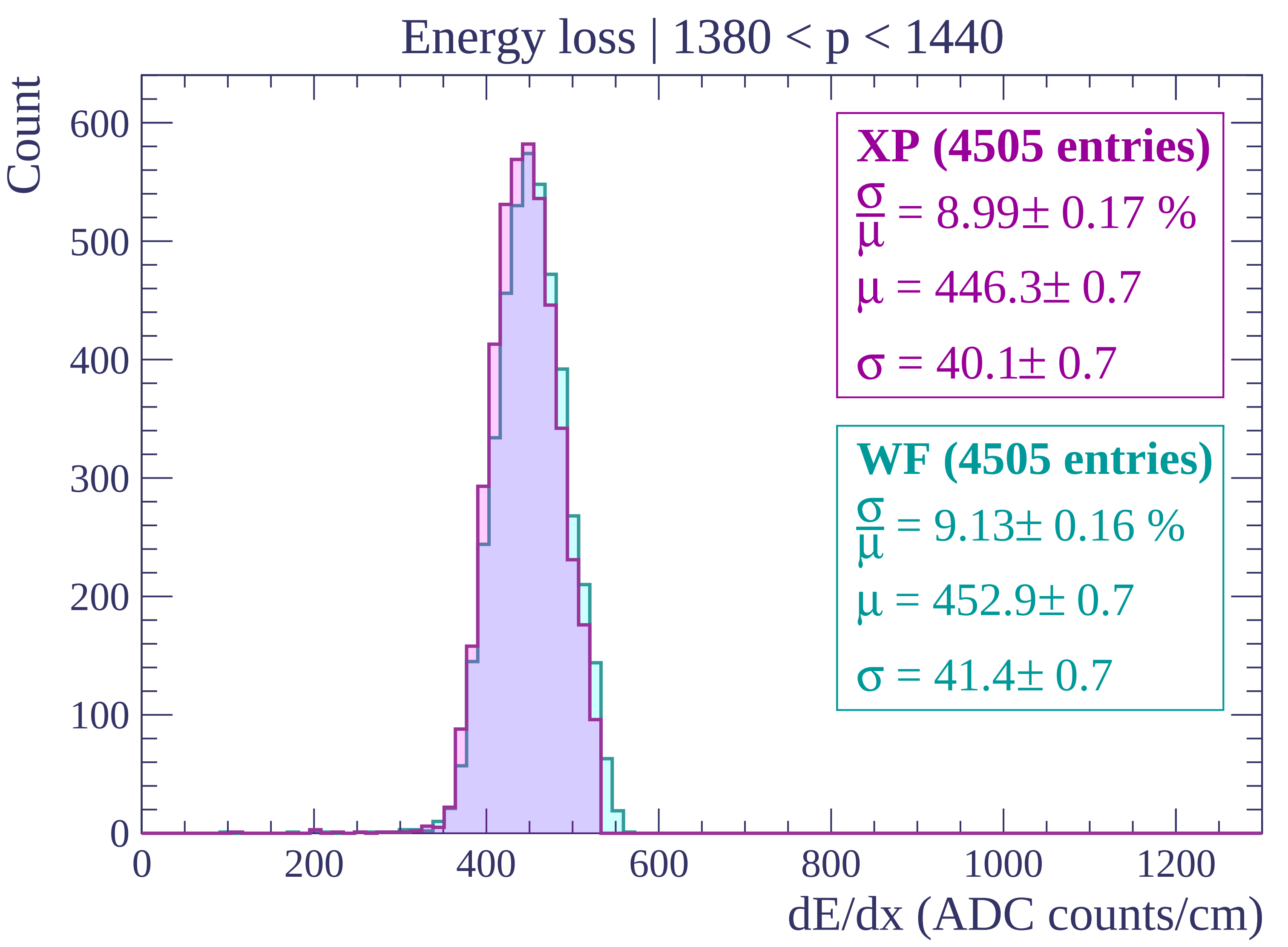


Energy loss | 1200 < p < 1260
Count
dE/dx (ADC counts/cm)
XP (5141 entries)
$\frac{\sigma}{\mu}$ = 9.31± 0.18 %
μ = 438.6± 0.7
σ = 40.8± 0.7
WF (5141 entries)
$\frac{\sigma}{\mu}$ = 9.33± 0.14 %
μ = 445.3± 0.7
σ = 41.5± 0.6
0
100
200
300
400
500
600
700
0
200
400
600
800
1000
1200

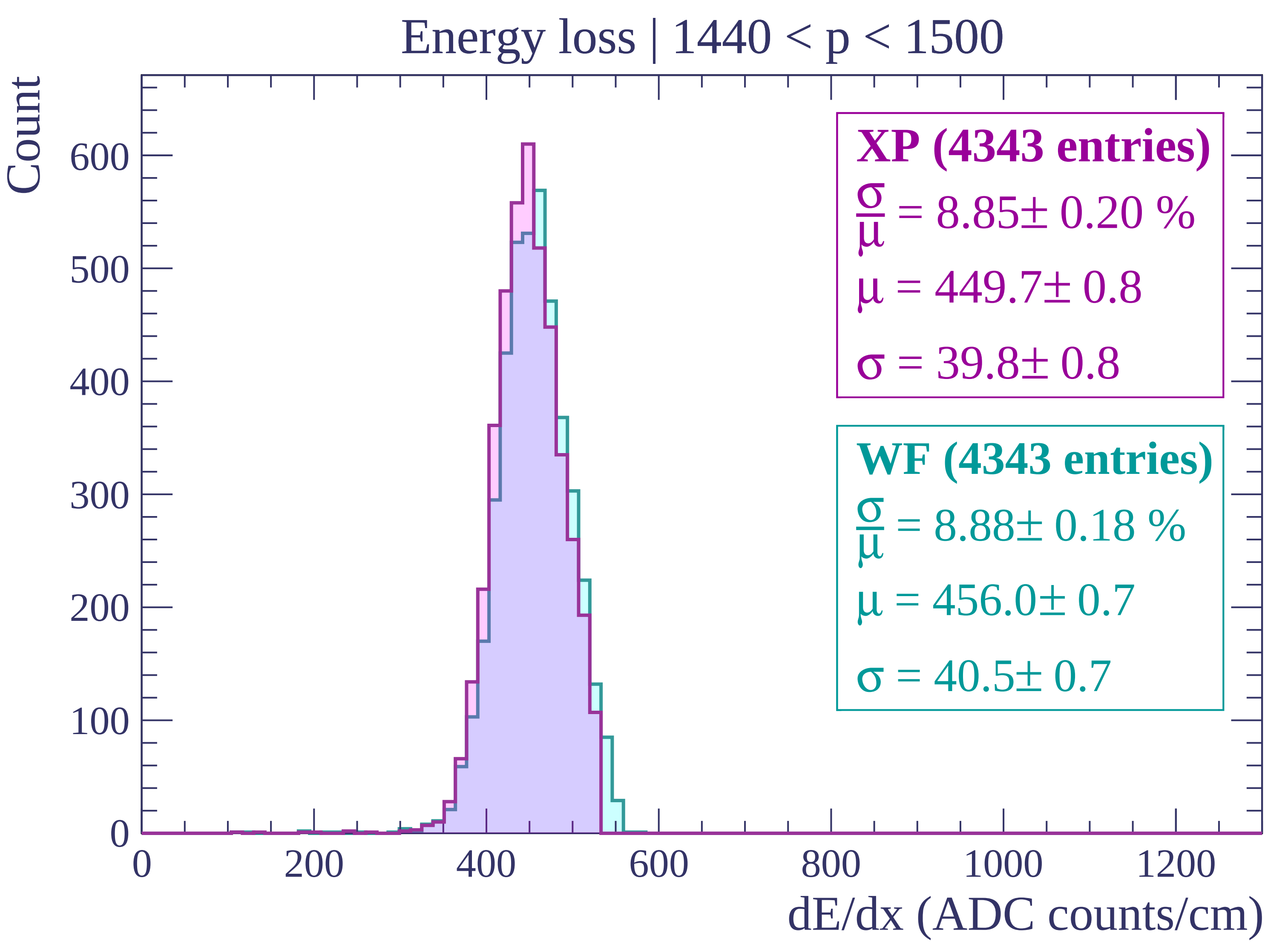

Energy loss | 1260 < p < 1320
Count
700
600
500
400
300
200
100
0
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP (4927 entries)
$\frac{\sigma}{\mu}$ = 8.89± 0.15 %
μ = 441.2± 0.7
σ = 39.2± 0.6
WF (4927 entries)
$\frac{\sigma}{\mu}$ = 9.02± 0.17 %
μ = 447.0± 0.7
σ = 40.3± 0.7

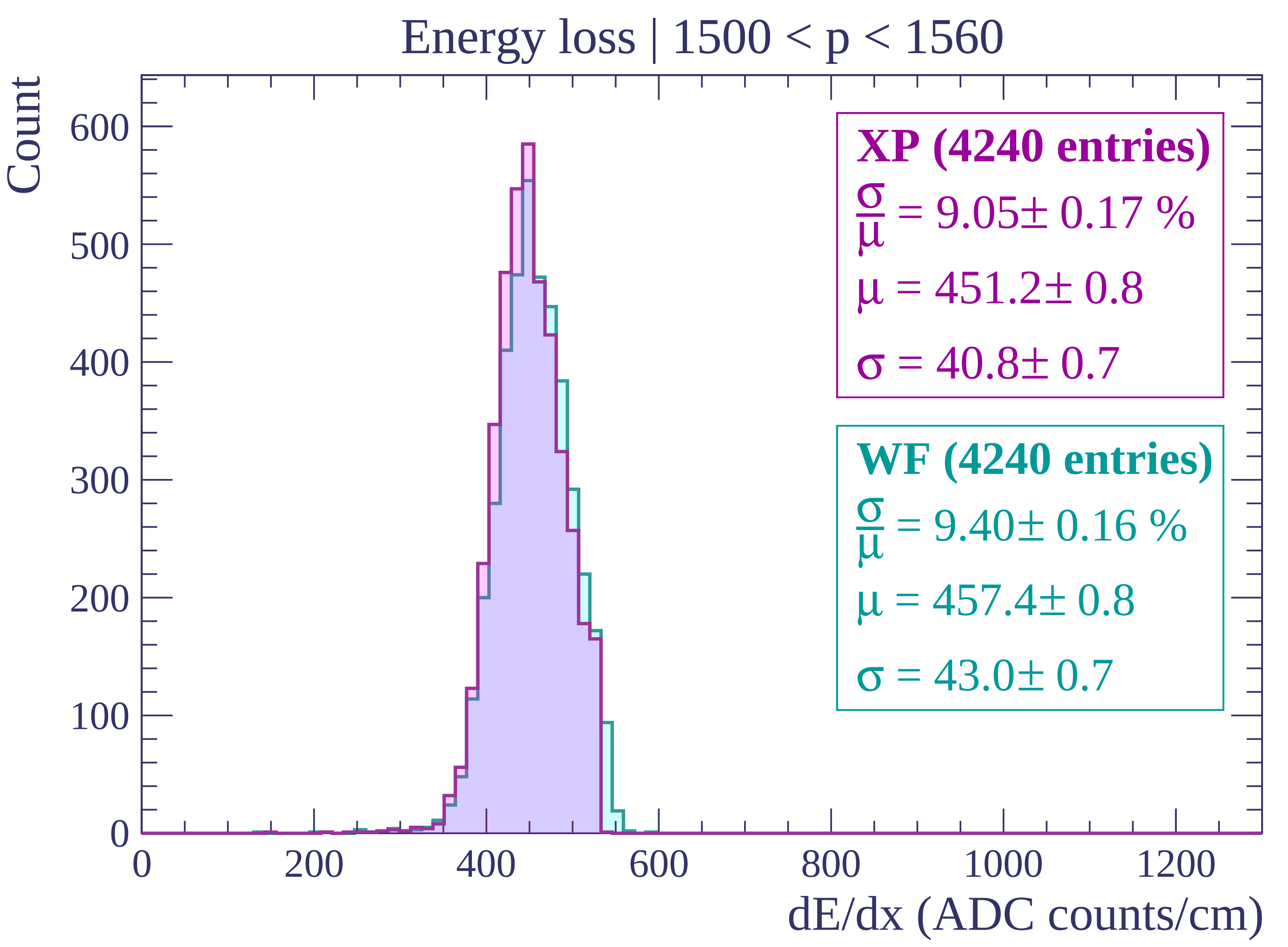


Energy loss | 1320 < p < 1380
Count
dE/dx (ADC counts/cm)
0
100
200
300
400
500
600
700
0
200
400
600
800
1000
1200
XP (4726 entries)
$\frac{\sigma}{\mu}$ = 8.94± 0.17 %
μ = 443.8± 0.7
σ = 39.7± 0.7
WF (4726 entries)
$\frac{\sigma}{\mu}$ = 9.13± 0.16 %
μ = 450.7± 0.7
σ = 41.1± 0.7

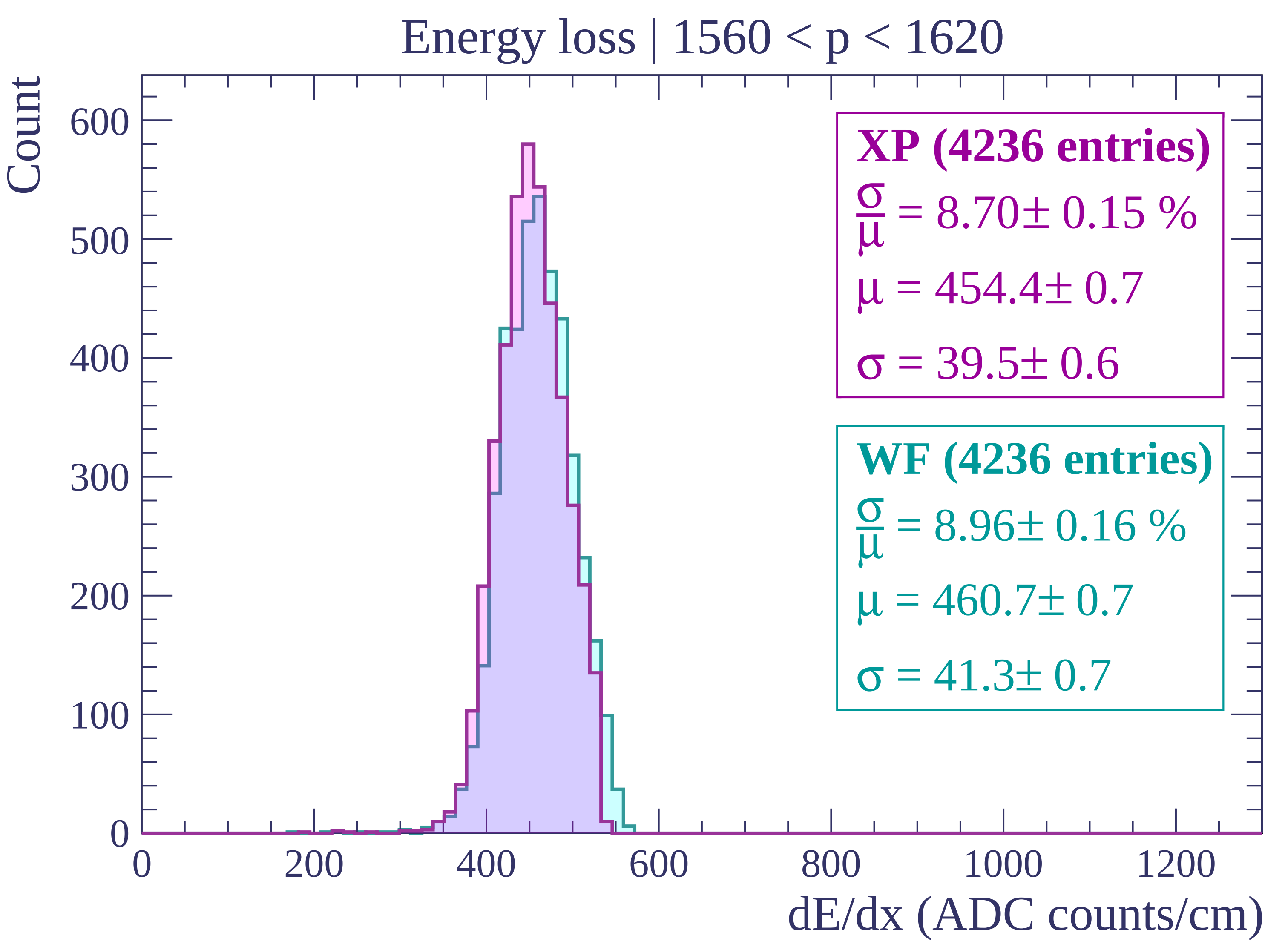


Energy loss | 1380 < p < 1440
Count
dE/dx (ADC counts/cm)
0
100
200
300
400
500
600
0
200
400
600
800
1000
1200
XP (4505 entries)
σ/μ = 8.99± 0.17 %
μ = 446.3± 0.7
σ = 40.1± 0.7
WF (4505 entries)
σ/μ = 9.13± 0.16 %
μ = 452.9± 0.7
σ = 41.4± 0.7

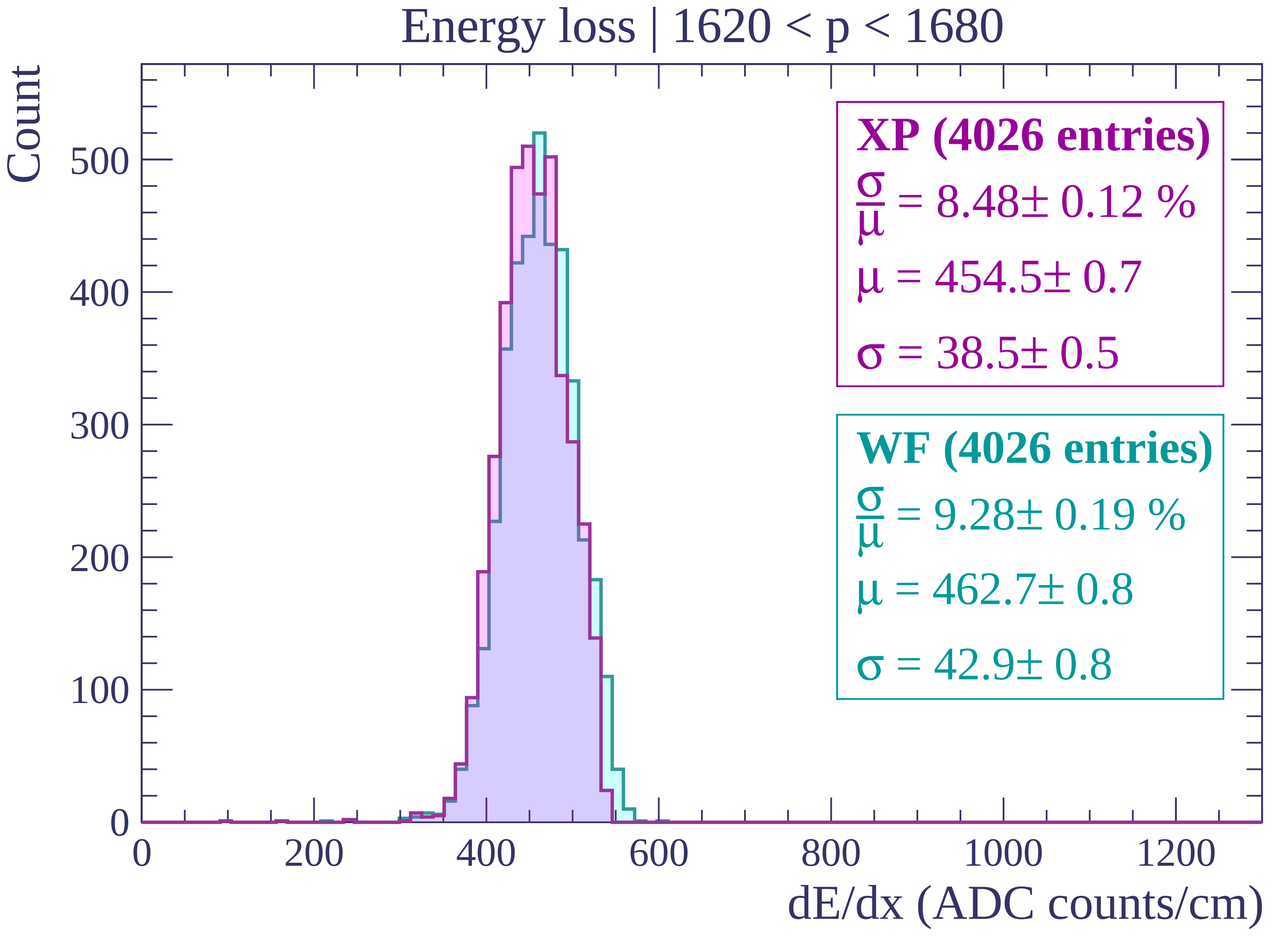

Energy loss | 1440 < p < 1500
Count
dE/dx (ADC counts/cm)
0
100
200
300
400
500
600
0
200
400
600
800
1000
1200
XP (4343 entries)
$\frac{\sigma}{\mu}$ = 8.85± 0.20 %
μ = 449.7± 0.8
σ = 39.8± 0.8
WF (4343 entries)
$\frac{\sigma}{\mu}$ = 8.88± 0.18 %
μ = 456.0± 0.7
σ = 40.5± 0.7



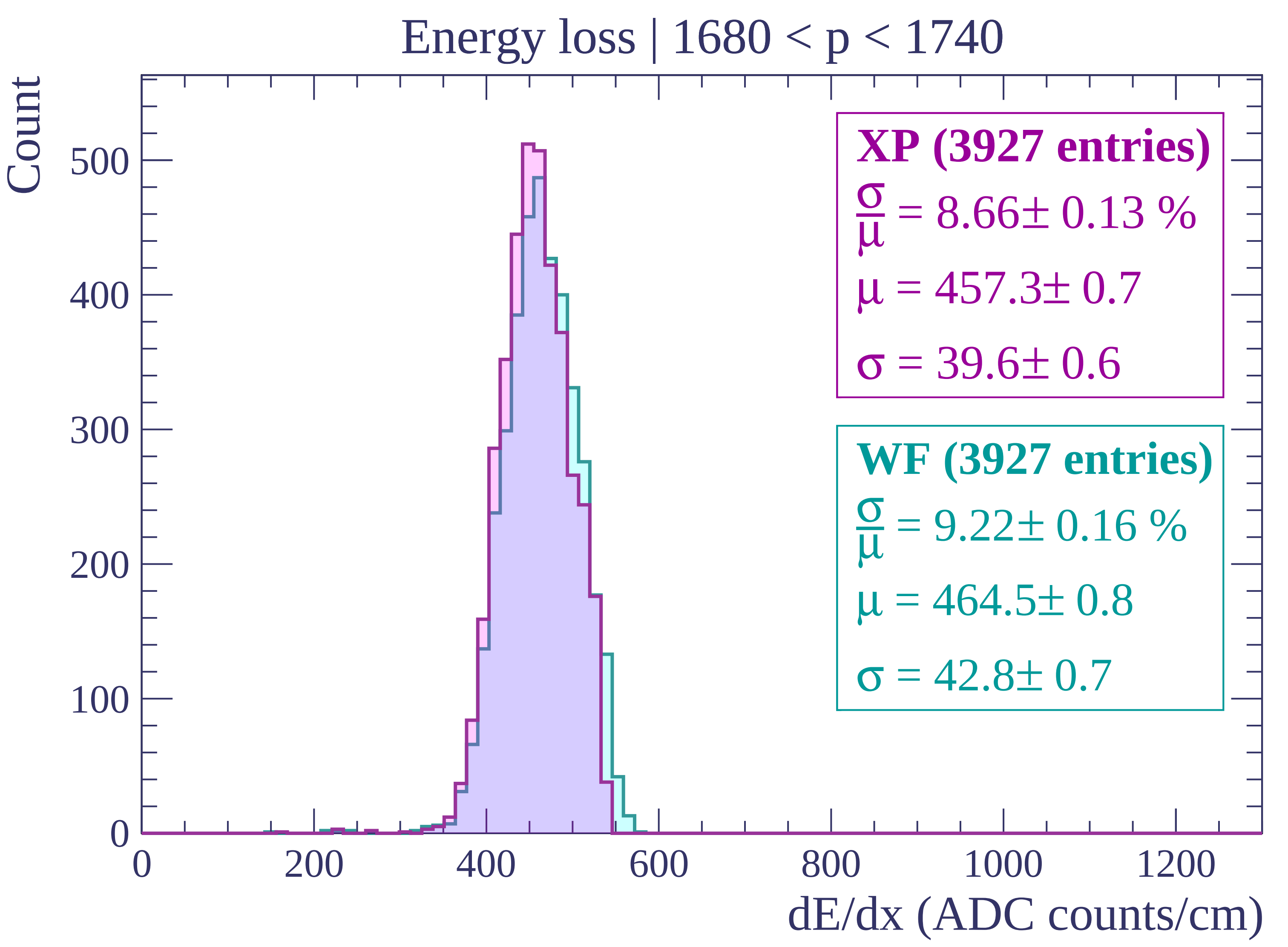
Energy loss | 1500 < p < 1560
Count
600
500
400
300
200
100
0
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP (4240 entries)
$\frac{\sigma}{\mu}$ = 9.05± 0.17 %
μ = 451.2± 0.8
σ = 40.8± 0.7
WF (4240 entries)
$\frac{\sigma}{\mu}$ = 9.40± 0.16 %
μ = 457.4± 0.8
σ = 43.0± 0.7

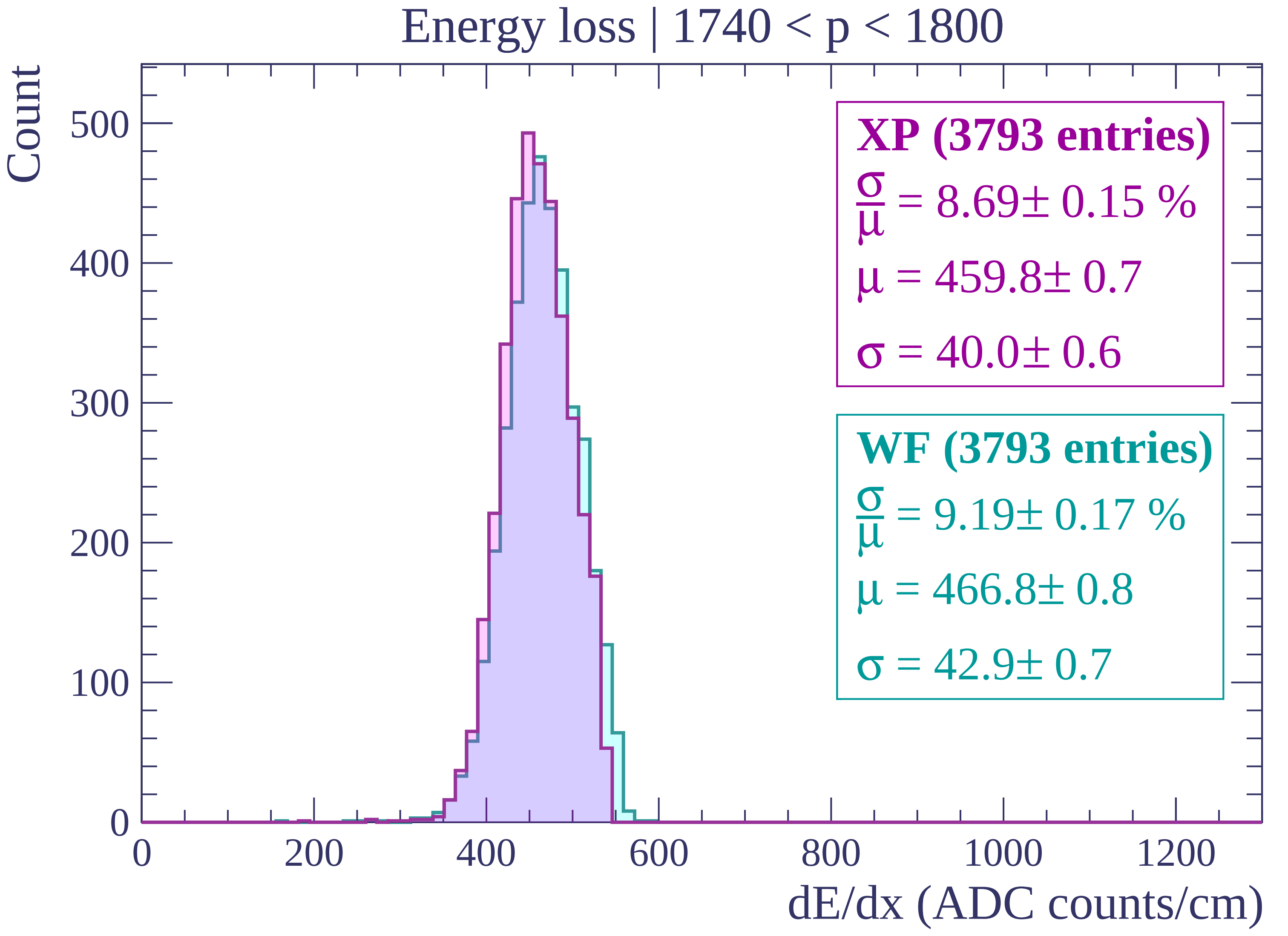


Energy loss | 1560 < p < 1620
Count
0
100
200
300
400
500
600
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP (4236 entries)
$\frac{\sigma}{\mu}$ = 8.70± 0.15 %
μ = 454.4± 0.7
σ = 39.5± 0.6
WF (4236 entries)
$\frac{\sigma}{\mu}$ = 8.96± 0.16 %
μ = 460.7± 0.7
σ = 41.3± 0.7

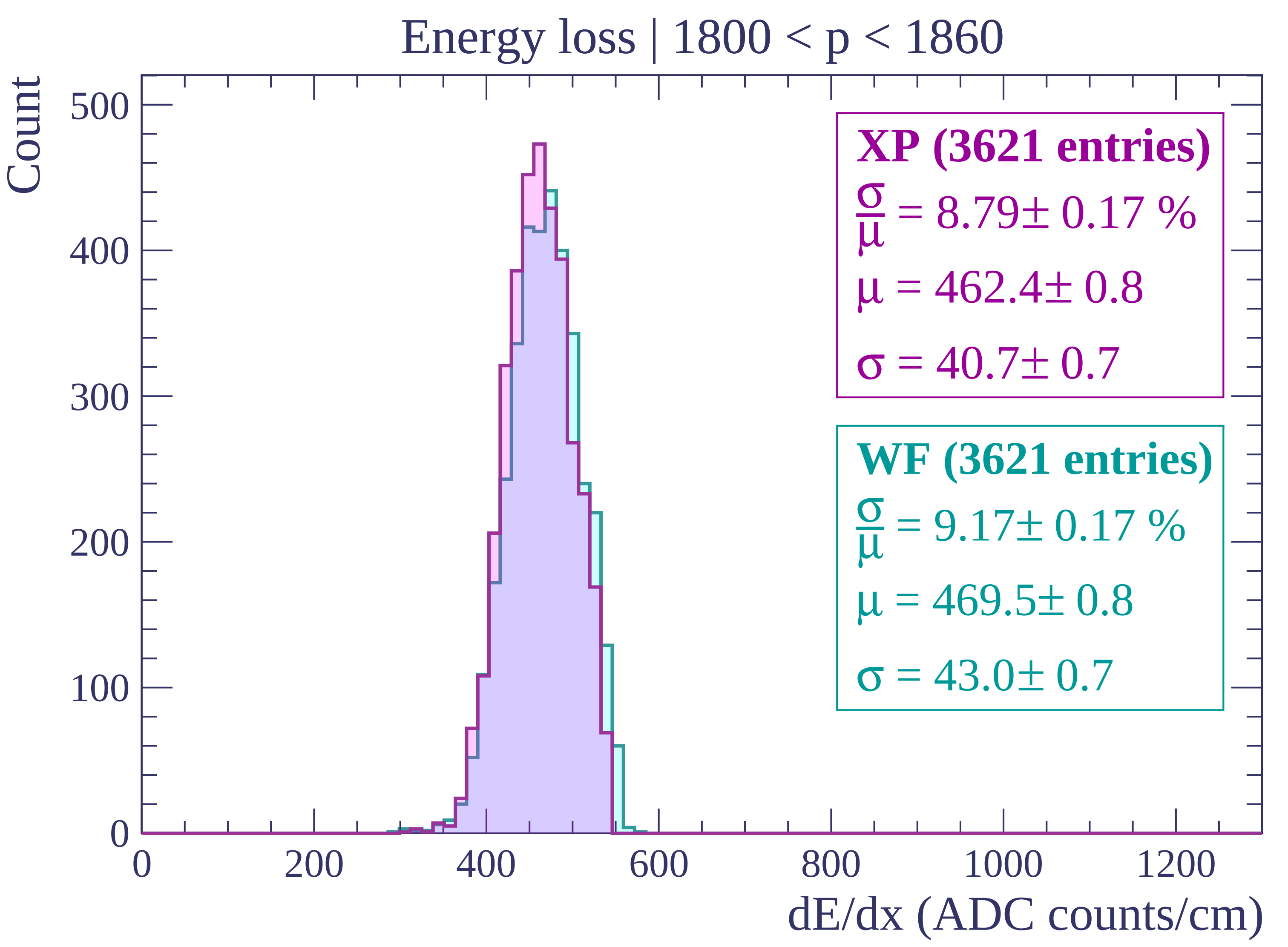


Energy loss | 1620 < p < 1680
Count
dE/dx (ADC counts/cm)
0
100
200
300
400
500
0
200
400
600
800
1000
1200
XP (4026 entries)
$\frac{\sigma}{\mu}$ = 8.48± 0.12 %
μ = 454.5± 0.7
σ = 38.5± 0.5
WF (4026 entries)
$\frac{\sigma}{\mu}$ = 9.28± 0.19 %
μ = 462.7± 0.8
σ = 42.9± 0.8

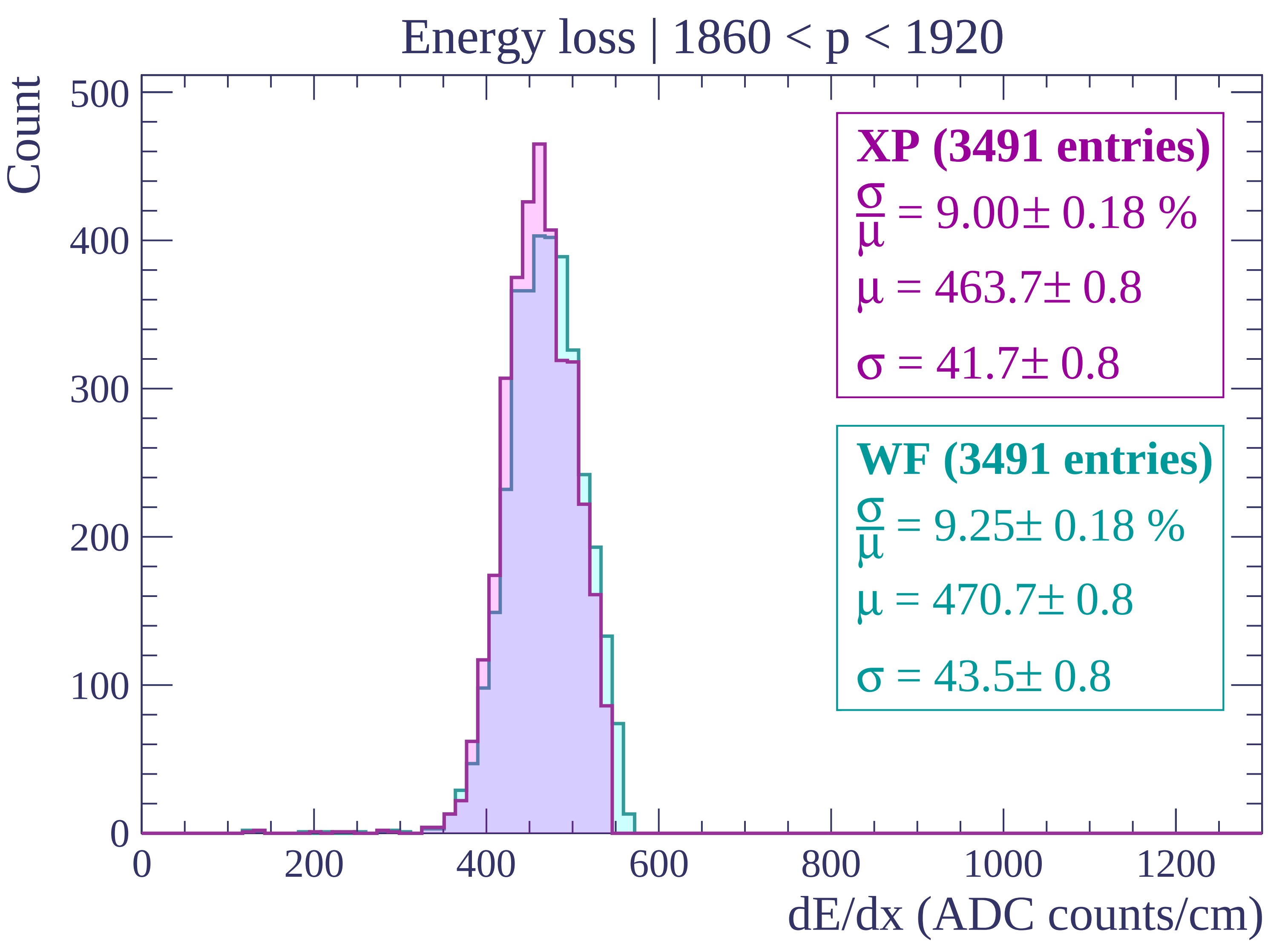

Energy loss | 1680 < p < 1740
Count
dE/dx (ADC counts/cm)
XP (3927 entries)
σ/μ = 8.66± 0.13 %
μ = 457.3± 0.7
σ = 39.6± 0.6
WF (3927 entries)
σ/μ = 9.22± 0.16 %
μ = 464.5± 0.8
σ = 42.8± 0.7
0
100
200
300
400
500
0
200
400
600
800
1000
1200



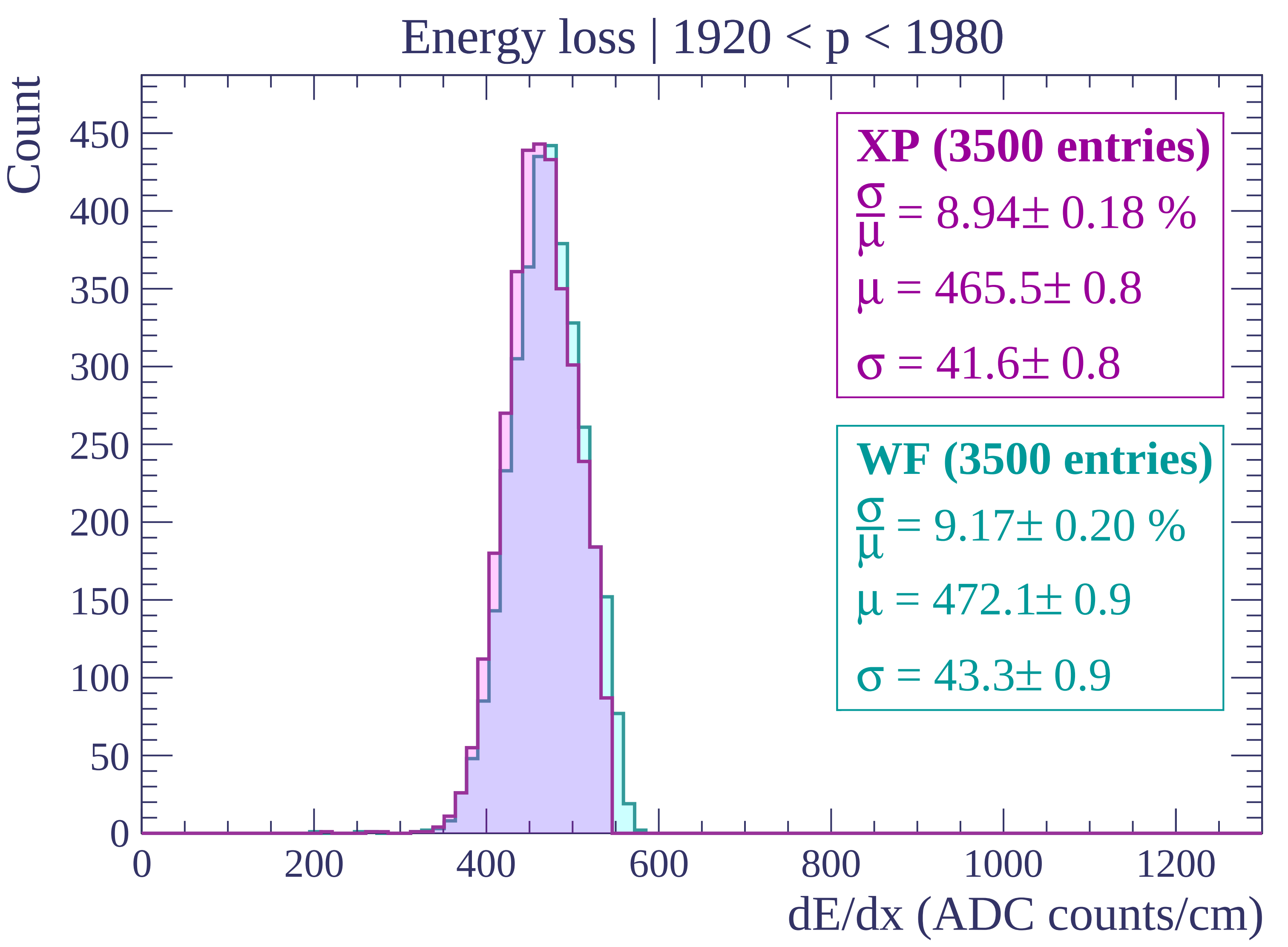
Energy loss | 1740 < p < 1800
Count
500
400
300
200
100
0
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP (3793 entries)
σ/μ = 8.69± 0.15 %
μ = 459.8± 0.7
σ = 40.0± 0.6
WF (3793 entries)
σ/μ = 9.19± 0.17 %
μ = 466.8± 0.8
σ = 42.9± 0.7

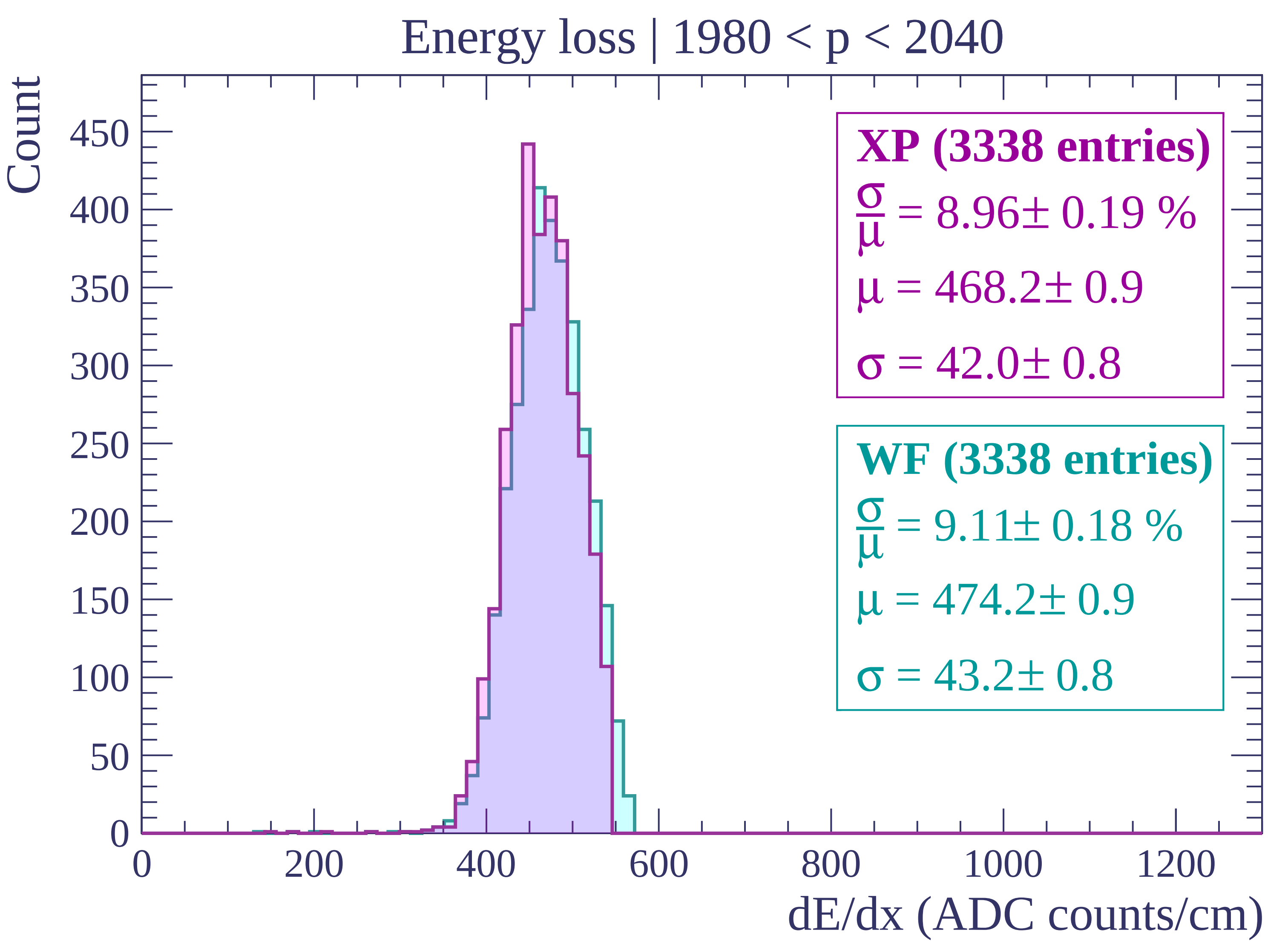


Energy loss | 1800 < p < 1860
Count
500
400
300
200
100
0
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP (3621 entries)
σ/μ = 8.79± 0.17 %
μ = 462.4± 0.8
σ = 40.7± 0.7
WF (3621 entries)
σ/μ = 9.17± 0.17 %
μ = 469.5± 0.8
σ = 43.0± 0.7

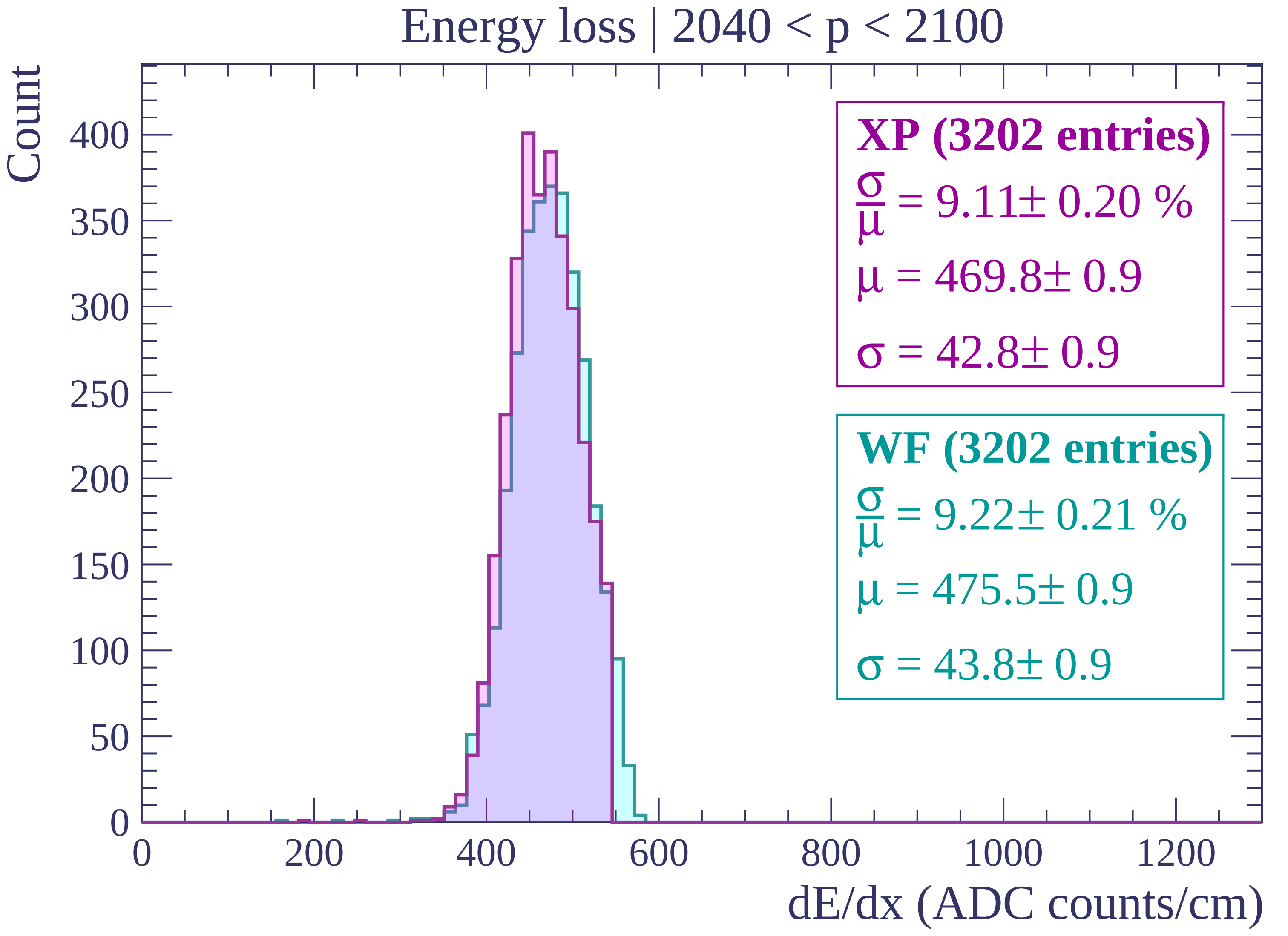


Energy loss | 1860 < p < 1920
Count
dE/dx (ADC counts/cm)
0
100
200
300
400
500
0
200
400
600
800
1000
1200
XP (3491 entries)
$\frac{\sigma}{\mu}$ = 9.00± 0.18 %
μ = 463.7± 0.8
σ = 41.7± 0.8
WF (3491 entries)
$\frac{\sigma}{\mu}$ = 9.25± 0.18 %
μ = 470.7± 0.8
σ = 43.5± 0.8

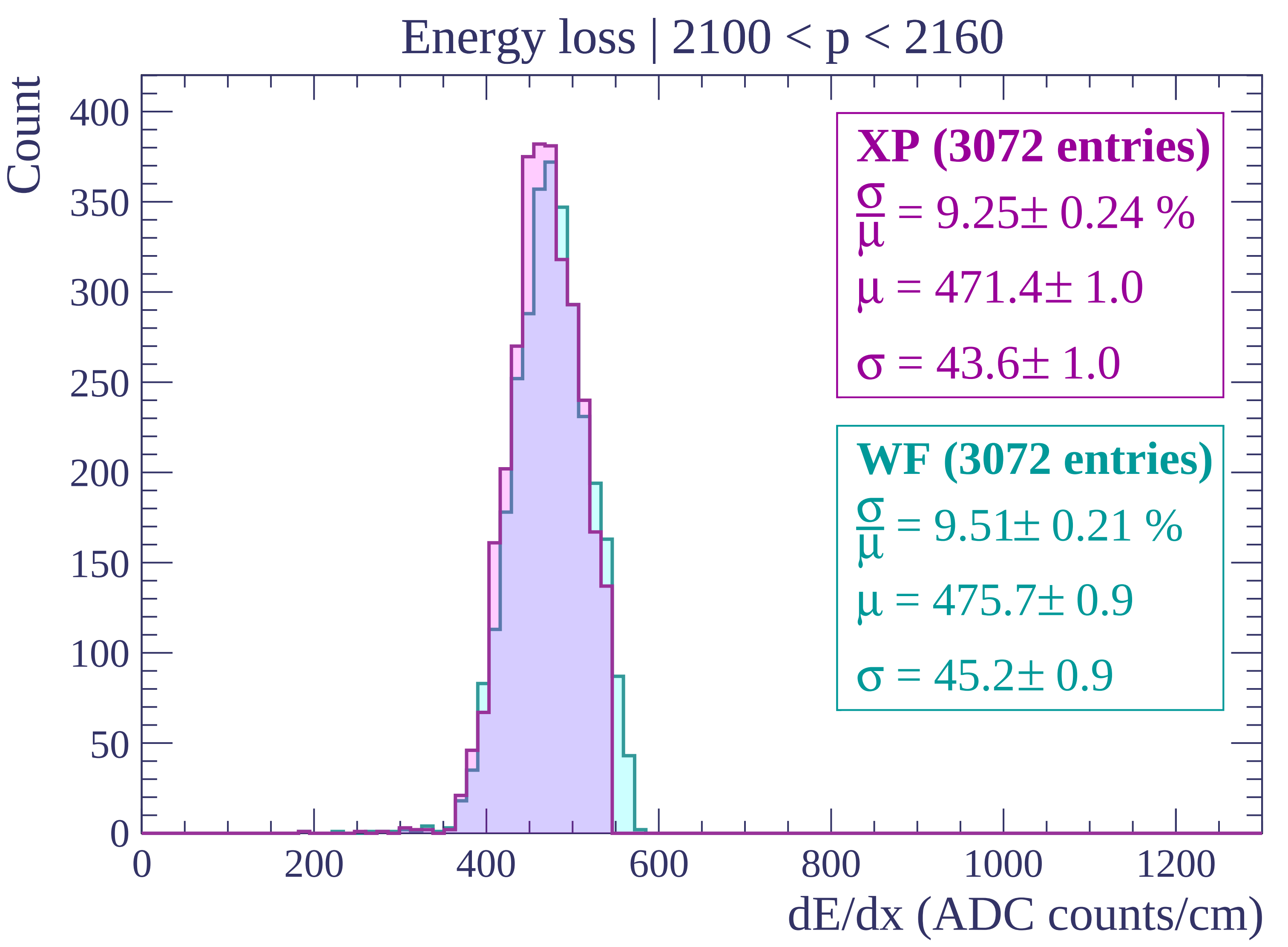

Energy loss | 1920 < p < 1980
Count
dE/dx (ADC counts/cm)
0
50
100
150
200
250
300
350
400
450
0
200
400
600
800
1000
1200
XP (3500 entries)
σ/μ = 8.94± 0.18 %
μ = 465.5± 0.8
σ = 41.6± 0.8
WF (3500 entries)
σ/μ = 9.17± 0.20 %
μ = 472.1± 0.9
σ = 43.3± 0.9



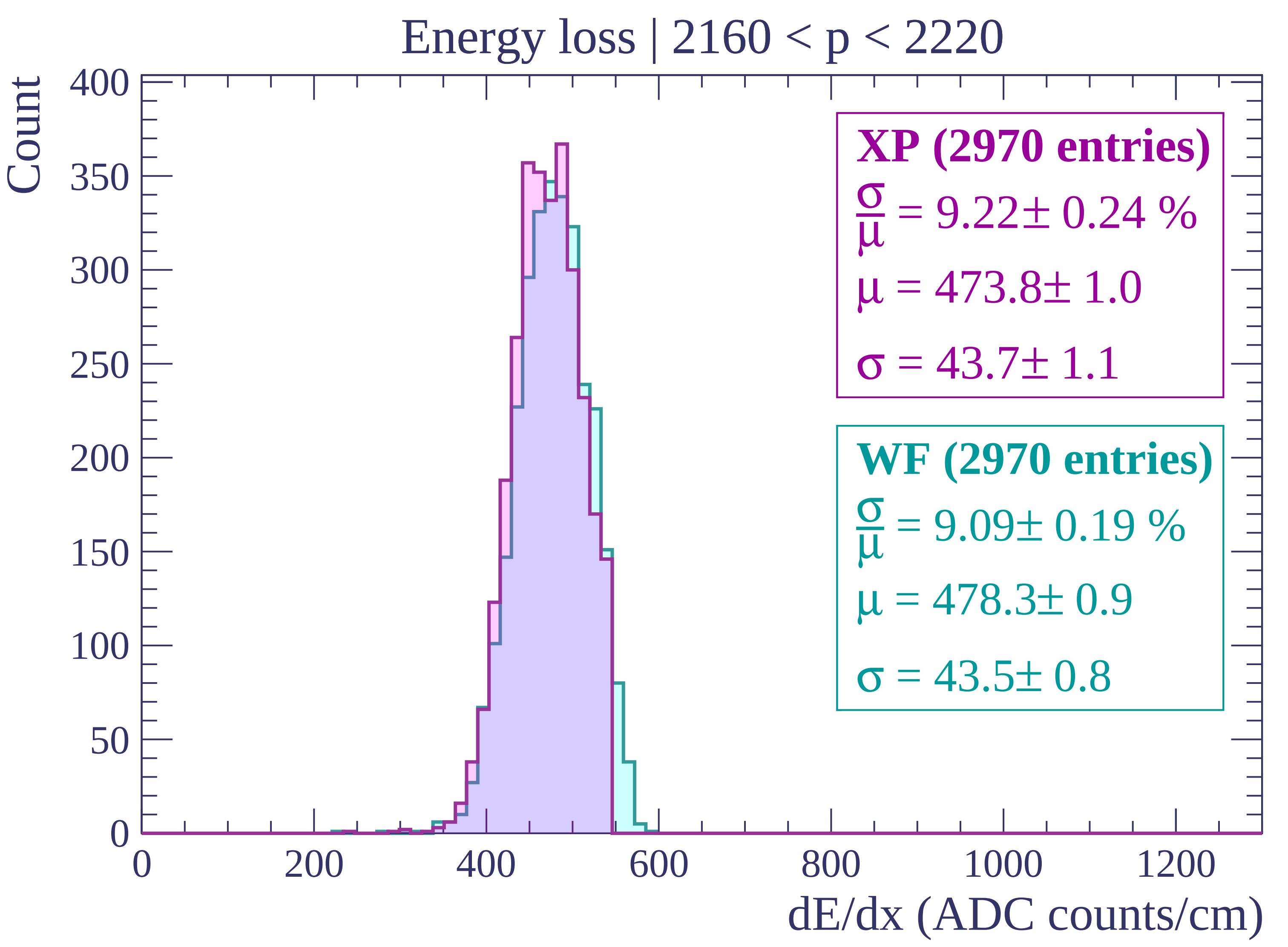
Energy loss | 1980 < p < 2040
Count
0
50
100
150
200
250
300
350
400
450
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP (3338 entries)
σ/μ = 8.96± 0.19 %
μ = 468.2± 0.9
σ = 42.0± 0.8
WF (3338 entries)
σ/μ = 9.11± 0.18 %
μ = 474.2± 0.9
σ = 43.2± 0.8

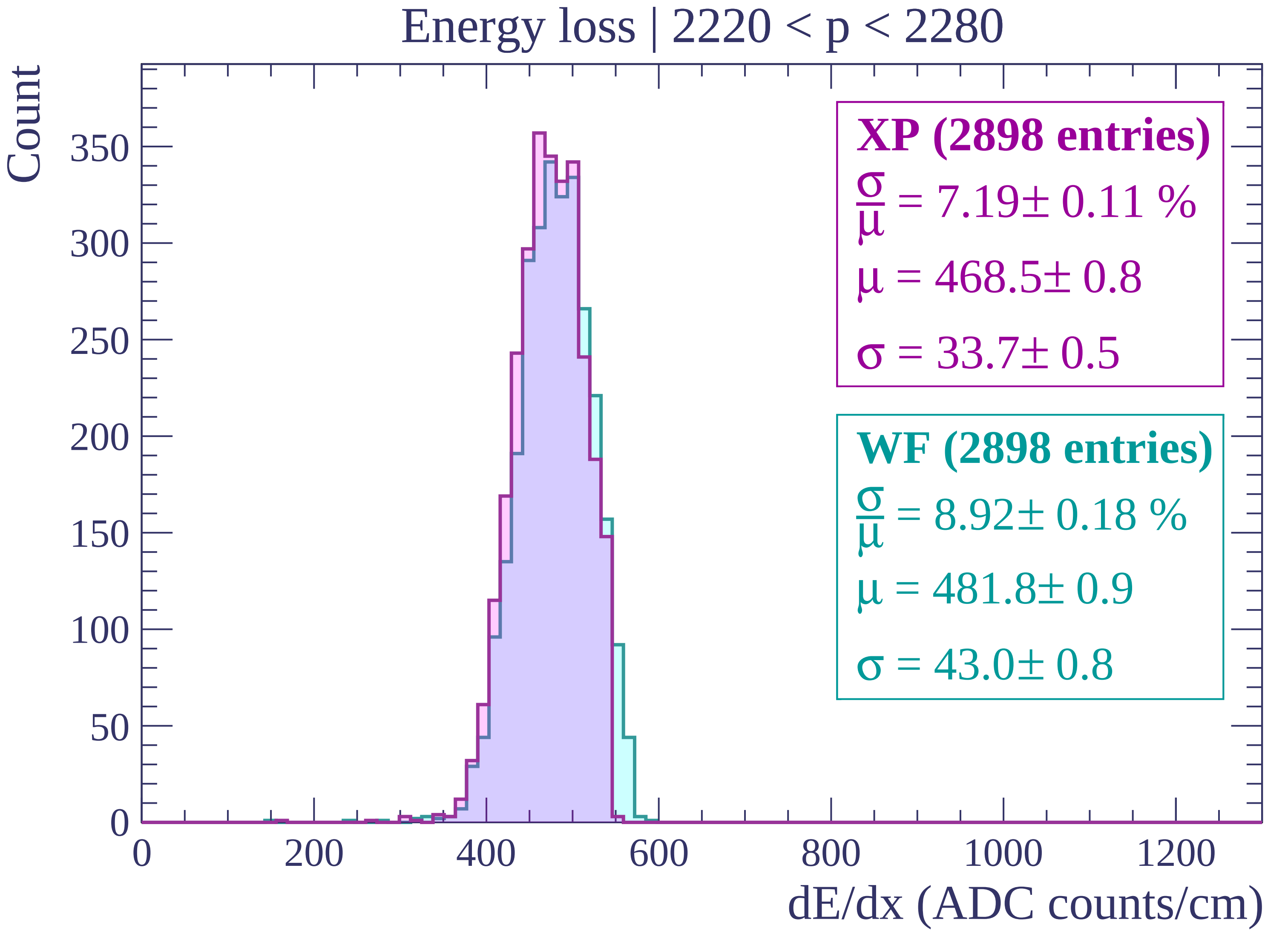


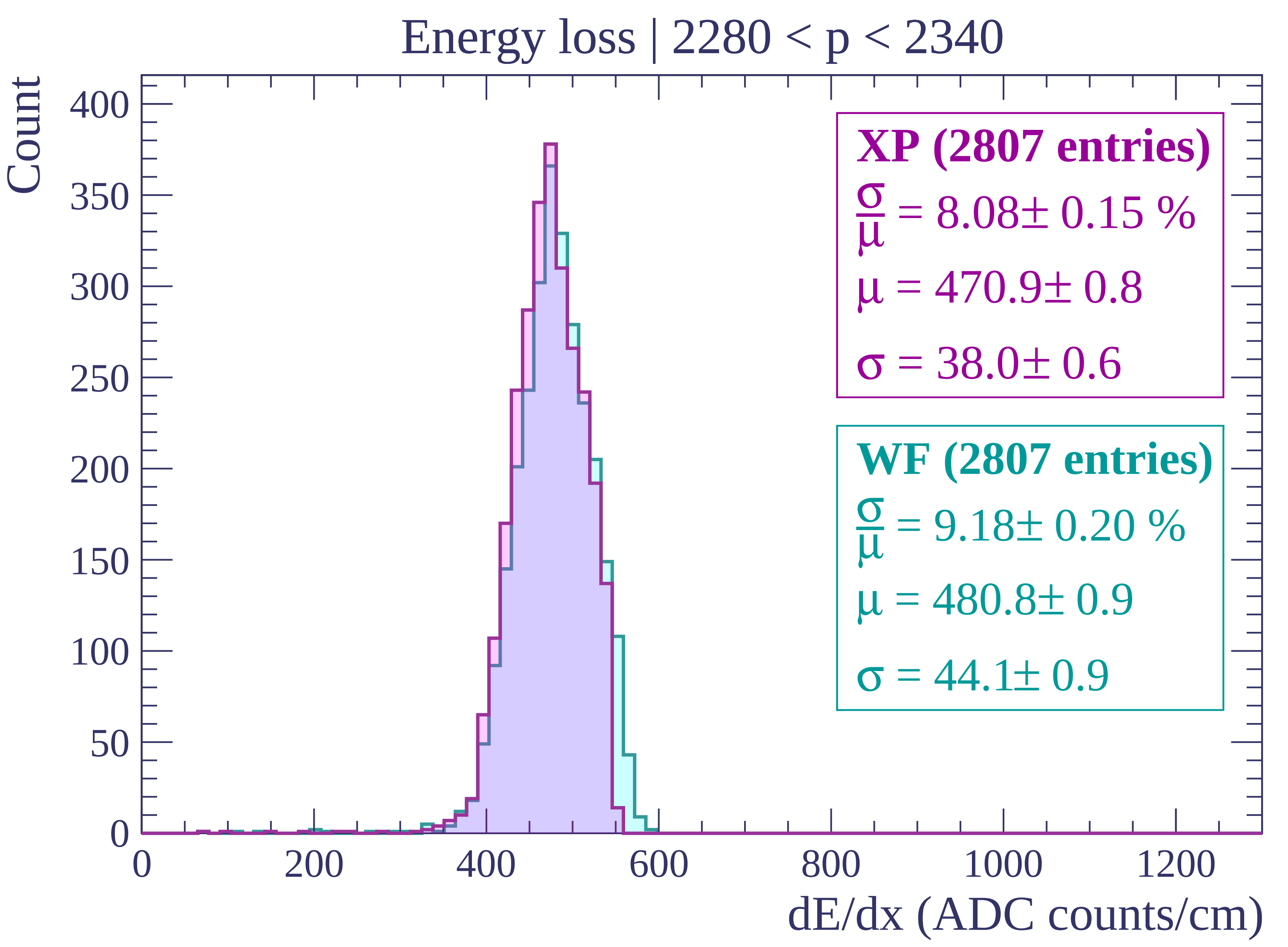
Energy loss | 2040 < p < 2100
Count
dE/dx (ADC counts/cm)
0
50
100
150
200
250
300
350
400
0
200
400
600
800
1000
1200
XP (3202 entries)
$\frac{\sigma}{\mu}$ = 9.11± 0.20 %
μ = 469.8± 0.9
σ = 42.8± 0.9
WF (3202 entries)
$\frac{\sigma}{\mu}$ = 9.22± 0.21 %
μ = 475.5± 0.9
σ = 43.8± 0.9



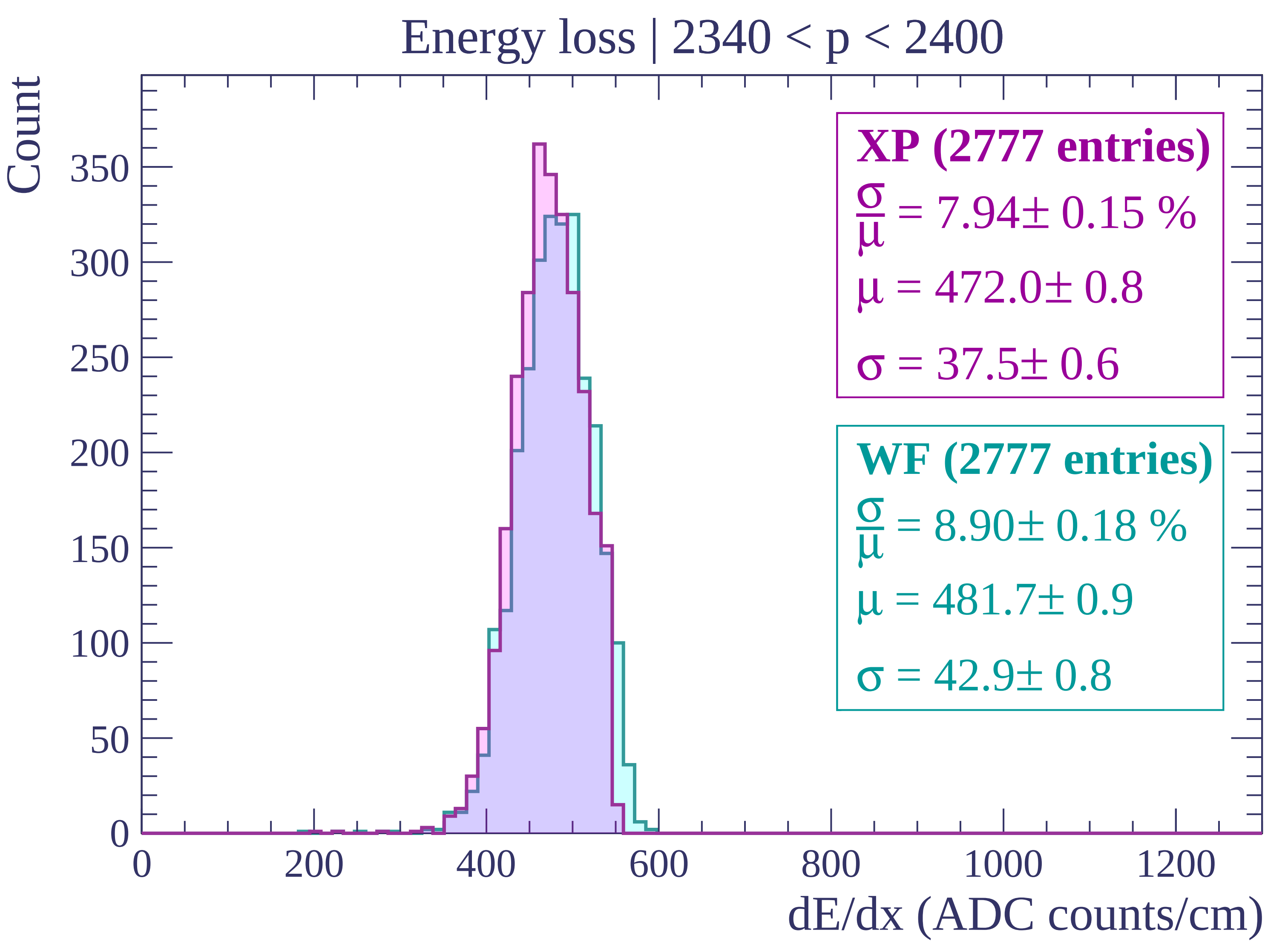
Energy loss | 2100 < p < 2160
Count
dE/dx (ADC counts/cm)
0
50
100
150
200
250
300
350
400
0
200
400
600
800
1000
1200
XP (3072 entries)
σ/μ = 9.25± 0.24 %
μ = 471.4± 1.0
σ = 43.6± 1.0
WF (3072 entries)
σ/μ = 9.51± 0.21 %
μ = 475.7± 0.9
σ = 45.2± 0.9


Energy loss | 2160 < p < 2220
Count
dE/dx (ADC counts/cm)
0
50
100
150
200
250
300
350
400
0
200
400
600
800
1000
1200
XP (2970 entries)
σ/μ = 9.22± 0.24 %
μ = 473.8± 1.0
σ = 43.7± 1.1
WF (2970 entries)
σ/μ = 9.09± 0.19 %
μ = 478.3± 0.9
σ = 43.5± 0.8

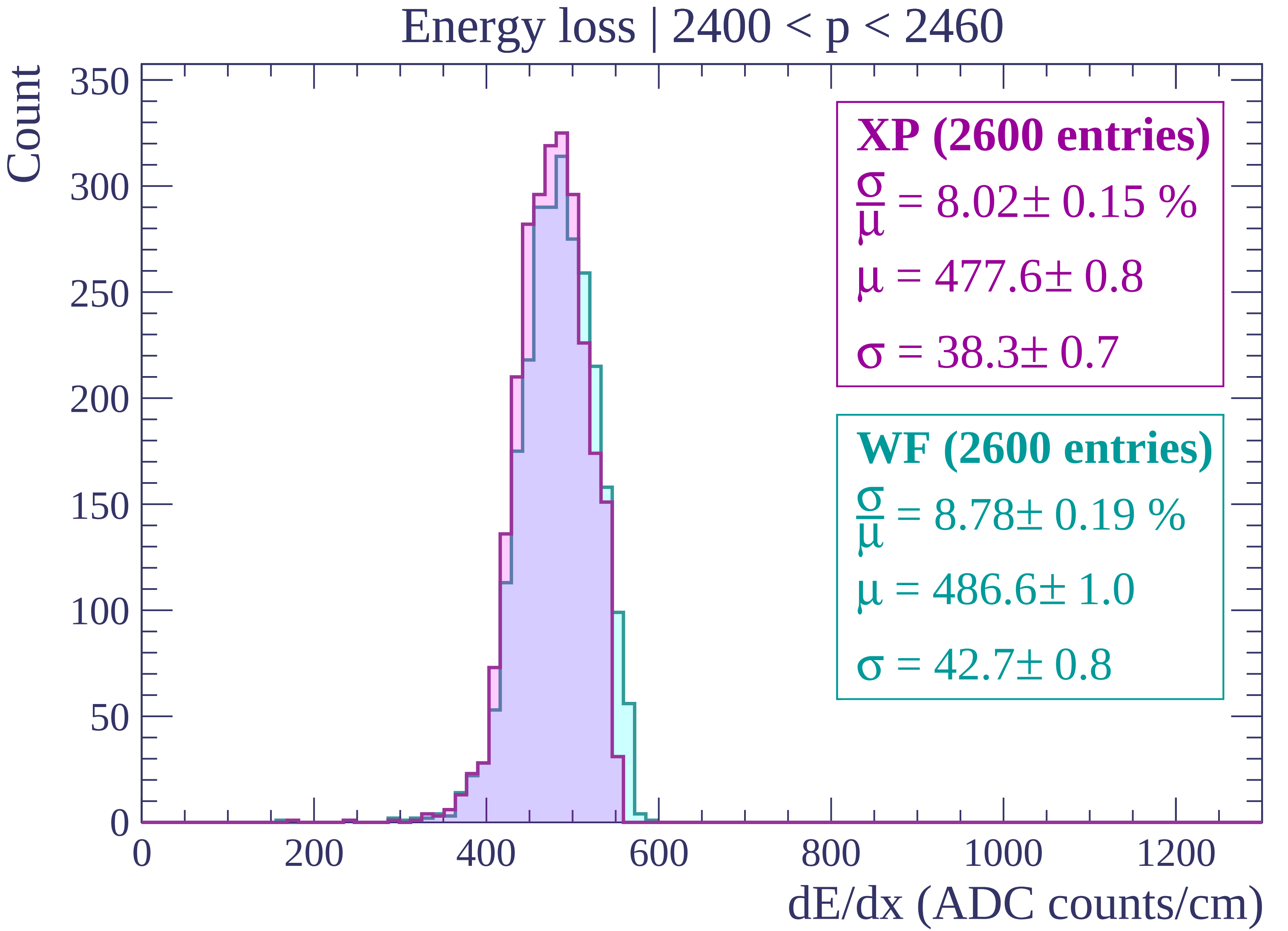

Energy loss | 2220 < p < 2280
Count
dE/dx (ADC counts/cm)
0
50
100
150
200
250
300
350
0
200
400
600
800
1000
1200
XP (2898 entries)
σ/μ = 7.19± 0.11 %
μ = 468.5± 0.8
σ = 33.7± 0.5
WF (2898 entries)
σ/μ = 8.92± 0.18 %
μ = 481.8± 0.9
σ = 43.0± 0.8

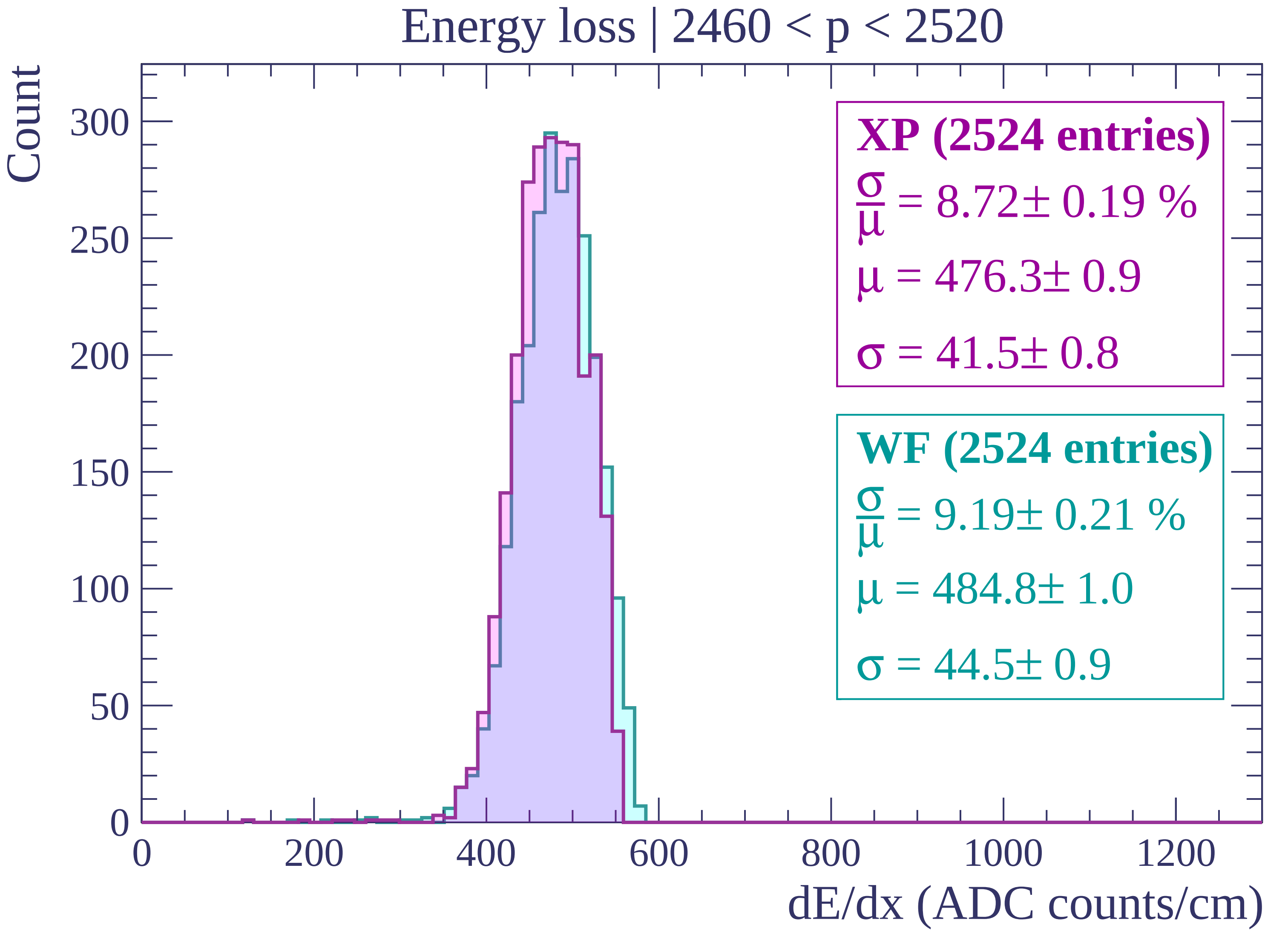


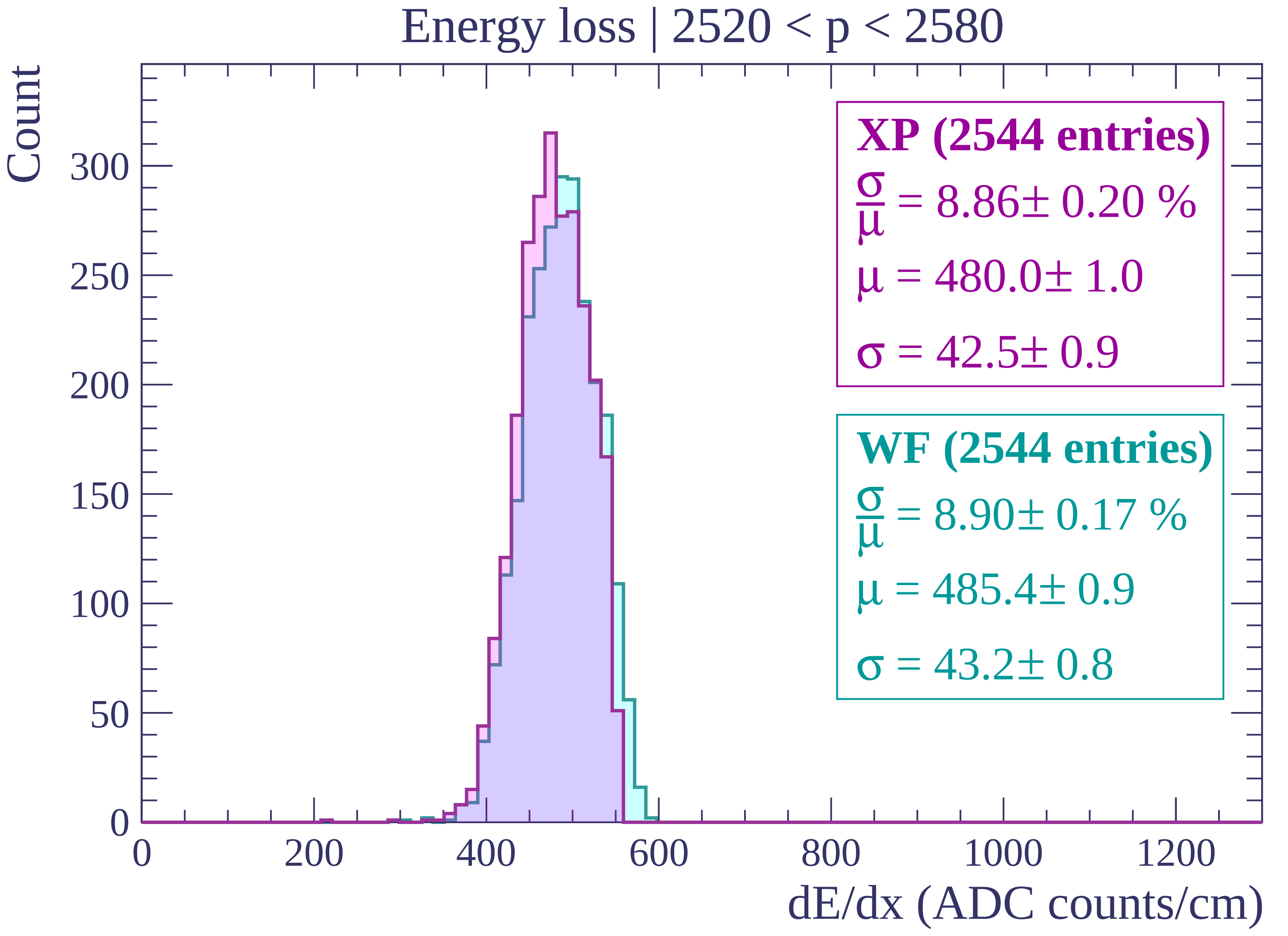
Energy loss | 2280 < p < 2340
Count
dE/dx (ADC counts/cm)
0
50
100
150
200
250
300
350
400
0
200
400
600
800
1000
1200
XP (2807 entries)
σ/μ = 8.08± 0.15 %
μ = 470.9± 0.8
σ = 38.0± 0.6
WF (2807 entries)
σ/μ = 9.18± 0.20 %
μ = 480.8± 0.9
σ = 44.1± 0.9

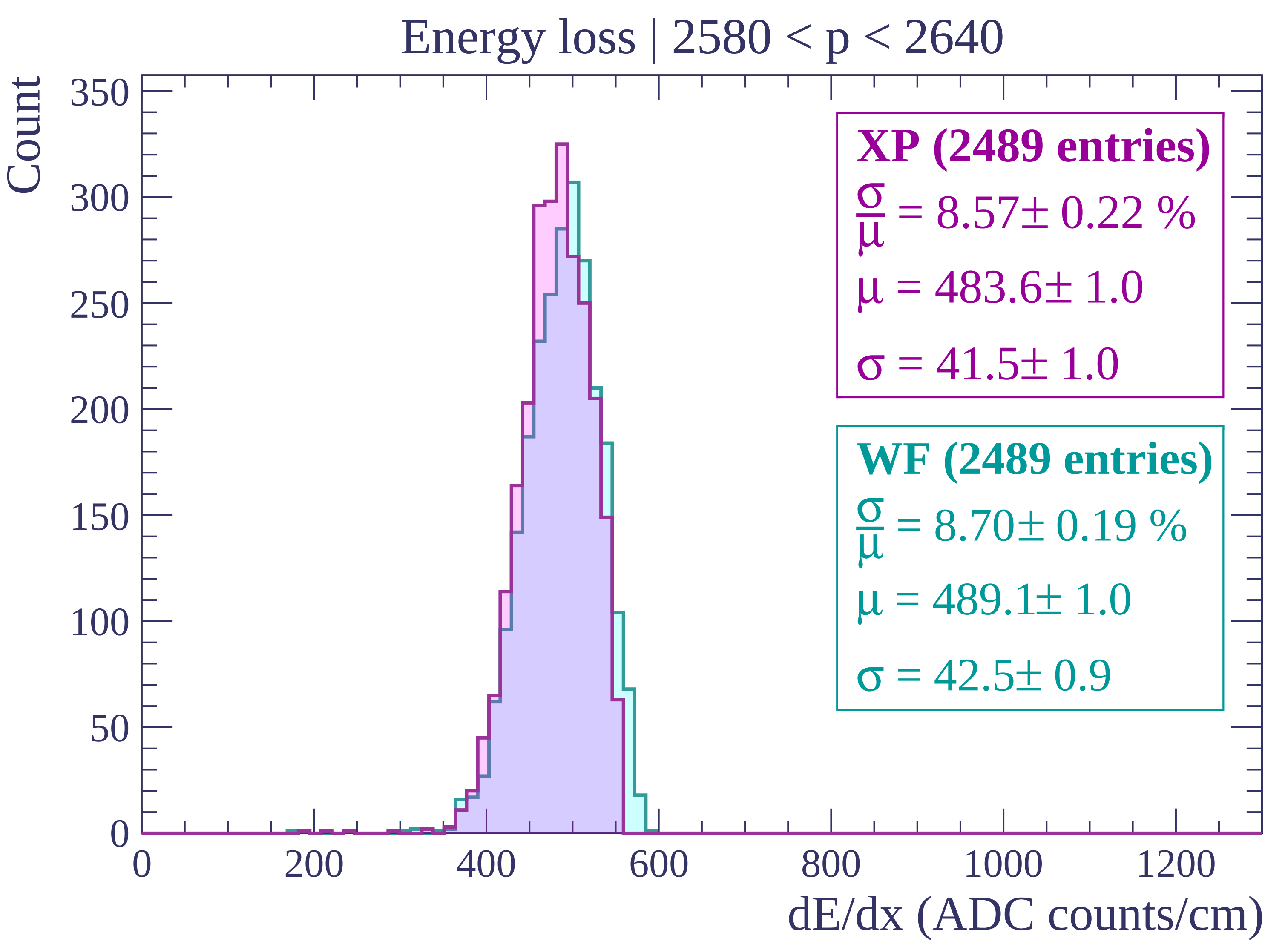


Energy loss | 2340 < p < 2400
Count
dE/dx (ADC counts/cm)
0
200
400
600
800
1000
1200
50
100
150
200
250
300
350
XP (2777 entries)
$\frac{\sigma}{\mu}$ = 7.94± 0.15 %
μ = 472.0± 0.8
σ = 37.5± 0.6
WF (2777 entries)
$\frac{\sigma}{\mu}$ = 8.90± 0.18 %
μ = 481.7± 0.9
σ = 42.9± 0.8

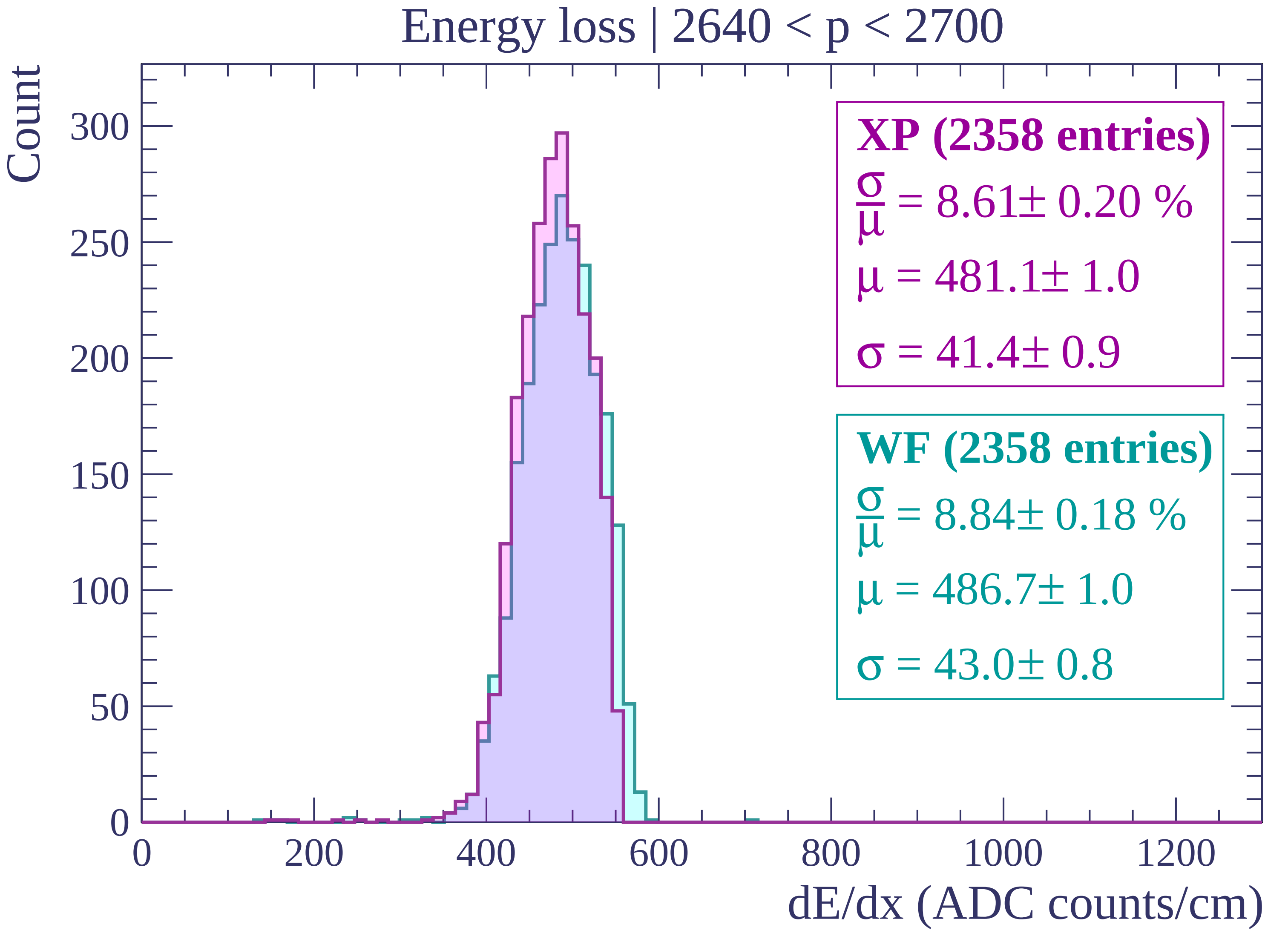

Energy loss | 2400 < p < 2460
Count
dE/dx (ADC counts/cm)
0
50
100
150
200
250
300
350
0
200
400
600
800
1000
1200
XP (2600 entries)
$\frac{\sigma}{\mu}$ = 8.02± 0.15 %
μ = 477.6± 0.8
σ = 38.3± 0.7
WF (2600 entries)
$\frac{\sigma}{\mu}$ = 8.78± 0.19 %
μ = 486.6± 1.0
σ = 42.7± 0.8


Energy loss | 2460 < p < 2520
Count
300
250
200
150
100
50
0
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP (2524 entries)
σ/μ = 8.72± 0.19 %
μ = 476.3± 0.9
σ = 41.5± 0.8
WF (2524 entries)
σ/μ = 9.19± 0.21 %
μ = 484.8± 1.0
σ = 44.5± 0.9

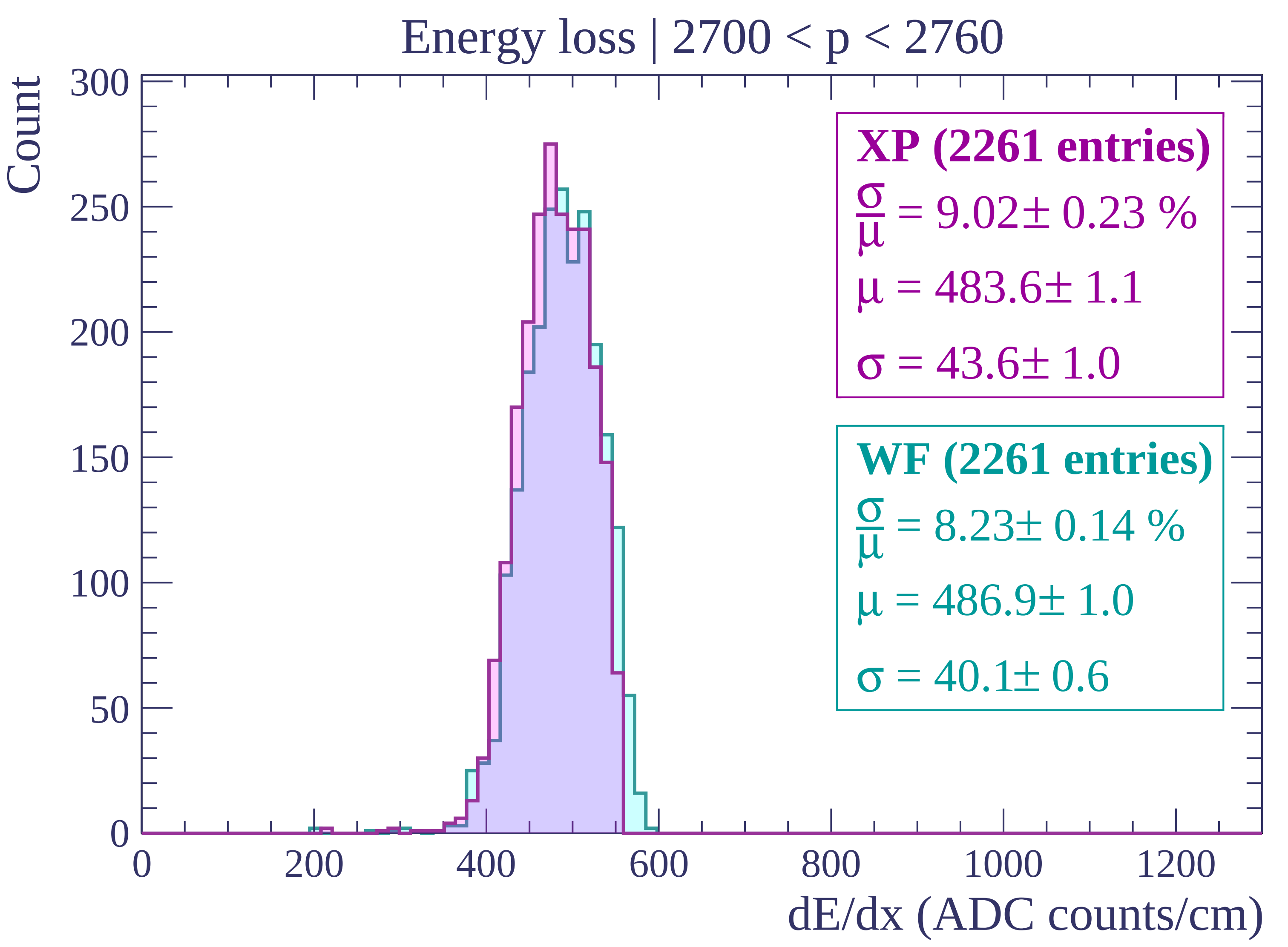

Energy loss | 2520 < p < 2580
Count
dE/dx (ADC counts/cm)
0
50
100
150
200
250
300
0
200
400
600
800
1000
1200
XP (2544 entries)
$\frac{\sigma}{\mu}$ = 8.86± 0.20 %
μ = 480.0± 1.0
σ = 42.5± 0.9
WF (2544 entries)
$\frac{\sigma}{\mu}$ = 8.90± 0.17 %
μ = 485.4± 0.9
σ = 43.2± 0.8



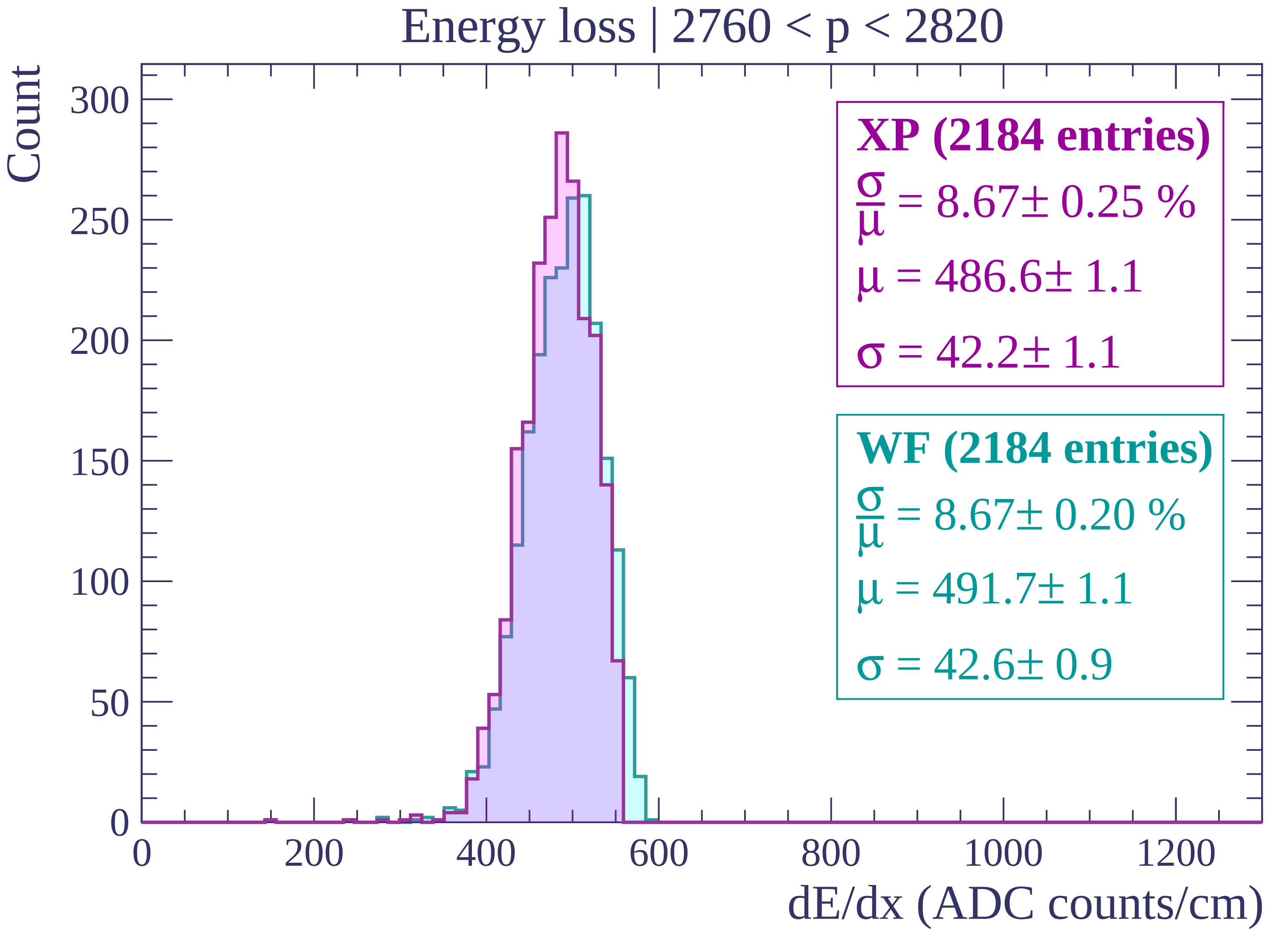
Energy loss | 2580 < p < 2640
Count
dE/dx (ADC counts/cm)
0
50
100
150
200
250
300
350
0
200
400
600
800
1000
1200
XP (2489 entries)
σ/μ = 8.57± 0.22 %
μ = 483.6± 1.0
σ = 41.5± 1.0
WF (2489 entries)
σ/μ = 8.70± 0.19 %
μ = 489.1± 1.0
σ = 42.5± 0.9

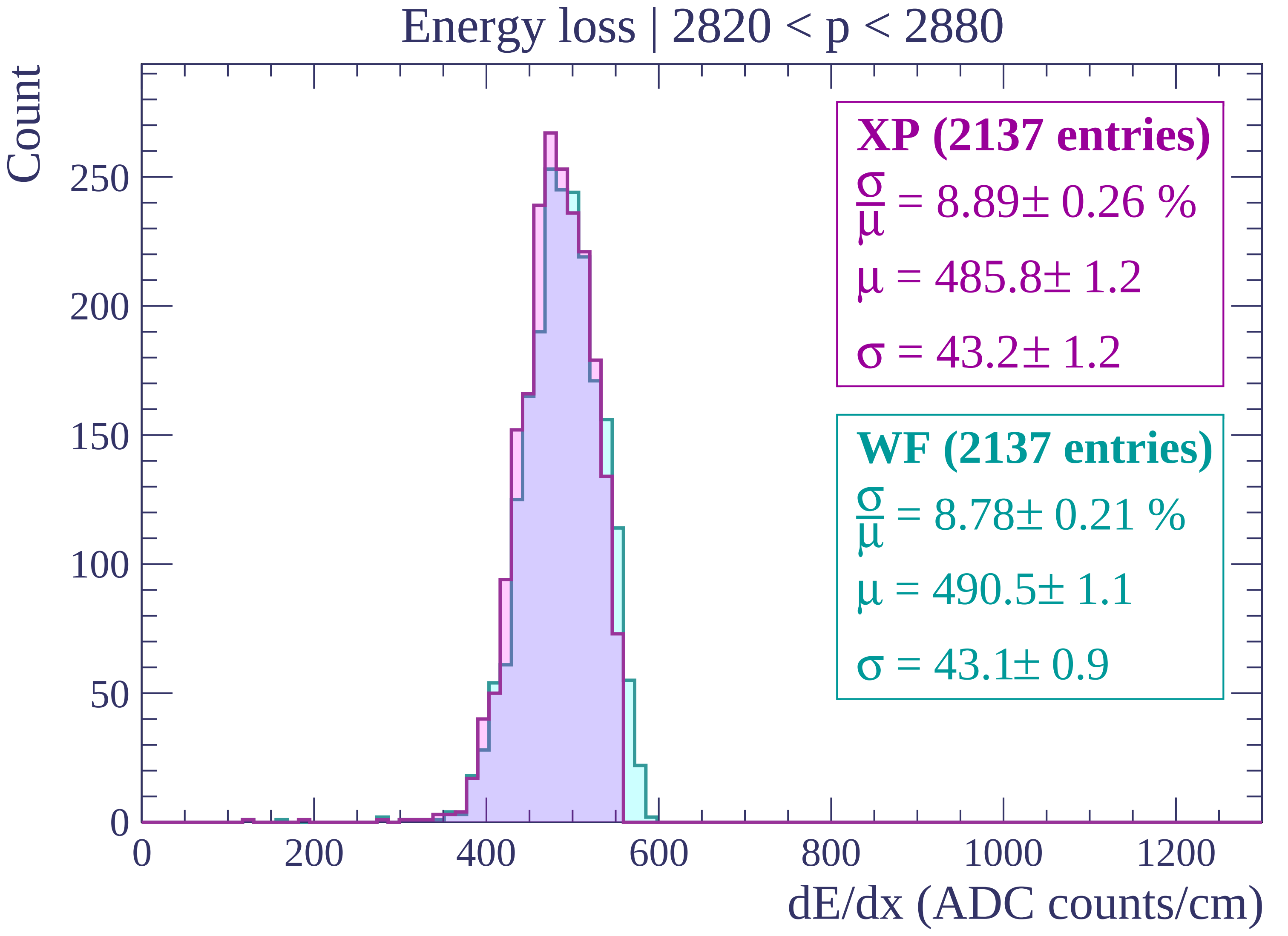


Energy loss | 2640 < p < 2700
Count
XP (2358 entries)
σ/μ = 8.61± 0.20 %
μ = 481.1± 1.0
σ = 41.4± 0.9
WF (2358 entries)
σ/μ = 8.84± 0.18 %
μ = 486.7± 1.0
σ = 43.0± 0.8
dE/dx (ADC counts/cm)
0
50
100
150
200
250
300
0
200
400
600
800
1000
1200

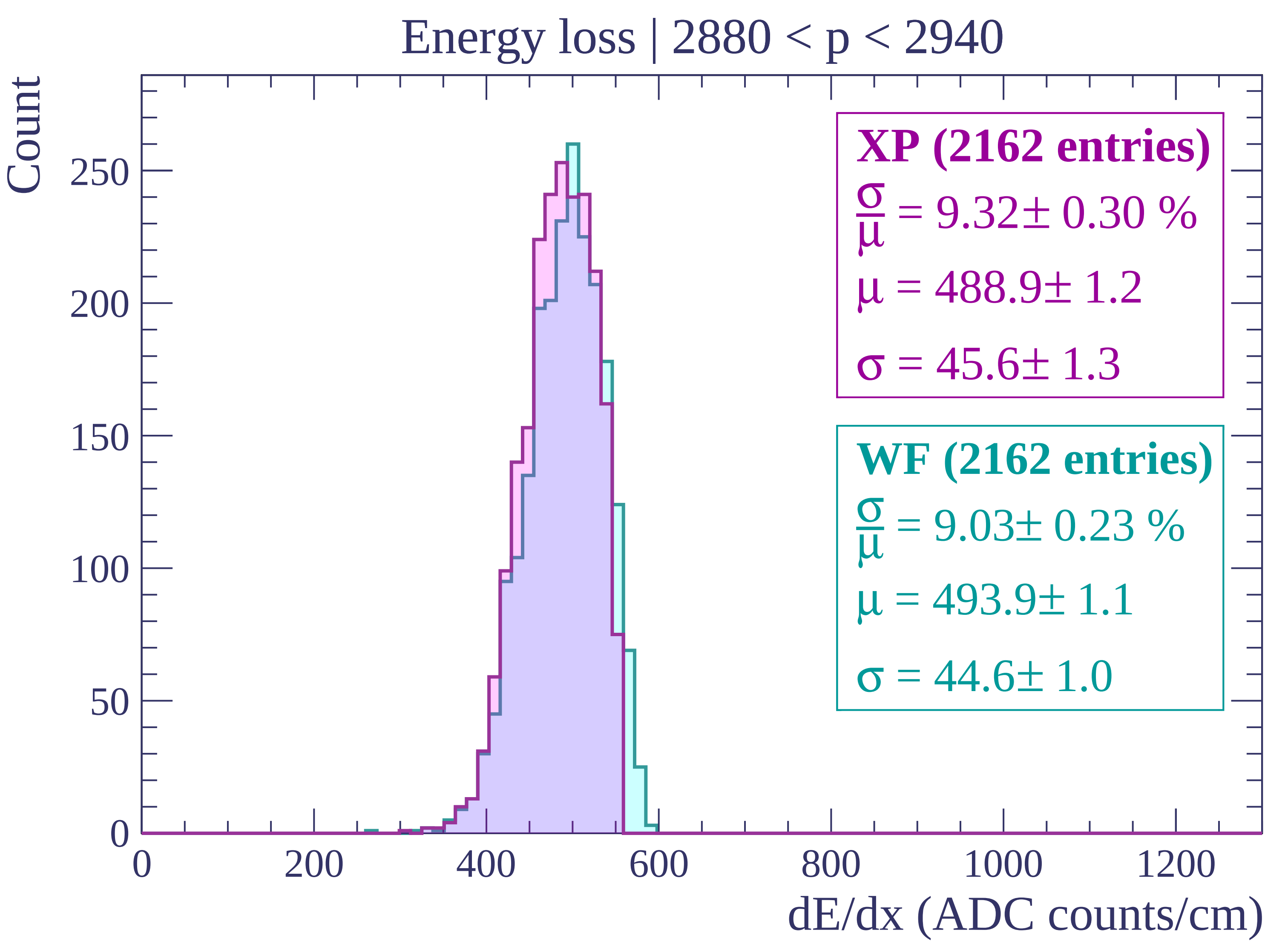


Energy loss | 2700 < p < 2760
Count
dE/dx (ADC counts/cm)
0
50
100
150
200
250
300
0
200
400
600
800
1000
1200
XP (2261 entries)
$\frac{\sigma}{\mu}$ = 9.02± 0.23 %
μ = 483.6± 1.1
σ = 43.6± 1.0
WF (2261 entries)
$\frac{\sigma}{\mu}$ = 8.23± 0.14 %
μ = 486.9± 1.0
σ = 40.1± 0.6

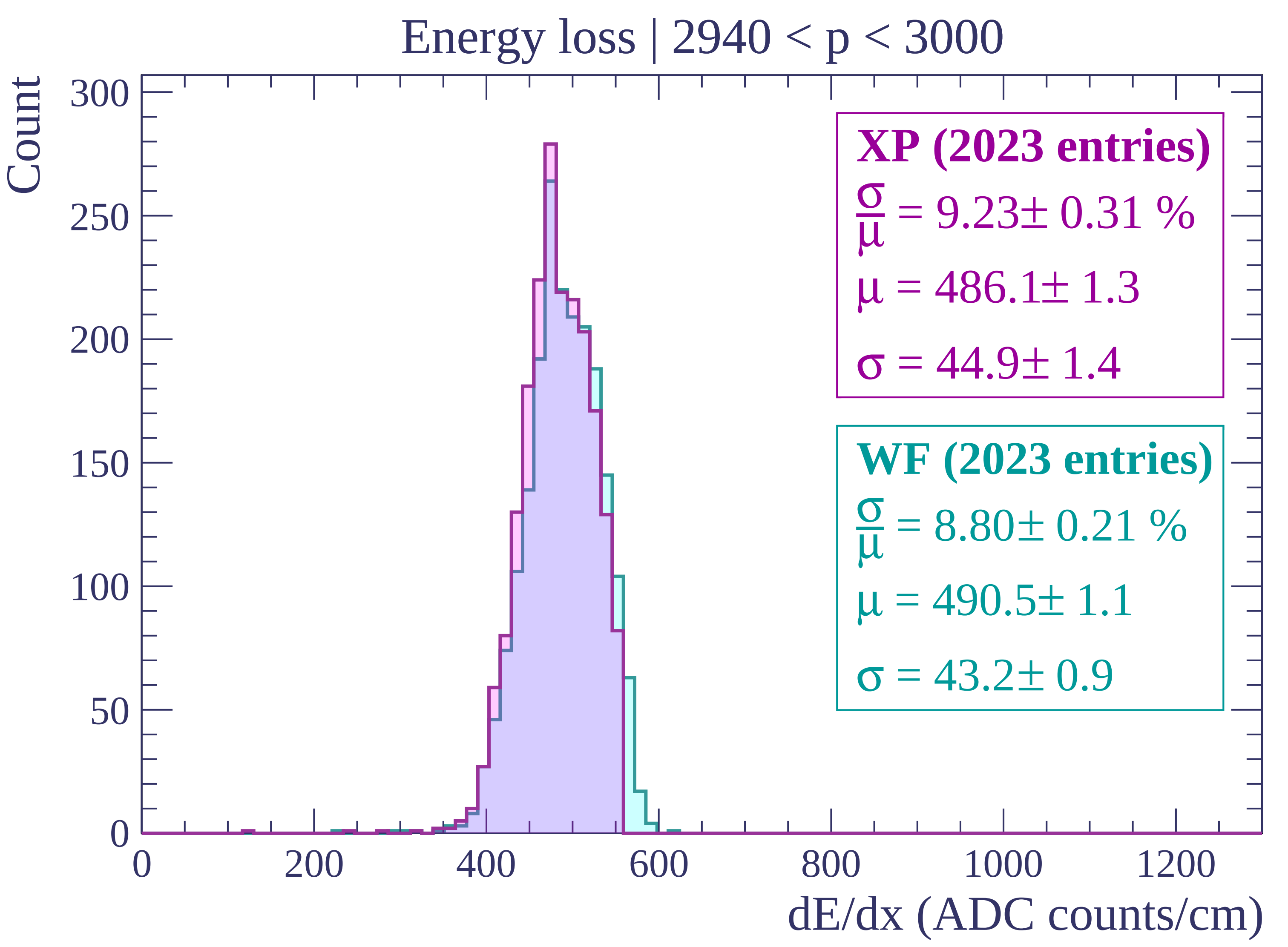

Energy loss | 2760 < p < 2820
Count
dE/dx (ADC counts/cm)
0
50
100
150
200
250
300
0
200
400
600
800
1000
1200
XP (2184 entries)
$\frac{\sigma}{\mu}$ = 8.67± 0.25 %
μ = 486.6± 1.1
σ = 42.2± 1.1
WF (2184 entries)
$\frac{\sigma}{\mu}$ = 8.67± 0.20 %
μ = 491.7± 1.1
σ = 42.6± 0.9

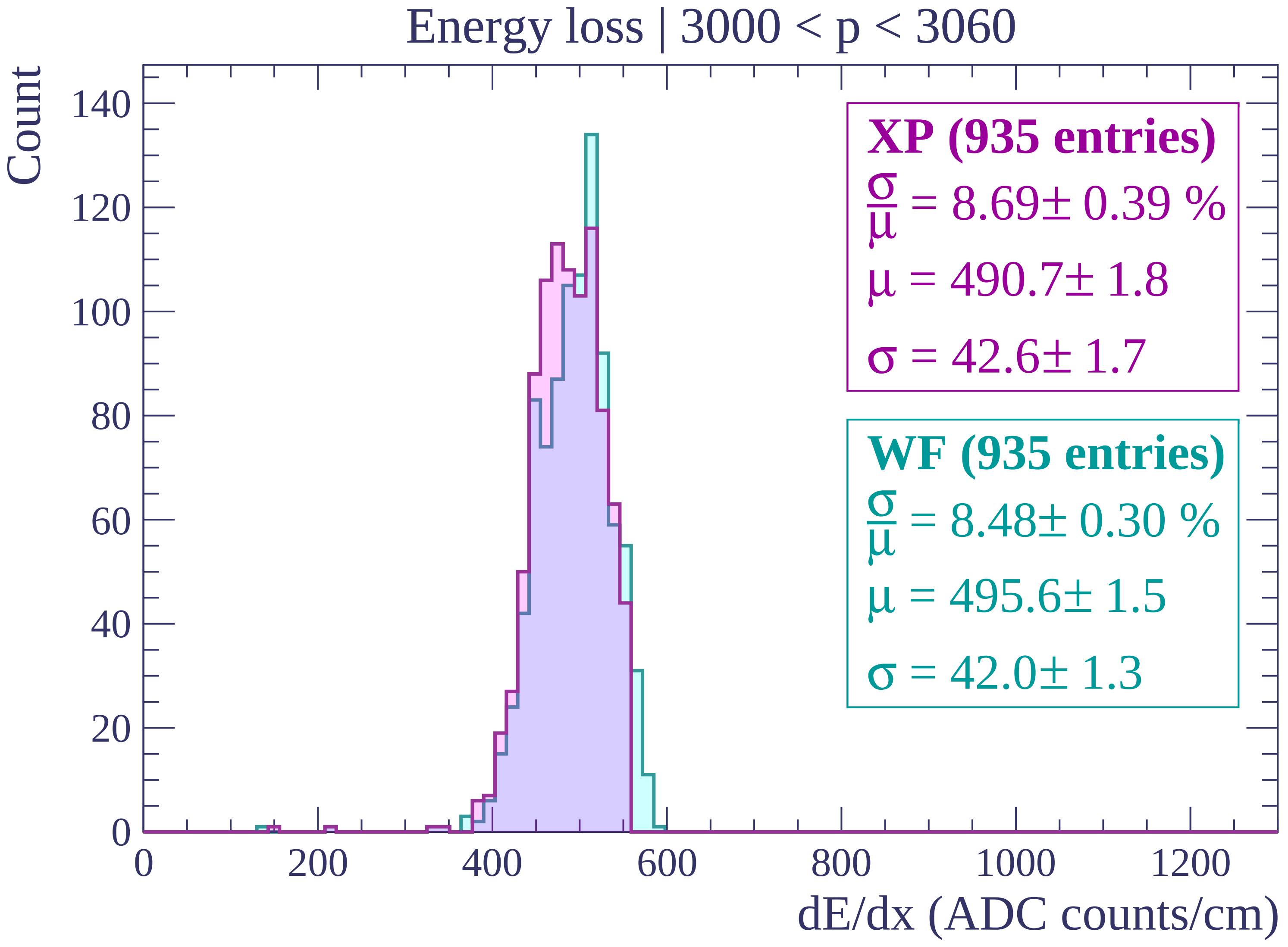


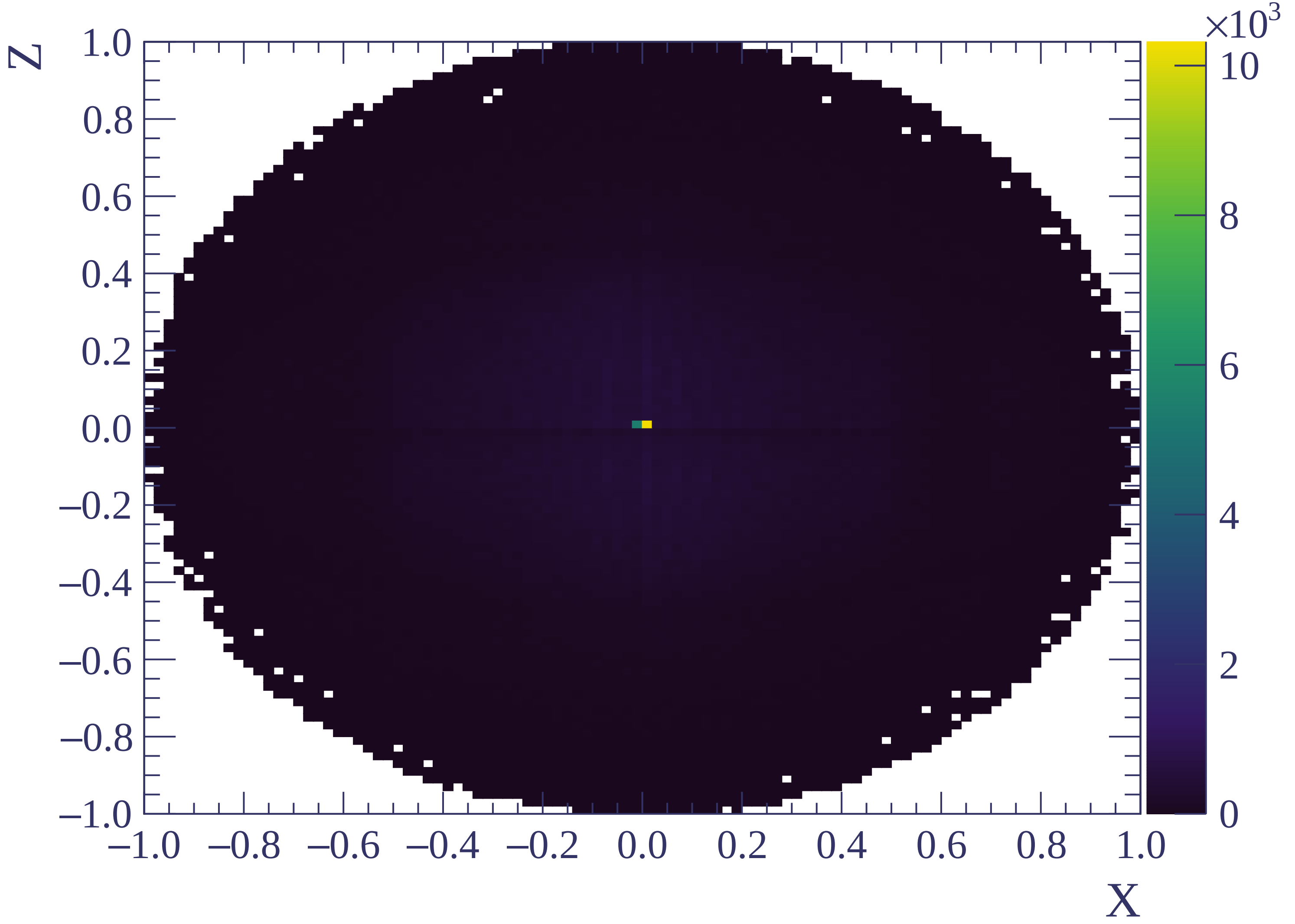
Energy loss | 2820 < p < 2880
Count
0
50
100
150
200
250
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP (2137 entries)
$\frac{\sigma}{\mu}$ = 8.89± 0.26 %
μ = 485.8± 1.2
σ = 43.2± 1.2
WF (2137 entries)
$\frac{\sigma}{\mu}$ = 8.78± 0.21 %
μ = 490.5± 1.1
σ = 43.1± 0.9

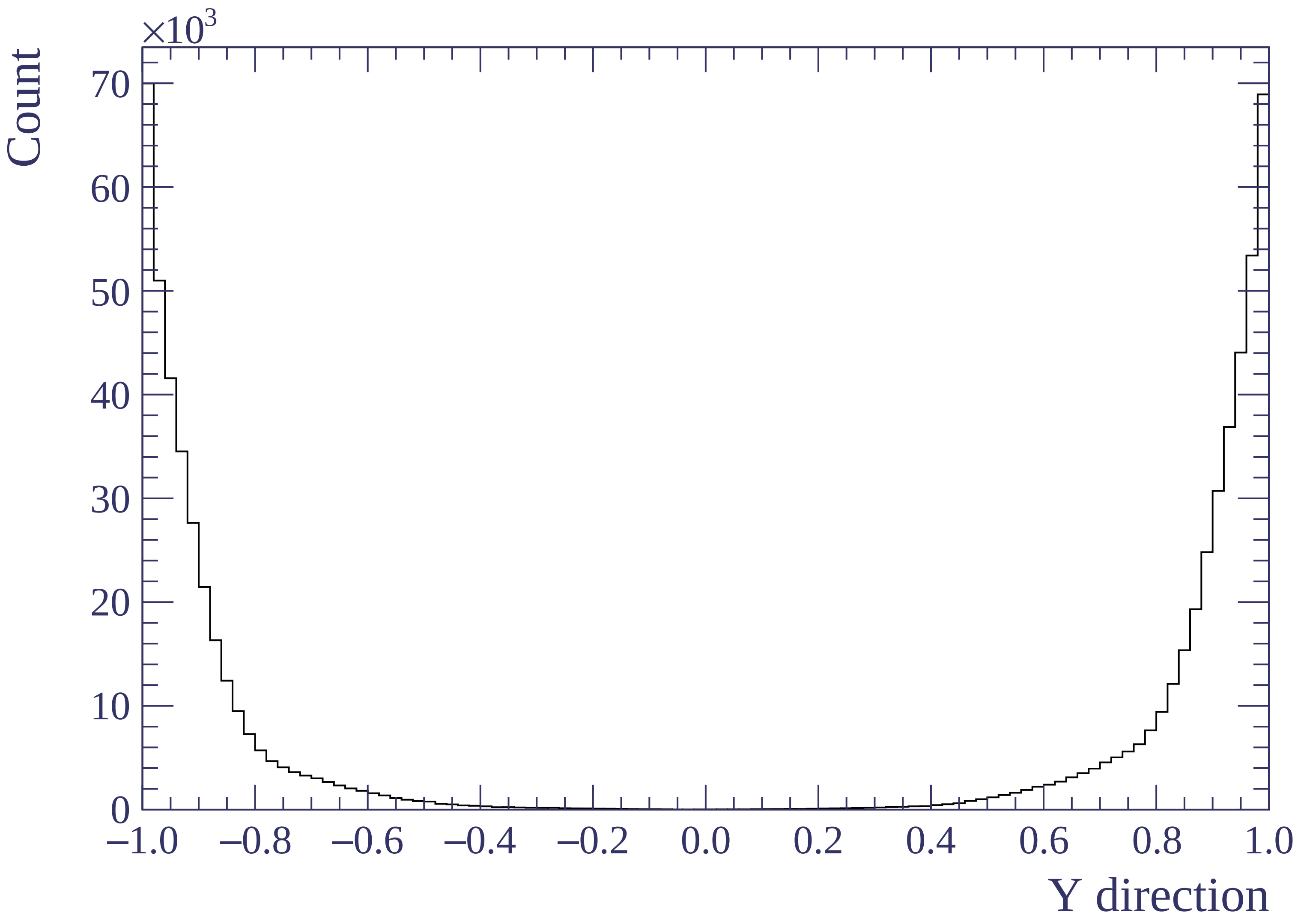


Energy loss | 2880 < p < 2940
Count
dE/dx (ADC counts/cm)
XP (2162 entries)
$\frac{\sigma}{\mu}$ = 9.32± 0.30 %
μ = 488.9± 1.2
σ = 45.6± 1.3
WF (2162 entries)
$\frac{\sigma}{\mu}$ = 9.03± 0.23 %
μ = 493.9± 1.1
σ = 44.6± 1.0
0
50
100
150
200
250
0
200
400
600
800
1000
1200

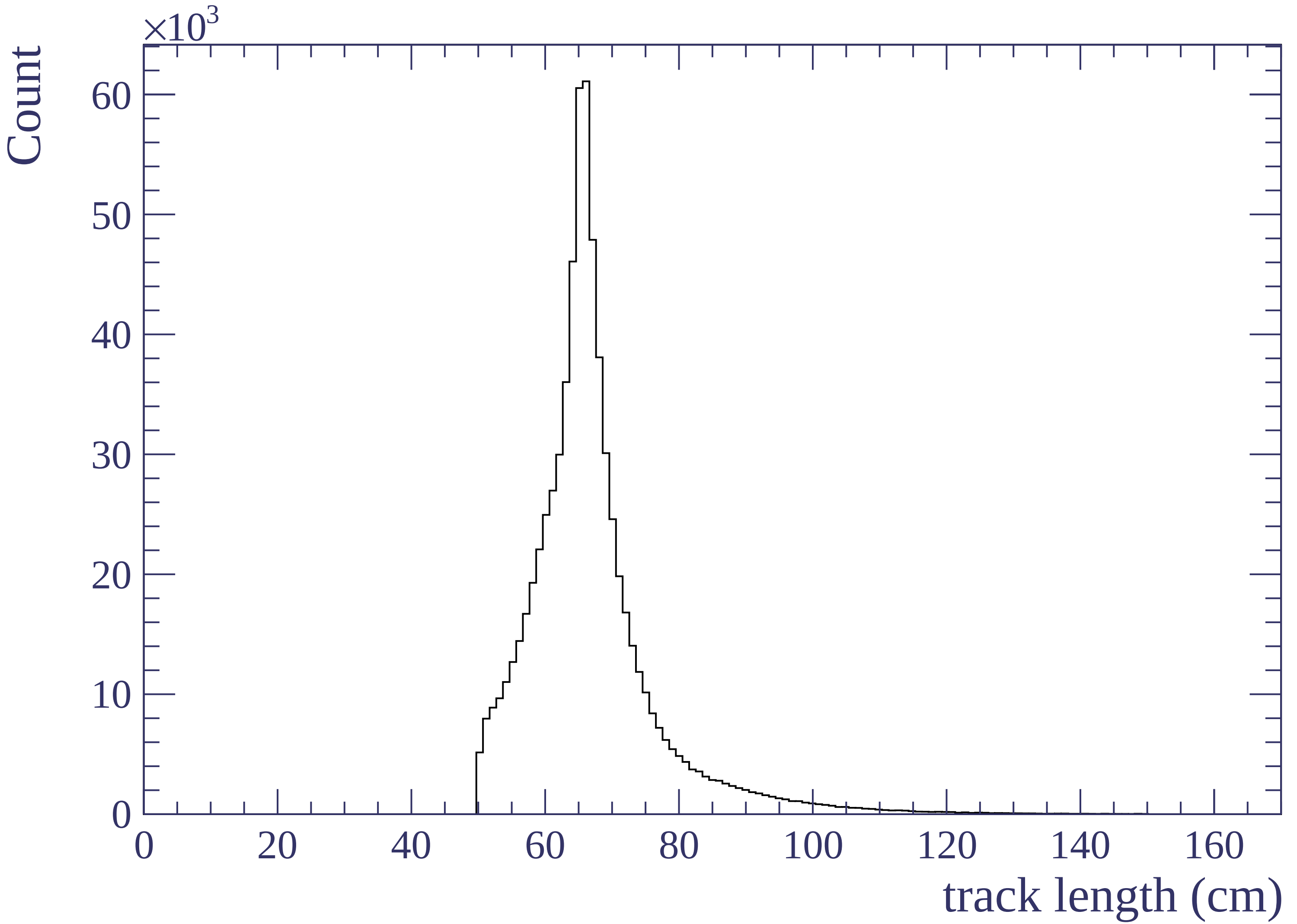

Energy loss | 2940 < p < 3000
Count
dE/dx (ADC counts/cm)
XP (2023 entries)
σ/μ = 9.23± 0.31 %
μ = 486.1± 1.3
σ = 44.9± 1.4
WF (2023 entries)
σ/μ = 8.80± 0.21 %
μ = 490.5± 1.1
σ = 43.2± 0.9
0
50
100
150
200
250
300
0
200
400
600
800
1000
1200

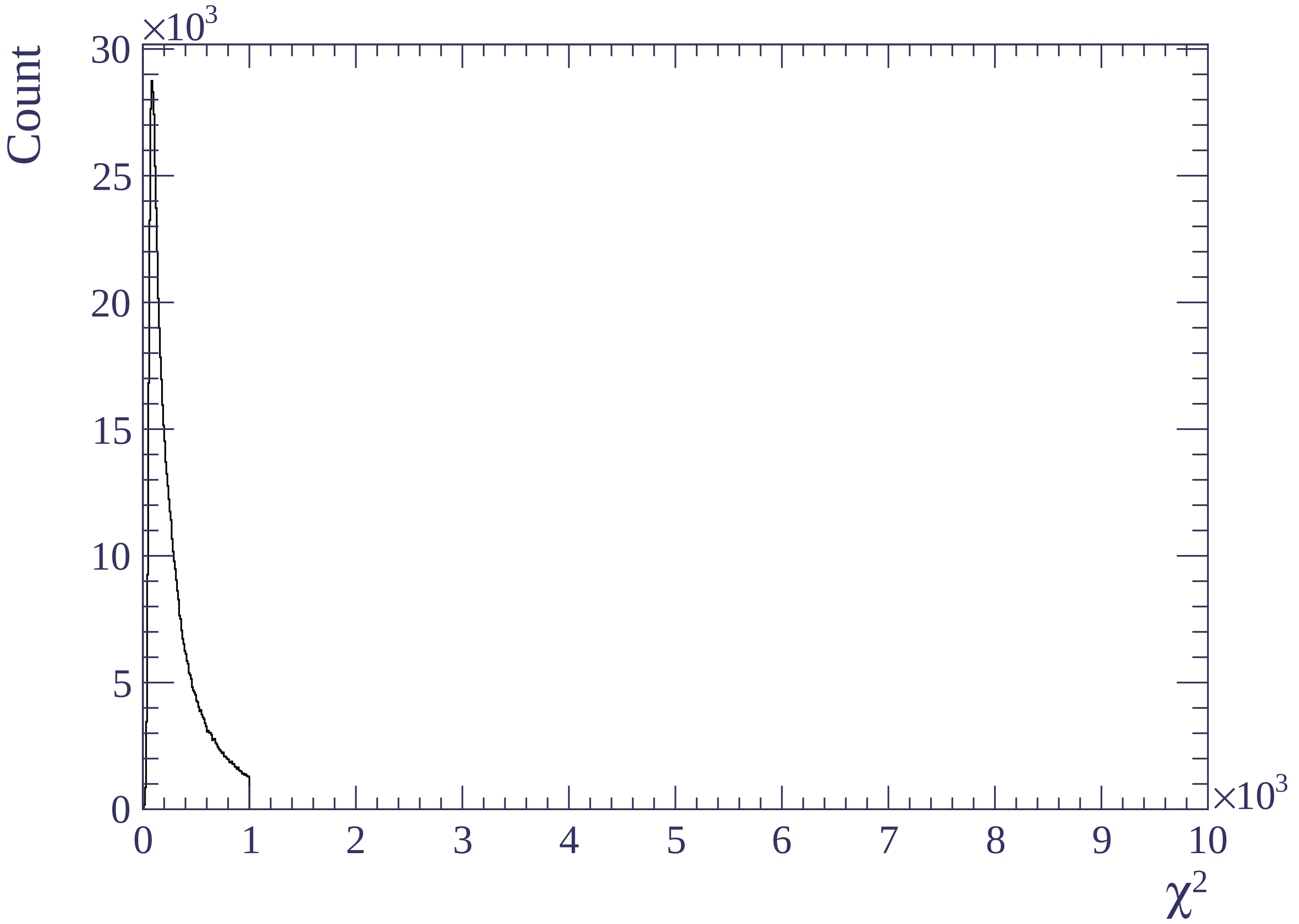

Energy loss | 3000 < p < 3060
Count
dE/dx (ADC counts/cm)
XP (935 entries)
σ/μ = 8.69± 0.39 %
μ = 490.7± 1.8
σ = 42.6± 1.7
WF (935 entries)
σ/μ = 8.48± 0.30 %
μ = 495.6± 1.5
σ = 42.0± 1.3

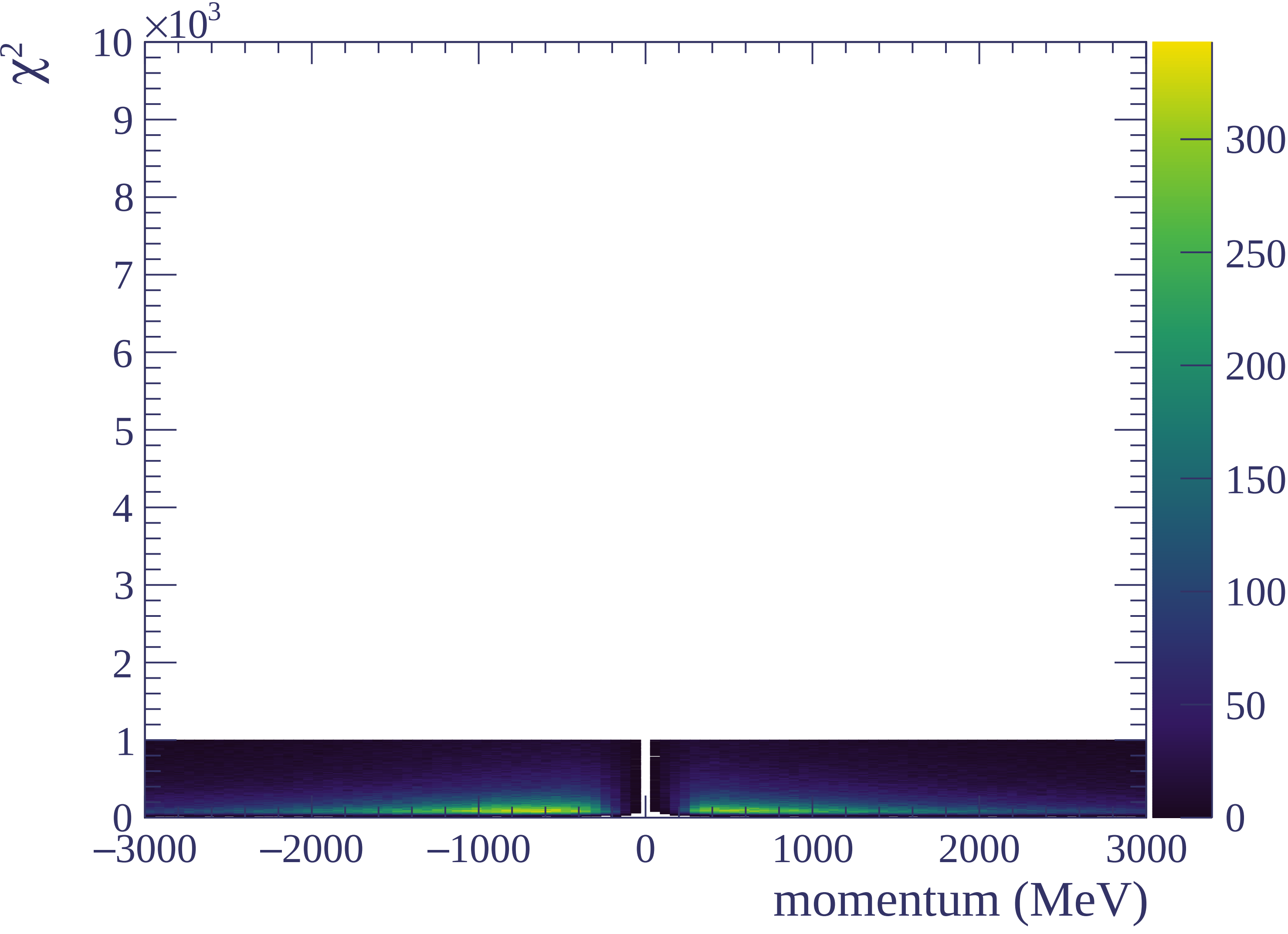

Z
1.0
0.8
0.6
0.4
0.2
0.0
−0.2
−0.4
−0.6
−0.8
−1.0
−1.0
−0.8
−0.6
−0.4
−0.2
0.0
0.2
0.4
0.6
0.8
1.0
X
×10^3
10
8
6
4
2
0



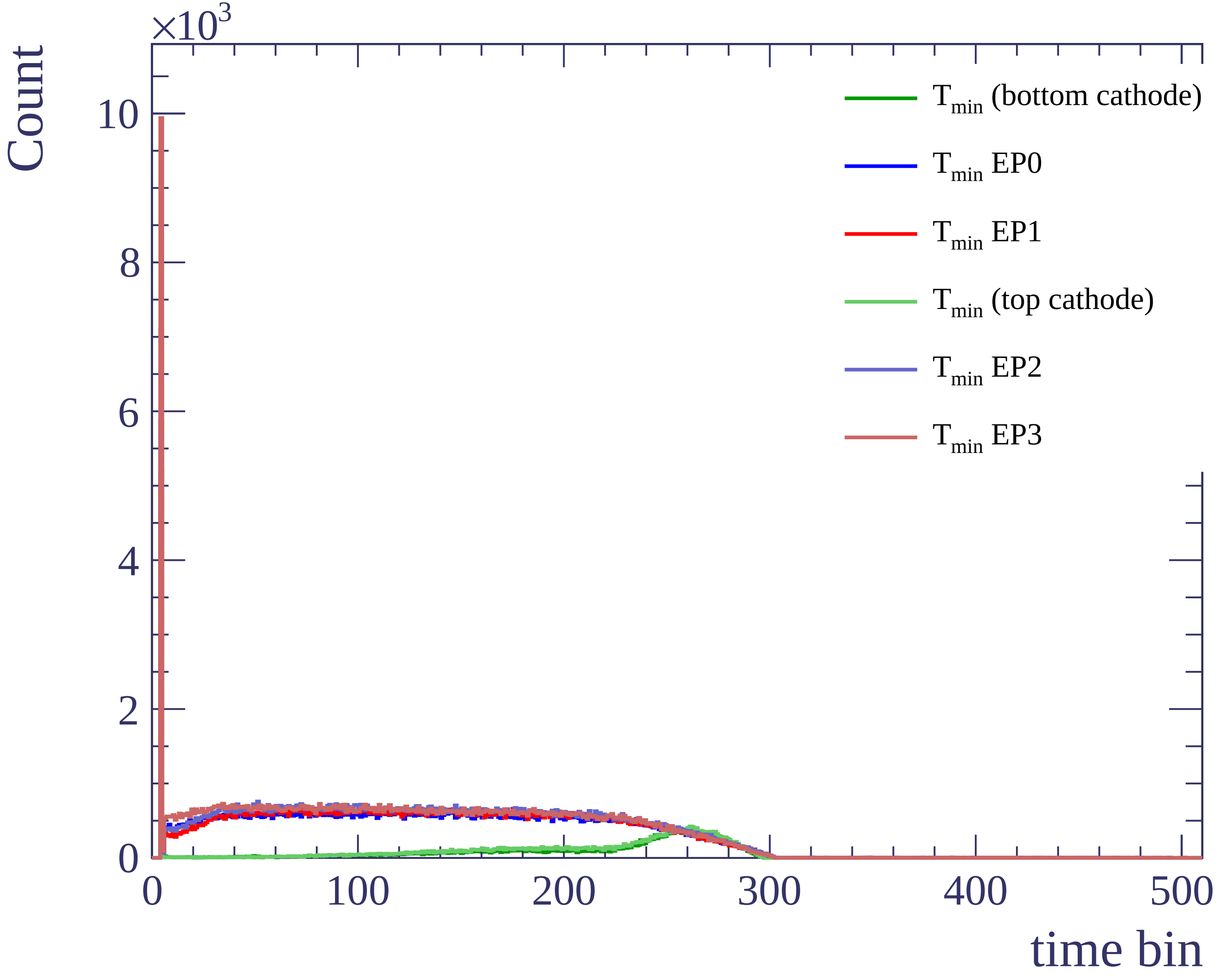
×10^3
Count
70
60
50
40
30
20
10
0
−1.0
−0.8
−0.6
−0.4
−0.2
0.0
0.2
0.4
0.6
0.8
1.0
Y direction


×10³
Count
0
10
20
30
40
50
60
0
20
40
60
80
100
120
140
160
track length (cm)

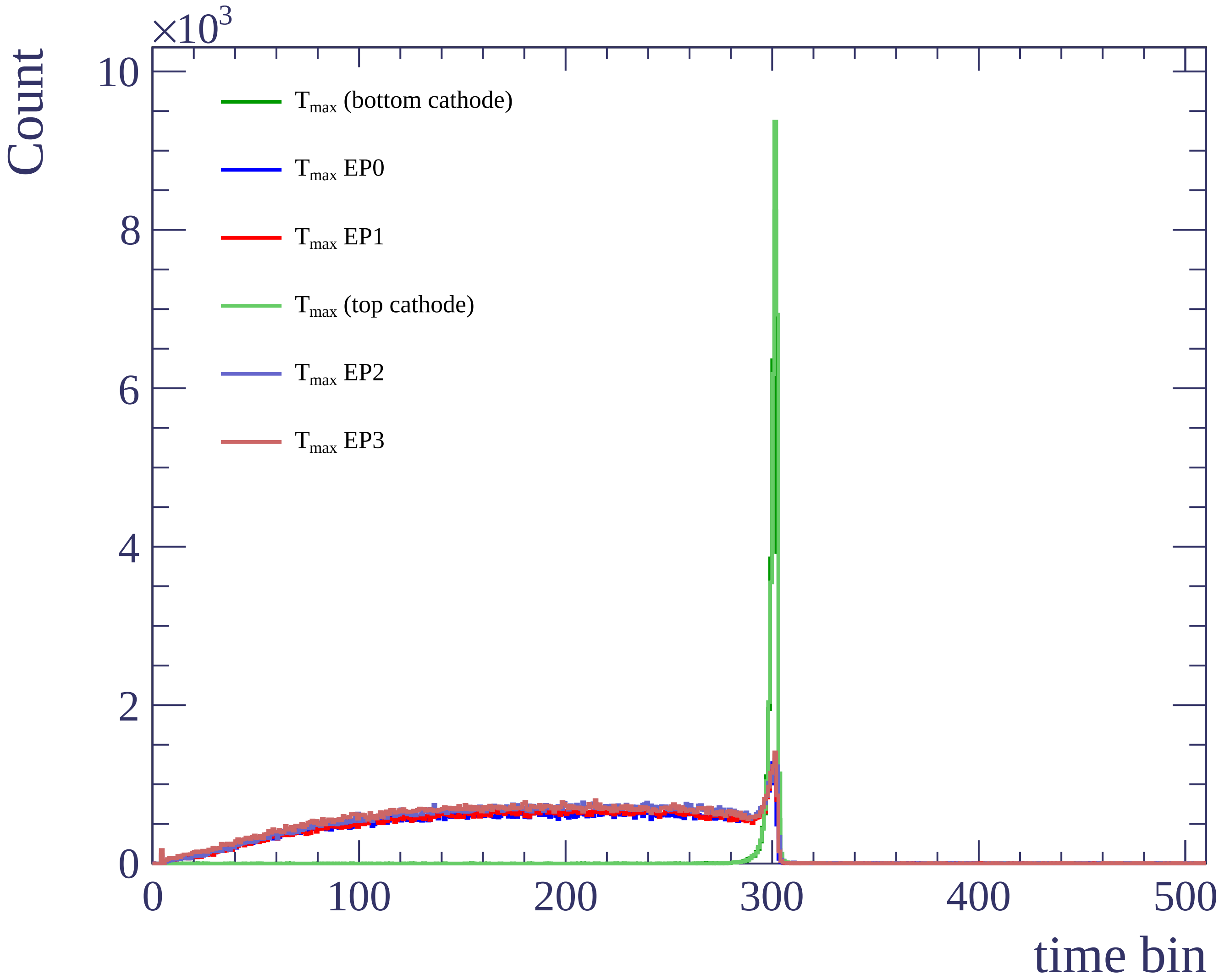

Count
×10^3
30
25
20
15
10
5
0
0
1
2
3
4
5
6
7
8
9
10
×10^3
χ^2



χ^2
$\times 10^3$
10
9
8
7
6
5
4
3
2
1
0
−3000
−2000
−1000
0
1000
2000
3000
momentum (MeV)
300
250
200
150
100
50
0



×10^3
Count
10
8
6
4
2
0
0
100
200
300
400
500
time bin
T_min (bottom cathode)
T_min EP0
T_min EP1
T_min (top cathode)
T_min EP2
T_min EP3



×10^3
Count
10
8
6
4
2
0
0
100
200
300
400
500
time bin
T_max (bottom cathode)
T_max EP0
T_max EP1
T_max (top cathode)
T_max EP2
T_max EP3